

Custom Compiler™ Design Checking and Physical Verification User Guide

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About This User Guide

The *Custom Compiler Design Checking and Physical Verification User Guide* describes how to use the available tools in the Custom Compiler environment to perform in-design rule checks and physical verification.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Synopsys® Custom Compiler™

Synopsys IC Validator™

Synopsys StarRC™

Synopsys SolvNetPlus support site

Mentor Graphics® Calibre®

Mentor Graphics Calibre DFM

Mentor Graphics Calibre nmDRC

Mentor Graphics Calibre Interactive™

Mentor Graphics Calibre LVS

Mentor Graphics Calibre PERC™

Mentor Graphics Calibre RVE™

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Introduction

In-design quality checking and physical verification ensure that your design meets various criteria for performance and reliability. The Custom Compiler environment offers comprehensive tools that you can use for these procedures.

This chapter describes the following:

- [In-Design Quality Checking and Physical Verification](#)
- [Tools to Use for In-Design Quality Checks](#)
- [Products to Use for Physical Verification](#)
- [PVE Toolbar](#)

In-Design Quality Checking and Physical Verification

In-design quality checking and physical verification are processes you use to make sure the designs are correctly constructed to meet manufacturing requirements. You can perform the checks iteratively at various stages of the design process.

Tools to Use for In-Design Quality Checks

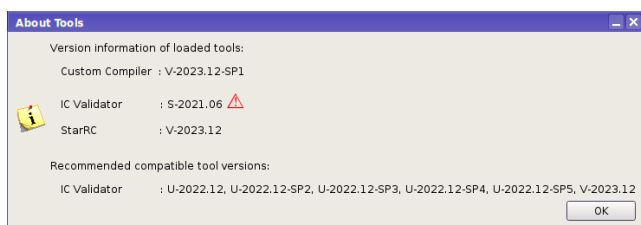
In-design quality checks let you look for missing vias or non-compliant issues as you edit the design. The following table lists the in-design quality checks you can perform and the corresponding tools you can use.

Quality Check	Tool	Description
Design rules and constraints	DRD	Applies rules and constraints and interactively checks for violations in the entire design or in the viewport of your design. For more information, see Checking for Violations as You Edit (DRD) and Introduction .

Quality Check	Tool	Description
Rule violations	ICV Live DRC	Incrementally checks for violations as you create and edit the design. For more information, see Checking for Violations as You Edit (ICV Live DRC) .
Missing or suboptimal vias	Via Checker	Checks for missing or suboptimal vias in your design. For more information, see Checking Vias .
Metal density rules and constraints	Metal Density Analyzer	Checks the metal density of your design by layer or by groups that you define. For more information, see Checking Density .
Color decomposition	Color Decomposition Setup and Run	Automates the process in which colorable shapes across the hierarchy are assigned colors for multiple patterning. For more information, see Automating Color Decomposition .

To know the current version of the selected tool, click **Verification > About Tools**.

If the selected PVE tool is of a previous version, a warning message appears and lists the versions that are compatible with the Custom Compiler tool as shown in the following example:



Products to Use for Physical Verification

Physical verification consists of design rule checks, layout-versus-schematic comparison, and parasitic extraction. You typically perform physical verification on completed designs.

The following table lists the physical verification processes and the corresponding products you can use. Although they are other Synopsys or third-party products, you can access them from within the Layout Editor.

Physical Verification Process	Tool	Description
Design rules and constraints	Mentor Graphics Calibre Hercules IC Validator	Inspects the design for rule violations to ensure layout fabrication compliance. For more information, see Design Rule Checking Using Integrated Tools and DRC and LVS Using the Calibre Interactive Tool .
Layout-versus-schematic comparison	Mentor Graphics Calibre Hercules IC Validator	Verifies that the layout accurately represents the electronic equivalent of the schematic. For more information, see Comparing Layout Versus Schematic (LVS) and DRC and LVS Using the Calibre Interactive Tool .
Parasitic extraction	StarRC	Inspects and extracts unintended resistance, capacitance, and inductance. For more information, see Extracting Interconnect Parasitics .

Click **About Tools** option under the **Verification** menu to know the current version of the selected tool.

Physical verification also includes electrical checks. For information about checking and reporting electromigration, resistance, capacitance, voltage drops, and shield coverage, see In-Design Electrical Checking and Reporting.

PVE Toolbar

The **PVE** toolbar contains the default DRC, LVS, and RCX icons. You can add icons for custom jobs and control icons and tooltips from a `pdk.xml` file.



The **PVE** Toolbar contains these options:

Option	Description
	Runs IC Validator DRC job.
	Runs IC Validator LVS job.
	Runs StarRC LPE job.
	Runs FILL job.

2

Checking for Violations as You Edit (DRD)

Instead of waiting until the design is finished, you can use the Design Rule Driven (DRD) tool to interactively check for violations as you create and edit the design.

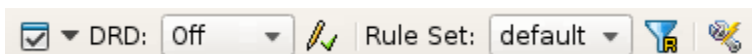
This chapter contains the following topics:

- [About the DRD Tool](#)
- [Setting DRD Options](#)
- [DRD Toolbar](#)
- [AutoFix Toolbar](#)
- [Checking for Violations in the Viewport](#)
- [Assigning Colors to Shapes in the Viewport](#)
- [Fixing Violations With Guidance](#)
- [Fixing Violations in a Region of the Top-Level Design](#)
- [Fixing Violations in the Top-Level Design](#)
- [Displaying Violation Markers After Running DRD AutoFix](#)
- [Fixing Violations Across the Design Hierarchy](#)

About the DRD Tool

The Design Rule Driven (DRD) tool in the Layout Editor applies rules and constraints and quickly checks for violations as you edit your design. The DRD tool checks for fabrication rules, such as width, spacing, enclosure, and other requirements. You also have the option of selecting the constraints and an external rule file.

You can access the DRD toolbar in the Layout Editor by choosing **Window > Toolbars > DRD**.



On the **DRD** toolbar, you can choose one of the following modes: **Off**, **Visual**, and **Assist**.

When **Visual** or **Assist** is enabled, the tool:

- Displays a dynamic ruler on the canvas as you interactively edit layout objects, based on the clearance constraints between layers.
- Displays a DRD marker when the action actually results in a minimum rule or recommended rule violation.

For the description of the DRD toolbar options, see [DRD Toolbar](#).

Setting DRD Options

Before you use the DRD tool, you should set the required DRD preferences. The **DRD Options** dialog box lets you manage DRD related preferences.

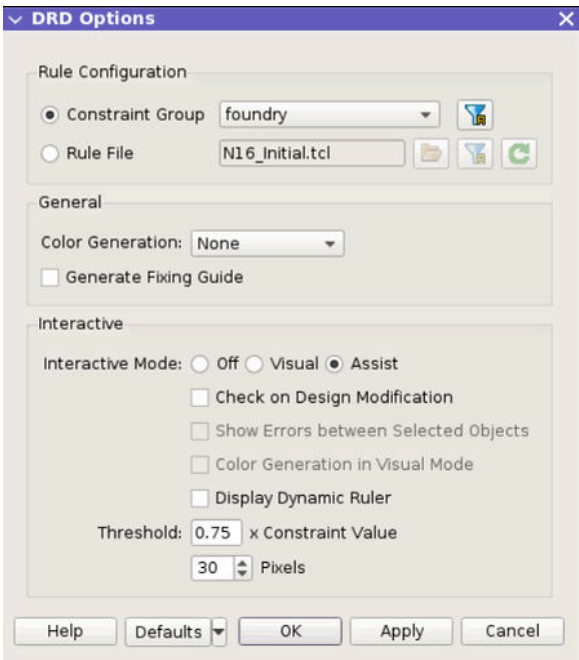
This section contains the following topics:

- [Setting Up for DRC Checks](#)
- [Selecting Constraints in the Technology Library for DRD](#)
- [Selecting DRD Rule Deck Files](#)

Setting Up for DRC Checks

To display the **DRD Options** dialog box:

- In the Layout Editor, choose **Options > DRD**. The **DRD Options** dialog box opens.

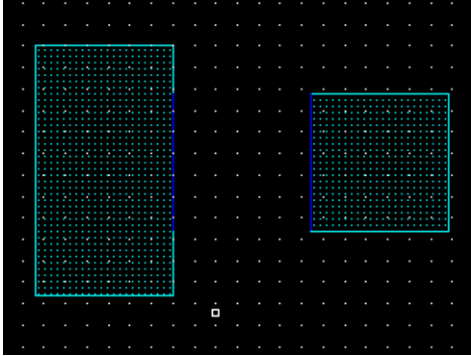


The **DRD Options** dialog box contains the following options:

Option	Description
<i>Rule Configuration</i>	Specifies the rule source.
Constraint Group	The process-level constraints (defined in the technology database) to apply. The default constraint group is foundry . For details, see Selecting Constraints in the Technology Library for DRD .
Rule File	The rule deck file (in Tcl format) that contains DRC rules to apply. If nothing is selected, the command <code>unselectRuleByLayer</code> or <code>unselectRule</code> in the rule deck file is followed. If these commands do not exist in the rule deck file, all rules are checked by default. For details, see Selecting DRD Rule Deck Files .

General

Option	Description
Color Generation	The option to automatically assign colors to colorable shapes for multiple patterning. Select one of the following: ◦ None: Does not assign colors to the colorable shapes in the viewport. ◦ Top Level Only: Assigns colors only to top-level colorable shapes in the viewport. ◦ Hierarchy: Creates a new cell, Color, that contains the newly colored shapes. During color propagation, the colors from these shapes will propagate upward to the 0x shapes. Be aware that editing or deleting 0x shapes automatically deletes the color instance. For more information, see Assigning Colors to Shapes in the Viewport.
Generate Fixing Guide	Generates fixing guides on violations after running window checking. For more information, see Fixing Violations With Guidance. Tcl preferences: <code>leDRDGenerateFixingGuide</code> and <code>leDRDGenerateFixingGuideInInstance</code>
<i>Interactive</i>	Controls whether interactive design rule checking is enabled and how rule violations are displayed when you create or edit objects.

Option	Description
Interactive Mode	- Select one of the following to specify the interactive mode for design rule checking: Off: Disables design rule checking and ignores design violations. Visual: Enables design rule checking on active objects. A X displays to let you know that you are creating a violation. Dynamic rulers display the current distance between two shapes. Dynamic markers also show the violated design rules. Design rule checking only occurs on visible objects. The checking will stop if it exceeds 90ms or if the mouse moves. In these instances, you might see only a partially checked result. Assist: Enables design rule checking on active objects. The guide lines display to indicate a blockage area to let you know that you are reaching the minimum spacing. The predefined color for the guide line is blue and the color packet for the guide line is makerAno.  Create Interconnect: Specifies the Truncate, Extend, or Variable styles in Assist mode. Only moving edges that are perpendicular to the center line of the created path are supported. If the center line of the created path is not orthogonal, Assist mode is inactive. Assist mode works only with orthogonal, X-first, and Y-first angles.
Check on Design Modification	When enabled, checks design violations when a design is modified. When disabled (default), the tool does not check for design violations when a design is modified. Window checking is always available, even if the DRD option is set to Off on the DRD toolbar. Tcl preference: <code>leDRDCheckOnModification</code>
Show Errors between Selected Objects	Shows violations between active objects, which are the objects you create or edit. Tcl preference: <code>leDRDShowErrorInSelectedObject</code>

Option	Description
Color Generation in Visual Mode	Shows the color generation result in visual mode. This option is enabled when specifying Interactive Mode to Visual and Color Generation to Top Level Only .
Display Dynamic Ruler	Shows the relevant distance with space ruler when active shapes are approaching the environment shapes. The dynamic rulers display only the distance which is larger than the minimum spacing. Tcl preference: <code>leDRDDynamicRulerThreshold</code> . Specifies the maximum multiples of constraint value, and defines the scope that displays the dynamic rules in dynamic mode. When the preference value is less than or equal to 2, it maintains the original usage of 2x the constraint value.
Threshold	Specifies the threshold in the blockage area in Assist mode. After the mouse exceeds the threshold point, you can edit the objects as editing in Visual mode. This option is enabled when specifying Interactive Mode to Assist . Use the ratio Constraint Value or number of Pixels to represent the threshold value.
Defaults	Click this button to restore all rule source settings to default preference settings.

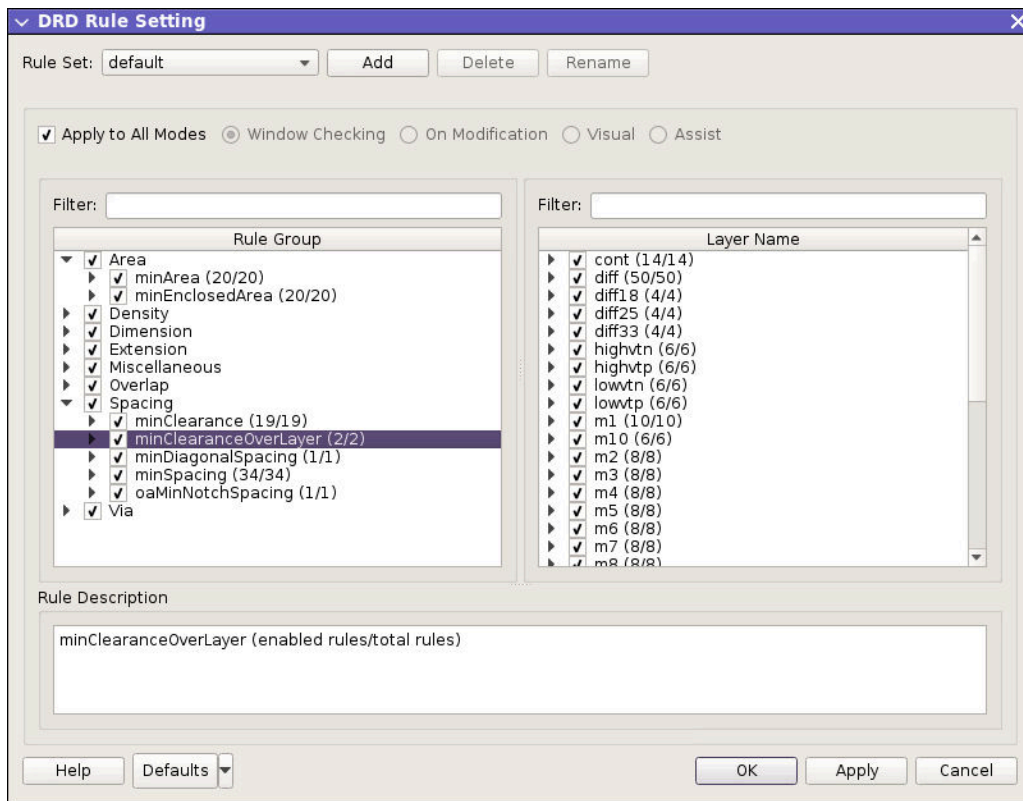
Selecting Constraints in the Technology Library for DRD

You can specify that the DRD tool uses selected OpenAccess process constraints defined in the technology library when checking the design in the viewport. Window checking that uses process constraints supports the following OpenAccess objects:

- Rectangles
- Polygons
- Paths
- Routes that contain path segments and vias
- Path segments and vias that are outside of routes
- Scalar and array instances
- Blockages and boundaries (if rules are associated with them)
- MCell guard rings
- Multipart Path (MPP) objects

- Object types supported by foundry rules:
 - Objects that are currently not visible on the canvas because their **Visibility** options are turned off in the Object/Layer Panel
 - Objects within the hierarchy level specified by the **Start Level** (`leStartLevel`) and **Stop Level** (`leStopLevel`) boxes on the **Hierarchy** toolbar

To select specific constraints within the constraint group, click . The **DRD Rule Setting** dialog box opens.



The **Rule Set** list contains both the default rule set and the rule sets you defined.

The selected rule set is the active rule set used for DRD checking. This is controlled by the `leLiveDRDCheckRuleSet` Tcl preference. This selection file corresponds to the **Rule Set** selection field on the **DRD** toolbar. For details, see [DRD Toolbar](#).

You use the following buttons to define multiple rule sets with selected rules for different checking purposes and modify the user-defined rule sets as needed:

- **Add:** Adds a user-defined rule set through the **Add Rule Set** dialog.



All the user-defined rule sets correspond to a single rule source. They are controlled by the `leLiveDRDRuleSets Tcl` preference.

If the associated rule source is:

- From a foundry constraint group, the preference value can be:

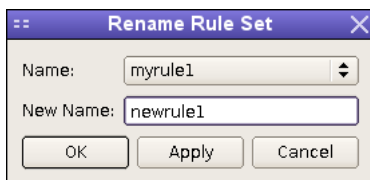
```
{constraintGroup foundry {myRuleSet1 checkRules m1.s.1.1R  
m1.w.1 ...} {myRuleSet2 skipRules m2.w.1 m2.w.2 m2.a.1 ...} ... }
```
- From a rule file (for example: `/home/tmp/ruleFileA.tcl`), the preference value can be:

```
{ruleFile /home/tmp/ruleFileA.tcl {myRuleSet1 checkRules M1.S.1  
M2.S.1} {myRuleSet2 skipRules M1.G0.COLOR M2.G0.COLOR} ... }
```

- **Delete:** Removes the selected user-defined rule set through the **Delete Rule Set** dialog.



- **Rename:** Changes the name of the selected rule set through the **Rename Rule Set** dialog.



You can set the rule configuration by selecting any of the following options:

- **Apply to All Modes:** Applies rule set to all interactive DRD checking modes. When checked, **Window Checking** is automatically selected and the **On Modification** and **Visual** options cannot be selected. Checked by default when the rule source is from the rule file.

- **Window Checking:** Checks the target objects according to the current active rule set and marks violations with a red X.
- **On Modification:** Checks design violations that are related to active layers only. Active layers are those that are relevant to the layers that are being modified. For details, see [DRD Toolbar](#).
- **Visual:** Checks design rules dynamically. As you edit the design, a red X displays to let you know that you are creating a violation. Dynamic rulers display the current distance between two shapes. Dynamic markers also show the violated design rules. Design rule checking only occurs on visible objects. The checking will stop if it exceeds 90ms or if the mouse moves. In these instances, you might see only a partially checked result.
- **Assist:** Checks design rules dynamically. As you edit the design, a red X displays to let you know that you are creating a violation. The guide lines display to indicate a blockage area to let you know that you are reaching the minimum spacing.

You can select the desired constraints from one of the following lists:

- **Rule Name:** Lists all rules defined in the selected constraint group. By default, all rules are selected. Each rule can be deselected or selected separately.

You can access the following actions by right clicking the **Rule Name** list heading:

- **Enable All Rules:** Enables all of the visible rules in the **Rule Name** list to get checked by interactive DRD. Does not enable rules that have been filtered out.
- **Disable All Rules:** Disables all of the visible rules in the **Rule Name** list to get checked by interactive DRD. Does not disable rules that have been filtered out.
- **Sort by Name:** Sorts the rules by their name.
- **Sort by File:** Sorts the rules by their original order in the rule file.

This menu is accessible only when the rule source is from the rule file.

- **Rule Group:** Lists all predefined rule groups by constraint type. Rules that belong to the same constraint type belong in the same group. By default, all rule groups are selected. This list appears when the rule source is from OpenAccess constraints.
- **Layer Name:** Lists all layers. Each layer is associated with a set of rules. Layers that do not have associated rules do not appear in the list. By default, all layers are selected.

Selecting DRD Rule Deck Files

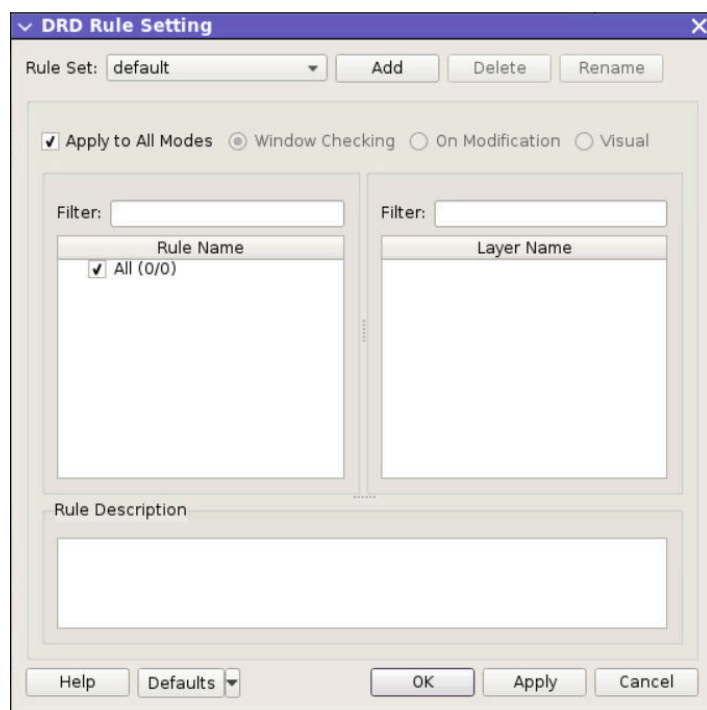
You can specify that the window checking tool use the rules in a selected DRD rule deck file when checking the design in the viewport. The DRD rule deck file has the .tcl file name extension.

If the DRD rule deck file contains rules for checking the color balance in the design, the window checking tool will do so during the design check. For more information about color balancing, see the *Custom Compiler Technology and Display Resource File Reference Manual* for more information.

You can select only one DRD rule deck file. If you want to use rules that are defined across multiple DRD rule deck files, create a .tcl file that references all the DRD rule deck files.

In the **DRD Options** dialog box, select **Rule File** and then click  to select the rule deck file (*.tcl).

To select specific rules within the rule deck file, click . The **DRD Rule Setting** dialog box opens.



The rule set appearing in the **Rule Set** selection field includes a default rule set and user-defined rule sets. The selected rule set is the active rule set used for DRD checking. You can define multiple rule sets with selected rules and modify the user-defined rule sets

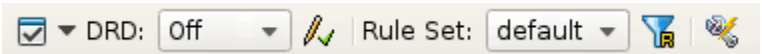
using the **Add**, **Delete**, and **Rename** buttons. For details, see [Selecting Constraints in the Technology Library for DRD](#).

You can select the desired rules from one of the following lists:

- **Rule Name:** Lists all rules defined in the rule deck file. By default, all rules are selected.
- **Rule Group:** Lists the rule groups that are defined in the DRD rule deck file. This list is hidden if no rule group exists in the rule deck file.
- **Layer Name:** Lists all layers. Each layer is associated with a set of rules. Layers that do not have associated rules do not appear in the list. By default, all layers are selected.

DRD Toolbar

To show or hide the **DRD** toolbar, see [About the DRD Tool](#).



The **DRD** toolbar contains these options:

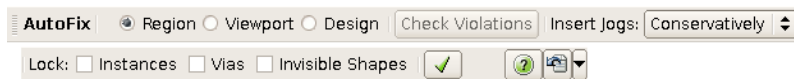
Option	Description
Check DRC in current viewing region 	Performs window check in the current viewing region. For more information, see Checking for Violations in the Viewport . Note: Do not click this button until you are ready to start the window checking process in the viewport.
DRD	Turns on or off interactive design rule checking on active objects. Select one of the following: • Off: Turns off interactive design rule checking. • Visual: Turns on interactive design rule checking when you create or edit objects. A red X displays to let you know that you are creating a violation. Dynamic rulers display the current distance between two shapes. Dynamic markers also show the violated design rules. • Assist: Turns on interactive design rule checking when you create or edit objects. The guide lines display to indicate a blockage area to let you know that you are reaching the minimum spacing. For information about the DRD Options dialog box, see Setting DRD Options .

Option	Description
 On Modification	When enabled, checks design violations when a design is modified. When disabled (default), the tool does not check for design violations when a design is modified. Window checking is always available, even if the DRD option is set to Off on the DRD toolbar.
Rule Set	Switches between default or user-defined rule sets. For details, see Selecting DRD Rule Deck Files .
 Show the DRD Rule Selection Dialog	Opens the DRD Rule Setting dialog based on current rule source and rule configuration settings. For details, see Selecting Constraints in the Technology Library for DRD .
 DRD AutoFix	Displays the AutoFix toolbar. For the description of the AutoFix toolbar options, see AutoFix Toolbar . AutoFix finds DRD violations in the viewport or design and displays corresponding markers. See Displaying Violation Markers After Running DRD AutoFix .

AutoFix Toolbar

For information on how to display the **AutoFix** toolbar and use AutoFix, see:

- [Fixing Violations in a Region of the Top-Level Design](#)
- [Fixing Violations in the Top-Level Design](#)
- [Displaying Violation Markers After Running DRD AutoFix](#)



The **AutoFix** toolbar contains the following options:

Option	Description
Auto Fix	• Region: Lets you click and drag to define a region of the design that you want checked and fixed. • Viewport: Checks and fixes only the current viewport. • Design: Checks and fixes the entire design.
Check Violations	Finds DRD violations in the viewport or design and displays corresponding markers.

Option	Description
Insert Jogs	A jog is a split introduced in a path segment during legalization to more efficiently remove spacing and clearance violations. • Never: No jogs are inserted. • Conservatively: (Default) Inserts jogs only at DRC violations determined to be otherwise unfixable. • Aggressively: Inserts jogs on all path segments involved in DRC violations.
Lock	Specifies to keep the following designated object sets in their current positions: • <i>Instances</i> • <i>Vias</i> • <i>Invisible Shapes</i>¹
Run DRD AutoFix	Runs DRD AutoFix.
Defaults	Resets all of the options to your Custom Compiler environment default values.

Checking for Violations in the Viewport

When performing design rule checking in the viewport, you can apply constraint rules and quickly check violations in the viewport (viewing region) of your design.

Click  (**Check DRC in current viewing region**) on the **DRD** toolbar to start the window checking process and view the results in the Console.

Press Ctrl+C to stop the window checking process if it takes too long to wait.

For more information about setting the constraint rules for window checking, see [Setting DRD Options](#).

Assigning Colors to Shapes in the Viewport

When performing design rule checking in the viewport, the window checking tool can automatically assign colors to colorable shapes if you enabled this option in the **DRD Options** dialog box. The window checking tool assigns colors to shapes as follows:

- If you choose to assign colors to top-level (0x) shapes only, the tool will assign colors to shapes as described in Rules for Overlapping Shapes on the Same Layer.
- If you choose to assign colors to shapes across the hierarchy, the tool will create a Color cell that contains the color results, and then the Layout Editor propagates the

1. Use the Object/Layer Panel Assistant to specify whether an LPP is visible. See the Custom Compiler Environment User Guide for information on using the Object/Layer Panel Assistant.

colors from the Color cell to the top-level cell. The Color cell is another cell view and is stored with the design.

The hierarchical color assignment is described in *Rules for Overlapping Shapes Across the Hierarchy*.

As the window checking tool assigns colors, it also uses the color density rules to ensure that the color distribution in the design meets the specified requirements. Color density specifications are derived from the `vr::set` Tcl command in the DRD rule deck file. For more information about the command, see the Design Rule Verification Tcl topic in the *Custom Compiler Technology and Display Resource File Reference Manual*.

To assign colors automatically during viewport-level design rule checking:

1. In the Layout Editor, open the **Layout Design Options** dialog box, and then set the **Precoloring Depth** value in **General** tab to ensure that colors propagate from the desired level. Click **OK** or **Apply** to save the change.
2. In the Layout Editor, choose **Options > DRD** to open the **DRD Options** dialog box.
3. Select **Constraint Group** or **Rule File** you want to apply during violation checking.
4. Set the desired options.
5. Click **Apply** to save the changes, and then click **OK** to close the dialog box.
6. To start checking for violations in the viewport, on the **DRD** toolbar, click  (**Check DRC in current viewing region**).

The tool starts to check all supported objects within the viewport. After checking all target objects, the Layout Editor displays markers on the canvas to indicate the locations of the violations.

Fixing Violations With Guidance

When fixing violations in the design, the window checking tool displays fixing guides with information such as the violated edges with design rules, the required distance, and the possible direction (in horizontal or vertical) to fix violations.

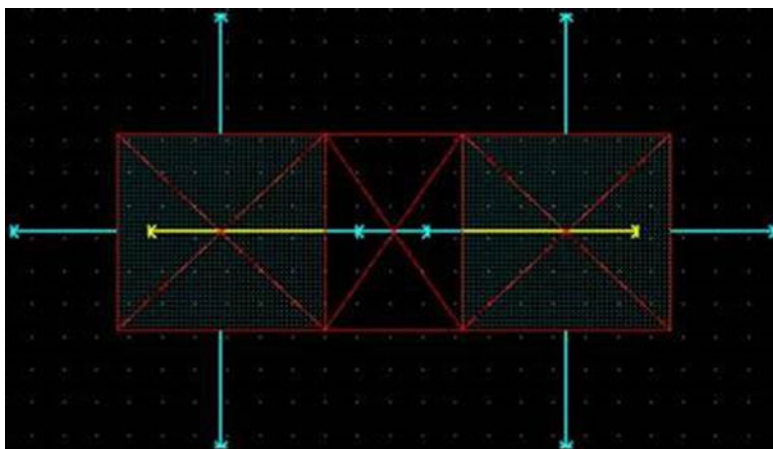
The tool uses oaMarkers to display fixing guides on the canvas. Fixing guides are cleared after the Undo action, and they will be saved when the design is saved. You can manage the colors of fixing guide with color packets. The color packets can be edited in the Display Resource Editor. See Editing Display Attributes in the *Custom Compiler Environment User Guide*. [Table 1](#) shows the default colors of fixing guide and three types of fixing guides.

Table 1 *Predefined Colors and Color Packets for Fixing Guide and Fixing Guide Types*

Predefined Color	Color Packets to Fixing Guide	Fixing Guide Types
yellow	fixingGuide	Spacing
red	fixingGuide1	
green	fixingGuide2	Enclosure
blue	fixingGuide3	
purple	fixingGuide4	
cyan	fixingGuide5	Width
white	fixingGuide6	
gold	fixingGuide7	
maroon	fixingGuide8	
gray	fixingGuide9	

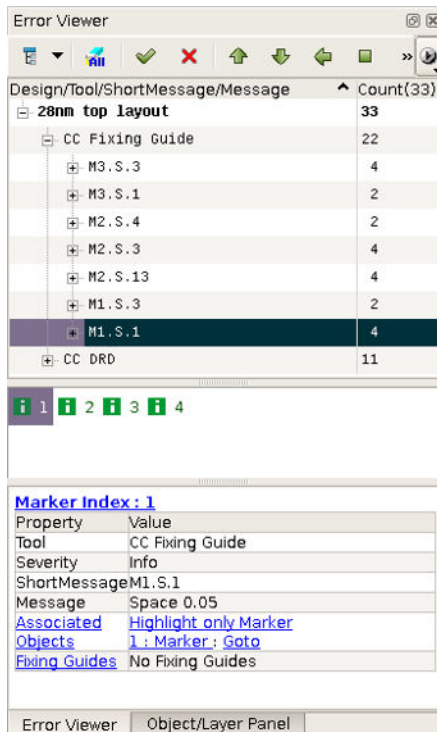
Figure 1 shows the fixing guide types. The yellow fixing guides are the spacing fixing guides and the other fixing guides in cyan color are width fixing guides.

Figure 1 *Fixing Guide Types*



All fixing guides are listed in the Error Viewer. When selecting a violation maker, the details for the fixing guide are also listed in the Error Viewer. The fixing guides can be highlighted on the canvas. See Manipulating Error Markers in the *Custom Compiler Environment User Guide*.

Figure 2 Listing Fixing Guides in Error View

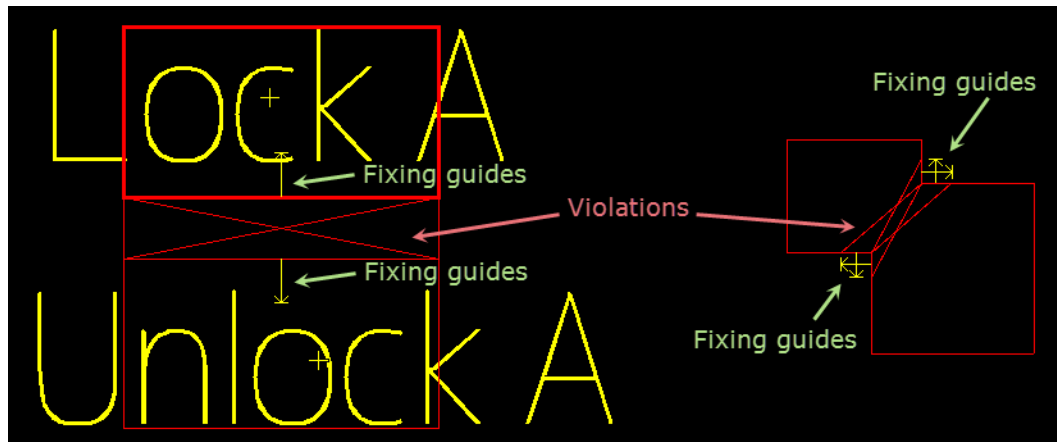


Fixing guides are created with the `vr::internal`, `vr::external`, and `vr::enclosure` Tcl commands. For more information about the commands, see the Design Rule Verification Tcl topic in the *Custom Compiler Technology and Display Resource File Reference Manual*.

To fix violations with guidance:

1. In the Layout Editor, choose **Options > DRD** to open the **DRD Options** dialog box.
2. Select **Generate Fixing Guide** to show fixing guides on violations.
3. Click **Apply** to save the changes, and then click **OK** to close the dialog box.
4. To start checking for violations in the viewport, on the **DRD** toolbar, click  (**Check DRC in current viewing region**).

The tool starts to check all supported objects within the viewport. After checking all target objects, the Layout Editor displays markers and fixing guides on the canvas to indicate the locations of the violations.



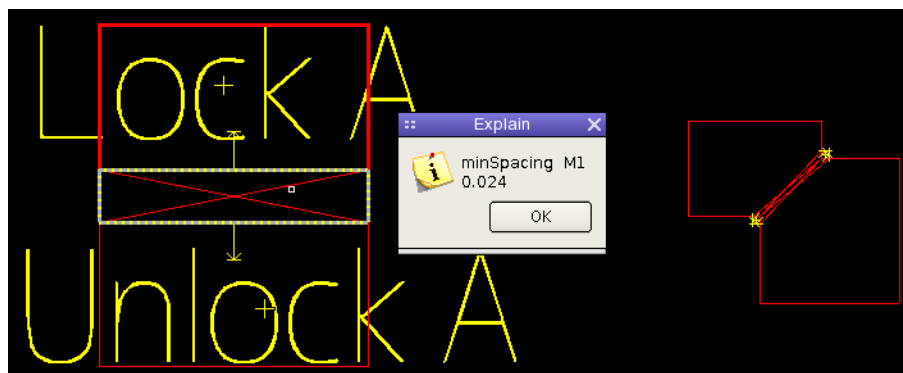
Note:

When `leDRDGenerateFixingGuide` is set to *true*, shows fixing guides on violations after running window checking.

When `leDRDGenerateFixingGuideInInstance` is set to *true*, shows fixing guides for violations in an instance.

5. Select one of the markers and click-right to show a shortcut menu.

Choose **Explain** to display the information of the selected marker in the following **Explain** dialog box. Click **OK** to close the dialog box.



(Optional) Click-right on the same marker and then choose **Select in Error Viewer** from the command menu.

6. Move the objects (on the left) or stretch the edges of the polygon (on the right) according to the suggested directions, and then click  (**Check DRC in current viewing region**) on the **DRD** toolbar to verify the design rules.

Repeat this step until all violations are fixed.

Fixing Violations in a Region of the Top-Level Design

The **DRD AutoFix** tool lets you apply constraint rules and automatically fixes violations in a region of your edit design or in the entire edit design. This topic describes how to apply the constraint rules to a region of a top-level design.

Depending on the selection in the **Command** tab of the **Layout Design Options** dialog box, DRD AutoFix applies net class-to-class spacing and nocross rules from the foundry only or from both the foundry and design-level specifications.

Note:

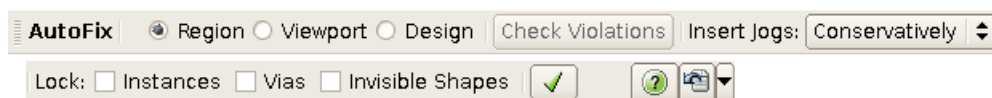
The **DRD AutoFix** tool does not replace industry standard DRC tools.

To apply constraints to a region of your design and automatically fix violations:

1. Display the **AutoFix** toolbar using one of these methods:

- Choose **Edit > Other > DRD AutoFix**.
- Use the shortcut key Alt+L.
- Click the **DRD AutoFix** button on the **DRD** toolbar.

The **AutoFix** toolbar appears.



2. Select **Region**.
3. Specify the options you want to apply.
4. Define the region using one of the following methods:
- Click and drag
 - Click twice to define the two corners of a rectangle

Only geometry within the region is modified.

5. Press Enter or click **Apply**.
- The cursor and the status bar reflect progress as shapes are modified.

Note:

During the modification operation, the command tries to remove overlaps across same layer shapes that belong to different nets.

- You can use Ctrl+C to interrupt the command and save the current state.
 - After shape modification, the command remains active.
6. Press the Esc key to end the command.

For more information about the **AutoFix** toolbar, see [AutoFix Toolbar](#).

Fixing Violations in the Top-Level Design

The **DRD AutoFix** tool lets you apply constraint rules and automatically fix violations in a region of your top-level design or in the entire top-level design. This section describes how to apply the constraint rules to the entire top-level design.

Depending on the selection in the **Command** tab of the **Layout Design Options** dialog box, DRD AutoFix applies net class-to-class spacing and nocross rules from the foundry only or from both the foundry and design-level specifications.

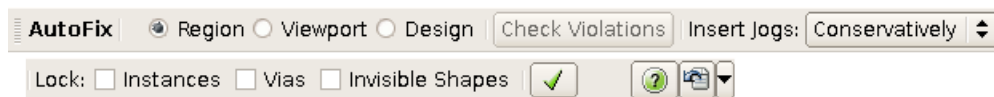
Note:

The **DRD AutoFix** tool does not replace industry standard DRC tools.

To apply constraints to your entire top-level design and automatically fix violations:

1. Display the **AutoFix** toolbar using one of these methods:
 - Choose Edit > Other > DRD AutoFix.
 - Use the shortcut key Alt+L.
 - Click the **DRD AutoFix** button on the **DRD** toolbar.

The **AutoFix** toolbar appears.

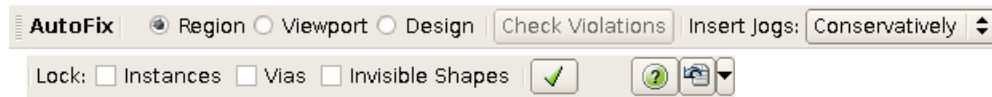


2. Select **Design**.
3. Specify the options you want to apply.
4. Press Enter or click **Apply**.
 - The cursor and the status bar reflect progress as shapes are modified.
 - You can use Ctrl+C to interrupt the command and save the current state.
 - After shape modification, the command remains active.
5. Press the Esc key to end the command.

For more information about the **AutoFix** toolbar, see [AutoFix Toolbar](#).

Displaying Violation Markers After Running DRD AutoFix

The **AutoFix** toolbar includes the **Check Violations** feature, which lets you find and display violation markers before you run DRD AutoFix.



The **Check Violations** feature is available when the following two conditions are true:

- The DRD mode is set to **Visual** on the **DRD** toolbar and the mode is set to either **Viewport** or **Design** in the AutoFix options toolbar.
- The DRD mode is set to **Visual** on the **DRD** toolbar, the mode is set to **Region**, and you have drawn the region.

When the above conditions are true and you press the **Check Violations** button, DRD markers appear in the **Viewport** or **Design** area that you specified.

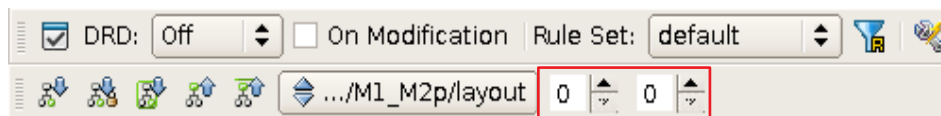
When the **On Modification** option is enabled, markers are recomputed after the AutoFix layout modification and made available in the Marker Browser Assistant.

For more information about the **AutoFix** toolbar, see [AutoFix Toolbar](#).

Fixing Violations Across the Design Hierarchy

The AutoFix command respects design hierarchy while performing layout modifications. To fix violations, AutoFix can modify or add top-level (0x level) shapes, as well as transform instances, PCells, or vias. But AutoFix cannot modify the geometry inside instances, PCells, and vias. This restriction avoids the unintended consequences that can occur when sub-cells are instantiated in other designs. Also, in Region mode, AutoFix modifies only those geometries (including partial shapes) that are inside the specified region.

You can specify the Start and Stop levels from the **General** tab of the **Layout Design Options** dialog box (see Setting Layout Design Options) or from the **Start Level** and **Stop Level** fields on the **Hierarchy** toolbar.



AutoFix reads in all shapes from the Start level to the Stop level and fixes design rule violations within that geometry. AutoFix does not open instances beyond the

Stop level and is not aware of the geometry inside those instances. AutoFix treats the instances at the Stop level as rectangular bounding boxes, but can apply either a conservative or aggressive approach to preserving shapes, depending upon the value of the `leAutoFixInstStopLevelMode` preference.

- If `leAutoFixInstStopLevelMode` is set to *opaque*, AutoFix conservatively preserves all shapes that overlap with the Stop level instances, either partially or completely if the shape is contained inside the instance bounding box.
- If set to *clear*, there is no such restriction.

In both modes of operation, however, shapes that do not overlap with Stop level instances are prevented from forming new overlaps. Because of these restrictions and because AutoFix is not aware of the geometry inside Stop level instances, some design rule violations might not be fixed. In that case, you need to increase the Stop level value and reapply the AutoFix command.

You can also use AutoFix during an in-context Edit-In-Place operation. In this case, AutoFix is aware of all the shapes from the view design to the edit design, subject to all the Stop level preferences for designs opened for descend-edit from view design to edit design. However, only the top-level (0x level) geometry of the edit design is modified after running AutoFix.

For more information about the **AutoFix** toolbar, see [AutoFix Toolbar](#).

3

Checking for Violations as You Edit (ICV Live DRC)

Instead of waiting until the design is finished, you can use the ICV Live DRC tool to incrementally check for violations as you create and edit the design.

This chapter contains the following topics:

- [About the ICV Live DRC Tool](#)
- [Workflow for Running ICV Live DRC](#)
- [Setting Up the IC Validator Environment](#)
- [Enabling ICV Live DRC](#)
- [Setting Up ICV Live DRC](#)
- [Managing ICV Live DRC License Use](#)
- [Checking for Violations in the Viewport](#)
- [Terminating ICV Live DRC Jobs](#)
- [Viewing and Fixing Violations](#)
- [Creating and Modifying Recipes](#)
- [Runset Parsing Time](#)

About the ICV Live DRC Tool

Instead of running design rule checks (DRC) after you have completed the design, you can check for rule violations incrementally as you create and edit the design. Catching violations early in the process reduces signoff DRC runtime and the number of debugging sessions.

You can use the IC Validator Live DRC (ICV Live DRC) tool in the Layout Editor to run incremental DRC. ICV Live DRC uses the IC Validator signoff rule file so that the checks

are comprehensive. However, because you are limiting the checks to the viewport and visible layers only, the runtime is significantly faster.

Workflow for Running ICV Live DRC

To run ICV Live DRC:

Step		See...
1	Set up the IC Validator environment.	Setting Up the IC Validator Environment
2	Enable ICV Live DRC.	Enabling ICV Live DRC
3	Set up ICV Live DRC.	Setting Up ICV Live DRC
4	Run DRC within the viewport.	Checking for Violations in the Viewport
5	View and fix the violations.	Viewing and Fixing Violations

Setting Up the IC Validator Environment

Some IC Validator environment variable settings are required before starting Custom Compiler. To load the IC Validator Live DRC menus, you must set two environment variables:

- Set `ICV_HOME_DIR` to the IC Validator installation directory
- Set `PATH` to include `$ICV_HOME_DIR/bin`

For example:

```
setenv ICV_HOME_DIR </path/to/icv>
setenv PATH $ICV_HOME_DIR/bin:$PATH
```

To make the IC Validator Live DRC toolbar visible, select **Window > Toolbars > DRD**.

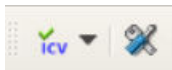
Enabling ICV Live DRC

You can specify an ICV environment variable before enabling ICV Live DRC. To enable ICV Live DRC:

- [Window > Toolbars](#)
- [Verification > ICV Live DRC](#)

Window > Toolbars

In the Layout Editor, choose **Window > Toolbars > DRD** to show the **ICV Live DRC** toolbar.



The **ICV Live DRC** toolbar contains the following buttons:

- : Runs DRC based on the ICV Live DRC settings. See [Checking for Violations in the Viewport](#).
- : Displays the ICV Live DRC Setup dialog box. See [Setting Up ICV Live DRC](#).
- Rule Recipe: >> Selects the **Rule Recipe**.

The **Rule Recipe** option contains the last recipes previously used. If the `xtICVLiveRecipeDirectory` preference is specified in the `in_pdk.xml` file, the **Rule Recipe** option instead contains all of the recipe options included in the all recipe files of the specified directory. You can set the `xtICVLiveRecipeHistoryMaxCount` preference to specify the maximum number of history recipes to display. By default, the `xtICVLiveRecipeHistoryMaxCount` is 0, which indicates no maximum count, that is unlimited. The `xtICVLiveRecipeHistoryMaxCount` preference is ignored when using `xtICVLiveRecipeDirectory`. See [Creating and Modifying Recipes](#) for more information.

Verification > ICV Live DRC

In the Layout Editor, choose **Tools > Verification**. The Verification menu appears.

The **Verification > ICV Live DRC** menu contains the following commands:

- **Setup**: Displays the **ICV Live DRC Setup** dialog box. See [Setting Up ICV Live DRC](#).
- **Run**: Runs DRC based on the ICV Live DRC settings. See [Checking for Violations in the Viewport](#).

Setting Up ICV Live DRC

Before you run incremental DRC, you must supply the required information in the **ICV Live DRC Setup** dialog box. To display the dialog box, do one of the following:

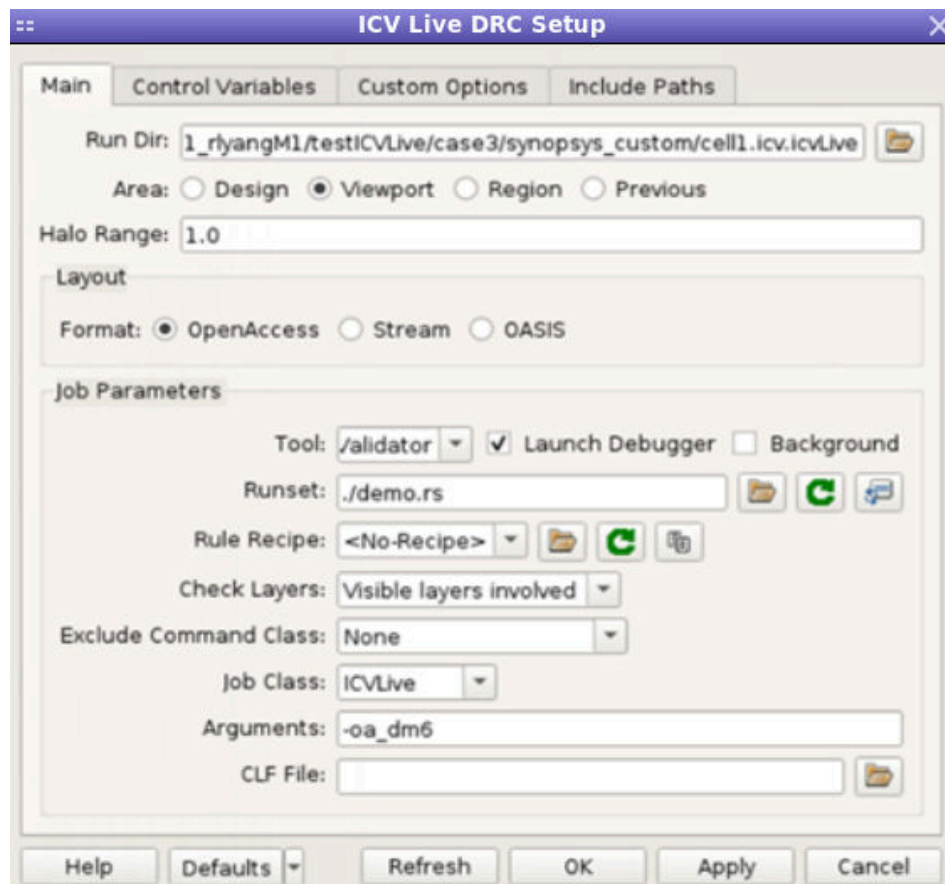
- On the **ICV Live DRC** toolbar, click :
- Choose **Verification > ICV Live DRC > Setup**.

In the **ICV Live DRC Setup** dialog box, specify the following:

Setup Information	See...
Design rule file Rules to use Layout information Command-line arguments	Main Tab
Runset control variables	Control Variables Tab
Layer map file	Custom Options
Additional file inclusions	Include Paths

Main Tab

Set the following in the **Main** tab:

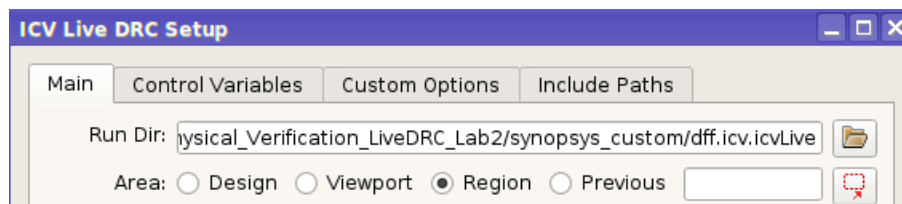


Settings	Description
Run Dir	Select the directory in which you will run ICV Live DRC. The output files are stored in this directory.

Settings	Description
----------	-------------

Area

Area on which to run the DRC check as **Design**, **Viewport**, **Region** or **Previous**.



- When **Area** is set to **Design**, checks include all of the supported visible objects within the bounding box and enclosing the entire design. The **Keep Hierarchy** check box, as displayed in the **Main** tab, allows you to run ICV Live DRC on all levels (hierarchies) of the design.
- When **Area** is set to **Viewport** the viewing range of the layout canvas is represented.
- When **Area** is set to **Region**, the DRC check is specified by the coordinates of its rectangle, for example: {-20, -20, 20, 20}. To automatically insert coordinates here for a selection area, click the interactive selection button, and then select the area in the design.
- When **Area** is set to **Previous**, the last checked area for the run is considered. ICV Live DRC saves the current bounding box for every run, and it uses the last checked area when **Area** is set to **Previous**. When the first run is set to **Previous**, no saved data is present and **Viewport** coordinates are considered for the run.

Halo Range

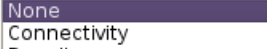
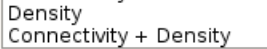
Specifies the design objects within a halo range. The default is derived from the `xtICVLiveHaloRange` preference.

Layout Format

For the **OpenAccess** format, verification runs on the current cell view.

For the **Stream** and **OASIS** formats, the tool exports **Stream** or **OASIS** and runs ICV Live DRC on the generated GDSII or OASIS file. Set the following:

- **Export Stream** or **Export OASIS**: Exports the layout cellview used as input to the **File** you specify.
- **Config**: (Available only if you selected **Export Stream** or **Export OASIS**) Opens the Export Stream or Export OASIS dialog box. To set the options in the dialog box, see Exporting Design Data in GDSII Stream Format or Exporting Design Data in OASIS Format. The options you specify are integrated with PVE (preferences in the `pdk.xml` configuration).

Settings	Description
Job Parameters	Select the following: Tool is IC Validator. Launch Debugger launches (or updates existing) Error Viewer when the process completes. In Background mode, you can continue using Custom Compiler while ICV Live DRC checks runs in the background. Use the <code>xtICVLiveNotifyWhenComplete</code> preference to get a notification dialog box when ICV Live DRC checking completes. Runset: The design rule (runset) file used for incremental DRC. If you want to use a rule file that was modified after it was already loaded, reopen the dialog box and click  to reload the modified file. Click  to open the runset file with a text viewer. Rule Recipe: Lists all of the recipes and their corresponding recipe files. Set the <code>xtICVLiveRecipeRunHistory</code> preference to see all of the recipes. Recipe files must have the <code>.icvrp</code> suffix. The format of the item in the Rule Recipe is the Recipe file name excluding the file path and suffix (<code>.icvrp</code>), a separator(<code>:</code>), and the recipe name. For example, <code>first:Metal1</code>. Click  to select an existing recipe file. Click  to reload all of the used recipes. Set the <code>xtICVLiveReloadRecipeFiles</code> preference to see the used recipe files. Click  to view and edit selected recipe. This opens the Recipe Editor. Recipe files can be created or modified using the Recipe Editor. See Creating and Modifying Recipes for more information. When you choose a recipe from the Rule Recipe list, click  to edit the recipe in the Recipe Editor. Check Layers: Use the <code>xtICVLiveCheckLayers</code> preference to set the following: All layers: Selects rules containing any layer in the active window (including invisible). Visible layers involved: Selects rules related to the visible layers (the default). Visible layers only: Selects rules containing only the visible layers. An information message is printed when Check Layers is set to Visible layers involved or Visible layers only, and when you choose the specified set of rules, or use recipes. Exclude Command Class: Set None, Connectivity, Density, or Connectivity + Density to specify the sets of command types to be excluded based on the options available in the <code>CommandClass</code> enumerator (<code>CONNECTIVITY</code>, <code>DENSITY</code>). Equivalent to <code>-lxf</code>. Exclude Command Class:  Job Class:  Arguments:  Arguments: The command-line arguments that control the behavior of the interactive DRC. CLF File: Text file representing the IC Validator command-line options. The order of concatenation starts from the CLF file and then from the settings in the ICV Live DRC Setup dialog box. The merged CLF file removes the <code>-svc</code>, <code>-uvc</code>, <code>-lxf</code> and <code>-recipe</code> command-line options to ensure the complete set of rules is present in the specified runset file. You do not need to select the Tool and the Job Class.

Note:

When the ICV Live runset preference, `xtICVLiveICVRunsetFile` is not set, the IC Validator DRC runset preference, `xtDRCICVRunsetFile` is used for the ICV Live DRC run, and a warning message appears in the Custom Compiler console. The runset box in the **ICV Live DRC Setup** dialog box remains empty if you do not set the `xtICVLiveICVRunsetFile` preference.

The IC Validator command-line arguments you can specify are:

- -D
- -I
- -U
- -bct
- -clf
- -m
- -oa_dm5
- -fn
- -svc
- -svn
- -ufn
- -uvc
- -uvn
- -lxf

For the description of the arguments, see the [IC Validator user documentation](#).

Caution:

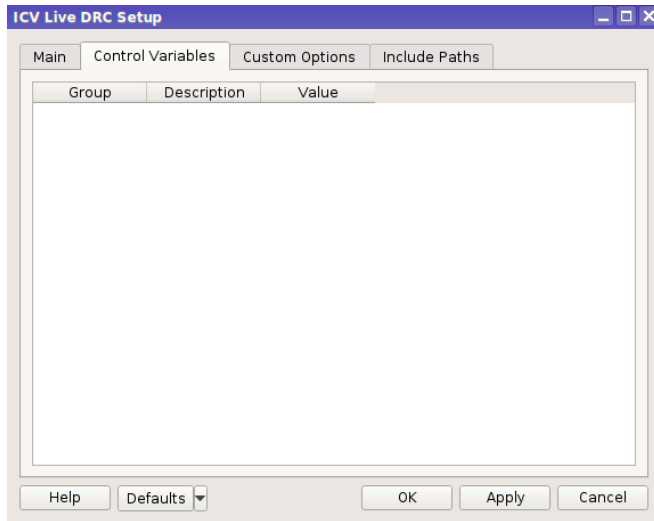
You can also use the -svn, -uvn, -svc, -uvc, -sfn, and -ufn arguments to filter rules. If these arguments conflict with the rule selections, either errors result or DRC produces an empty violation list.

Control Variables Tab

In the **Control Variables** tab, review, and optionally modify the current values for DRC runset control variables as preferred for this DRC run. The control variables are preset for

a PDK. To modify a preset value, click with the alternate (non-selection) mouse button, and edit the entry field.

The control variable setup in this dialog box is similar to the setup in the **DRC Setup and Run** dialog box. See [Using the Integrated Tools for Design Rule Checking](#).



Custom Options

The layer mapping file maps OpenAccess (OA) layer-purpose names to GDSII or OASIS layer number and data types. The file is used by ICV Live DRC for incremental DRC.

You can use the same layer map file as is used for streaming out GDS or OASIS files. Often these can be found from previously defined Custom Compiler preferences by typing:

```
db::getPrefValue dbExportStreamLayerMapFile
```

or

```
db::getPrefValue dbExportOasisLayerMapFile
```

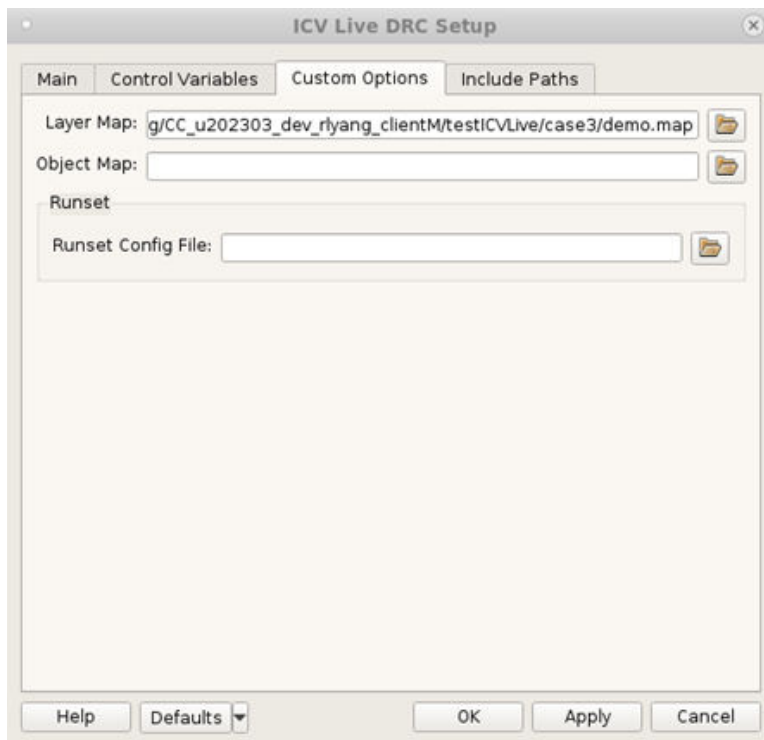
in the Custom Compiler console.

You can specify the **Layer Map** file in the **Custom Options** tab.

To perform DRC checking when the input layer format is OpenAccess and there are boundary or blockage rules related to OpenAccess objects, use the `xtICVLiveICVObjectMapFile` preference to specify the **Object Map** file.

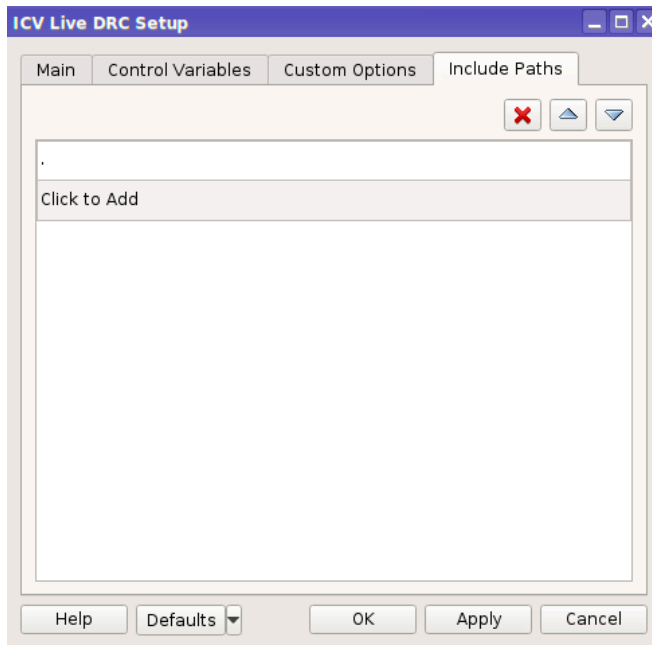
You can also specify the **Object Map** file by adding the `-oa_object_map` argument to the **Arguments** list in the **ICV Live DRC Setup Main**, tab or by specifying the **Object Map** file in the **Custom Options** tab.

To set ICV Live DRC Runset, you must specify a runset configuration file.



Include Paths

In the **Include Paths** tab, you can specify the folder that contains additional files to pass to ICV Live DRC during the DRC operation.



Managing ICV Live DRC License Use

The first ICV Live DRC check opens a job manager that remains active and waits for other checks. As a result, runtime for subsequent checks is faster because the setup (runset, options, and so on) remains in cache memory. The ICV Live DRC Live license remains checked out as long as this job manager is active.

This section contains the following topics:

- [Methods for Stopping the ICV Live DRC Job Manager](#)
- [Licensing for Large Designs](#)
- [Switching From ICV Live DRC to ICV Signoff](#)

Methods for Stopping the ICV Live DRC Job Manager

You can stop the ICV Live DRC job manager using the following methods:

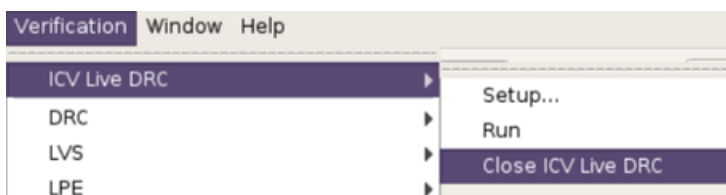
- Close the Custom Compiler session
- From the **Job Monitor**

You can use the **Job Monitor**, as described in [Terminating ICV Live DRC Jobs](#) to stop the ICV Live Manager job. When the job stops, IC Validator releases license keys and removes the runset from the cache.

Job	Status	Host	Run Description
▼ PVE	1 of 1 done (1 failed)		
▼ ICV Live DRC	1 of 1 done (1 failed)		
ICV Live Manager	KILLED	localhost	ICV Live manager waiting for checking request

- From the **Verification** menu

Choose **Close ICV Live DRC** from the **Verification > ICV Live DRC** menu to stop the ICV Live DRC job manager. If you run ICV Live DRC in the background and then close the job manager before the check is completed, Custom Compiler stops the ongoing check and the job manager (which releases cache memory and license keys).



- Through Custom Compiler timeout

To release the ICV Live DRC license after Custom Compiler times out, set the `dbIdleLicenseTimeout` preference. If Custom Compiler times out, the ICV Live DRC job stops and the **Close ICV Live DRC** selection from the **Verification** menu in the Layout Editor is unavailable.

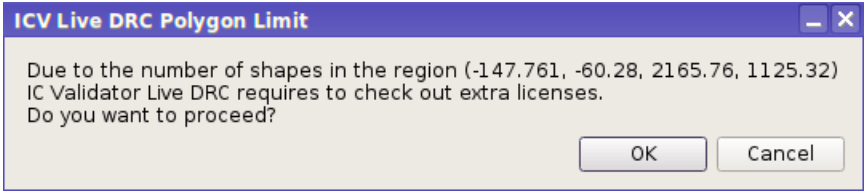
Licensing for Large Designs

IC Validator checks out additional license keys to run ICV Live DRC on a layout with the number of shapes that exceeds the polygon requirement. Set the `xtICVLivePolygonLicense` preference to confirm if you want to run with the additional license keys.

Table 2 *xtICVLivePolygonLicense Preference Values*

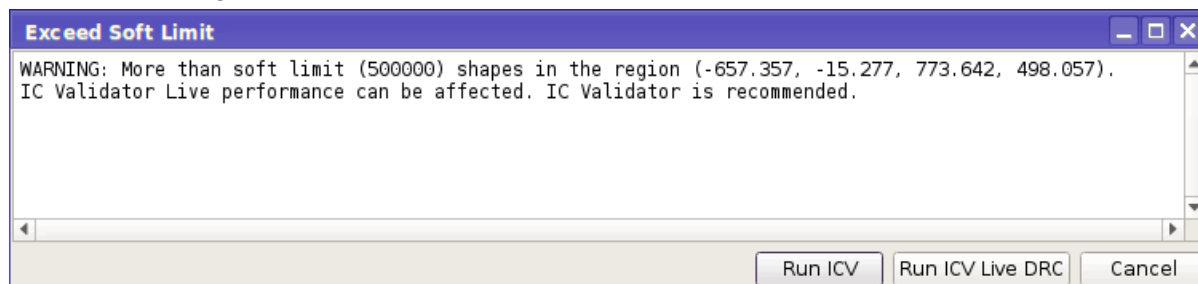
Preference value	Description
Always (default)	Extra license keys are checked out automatically when the layout exceeds the polygon count limit without asking, and the check is then performed.

Table 2 *xtlCVLivePolygonLicense Preference Values (Continued)*

Preference value	Description
Ask	You can choose to: • Check out extra license keys and perform the check. • Do not check out extra license keys, but adjust the window size before invoking ICV Live DRC again.  When you click OK, IC Validator checks out additional license keys to perform DRC checking and does ask again for additional license keys until the end of the session. When you click Cancel, Custom Compiler stops the ICV Live DRC job.
Never	No extra license keys are checked out when the layout exceeds the polygon count limit without asking, and the check is not performed.

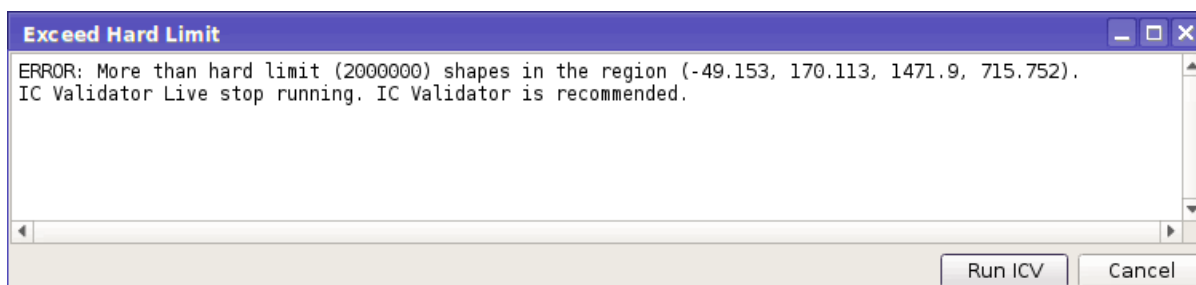
Switching From ICV Live DRC to ICV Signoff

The icvl_manager (ICVL engine) uses soft and hard polygon limits to manage the number of shapes needed to meet polygon requirements. If the number of shapes is greater than the soft limit, Custom Compiler displays the following dialog box to notify you that performance might be affected.



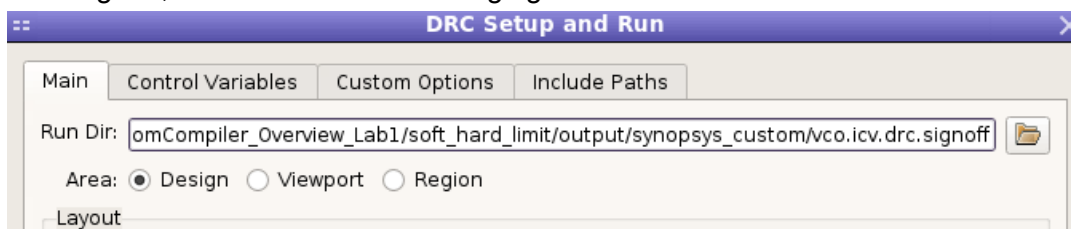
When this occurs, you can press either **Run ICV** or **Run ICV Live DRC**. The default is to run ICV signoff to improve performance when the number of shapes exceeds the soft limit. Therefore, you can simply press Enter to use the default behavior.

If the number of shapes is greater than the hard limit, Custom Compiler displays the following dialog box to notify you that ICV Live DRC cannot handle the current number of shapes and recommends running ICV signoff.



The default is to run ICV signoff when the number of shapes exceeds the hard limit. Therefore, you can simply press Enter to use the default behavior.

When running ICV signoff, the original run directory used in the ICV run is appended with .signoff, as shown in the following figure.



While running ICV signoff, settings from the **pdk.xml** file are loaded first, and then all of the following settings – **Area**, **Layout Format**, **Launch Debugger**, **Runset**, **Layer Map**, **Object Map**, and **Include Paths** – are copied from the **ICV Live DRC Setup** dialog box to the **DRC Setup and Run** dialog box.

You do not need to load the values from the existing **prefs.xml** file inside the specified **Run Dir** into the **DRC Setup and Run** dialog box. It is automatically invoked.

You can review job settings and set a queue engine in the **Job Management Options** dialog box. Currently, only the LSF and SGE queue engines are supported for this behavior. For details, see Managing Job Options in the *Custom Compiler Environment User Guide*.

When **Area** is set to **Previous** in the **ICV Live DRC Setup** dialog box, **Area** is set to **Region** in the **DRC Setup and Run** dialog box, the previous checking region is used when switching to ICV signoff. If it is a first time run, no saved data is present in ICV Live DRC, and Viewport coordinates are considered for the run. In addition, the shape count information is removed from the **Exceed Soft Limit** and **Exceed Hard Limit** dialog boxes.

Checking for Violations in the Viewport

Note:

You can define the halo range using ICV Live DRC by specifying the Tcl Preferences. You can control if ICV Live DRC writes out two OA markers to

indicate the checking range by specifying the Tcl Preferences, in which case one marker shows the checked viewport, and the other marker considers the halo range.

The OA markers are displayed in the Error Viewer (choose **Windows > Assistants > Error Viewer**). See Error Viewer Assistant.

You can control if ICV Live DRC displays a notification dialog when checking completes by specifying the Preferences.

To run DRC using ICV Live DRC:

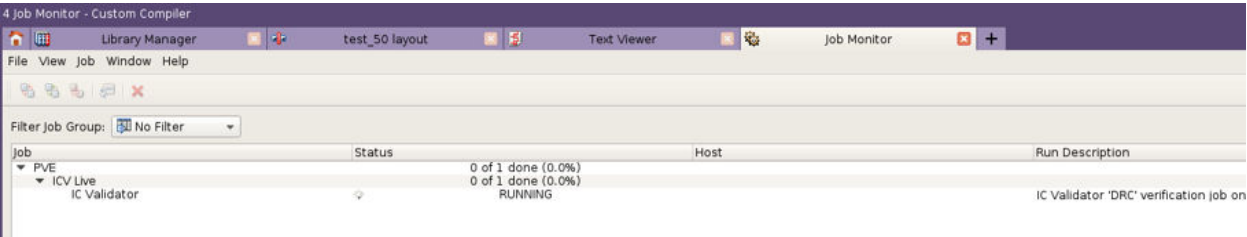
1. In the Layout Editor, arrange the layout so that the region you want to check is within the viewport.
2. Make sure the layers you want to check are visible. You can change the visibility of layers using one of the following methods:
 - On the Hierarchy Toolbar, set the Stop Level.
 - In the Making LPP Sets Visible and Selectable, display or hide layers.

3. On the **ICV Live DRC** toolbar, click :

The DRC process starts. When it is finished, the violation markers appear on the canvas. The violations are also listed in the Error Viewer.

Terminating ICV Live DRC Jobs

The ICV Live DRC job is displayed in the Job Monitor with correct **Status** and **Run Description**.



When selecting and right-clicking the ICV Live DRC job, you can now terminate a running ICV Live DRC job using the **Kill** command in the menu and view the output (icvl_manager.log) file.

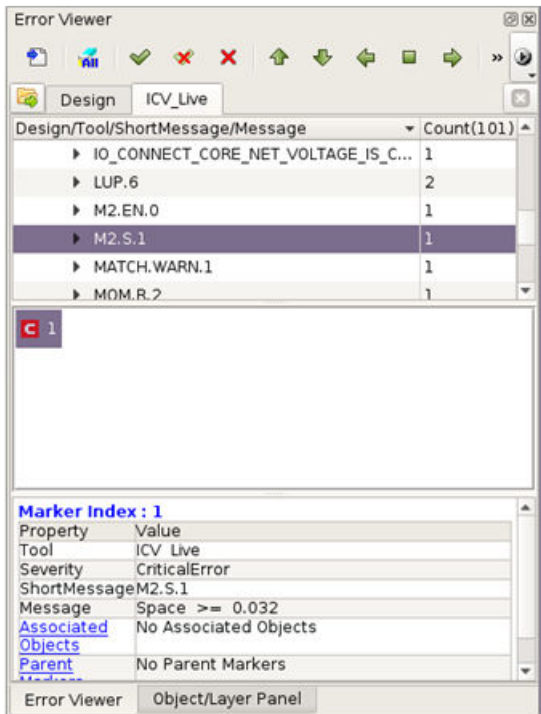


Viewing and Fixing Violations

To view the DRC violations:

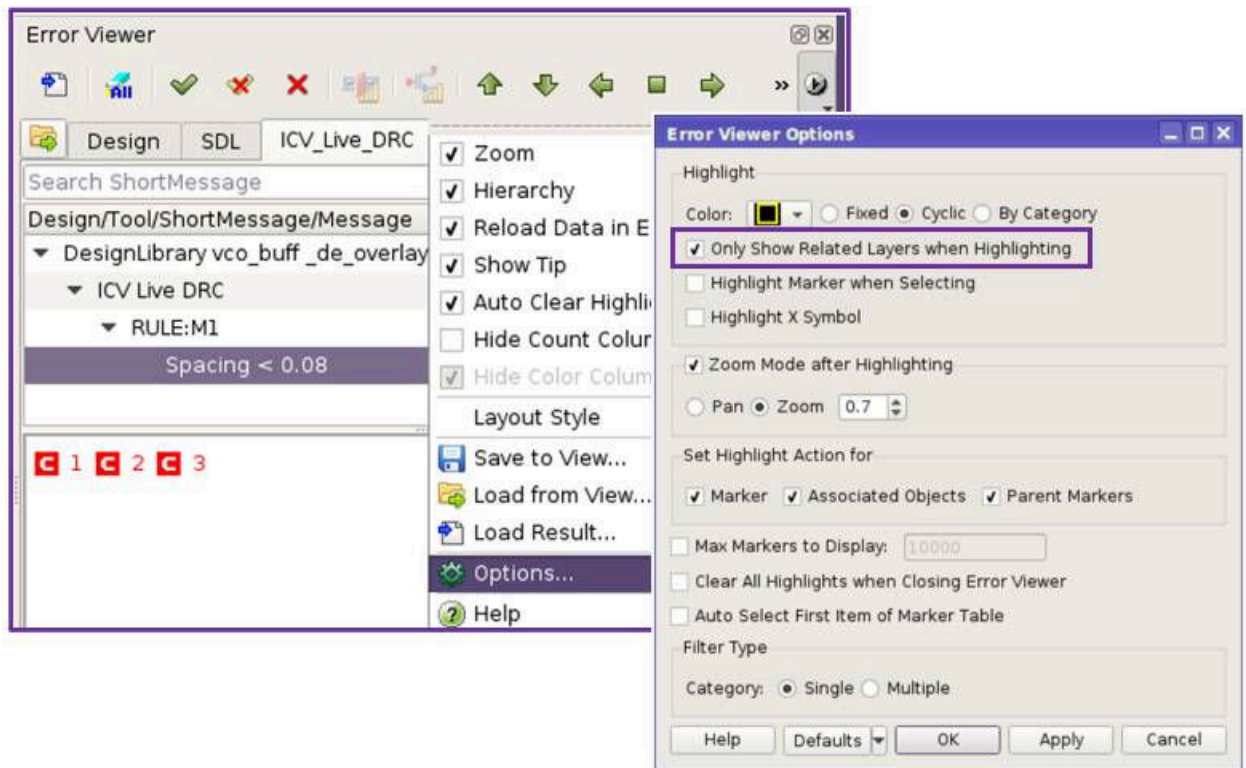
1. Display the Error Viewer.
2. Click the **ICV_Live_DRC** tab to view the list of DRC errors.

Figure 3 Error Viewer



From the **Error Viewer Options** dialog box, you can choose **Only Show Related Layers when Highlighting**. When this selection is enabled, the ICV Live DRC error highlight shows only those layers that are related to the violation and hides all other layers.

Figure 4 Error Viewer Options



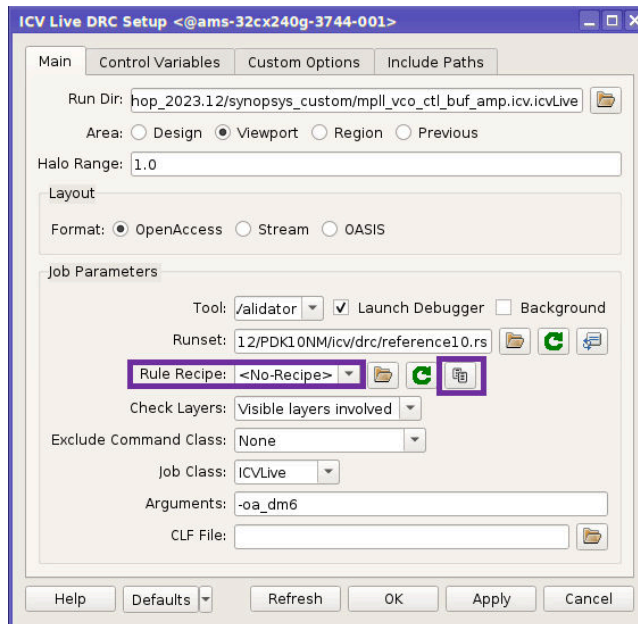
Creating and Modifying Recipes

You can create, edit and manage recipe files. This topic describes the following:

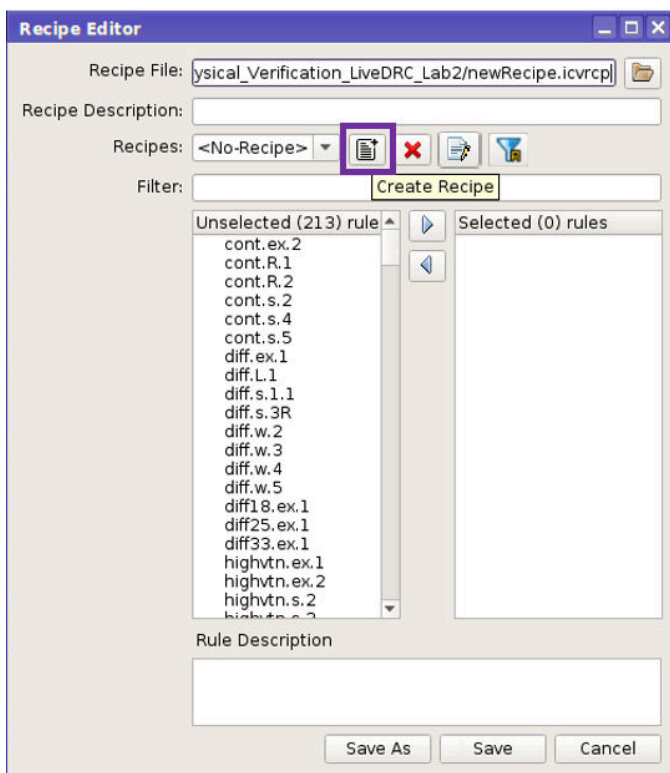
- [Create a recipe](#)
- [Rename a recipe](#)
- [Remove a recipe](#)
- [Save a recipe](#)
- [Managing Recipes](#)

Create a recipe

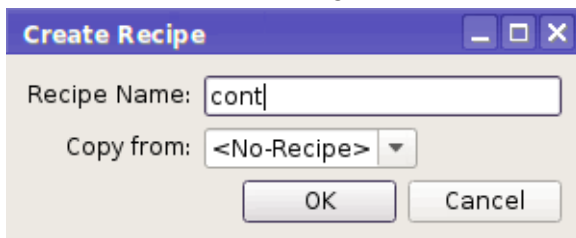
1. If existing recipes do not exist, as indicated by **Rule Recipe** > **<No-Recipe>** in the **ICV Live DRC Setup** dialog box, click **Create new recipe**  to create a new recipe in the **Recipe Editor**.



2. In the **Recipe File** box, enter new recipe file name, or click  to choose an existing recipe file.



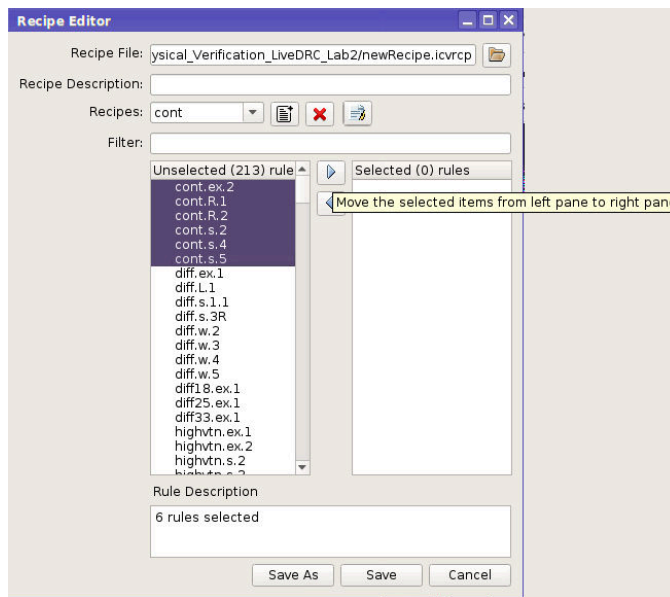
The **Create Recipe** dialog box appears.



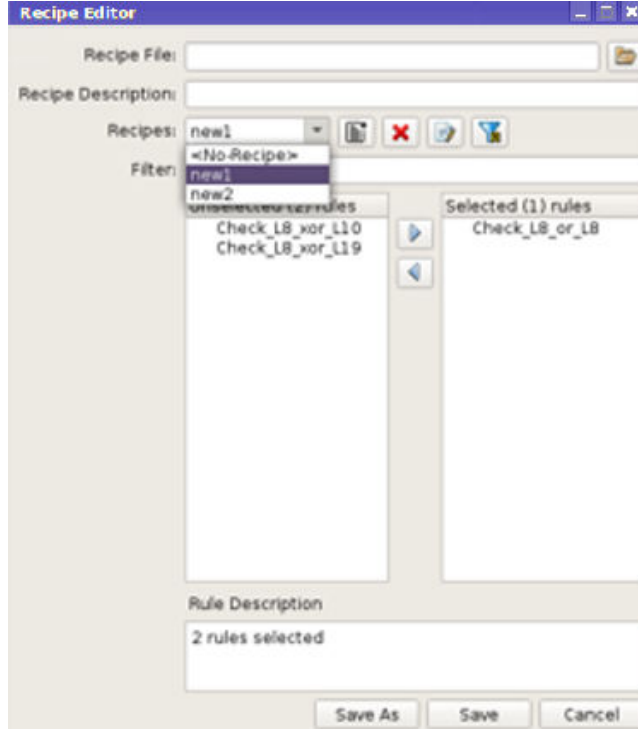
3. Enter the name of the new recipe in the **Recipe Name** box.
4. Click **OK** to save the recipe name and close the **Create Recipe** dialog box.
5. To add new rules to the new recipe in the **Recipe Editor**, select the required rules from the **Unselected rules** section and click  to move the selected rules to the **Selected rules** section.

To remove rules from the recipe in the **Recipe Editor**, select the rules from the **Selected rules** section and click  to move the rules to the **Unselected rules** section.

You can select multiple rules simultaneously, as shown in the following figure:



6. You can create multiple recipes and select rules for each recipe simultaneously. To check and verify the selected rules for each recipe, select the name of the recipe from the **Recipes** box.



The new recipes are temporarily stored in the memory until you save the recipes.

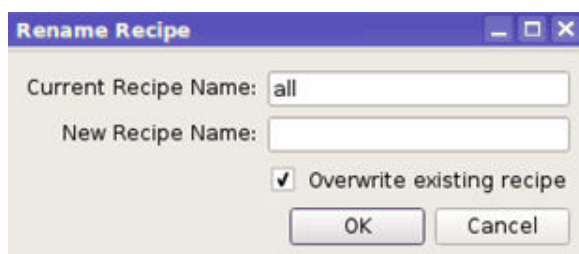
7. To save the new recipes, click **Save**.
8. To close **Recipe Editor**, click **Cancel**.

Note:

If you close the **Recipe Editor** dialog box without clicking **Save**, the recipes are not saved.

Rename a recipe

1. Click  in the **ICV Live DRC Setup** dialog box to choose an existing recipe file.
2. Click . The **Rename Recipe** dialog box appears.



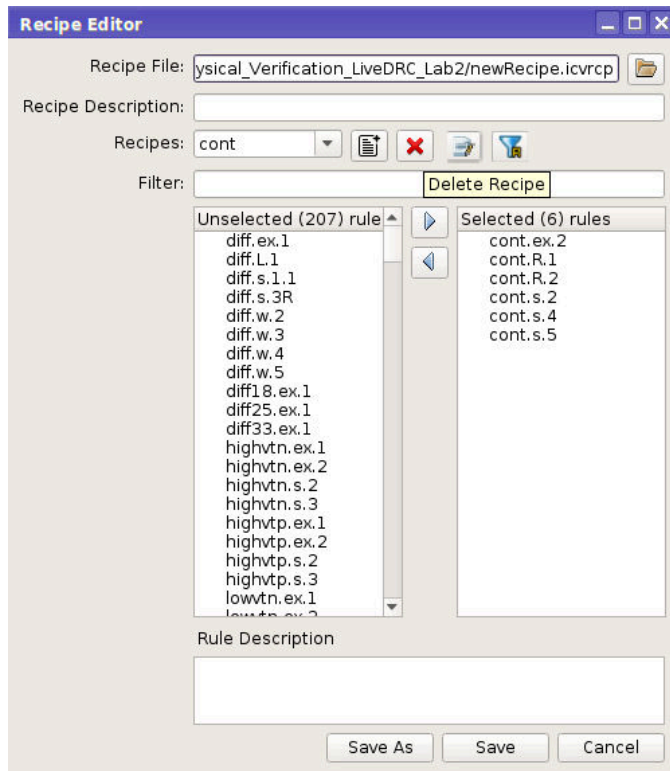
3. Enter the new recipe name in the **New Recipe Name** box.
4. The **Overwrite existing recipe** check box is selected by default. If you deselect this check box, the file is not renamed and a warning message appears, as shown in the following example:

"all" is the existed recipe so we don't change "all" due to turning off "Overwrite existing recipe"

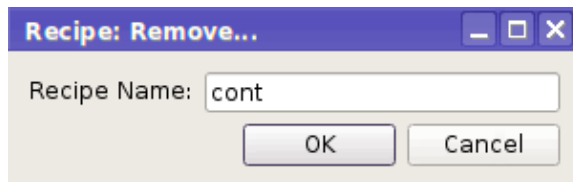
5. Click **OK** to rename the recipe.

Remove a recipe

1. Select the recipe to delete from the **Rule Recipe** menu in the **ICV Live DRC** dialog box.
2. Click **View/Edit selected recipe** . The **Recipe Editor** dialog box opens.



3. Click **Delete Recipe** . The **Recipe: Remove** dialog box appears.

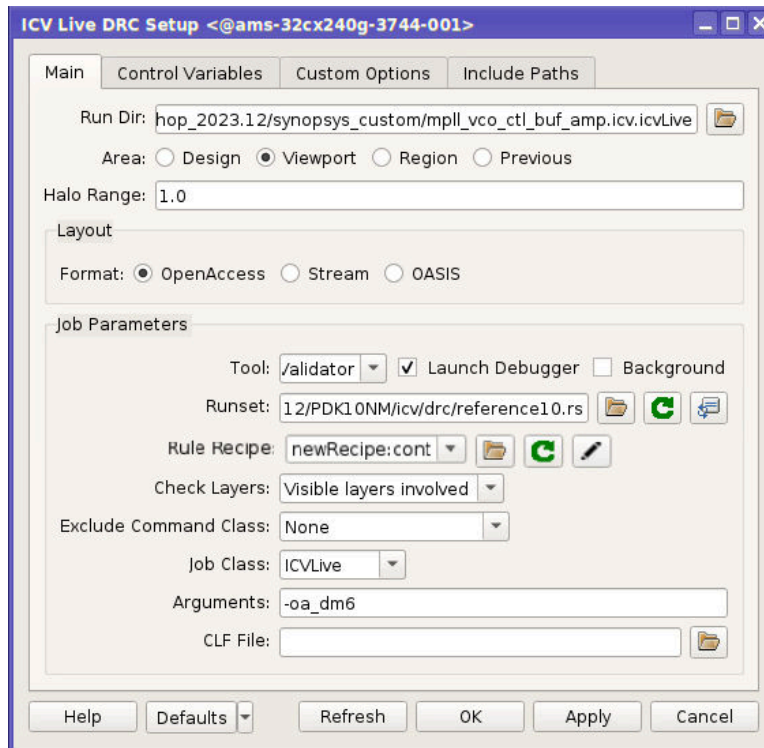


The name of the selected recipe appears in the **Recipe Name** box.

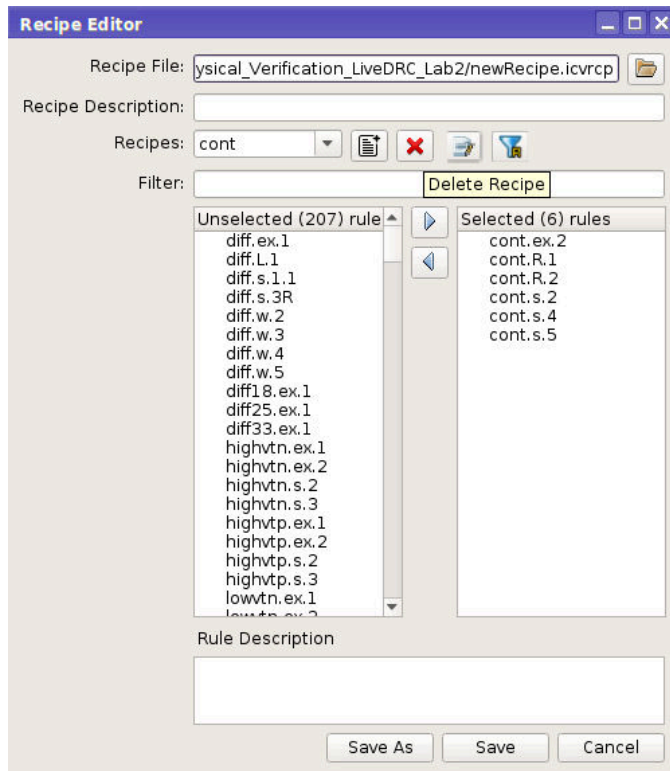
4. Click **OK**.
5. Click **Save** in the **Recipe Editor** dialog box to save the changes.
6. Click **Cancel** in the **Recipe Editor** to close the dialog box.

Save a recipe

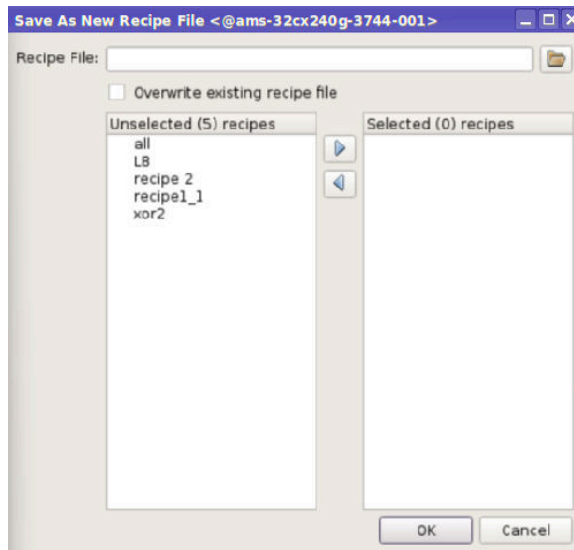
1. Click  in the **ICV Live DRC** dialog box to choose the recipe file.
2. Select the recipe from the **Rule Recipe** menu.



3. Click  to view and edit the selected recipe. The **Recipe Editor** dialog box opens.



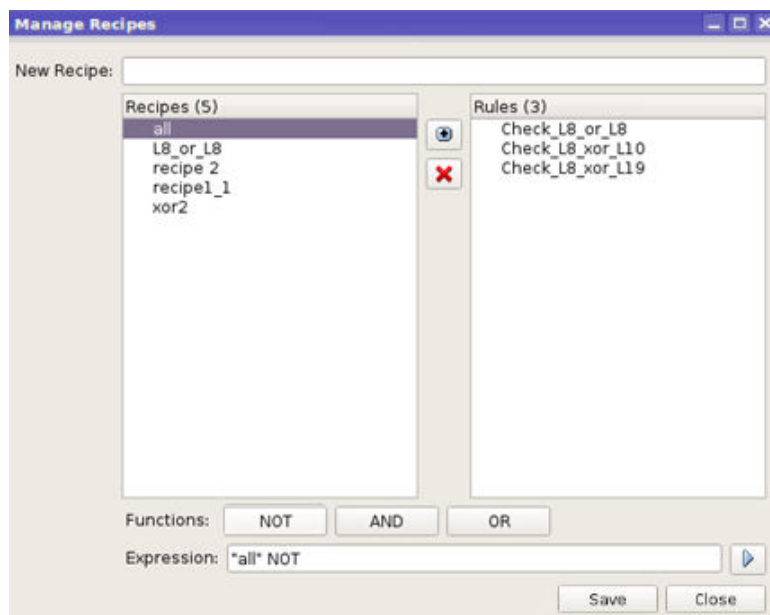
4. In the **Recipe Editor**, you can view the **Selected rules** for the selected recipe. Add or remove rules as required for the recipe.
5. Click **Save**.
6. To save all recipes from the given **Recipe File** to a new file, click **Save As**. The **Save As New Recipe File** dialog box opens.



- a. To add recipes to the new file, select the required recipes in the **Unselected recipes** section and click  to move the recipes to the **Selected recipes** section.
 - b. Optionally, select the required recipes in the **Selected recipes** section and click  to move the recipes to the **Unselected recipes** section.
 - c. After entering the new **Recipe File** name, click **OK**.
7. After saving all of the changes, click **Cancel** to close the **Recipe Editor**.

Managing Recipes

Click  in the **Recipe Editor** dialog box. The **Manage Recipe** dialog box opens.



You can perform the following edits to the recipe files in the **Manage Recipe** option:

- [Create Recipes Using Boolean Expressions](#)
- [Delete Multiple Recipes](#)

Create Recipes Using Boolean Expressions

1. Select the name of the required recipe in the **Recipes** box and click  or double-click on the name of the selected recipe to display the name in the **Expression** box.

You can also type the name of the recipe within quotation marks in the **Expression** box. If the name is not in quotation marks, the following error message appears:

Error: Please enter the recipe name in double quotes to create a New Recipe.

2. Select the required Boolean functions **NOT**, **AND**, **OR** from the **Functions** options. The selected function appears in the **Expression** box with the name of the recipe. You can select multiple recipe names and its corresponding Boolean functions to create a new recipe.

If the syntax of the expression is incorrect, the **Expression** box is highlighted, and the following error message appears:

```
Error: The entered Expression syntax in the Manage Recipes dialog is
incorrect. Hence, could not create new recipe.
```

3. Enter the name of the new recipe in the **New Recipe** box. If the **New Recipe** box is empty, the **New Recipe** box is highlighted and the following error message appears:

```
Error: New Recipe in Manage Recipes dialog is a required field. Please
enter recipe name to create a new recipe.
```

4. Click the **Evaluate** button . The new recipe name appears in the **Recipes** box, and the following message appears:

```
Information: new recipe = expression being created and evaluated
successfully
```

5. Click **Save** to save the recipe file in the **Recipe Editor**. The following message appears:

```
Information: Save recipes in the specific file.
```

If the new recipe created does not contain any rules, then the following error message appears:

```
Error: The evaluated Expression doesn't have any rules in common.
Hence, could not create a New Recipe <NewRecipeName>.
```

6. Click **Close** to close the **Manage Recipes** dialog box.

Delete Multiple Recipes

1. Select the required recipes in the **Recipes** box.
2. Click .
3. Click **Close** to close the **Manage Recipes** dialog box.

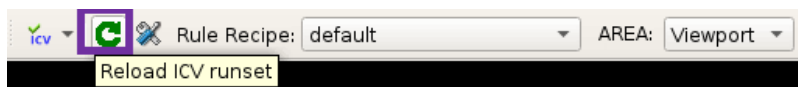
Runset Parsing Time

You can improve the time taken for design rule checks on the runset file or the time taken to check the rules in the recipe of the runset file.

In the Layout Editor, choose **Window > Toolbars > DRD** to display the **ICV Live DRC** toolbar. Use one of the following functions to improve the time for runset parsing:

- [Reload ICV Runset](#)
- [Reload Runset](#)
- [Create or Modify Recipe](#)

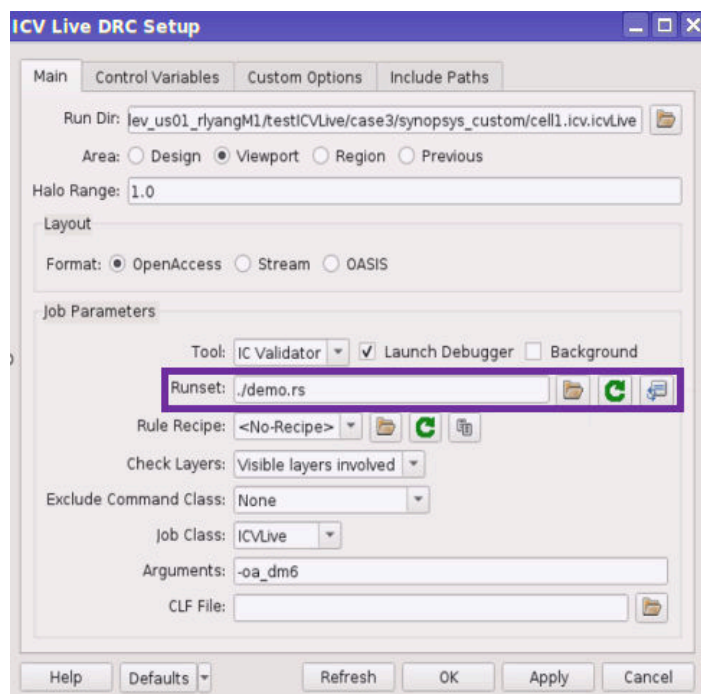
Reload ICV Runset



Click the Reload ICV Runset icon  in the **ICV Live DRC** toolbar when the `xtICVLiveICVRunsetFile` preference or the `xtDRCICVRunsetFile` preference is set to TRUE.

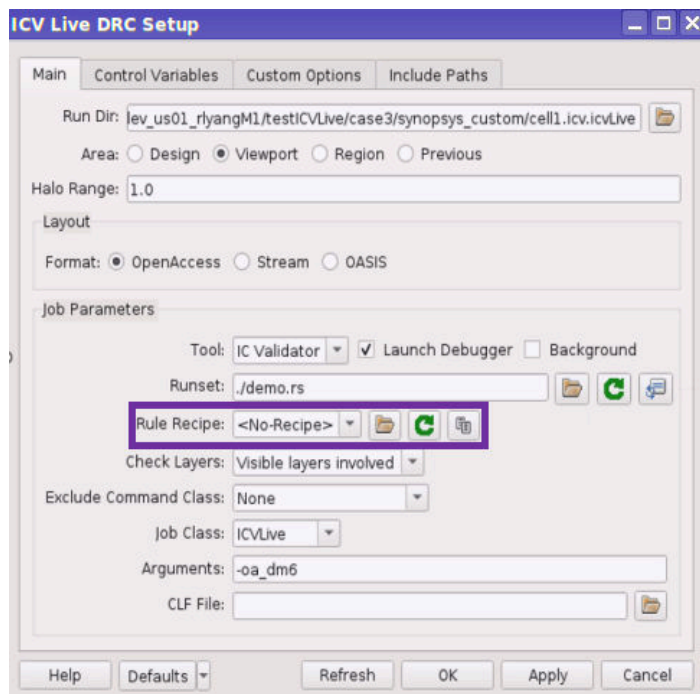
Reload Runset

Click the  icon in the **ICV Live DRC** toolbar to display the ICV Live DRC Setup dialog box.



1. Click the  icon in the **Runset** field to choose the runset file.
2. Click the  icon in the **Runset** field to run the **Reload runset** option.

Create or Modify Recipe



1. (Optional) Click the  icon in the **Rule Recipe** field to create a new recipe in the **Recipe Editor** dialog box.
2. (Optional) Click the  icon in the **Rule Recipe** field to view or edit the recipe in the **Recipe Editor** dialog box.

See: [Creating and Modifying Recipes](#)

Note:

- If you edit the runset file or select a different runset file, you must perform the runset parsing again.
- To load the runset file in the background, select the **Background** option. The following message is displayed after the runset file is loaded:

Complete ICV runset loading. It is ready to check DRC violations by ICV Live.

4

Checking Vias

This chapter explains how to check for missing and suboptimal vias in the design and how to perform non-default width and spacing checks.

This chapter contains the following topics:

- [Checking Vias With the LE Via Checker Dialog Box](#)
- [Checking Vias With the `le::checkVias` Tcl Command](#)
- [Fixing Missing and Suboptimal Vias](#)

Checking Vias With the LE Via Checker Dialog Box

You can use the Via Checker to look for missing or suboptimal vias in your entire design or in a subsection of the design. You use the **LE Via Checker** dialog box to perform via checking.

The Via Checker outputs error markers at the locations where vias are missing or undersized in the Error Viewer.

To check vias using the **LE Via Checker** dialog box:

1. In the Layout Editor, choose **Verification > Via Checker**. The **LE Via Checker** dialog box opens.
2. Set the options in the **LE Via Checker** dialog box.

Chapter 4: Checking Vias

Checking Vias With the LE Via Checker Dialog Box

Figure 5 *LE Via Checker - Reliability Tab*

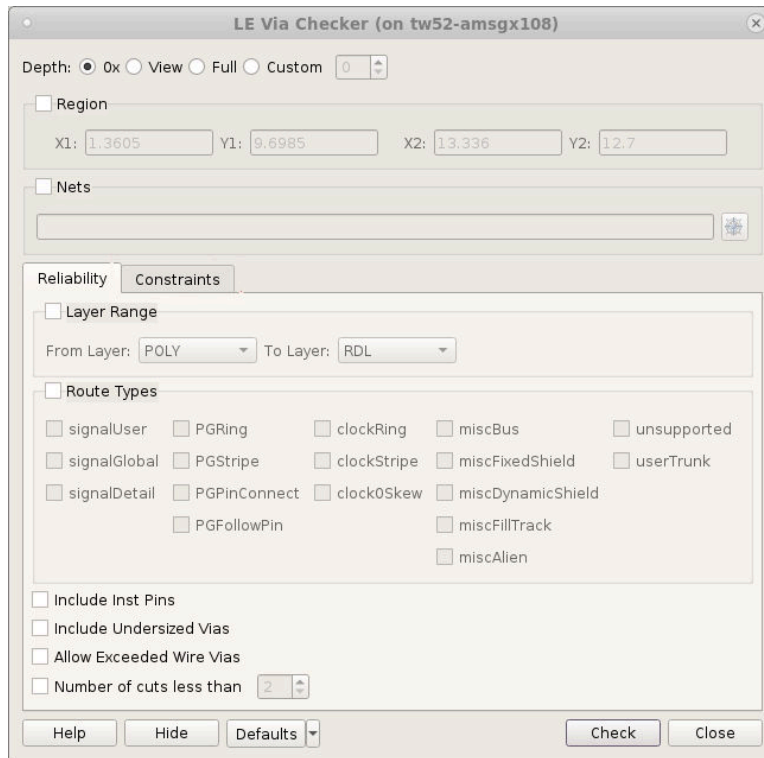


Figure 6 LE Via Checker - Constraints Tab

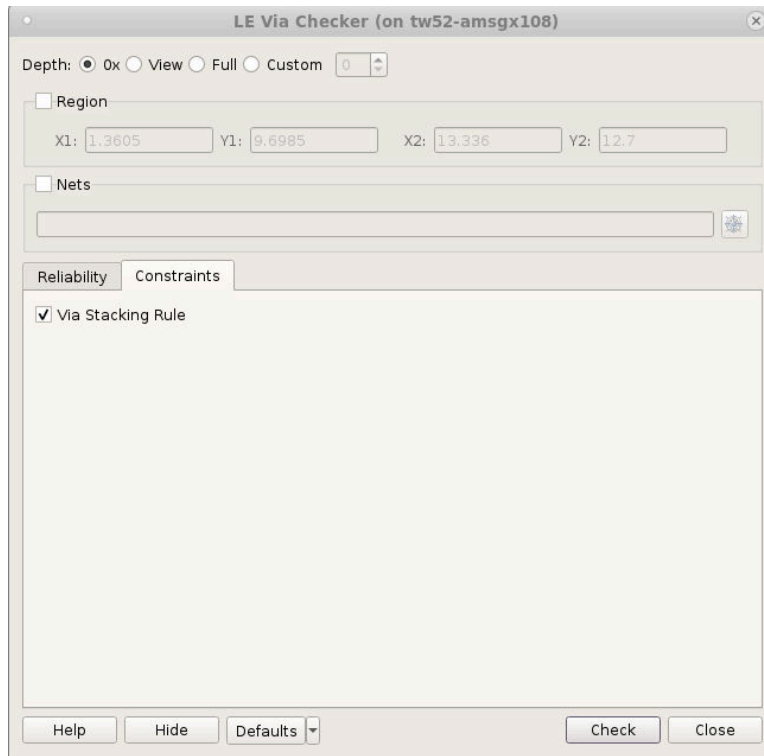
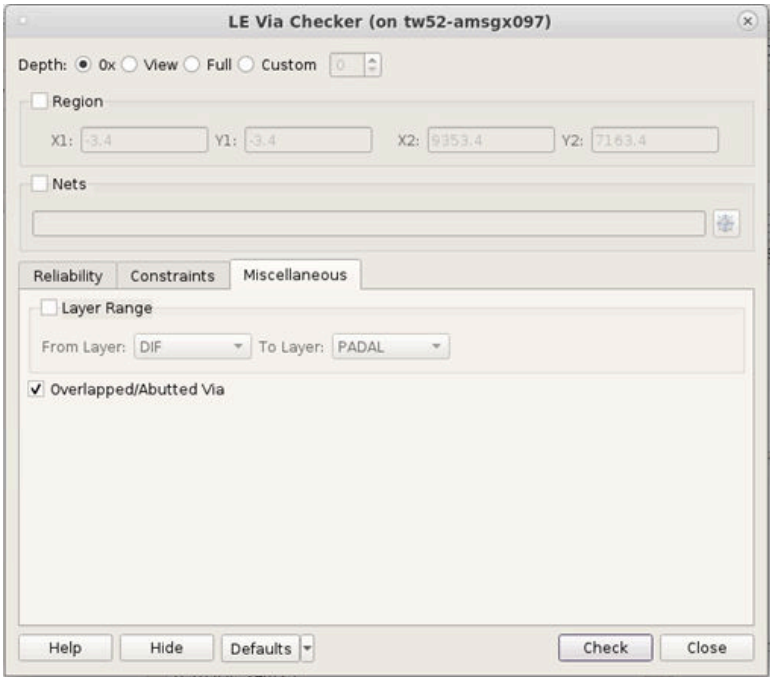


Figure 7 LE Via Checker - Miscellaneous Tab



Option	Description
Depth	The hierarchy levels that you want to check.
Region	The subsection of the design you want to check. Select Region, and then specify the coordinates of the subsection. X1 and Y1 specify the lower-left corner of the boundary box. X2 and Y2 specify the upper-right corner of the boundary box. You can specify the region coordinates using one of the following methods: ◦ Draw the rectangular region on the canvas. Based on your actions on the canvas, the X- and Y- coordinate values automatically populate the X1, X2, Y1, and Y2 boxes. ◦ Type the X- and Y- coordinate values in the X1, X2, Y1, and Y2 boxes.
Nets	The nets you want to search. Click  to open the Design Navigator to show the nets.
Reliability	
Layer Range	The layers you want to check.

Option	Description
Route Types	The route types you want to check. For the list of route types, see ◦ Fusion Compiler user documentation ◦ IC Compiler II user documentation ◦ IC Compiler user documentation Note here that the Custom Route route type is a property of shapes that are added by the Custom Router.
Include Inst Pins	The option is selected to check the instance pins, and is off by default.
Include Undersized Vias	The option is selected to check all undersized vias, and is off by default.
Allow Exceeded Wire Vias	Turns on or off the dropping of the via without full metal coverage. This option is off by default.
Number of cuts less than	Specifies the minimum number of cuts of the via. This option is off by default.
Constraints	
Via Stacking Rule	Checks stacking via rules from the existing design
Miscellaneous	
Layer Range	Specifies the range of layers you want to check.
Overlapped/Abutted Via	Checks the overlapped/abutted vias on the design.

3. When you are finished, click **Apply**, and then click **Close**. The missing via markers are created in the layout and displayed in the Error Viewer (choose **Windows > Assistants > Error Viewer**). See Error Viewer Assistant.

Checking Vias With the `le::checkVias` Tcl Command

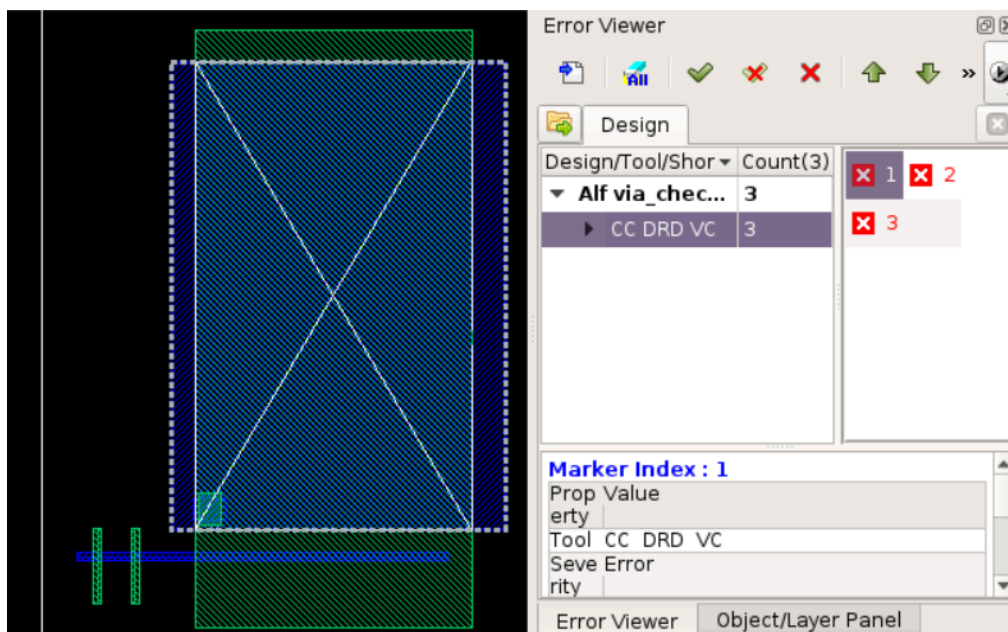
The `le::checkVias` Tcl command has the following features:

- Compares an existing via array with another via array created by auto via.
- Checks the no-via area on different, overlapping layers that have the same netName.
- Shows an error because there is not enough space to add more vias.
- Lets you set the net name, layers, and region of the no-via area you want to check.

For more information, see the *Custom Compiler Tcl Reference Manual*.

Fixing Missing and Suboptimal Vias

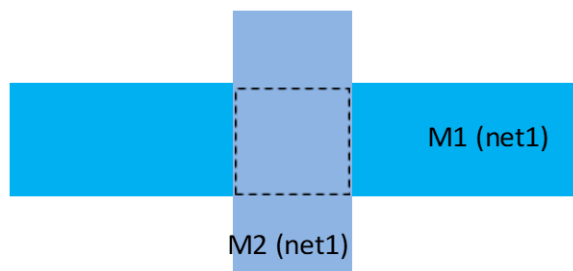
After checking for missing vias or suboptimal vias using the Via Checker, the missing via markers are created in the layout and displayed in the Error Viewer (choose **Windows > Assistants > Error Viewer**). See Error Viewer Assistant.



The details of error messages are shown in the Error Viewer. To fix the errors, you can create vias using the **Create > Vias** command. See Creating and Editing Vias.

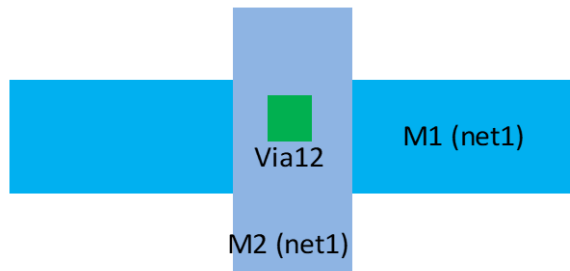
The Via Checker creates two types of via markers:

- Missing via: A via is expected when two paths that are on the same net intersect. Because the error marker is not associated with other objects, you must manually delete the marker after fixing the error.



- Undersized via: A via connects two intersecting paths that are on the same net. However, if the intersecting region can accommodate a larger via, an error marker is

created and associated with the undersized via. After making adjustments, the error marker is automatically deleted.



After fixing the errors, you must delete the missing via markers in the Error Viewer Assistant before using the Via Checker again. The undersized via markers are automatically cleared up when the associated object is modified or destroyed.

5

Net-Based Checkers

The following net-based checkers use IC Validator to provide these functions:

- Net-based checking. Performs non-default width and spacing checks based on a user-specified set of non-default dimensions by layer.
- Net-based via checking. Performs via coverage for overlapped top and bottom metal layers.
- Net-based shield checking. Performs closed loop and maximum spacing of shielding nets and forbidden regions.

Each Net-based check generates a separate file.

Note:

The net-based checkers are compatible only with the IC Validator U-2022.12-SP-4 and previous versions.

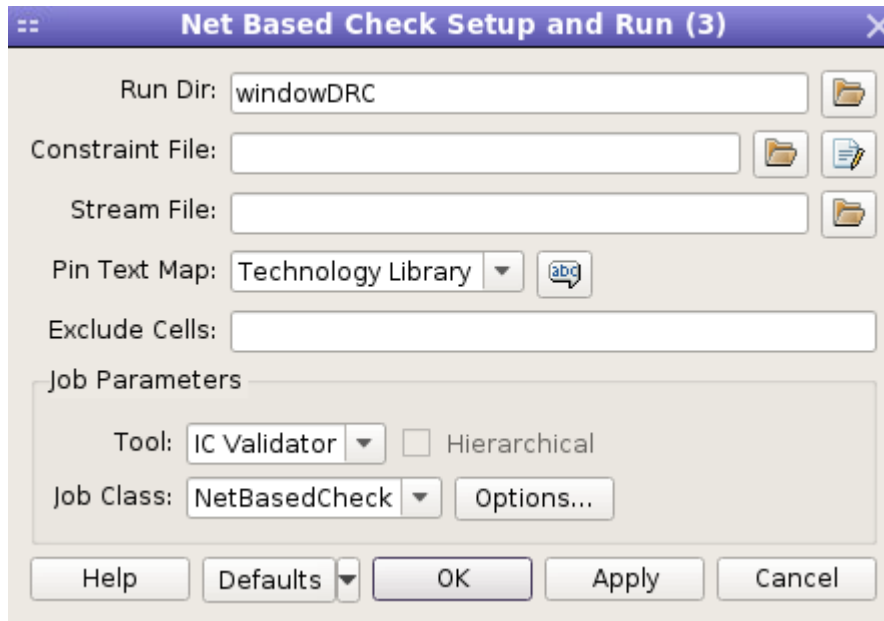
This chapter contains the following topics:

- [Setting Options for Net-Based Checking](#)
- [Setting Options for Net-Based Via Checking](#)
- [Setting Options for Net-Based Shield Checking](#)

Setting Options for Net-Based Checking

To set the options used for net-based checking:

1. In the Layout Editor, choose **Verification > Net Based Check**. The **Net Based Check Setup and Run** dialog box opens.
2. Set the options in the **Net Based Check Setup and Run** dialog box.



The dialog box titled "Net Based Check Setup and Run (3)" contains the following fields and controls:

- Run Dir:** A text field containing "windowDRC" with a folder icon to its right.
- Constraint File:** A text field with a folder icon and a document icon to its right.
- Stream File:** A text field with a folder icon to its right.
- Pin Text Map:** A dropdown menu showing "Technology Library" and a small icon to its right.
- Exclude Cells:** A text field.
- Job Parameters:** A section containing:
 - Tool:** A dropdown menu showing "IC Validator" and an unchecked checkbox labeled "Hierarchical".
 - Job Class:** A dropdown menu showing "NetBasedCheck" and an "Options..." button.
- At the bottom are buttons for "Help", "Defaults" (with a dropdown arrow), "OK", "Apply", and "Cancel".

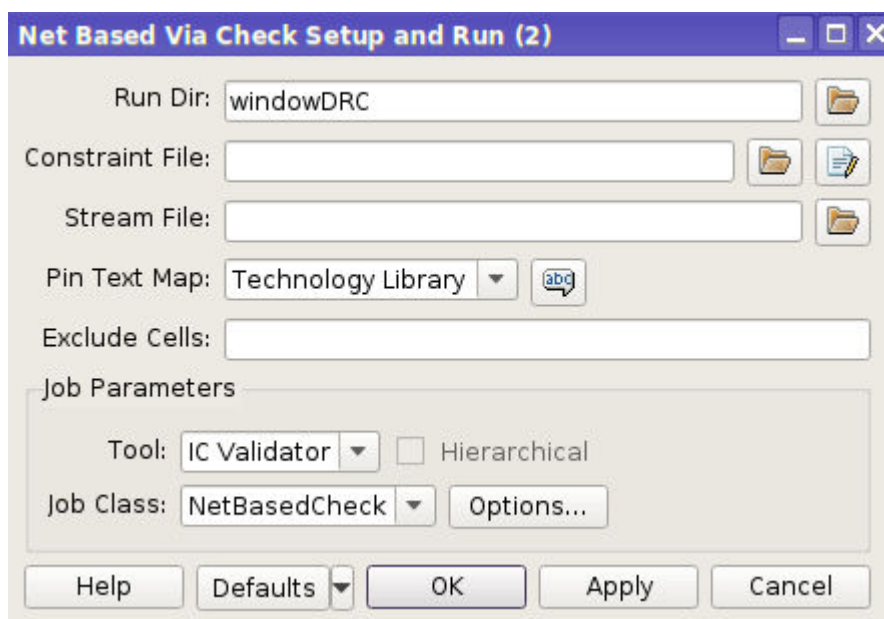
Option	Description
Run Dir	Set the <code>xtNetBasedCheckRunDir</code> Tcl preference to specify the run directory. The default directory is windowDRC .
Constraint File	Defines net, layer, width, and spacing values: <pre>#Net Layer Width SpacingAl M1 0.044 0.022</pre> Set the <code>xtNetBasedCheckConstraintFile</code> Tcl preference to specify the constraint file. The default directory is "".
Stream File	Selects the GDS file for the net-based check. If the field is empty, the option exports the design to a GDS file and uses the file for the net-based check.
Pin Text Map	Supports both the Technology Library and User Specified File, which contains the following sequence of shape, label, and pin layer purpose layers: <pre>M1 drawing M1 pin M1 pin</pre> Set the <code>xtNetBasedCheckPinTextMap</code> for the Technology Library and User Specified file. The default is Technology Library. Set the <code>xtNetBasedCheckPinTextMapFile</code> for the pin text map file when <code>xtNetBasedCheckPinTextMap</code> is User Specified File.
Exclude Cells	Specifies layout cells that are not to be processed by the net-based check. You can input the cells that are to be excluded. For example, <code>Res* AA</code> .
Job Parameters	Tool is IC Validator or Calibre. Set the <code>xtNetBasedCheckTool</code> Tcl Preference to choose one of these tools. The default is IC Validator.

Option	Description
	Set the <code>xtNetBasedCheckHierMode</code> Tcl preference in the Net Based Check Setup and Run dialog to run in Hierarchical Mode. Supports queue by Job Class > Options. Combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see Adding Job Classes With <code>xt::createJobClass</code> Tcl in the <i>Custom Compiler Environment User Guide</i>. Automatically loads results in Error Viewer when ◦ Highlighting net distribution ◦ Highlighting net-based width and spacing violations

Setting Options for Net-Based Via Checking

To set the options used for net-based via checking:

1. In the Layout Editor, choose **Verification > Net Based Via Check**. The **Net Based Via Check Setup and Run** dialog box opens.
2. Set the options in the **Net Based Via Check Setup and Run** dialog box.

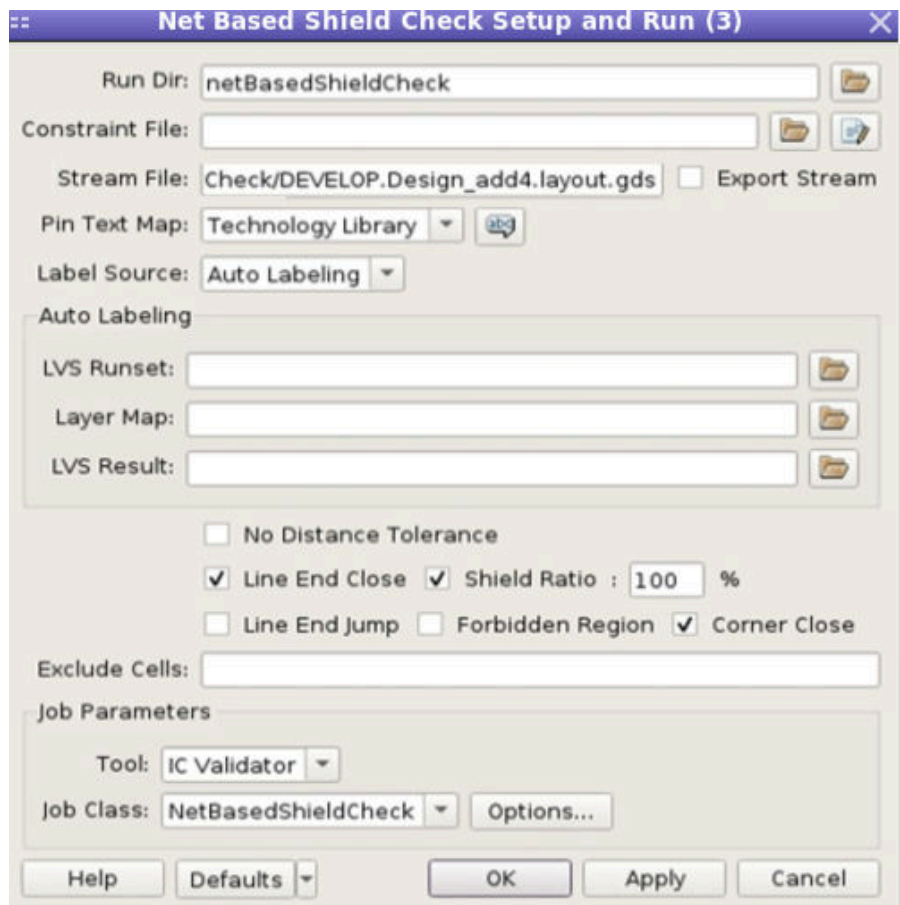


Option	Description
Run Dir	Set the <code>xtNetBasedViaCheckRunDir</code> Tcl preference to specify the run directory. The default directory is windowDRC .
Constraint File	Defines net, bottom/top/cut layers and cut ratio value: <pre>#Net Bottom_Layer Top_Layer Cut_Layer Cut_RatioA M1 M2 VIA12 0.15</pre> Set the <code>xtNetBasedViaCheckConstraintFile</code> Tcl preference to specify the constraint file. The default directory is "".
Stream File	Selects the GDS file for the net-based via check. If the field is empty, the option exports the design to a GDS file and uses the file for the net-based via check.
Pin Text Map	Supports both the Technology Library and User Specified File, which contains the following sequence of shape, label, and pin layer purpose layers: <pre>M1 drawing M1 pin M1 pin</pre> Set the <code>xtNetBasedViaCheckPinTextMap</code> for the Technology Library and User Specified file. The default is Technology Library. Set the <code>xtNetBasedViaCheckPinTextMapFile</code> for the pin text map file when <code>xtNetBasedViaCheckPinTextMap</code> is User Specified File.
Width Value	Supports the Width Value threshold to filter narrow wires. Set the <code>xtNetBasedViaCheckWidthValue</code> Tcl preference to check the width value of the region. The default is 0.
Exclude Cells	Specifies layout cells that are not to be processed by the net-based check. You can input the cells that are to be excluded. For example, <code>Res* AA</code> .
Job Parameters	Tool is IC Validator or Calibre. Set the <code>xtNetBasedViaCheckTool</code> Tcl Preference to choose one of these tools. The default is IC Validator. Supports queue by Job Class > Options. Combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see Adding Job Classes With <code>xt::createJobClass</code> Tcl in the <i>Custom Compiler Environment User Guide</i>. Automatically loads results in Error Viewer when ◦ Highlighting net distribution ◦ Highlighting via coverage violations ◦ Highlighting missing vias (<code>via_ration = 0</code>)

Setting Options for Net-Based Shield Checking

To set the options used for net based shield checking:

1. In the Layout Editor, choose **Verification >Net Based Shield Check**. The **Net Based Shield Check Setup and Run** dialog box opens.
2. Set the options in the **Net Based Shield Check Setup and Run** dialog box.



Option	Description
Run Dir	Set the <code>xtNetBasedShieldCheckRunDir</code> Tcl preference to specify the run directory. The default directory is <code>netBasedShieldCheck</code> .

Option	Description
--------	-------------

Constraint File Specifies the constraint file. Set the `xtNetBasedShieldCheckConstraintFile` Tcl preference to specify the constraint file. The default is "".

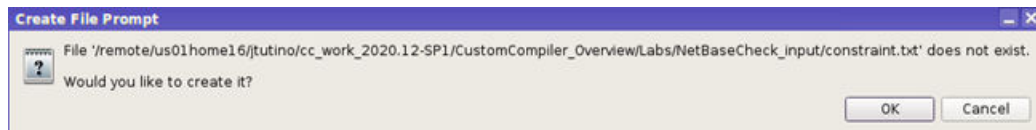
```
# Net Layer Width Spacing ShieldNets Forbidden_Region_Check ShieldPurposes
A0 m1 0.07 0.07 *
A0 m2 0.07 0.07 { shield_group_1 gnd! }
B0 m2 0.07 0.07 { I0/net62 } { forbidden_group_1 } { ShieldPurposes { drawing }
A1 m2 0.07 0.07 { I1/net62 } { ShieldPurposes { drawing }

#Forbidden_Region
forbidden_group_1 = { mtcap drawing }
forbidden_group_2 = { {mtcap drawing} {nat drawing} }

#ShieldingGroup
shield_group_1 = { I0/net62 Design_add4/I1/net62 }

#ShieldDesignRules
Shield m1 0.07 0.07
Shield m2 0.07 0.07
Shield m3 0.07 0.07
```

Click  to edit the net constraint file. If the constraint file does not exist, a **Create File Prompt** message appears:



When you click **OK**, the file is automatically created.

Specify the critical nets in this section:

- **Net:** Critical net name. Format is schematic net.
- **Layer:** Layer name of the critical net.
- **Width:** Width of the critical net.
- **Spacing:** Spacing of the critical net.
- **ShieldNets:** Shield nets for critical net. The wildcard character (*) is supported. All layer metal shapes can perform shielding. See [Setting Options for Net-Based Shield Checking](#).
- **Forbidden_Region_Check:** Optional. Forbidden region for a critical net. Group content are LPPs.
- **ShieldPurposes:** Optional. Specifies the layer purpose if the critical net is different from the layer of the shielding net. The format is { ShieldPurposes { <Purpose List> } }. For example, see line 4 of the constraint file figure. Critical net A1 and the shielding net could be in one of { m2 drawing }, { m2 dr8 }, { m2 dr9 } and { m2 dr10 }.
- **ShieldDesignRules:** Specify the shielding rules in this section. The prefix of each rules should be "Shield", and the rule format of shielding nets is "layer, width, spacing".

Stream File Set the `xtNetBasedShieldCheckLayStreamFile` preference to specify the stream file.
The **Export Stream** option exports the stream file without checking for existing stream files.

Option	Description
Pin Text Map	Specifies the Technology Library or User Specified File to use when choosing the pin text map. Set the <code>xtNetBasedShieldCheckPinTextMap</code> preference for the Technology Library and User Specified file. The scope is user and the default is Technology Library. When Pin Text Map is Technology Library, you must ensure all checked nets have labels in the design. Click  to shoe the pin text map content in this library. The content is shown in the Custom Compiler console. When Pin Text Map is User Specified File, you must ensure all checked nets have labels in the design, and that the mapping file is specified. Set the <code>xtNetBasedShieldCheckPinTextMapFile</code> preference. The scope is user and the default is "". Click  to edit or create a file. The Create File Prompt dialog box appears. Click OK to create a file with the following content with format: <code>genTextFromPin {text layer} {metal layer}</code>.

```

vi (on tw52-amsgrx109)
1 Use ; or # to comment out.
2 #Support below 3 formats:
3
4 #Format1:
5 #genTextFromPin { text layer } { shape layer }
6 #genTextFromPin { M1 pin } { M1 }
7
8 #Format2: (pin lpp is not used in below format)
9 #( (shape lpp) (text lpp) (pin lpp) )
10 #( (M1) (M1 text) (M1 pin))
11 #( (M1 drawing) (M1 text) )
12
13 #Format3: (pinLayer pinPurpose is not used in below format)
14 #shapeLayer shapePurpose textLayer textPurpose pinLayer pinPurpose
15 #M1 drawing M1 text M1 pin
16 #M1 drawing M1 text
17
~
~
~

```

Option	Description
Label Source	Specifies the label source. Preference name is <code>xtNetBasedShieldCheckLabelSource</code> and the scope is user scope. The default is Auto Labeling. When Label Source is Auto Labeling, the input box of the following three files appear. ◦ LVS Runset. Specifies the runset used for running LVS. Set the <code>xtNetBasedShieldCheckLVSRunset</code> preference to specify the LVS runset. The scope is user. ◦ Layer Map. Specifies the layer map file used for running LVS. Set the <code>xtNetBasedShieldCheckLVSLayerMap</code> preference to specify the layer map. The scope is user. ◦ LVS Result. Run before the net based shield check. Use <i>Block.vue</i> to perform IC Validator LVS database queries to transform the information in the runset. Set the <code>xtNetBasedShieldCheckLVSRresult</code> preference to specify the LVS results. The scope is user and the default is "". You must provide these three files for auto labeling.
Shield checking	Choose the following settings: ◦ Corner Close: Specifies that the net based shield follows the Corner Close rule. Set the <code>xtNetBasedShieldCheckCornerClose</code> preference. The scope is user and the default is true. ◦ Shield Ratio: Specifies that the net based shield follows the Shield Ratio rule. You can specify the shield ratio percentage in the neighboring text input option. By default, the shield ration percentage is 100%. A reasonable value is from 51.0% to 100.0%. Set the <code>xtNetBasedShieldCheckShieldRatio</code> preference. The scope is user and the default is 100%. ◦ Line End Close: Specifies that the net based shield follows the Line End Close rule. Set the <code>xtNetBasedShieldCheckLineEndClose</code> preference. The scope is user and the default is true. ◦ Line End Jump: Specifies that the net based shield follows the Line End Jump rule. Set the <code>xtNetBasedShieldCheckLineEndJump</code> preference. The scope is user and the default is false. ◦ Forbidden Region: Specifies that the net based shield follows the Forbidden Region rule. Set the <code>xtNetBasedShieldCheckForbiddenRegion</code> preference. The scope is user and the default is false. If No Distance Tolerance is checked, the shielding distance between the critical and shielding nets is equal to the value defined in the constraint file. Set the <code>xtNetBasedShieldCheckNoDistTolerance</code> preference. The scope is user and the default is false.
Exclude Cells	Specifies layout cells that are not to be processed by the net-based check. You can input the cells that are to be excluded. For example, <code>Res* AA</code>. Set the <code>xtNetBasedShieldCheckExcludeCells</code> preference. The scope is user and the default is "".
Job Parameters	Tool is IC Validator. Set the <code>xtNetBasedShieldCheckTool</code> preference. The scope is user and the default is IC Validator.

Option	Description
	Set the <code>xtNetBasedCheckHierMode</code> Tcl preference in the Net Based Check Setup and Run dialog to run in Hierarchical Mode .
	Supports queue by Job Class > Options . Combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see Adding Job Classes With <code>xt::createJobClass</code> Tcl in the <i>Custom Compiler Environment User Guide</i> .
	Automatically loads results in Error Viewer when ◦ Highlighting net distribution ◦ Highlighting net-based width and spacing violations

Note:

To run the Net Based Shield Check in the background set the `xtNetBasedShieldCheckRunInBackground` preference to TRUE.

Defining Shielding Nets

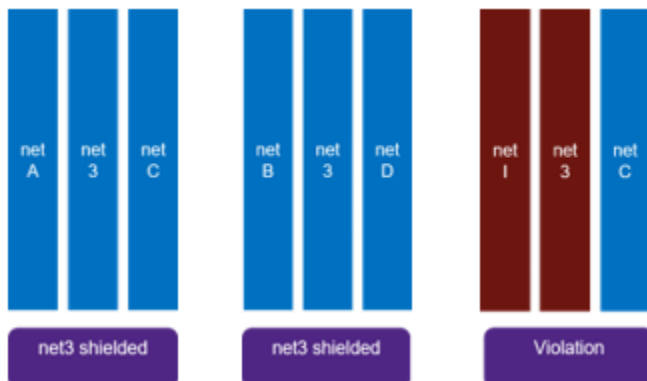
Shielding nets protect critical nets. Given the shielding distance, the distance between the critical nets and shielding nets is less than or equal to the shielding distance, and activates shielding by default. For example, if there is a shielding group defined as

netA, netB, netC > shielding group A (sh_groupA)

netB, netD, netH > shielding group B (sh_groupB)

netI, netG > shielding group C (sh_groupC)

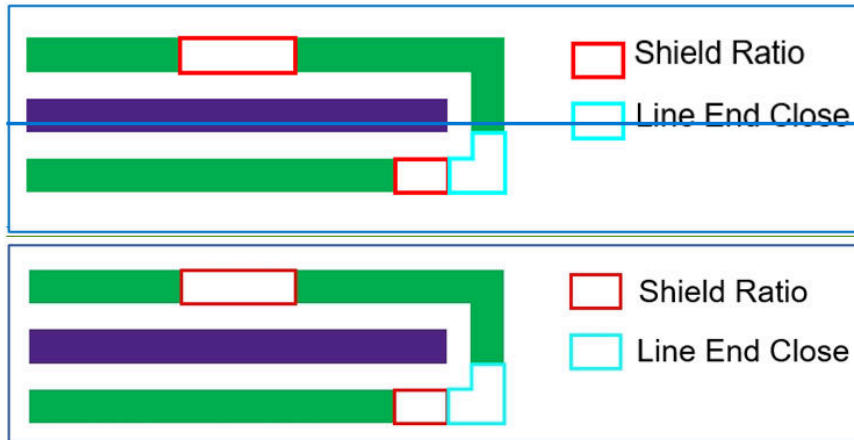
then, net3 uses sc_groupA/sh_groupB for shielding.



Violation Types

There are five violation types, as follows:

- **Line End Close:** Checks the shapes that come out of the critical net projection.
- **Shield Ratio:** Calculates only the projection direction. The formula is $(1 - (\text{sum of open length in both side shields}) / (\text{two side length of the critical net})) * 100\%$. The specific violations are shown in the following figure:



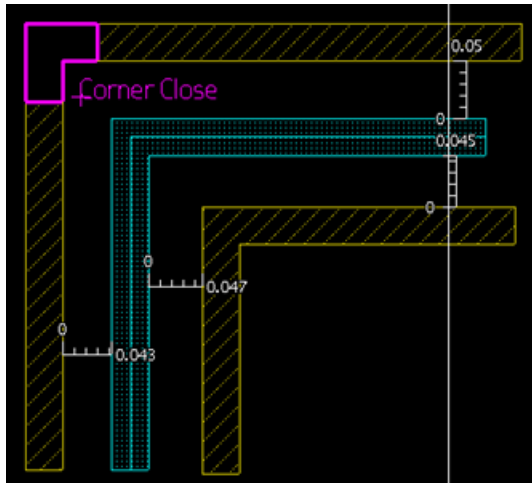
- **Line End Jump:** Specifies a VIA in the region 1 metal width from the line end.



- **Forbidden Region:** Not allowed for specific nets to be routed. For example, if there is a forbidden region {UM1 label} that is specified for critical net A0, it is reported as a forbidden region violation.



- **Corner Close:** Checks the corner opening of a critical net.



6

Checking Density

This chapter explains how to check color and density using the Density dialog box.

This chapter contains the following topics:

- [About the Density Analyzer](#)
- [Workflow for Checking the Density](#)
- [Preparing the Configuration File](#)
- [Setting Up the Density Analysis](#)
- [Manually Defining Density Methods](#)
- [Starting the Density Analysis](#)
- [Viewing the Density Analysis Results](#)

About the Density Analyzer

To improve the manufacturability of advanced node chips (such as those containing FinFETs), foundries require that designs meet certain density rules and constraints. Meeting these rules and constraints ensure uniform distribution of components across metal layers. Compliance with these rules and constraints can also result in better signal timing, improved power integrity, and reduced electromigration.

You can use the Layout Editor density analyzer to check the density of your design. The following sections describe how to use the analyzer:

- [Density Calculations](#)
- [Windows and Steps](#)
- [Analysis Results](#)

Density Calculations

The density analyzer calculates density based on one of the following:

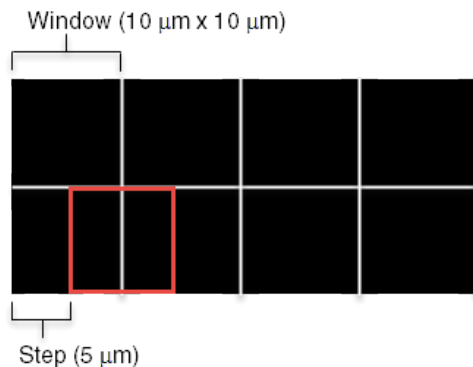
- Layer: By default, the analyzer calculates density by layer. Because each colorable layer can have more than one color, density is calculated for each of the colors, as well as for All Colors and No Color.
- Rules that you define: Instead of calculating the density by layer, you can define your own rules in a configuration file. For the configuration file requirements, see [Preparing the Configuration File](#).

Windows and Steps

The analyzer divides a design into subsections, or windows, and checks each window for density violations. You can specify the size of the windows. You can also specify the size of the step the analyzer takes toward the next window.

In [Figure 8](#), the window size is 10 mm x 10 mm, and the step size is 5 mm. After checking the density in the first window, the analyzer moves forward by 5 mm and performs the density check within the new 10 mm x 10 mm window. The analyzer continues this way until the entire design is checked.

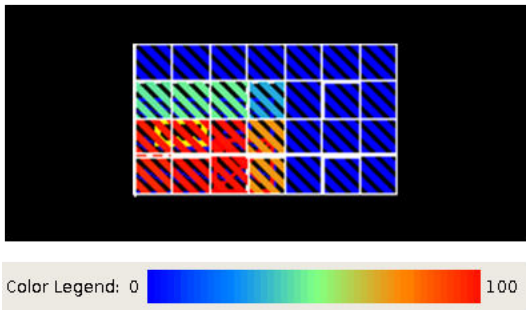
Figure 8 *Density Analyzer Window and Step*



Analysis Results

The analysis results are displayed on the canvas ([Figure 9](#)). A color scale is used indicate the density level. You can set viewing options to filter the results. For more information, see [Viewing the Density Analysis Results](#) and [Analyzing and Viewing Density Using Density Assistant](#).

Figure 9 *Density Analysis Results*



Workflow for Checking the Density

The following table shows the high-level workflow for checking the density.

[Show me](#) an example of how to check design density.

Step	Description
1	Prepare for density analysis. See Preparing the Configuration File .
2	Set up the density analysis. See Setting Up the Density Analysis .
3	Start the density analysis. See Starting the Density Analysis .
4	View the analysis results. See Viewing the Density Analysis Results .

Preparing the Configuration File

Preparing the configuration file is an optional step in the workflow that you need to perform only when checking density using the **Config File** calculation method.

Before you start the density analysis using this optional step:

- Determine whether you want the analyzer to calculate density by layer or by groups that you define.
- If you want to define custom groups, be sure to define the custom analysis groups in a configuration file.

The configuration file must be an ASCII file. Each line in the file describes one of the following density calculation rules:

- [Normal Density Calculation](#)
- [Same-Group Expression Calculation](#)
- [Multiple-Patterning Calculation](#)
- [Single-Patterning Calculation](#)

Normal Density Calculation

The normal density calculation computes the ratio of the shape areas on the specified layers to the check region areas.

The syntax of the normal density calculation is as follows:

```
groupName ruleName 1 {{checkLayerMap}...} {{excludeLayerMap}...}  
{{checkRegionLayerMap}{excludeRegionLayerMap}...} windowSize stepSize  
minDensity maxDensity
```

Field	Description
groupName	The name of the group. For example, if it is a layer, you would specify the layer name in this field. The name can consist of alphanumeric characters. The only special character permitted is the underscore (_).
ruleName	The name of the density calculation rule. For example, you might want to use the name M1 to indicate that the rule calculates density of shapes on the M1 layer. The name can consist of alphanumeric characters. The only special character permitted is the underscore (_).
1	The density calculation rule number. 1 indicates the rule is for normal density calculation.
checkLayerMap	The shapes to include in the density calculation. Use this field to filter for shapes to include. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status

Field	Description
excludeLayerMap	The shapes to exclude from the density calculation. Use this field to filter for shapes to exclude. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
checkRegionLayerMap	The region to include in the density calculation. Use this field to filter for shapes to include. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
excludeRegionLayerMap	The region to exclude from the density calculation. Use this field to filter for shapes to exclude. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
windowSize	The size of the Window. For more information, see Windows and Steps .
stepSize	The step size. For more information, see Windows and Steps .
minDensity	The minimum density value you want to include.
maxDensity	The maximum density value you want to include.

In the following example:

```
M1 M1 1 {{M1 drawing 31 250} {M1 drawing 31 255 color:colorA Locked}}
{{M1 dummyA 31 200}} {} {} 5.0 5.0 25.0 80.0
```

- **M1** is the group name. In this example, it is also the layer name.
- **M1** is the name of the rule. In this example, it indicates that the rule applies to the M1 layer.
- **1** indicates that this is the normal density calculation rule.
- **{{M1 drawing 31 250} {M1 drawing 31 255 color:colorA Locked}}** includes shapes on the M1 drawing layer, whose layer number is 31 and layer purpose numbers are

250 and 255. For shapes on layer M1 drawing 31 255, include only shapes who have colorA with the color status of Locked.

- `{{M1 dummyA 31 200}}` excludes shapes that are on the M1 dummyA 31 200 layer are excluded from the calculation.
- `5.0` is the window size.
- `5.0` is the step size.
- `25.0` is the minimum density value.
- `80.0` is the maximum density value.

Same-Group Expression Calculation

The same-group density calculation allows you to use an expression that operates on density data from the same group. You do not need to specify the window or step size.

The syntax of the same-group density calculation is as follows:

```
groupName ruleName 2 {expression} minDensity maxDensity
```

Field	Description
groupName	The name of the group. For example, if it is a layer, you would specify the layer name in this field. The name can consist of alphanumeric characters. The only special character permitted is the underscore (<code>_</code>).
ruleName	The name of the density calculation rule. For example, you might want to use the name M1 to indicate that the rule calculates density of shapes on the M1 layer. The name can consist of alphanumeric characters. The only special character permitted is the underscore (<code>_</code>).
2	The density calculation rule number. 2 indicates the rule is for same-group expression calculation.
minDensity	The minimum density value you want to include.
maxDensity	The maximum density value you want to include.

In the following example:

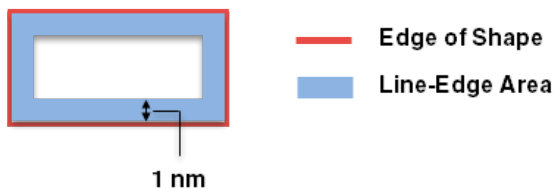
```
M1 A_B 2 {abs($ColorA-$ColorB)} 22.0 90.0
```

- `M1` is the group name. In this example, it is also the layer name.
- `A_B` is the name of the rule. In this example, the computed results from groups A and B are used in the expression.

- 2 indicates that this is the same-group expression calculation rule.
- $\{\text{abs}(\$ColorA-\$ColorB)\}$ calculates the absolute value of the difference between ColorA and ColorB density values.
- 22.0 is the minimum density value.
- 90.0 is the maximum density value.

Multiple-Patterning Calculation

The multiple-patterning calculation uses the area at the edges of shapes to compute the density values. The following figure shows how the line-edge area of each shape is determined. The density value is the ratio of the line-edge areas to the check region areas.



The syntax of the multiple-patterning calculation is as follows:

```
groupName ruleName 3 {{MxLayerMap} {MxCALayerMap}
{MxCBLayerMap}skipValue} {{TCDDMY_MXLayerMap}} windowSize stepSize
minDensity maxDensity
```

Field	Description
groupName	The name of the group. For example, if it is a layer, you would specify the layer name in this field. The name can consist of alphanumeric characters. The only special character permitted is the underscore (_).
ruleName	The name of the density calculation rule. For example, you might want to use the name M1 to indicate that the rule calculates density of shapes on the M1 layer. The name can consist of alphanumeric characters. The only special character permitted is the underscore (_).
3	The density calculation rule number. 3 indicates the rule is for multiple-patterning density calculation.

Field	Description
MxLayerMap	The No Color shapes to include in the density calculation. Use this field to filter for shapes to include. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
MxCALayerMap MxCBLayerMap	The ColorA and ColorB shapes to include in the density calculation. Use this field to filter for shapes to include. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
skipValue	The length of the line-edge area. If the length of any edge is larger than the specified value, omit the density data for that shape. skipValue lets you omit shapes that are too small.
TCDDMY_MxLayerMap	The shapes to exclude in the density calculation. Use this field to filter for shapes to exclude. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
windowSize	The size of the Window. For more information, see Windows and Steps .
stepSize	The step size. For more information, see Windows and Steps .
minDensity	The minimum density value for the rule. Together with maxDensity, the range defines the density region to display.
maxDensity	The maximum density value for the rule. Together with minDensity, the range defines the density region to display.

In the following example:

```
M2 D2_DN9 3 {{M2 drawing 32 250} {M2 drawing 32 255 color:ColorA Locked}
{M2 drawing 32 256 color:ColorB Locked} 0.08} {{M2 dummy1 32 200}} 10.0
10.0 25.0 80.0
```

- M2 is the group name. In this example, it is also the layer name.
- D2_DN9 is the name of the rule. In this example, it indicates that the rule applies to the M2 layer.

- 3 indicates that this is the multiple-patterning density calculation rule.
- {{M2 drawing 32 250} {M2 drawing 32 255 color:ColorA Locked} {M2 drawing 32 256 color:ColorB Locked}} includes the shapes that are on the M2 drawing layer, with the specified layer number, layer-purpose number, color, and color-lock status.
- 0.08 is the width threshold for omitting the shape in the density calculation.
- {{M2 dummy1 32 200}} excludes shapes on the M2 dummy1 layer, whose layer number is 32 and layer-purpose number is 200.
- 10.0 is the window size.
- 10.0 is the step size.
- 25.0 is the minimum density value.
- 80.0 is the maximum density value.

Single-Patterning Calculation

The single-patterning calculation uses the area at the edges of shapes to compute the density values. The following figure shows how the line-edge area of each shape is determined. The density value is the ratio of the line-edge areas to the check region areas.



The syntax of the rule is as follows:

```
groupName ruleName 4 {{MxLayerMap} skipValue} {{TCDDMY_MXLayerMap}}
windowSize stepSize minDensity maxDensity
```

Field	Description
groupName	The name of the group. For example, if it is a layer, you would specify the layer name in this field. The name can consist of alphanumeric characters. The only special character permitted is the underscore (_).
ruleName	The name of the density calculation rule. For example, you might want to use the name M1 to indicate that the rule calculates density of shapes on the M1 layer. The name can consist of alphanumeric characters. The only special character permitted is the underscore (_).

Field	Description
3	The density calculation rule number. 3 indicates the rule is for multiple-patterning density calculation.
MxLayerMap	The shapes to include in the density calculation. Use this field to filter for shapes to include. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
skipValue	The length of the line-edge area. If the length of any line-edge area is larger than the specified value, omit the density data for that shape. skipValue lets you omit shapes that are too small.
TCDDMY_MxLayerMap	The shapes to exclude in the density calculation. Use this field to filter for shapes to exclude. You can specify the following: • Layer name • Layer purpose • Layer number • Layer purpose number • Color name • Color lock status
windowSize	The size of the Window. For more information, see Windows and Steps .
stepSize	The step size. For more information, see Windows and Steps .
minDensity	The minimum density value you want to include.
maxDensity	The maximum density value you want to include.

In the following example:

```
M3 D3_DN9 4 {{M3 drawing 33 250} 0.08} {{M3 dummy1 33 200}} 20.0 20.0
25.0 80.0
```

- M3 is the group name. In this example, it is also the layer name.
- D3_DN9 is the name of the rule. In this example, it indicates that the rule applies to the M3 layer.
- 4 indicates that this is the single-patterning density calculation rule.
- {{M3 drawing 33 250}} includes the shapes that are on the M3 drawing layer, whose layer number is 33 and layer-purpose number is 250.
- 0.08 is the width threshold for omitting the shape in the density calculation.

- `{{M3 dummy1 33 200}}` excludes the shapes that are on the M3 dummy1 layer, whose layer number is 33 and the layer-purpose number is 200.
- `20.0` is the window size.
- `20.0` is the step size.
- `25.0` is the minimum density value.
- `80.0` is the maximum density value.

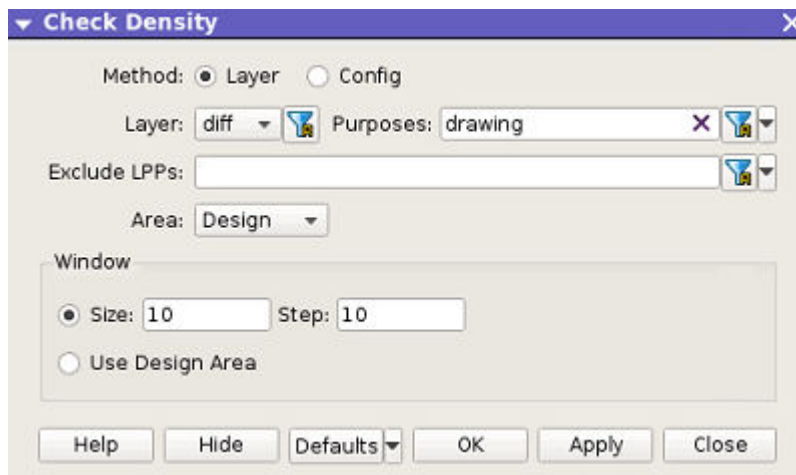
Setting Up the Density Analysis

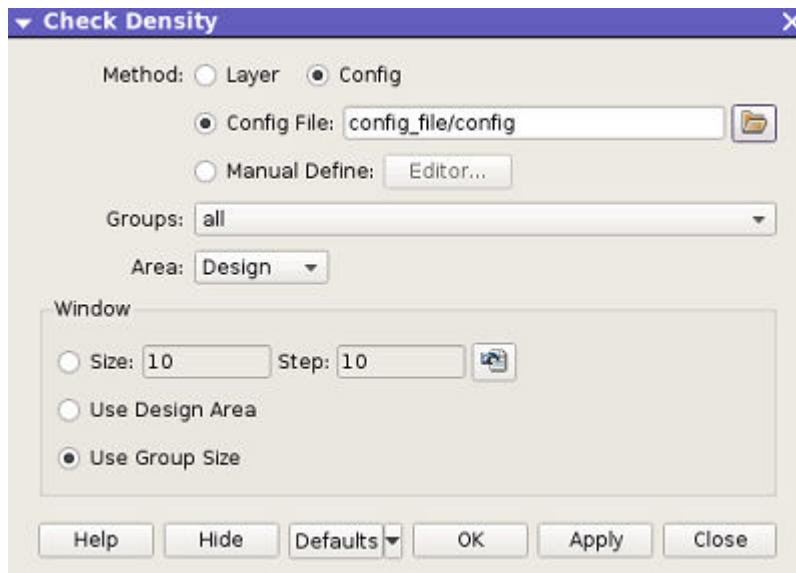
This topic describes how to set up density analysis using the **Density** dialog box. For information on using Tcl commands and preferences, see the *Custom Compiler Tcl Reference Manual*.

To set up the density analysis:

1. In the Layout Editor, choose **View > Check Density**. Alternatively, click **Check Density** in the Density Assistant. The **Check Density** dialog box opens.
2. Select the calculation **Method**:

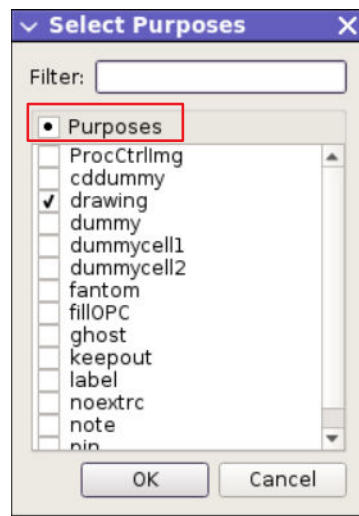
Calculation Method	Description
Layer	Performs the density calculation by layer.
Config	Performs the density calculation by the method you specify in the configuration file.





3. (Layer calculation method only) In the **Layer** list, choose the layer that you want to analyze.
4. (Layer calculation method only) In the **Purposes** field, specify purposes to calculate the density in layer mode. The wildcard character (*) and multiple purposes are supported. If this field is empty, all purposes are calculated. When you can click , the history of the specified purposes is displayed.

Or you also can click  to enable the **Select Purposes** dialog box. All purposes related to the layer are listed.



You can quickly click **Purposes** to select all purposes. In the **Filter** box, type the full or partial text to search for the items of interest.

5. (Layer calculation method only) In the **Excluded LPPs** field, specify the layer-purpose pairs (LPPs) that you want to exclude from the analysis using one of the following:

- Enter the LPPs in the field, using the following syntax:

```
{layer1 purpose1}...{layer $n$  purpose $n$ }
```

Enclose each LPP in braces, and separate multiple entries with a single character space.

Note:

You can quickly enter multiple LPPs in the field in the following ways:

- Enter only the layer in the Excluded LPPs field, without braces, when the layer purpose is drawing. For example, you can enter {M2 pin} M1 M3 to mean {M2 pin} {M1 drawing} {M3 drawing}.
- When applicable, specify a wildcard for the layer or purpose. For example, entering {M* pin} retrieves all LPPs that begin with “M” and have pin as their purpose, such as {M1 pin} {M2 pin}... {M n pin}. Entering {M1 drawing*} retrieves all M1 layers with the purpose drawing, such as {M1 drawing} {M1 drawing1}...{M1 drawing n }.

- Click  to select the LPPs in the **Select Excluded LPPs** dialog box.
- Click  to select an **Excluded LPPs** list that you have created before. The arrow displays a history of your last five lists.

6. (Config calculation method only) In the **Config** box, type the path and file name of the configuration file you want to use in the **Config** text field. Alternatively, click the browse button to locate the file. After specifying the configuration file, you can choose the metal that you want to analyze in the **Group** selection field.
7. (Layer calculation method only) In the **Window Size** box, type the value for the window size, in micrometers. The window is a square area within which the analyzer will check the density. The default and typical window size is 10 mm x 10 mm. For more information about the analysis windows, see [Windows and Steps](#).
8. (Config calculation method only) When **all** is selected in the **Group** field, the **Window Size**, **Step**, **Use Design Area**, and **Use Group Size** boxes are enabled. The original window and step sizes defined in the specified configuration file for all groups is used by default. When either the **Use Design Area** or **Use Group Size** box is unchecked,

type the value for the window size in the **Window Size** box and type the distance in the **Step** box.

When a single group (for example, M1) is selected in the **Group** field, the window and step sizes defined in the configuration file is used. You also can specify the window and step sizes in the **Window Size** and **Step** boxes.

You can click the  button to restore the window and step sizes defined in the configuration file.

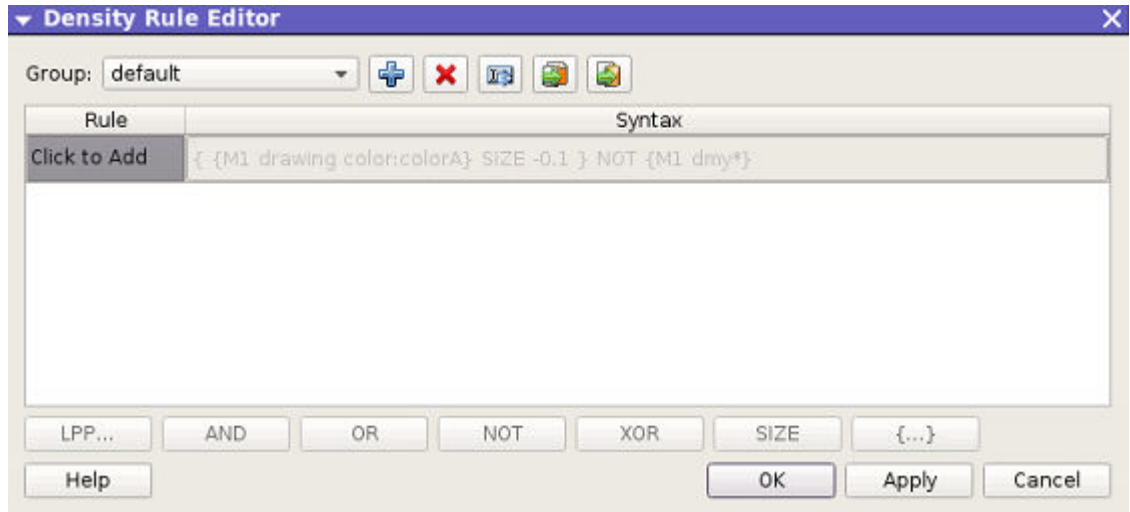
9. (Layer calculation method only) In the **Step** box, type the distance, in micrometers, the analyzer should advance after it finishes checking the current window. The default step is 10 mm. For more information about the analysis windows, see [Windows and Steps](#).
10. In the **Area** selection, select one of the following:

Area Selection	Description
Design	Calculates density across the entire cell (or design).
Viewport	Calculates density across the region that is visible in the viewport.
Region	Calculates density within the specified region. Click  , and then draw a rectangular region on the canvas. The coordinates of the lower left and upper right corners appear in the Specified Box field.

Manually Defining Density Methods

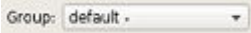
This topic describes how to manually define various density methods. For information on using Tcl commands and preferences, see the *Custom Compiler Tcl Reference Manual*.

1. From the **View** menu, select **Check Density**.
2. From the **Manual Define** method, click **Editor**. The **Density Rule Editor** appears.



About Density Rule Editor Commands

The topic describes the **Density Rule Editor** commands:

Toolbar Icons	Commands	Description
	Change current group	Group lists the default group and all other user-defined groups. The default group is empty when the editor is first opened. You can add rules to the group. When switching groups, the tool checks the validity of the rules syntax of the current group.
	Create group	Creates a group based on the Source Group . You can specify a source and Target Group name, and the tool creates a new group with the target group name based on the source group. The default of the source group is "default". The tool disables the OK button if the target group name is invalid.
	Remove group	Removes the group you created. The default is the current group in the Density Rule Editor dialog. The default group cannot be removed.
	Rename a group	Renames the group you created. The default is the current group in the Density Rule Editor dialog. The default group cannot be removed. The tool disables the OK button if the target group name is invalid.

Toolbar Icons	Commands	Description
	Export group to a file	Exports all groups to a file and selects the last specified file name by default. If the specified file already exists, a File Exists dialog appears to confirm the overwriting of the existing file. When saving groups, the tool does not check the rule syntax. Rather, it saves the groups as user-defined in the Density Rule Editor dialog.
	Import group from file	Imports all groups from a file and selects the last specified file name by default. The tool imports groups from a specified file when you click Open . When loading groups from the file, the tool confirms if the modified rules can be saved. See Defining Syntax of Manual Rules for more information.
	Rule Table	Shows all rules of the current group. Each rule includes two columns: Rule , used to specify the rule name, and Syntax , used to define the manual rule. Click to Add to add a new row for defining a new rule. For example, <pre>{{M1 drawing color:colorA} SIZE 0.1} NOT {M1 dmy*}</pre> To remove a user-defined rule, right-click the row containing the rule to open the Remove menu. You can manually key in the rule syntax, or you can use the quick insert buttons (LPP, AND, OR, NOT, XOR, SIZE, {...}) to easily complete the rule definition.
	Operator	 Edits rule syntax. These operators are enabled only when editing the rule syntax.

Defining Syntax of Manual Rules

The syntax for manual rules is defined as follows:

```
{LayerName PurposeName LayerNo PurposeNo color:colorName color states}  
BooleanOperator {LayerName PurposeName LayerNo PurposeNo color:colorName  
colorStates}
```

For example:

```
{ {M1 drawing 14 -1 color:colorA locked} SIZE -0.1 } AND {M2 drawing} XOR  
{ {M3} NOT {M4 drawing*} }
```

The format of the LPP setting is the same as that of the **Config** mode. Each LPP setting has at most, six entries: layer name, purpose name (optional, default “”), layer number (optional), purpose number (optional), color name (optional), lock property (optional). The Purpose field supports the wildcard, “drawing*”, which means all drawing purposes of the layer, including “drawing”, “drawing1” and so on.

Note:

The following characteristics are the same as those in **Config** mode.

- If the color name is not specified, only No Color shapes in this LPP are considered.
- Same Color is treated as a color. Therefore, if “same” is specified in the Config file, only shapes with the color “same” are calculated.
- Currently, the engine ignores the locked property.

Defining Syntax of the File

The syntax for a file is defined as follows:

```
GroupName RuleName RuleType RuleDefinition
```

For example:

```
G1 r1 5 {M1 drawing} AND {M2 drawing}
G1 r2 5 {M2 drawing} OR { {M3 drawing} SIZE 0.1 }
G2 nr1 5 {M3 *}
```

Note:

The manually-defined rule type is 5; to continue to 4, type it in **Config** mode.

The Boolean operator, string is not case-sensitive.

Rule Checking

When you switch the current group or create a group, click **OK/Apply** in the **Density Rule Editor** dialog for the tool to check the rule of the current group.

The tool checks the empty rule name first, and shows the following message on the console:

```
Specify name for rules of the group "GroupName."
```

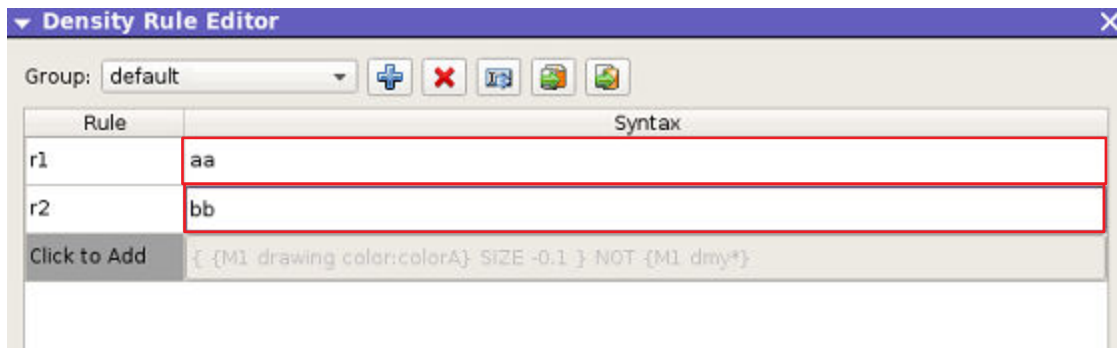
After all rules are named, the tool checks the rule syntax and shows the following message on the console. Each group has one message.

```
The Syntax of rules "GroupName: ruleName1, ruleName2" are incorrect.
```

If certain rules are invalid, the tool fails to switch or create the group and shows a red border on these invalid rules. It reports the following message on console (the tool clears the red border after rule name/syntax is modified).

```
Group operation is not permitted due to group "GroupName" has invalid rules.
```

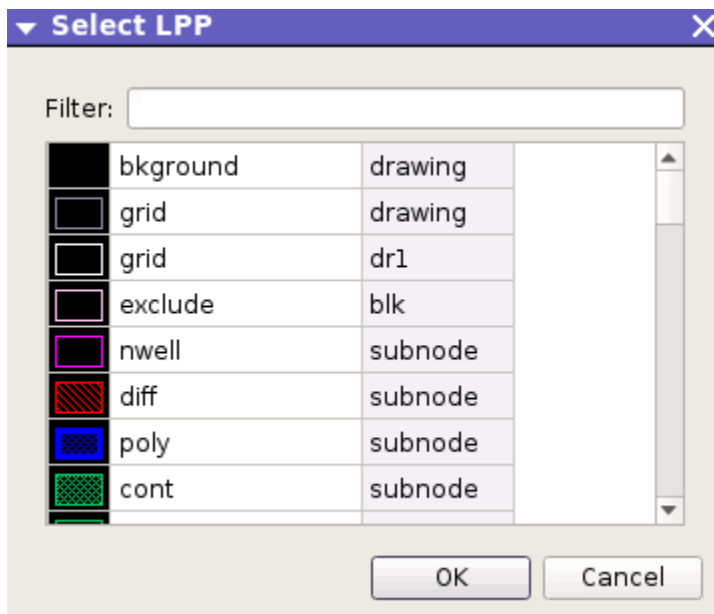
Click **OK** to show only the message on console. The **Density Rule Editor** dialog closes.



Using Operators

The **Density Rule Editor** dialog provides operators such as LPP..., AND, OR, NOT, XOR, SIZE and {...} to help you easily define rules. These operators are enabled only when editing rule syntax.

Click **LPP** to launch the **Select LPP** dialog.



This dialog provides a layer **Filter**, and it shows all layers. The table supports only a single selection. Click **OK** to populate the selected layer/purpose to the syntax column.

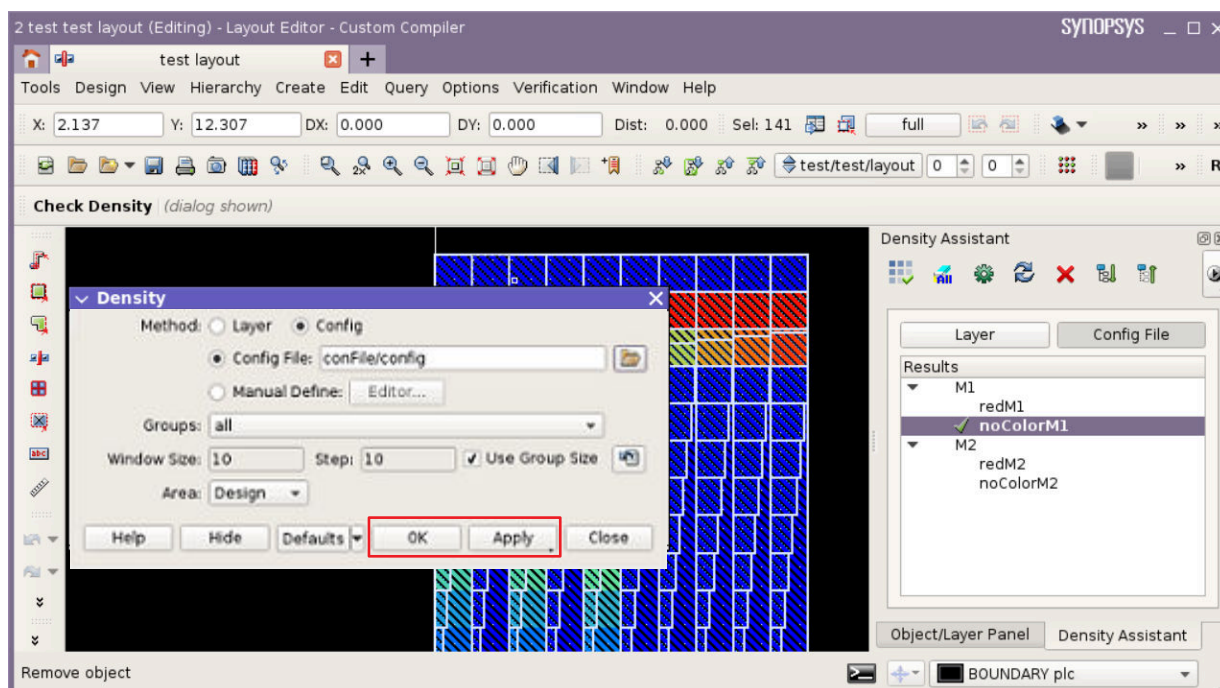
Click **AND**, **OR**, **NOT**, **XOR**, or **SIZE** to populate the related operator at the current cursor position of the syntax column, and automatically add a space before/after the operator, if needed.

Click {...} to populate "{}". If there is a selected string, {} is populated with the selected string and the cursor is moved to the position immediately following "{}". If there is no selected string, input the "{}" at the current cursor and move the cursor to the middle of "{}".

Starting the Density Analysis

To start the density analysis:

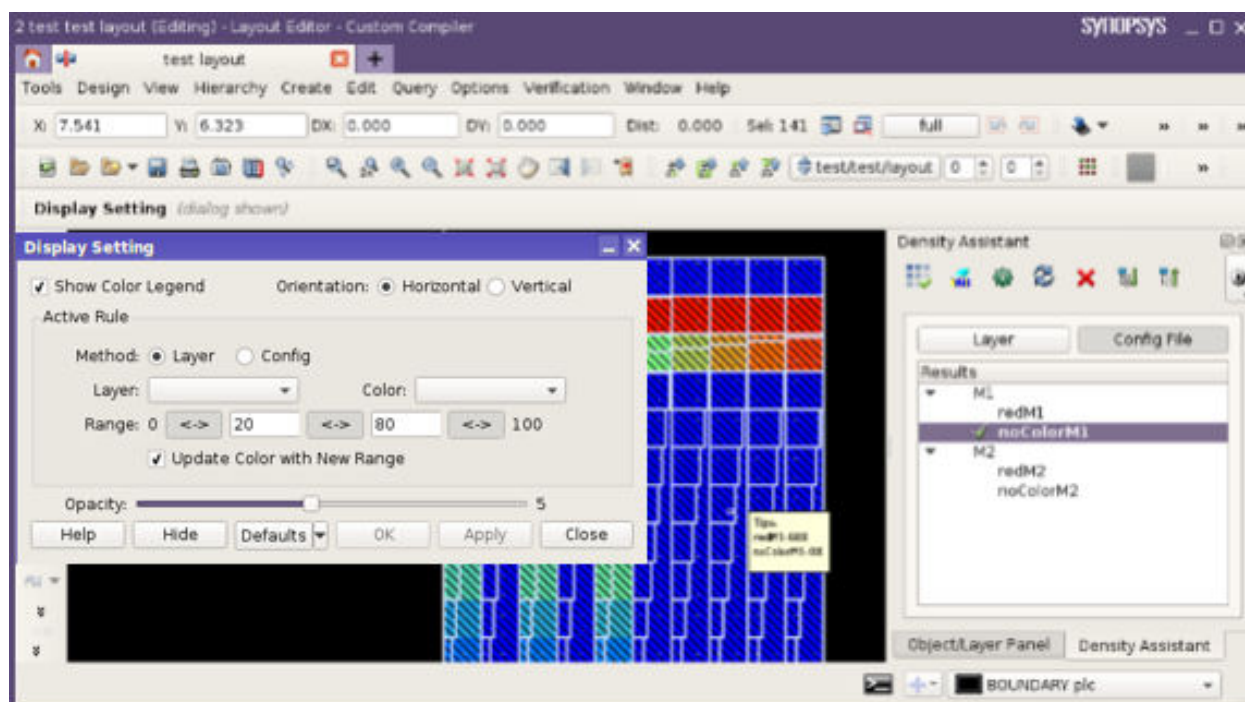
- In the **Density** dialog box, click **OK** or **Apply**. The density analysis results appear on the canvas, overlaying the design.



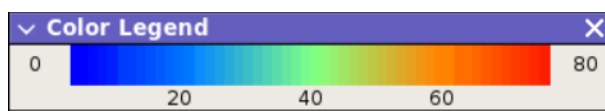
Viewing the Density Analysis Results

This topic describes how to change the display options when viewing the density analysis results. For information on using Tcl commands and preferences, see the *Custom Compiler Tcl Reference Manual*.

In the **Display Setting** dialog box:



1. On the canvas, rest the cursor on a density window. A tooltip appears and displays the density information for that window.
2. Review the analysis results using the **Color Legend**. To enable the **Color Legend**, You can select the **Show Color Legend** option. Windows that have blue and green colors indicate reasonable density. Windows that have orange and red colors indicate possible density problems that you might need to resolve. You can decide the direction of the **Color Legend** by choosing **Horizontal** or **Vertical**.



You can review the analysis results with multiple design windows. The **Color Legend** displays the analysis results of active design window. If the active design window doesn't have the analysis results, the Color Legend is invisible even if **Show Color Legend** is checked.

3. (Calculations based on layers only) In the **Layer** list, select the layer you want to view. The available layers you can select depend on the density calculations you have performed. For example, if you only calculated density for the m1 layer, then only m1 appears in the **Layer** list.

4. (Calculations based on layers only) In the **Color** list, select the analysis of the color for the selected layer.
5. (Calculations based on configuration files only) In the **Group** list, select the group you want to view. The available groups you can select depend on the density calculations you have performed. For example, if you only calculated density for group1, then only group1 appears in the **Group** list.
6. (Calculations based on configuration files only) In the **Rule** list, select the analysis of the category for the selected group.
7. In the **Range** boxes, type the minimum and maximum density values you want to view. By default, the range is 0 to 100. However, you can change the range so that only the densities within a certain range are displayed. For example, you can set the range so that only densities between 80 to 100 are displayed. Doing so lets you focus on high-density areas that require design adjustments.

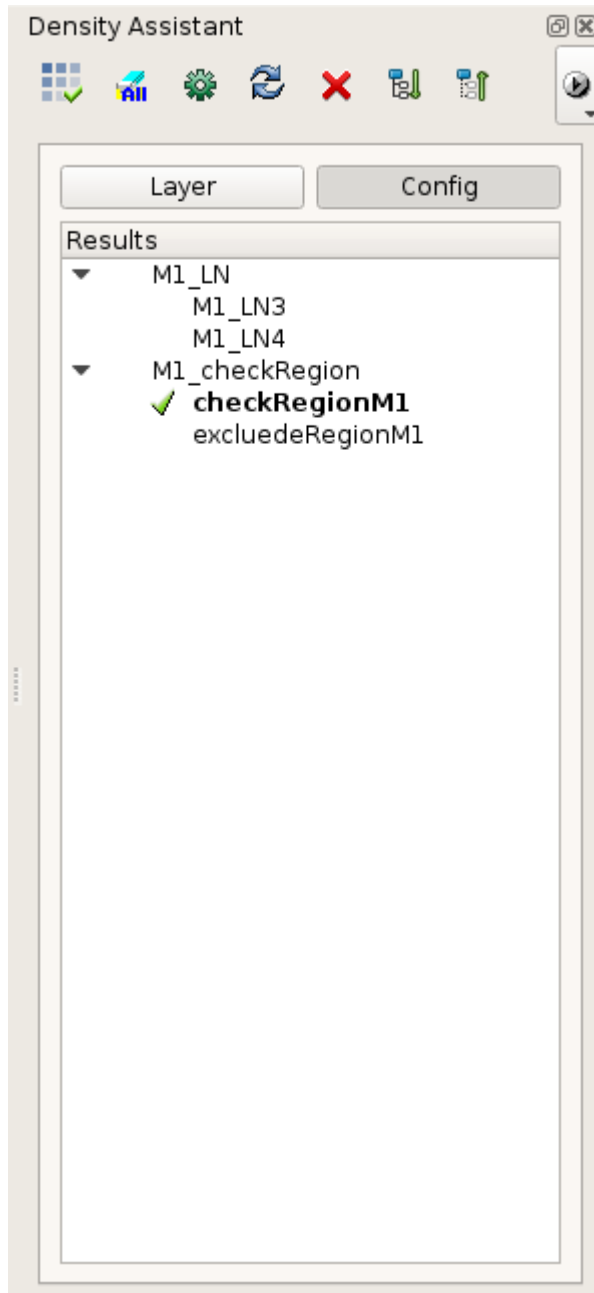
Check **Update Color with New Range** to control the minimum and maximum ranges of the color legend. If this setting is checked, the minimum and maximum ranges of the color legend are set based on a given range. If this setting is not checked, the maximum of color legend is set to 100 and the minimum is set to 0.
8. Use the **Opacity** slider to change the transparency of the results. Move the slider to the left if you want to be able to see the design under the results. Move the slider to the right if you want the results to be opaque and the design beneath is not visible.
9. To apply the display options, click **OK** or **Apply**. The canvas updates based on your display specifications.

Analyzing and Viewing Density Using Density Assistant

In the Density Assistant, you can set up the density analysis and view the density result.

To set up the density assistant:

1. In the Layout Editor, choose **Window > Assistants > Density Assistant**. The Density Assistant opens.



2. Click **Check Density** to open the **Density** dialog box. You can set up the Density Analysis. See [Setting Up the Density Analysis](#).

3. Click **OK** or **Apply** to display the results of density analysis in the Density Assistant. See [Starting the Density Analysis](#).
 - The results are shown in the individual **Results** tables by clicking the **Layer** and **Config** File buttons.
 - The analysis results are listed in the Density Assistant. The selected result is shown in the layout and displayed in bold with green check icon in the **Results** table. You can switch the analysis results shown in the layout by selecting the results in the **Results** table.
4. Click **Clear Density Display** to clear the displayed result in the layout.
5. Click **Density Display Setting** to open the **Display Setting** dialog box to modify the display options. See [Viewing the Density Analysis Results](#).
6. (Optional) Click **Refresh All Results** to update the analysis results after editing the layout design.

About Density Assistant Commands

The topic describes the Density Assistant commands:

Toolbar Icons	Commands	Description
	Check Density	Opens Density dialog box.
	Clear Density Display	Clear the selected density result in the layout. The analysis results are still listed in the Results table.
	Density Display Setting	Opens Display Setting dialog box.
	Refresh All Results	When modifying the layout design, the listed analysis results in the Results table are in italic with the wildcard character (*). Click the button to refresh all the results and the analysis results are updated.
	Delete All Results	Delete all the analysis density results.
	Expand All	Expand all the analysis density results.
	Collapse All	Collapse all the analysis density results.

7

Automating Color Decomposition

This chapter explains how to assign color to colorable shapes in your design for multiple patterning.

This chapter contains the following topics:

- [About Color Decomposition](#)
- [Workflow for Automating Color Decomposition](#)
- [Before You Start](#)
- [Setting Options for Color Decomposition](#)
- [Starting the Color Decomposition Process](#)
- [Monitoring the Color Decomposition Progress](#)

About Color Decomposition

Color decomposition is the process in which colorable shapes across the hierarchy are assigned colors for multiple patterning. The process consists of the following steps:

1. Export design data to the GDSII Stream format.
2. In IC Validator 2013.12 or later, import the Stream data, assign colors to uncolored shapes, and save the updated Stream data.
3. Import the updated Stream data into the Custom Compiler environment.
4. (Optional) Create an instance of the updated design in the original design and set the origin of the instance at (0, 0). Doing so propagates the color information from the colored design to the shapes of the original design. After the original design is colored, the newly created instance may be removed.

This steps colors the original design while preserving the original design's data integrity. Information about the hierarchy, parameterized cells, vias, and other objects are otherwise lost in the colored output data produced by IC Validator.

Instead of manually performing each step in this process, you can use the **Verification > Color Decomposition** command to automate all of the steps. During the process, you can view the progress of each step using the Job Monitor.

Workflow for Automating Color Decomposition

The following table shows the high-level workflow for automating the color decomposition process.

Step	Description
1	Prepare for color decomposition. See Before You Start .
2	Set options for color decomposition. See Setting Options for Color Decomposition .
3	Start the color decomposition process. See Starting the Color Decomposition Process .
4	Monitor the color decomposition progress. See Monitoring the Color Decomposition Progress .

Before You Start

This topic describes how to prepare for color decomposition:

- [Meeting the IC Validator Requirement](#)
- [Obtaining an ICV Color Decomposition Rule Deck](#)
- [Updating the pdk.xml File](#) for color decomposition
- [Obtaining the Layer Mapping File](#)

Meeting the IC Validator Requirement

Make sure you have IC Validator 2013.12 or later. Previous versions of IC Validator do not support advanced nodes and color decomposition.

Obtaining an ICV Color Decomposition Rule Deck

Make sure you have an IC Validator color decomposition rule deck from your foundry. Contact your foundry for details.

Updating the pdk.xml File

The **Verification > Color Decomposition** command is available only if you add the color decomposition runset information in the `pdk.xml` file.

The `pdk.xml` file is stored in the same directory as the PDK library. The file describes the locations of technology-specific data, such as the runset file needed for physical verification and extraction. The file also contains initial or common preference settings for runs.

Locate the `pdk.xml` file, and then add the following lines for color decomposition:

```
<pveJob name="colorDecomposition" jobType="DRC" title="Color  
Decomposition">  
  <tool name="IC Validator">  
    <preference name="RunsetFile">  
      <value>$PDK_INSTALL_DIR/example.rs</value>  
    </preference>  
    <preference name="Options">  
      <value>-turbo</value>  
    </preference>  
  </tool>  
</pveJob>
```

The values of `pveJob`, `jobType`, and `tool name` must be set as shown. The values of the remaining elements can be customized. For information about the `pdk.xml` file and the element descriptions, see [Configuration File Elements](#).

Make sure your design uses the correct PDK library with its updated `pdk.xml` file. To do this, before you start the Custom Compiler environment, set the `PDKLib` attribute to `true` for the PDK library:

```
DEFINE <lib_name> ./<lib_name>  
ASSIGN <lib_name> PDKLib true
```

If the `pdk.xml` file is not in the same directory as the PDK library, make sure the design references it:

```
ASSIGN <lib_name> PDKConfigPath /path/to/pdk.xml
```

For more information about ASSIGN statements, see [Controlling Library Availability](#).

Obtaining the Layer Mapping File

Make sure you have the layer mapping file from your foundry. Contact your foundry for details.

When the modified design is imported into the Custom Compiler environment, a layer mapping file is required to map OpenAccess (OA) layer/purpose pairs to GDSII layer

number/data type pairs. For color designs, each OA layer/purpose pair might map to multiple GDS layer/data type pairs.

The following figure shows an example layer mapping file. The OA M1 drawing layer maps to nine GDS layer/data type pairs. Each mask color (red, blue, and green) requires up to three GDS layer/data type pairs: one for the color, and two to indicate the color lock status.

#oaLayerName	oaLayerPurposeName	gdsLayer	gdsDataType	Color	ColorLock	oaMaterial	oaMaskNumber
M1	drawing	16	30	color:red		metal	
M1	drawing	16	51	color:red	locked	metal	
M1	drawing	16	52	color:red	unlocked	metal	
M1	drawing	16	31	color:blue		metal	
M1	drawing	16	61	color:blue	locked	metal	
M1	drawing	16	62	color:blue	unlocked	metal	
M1	drawing	16	32	color:green		metal	
M1	drawing	16	71	color:green	locked	metal	
M1	drawing	16	72	color:green	unlocked	metal	

For information about the format and content of layer mapping files, see Layer Mapping Files.

Setting Options for Color Decomposition

You use the **Color Decomposition Setup and Run** dialog box to set the color decomposition options. This topic describes the following:

- [Displaying the Color Decomposition Setup and Run Dialog Box](#)
- [Setting the Options in the Main Tab](#)
- [Setting the Options in the Control Variables Tab](#)
- [Setting the Options in the Custom Options Tab](#)
- [Setting the Options in the Include Paths Tab](#)

Displaying the Color Decomposition Setup and Run Dialog Box

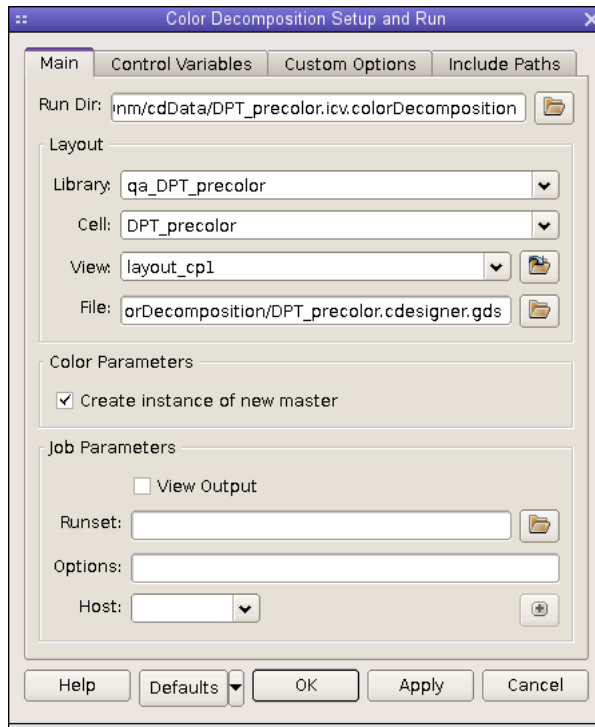
To display the **Color Decomposition Setup and Run** dialog box:

1. Open the design layout. For instructions, see Opening Designs.
2. In the Layout Editor, choose **Verification > Color Decomposition > Setup and Run**. The **Color Decomposition Setup and Run** dialog box opens.

Notice that the dialog box contains four tabs:

- **Main**
- **Control Variables**

- **Custom Options**
- **Include Paths**

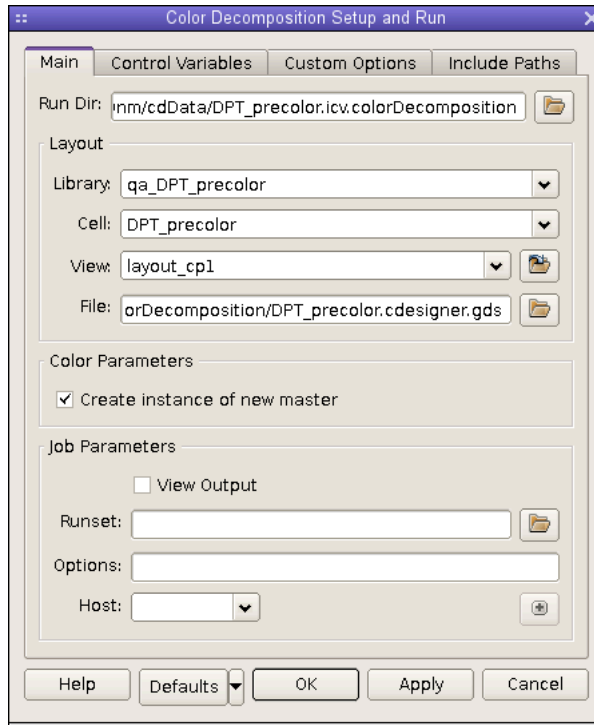


Setting the Options in the Main Tab

The **Main** tab contains typical options you would set to complete color decomposition.

In the **Main** tab:

- Set the following options:



Option	Description
Run Dir	The directory in which IC Validator will run. The directory also contains the input data for IC Validator, such as the .rc file and the rule deck that incorporates your GUI settings. IC Validator will create a log file in this directory, showing the commands that were executed and other details from the run.
Layout	The layout that you want to apply the color decomposition process. You can accept the current Library , Cell , and View . Alternatively, in the File field, browse to choose a different layout.
Color Parameters	The option to create an instance of the updated design in the original design, and set the origin of the instance at (0, 0). Select Create instance of new master to create the instance.
Job Parameters	View Output: The option to display the Text Viewer so that you can view the log during the color decomposition process. Runset: The directory of IC Validator rule deck, which contains IC Validator commands for the color decomposition process.

Option	Description
	Options: The command-line arguments for IC Validator. You can enter any argument supported by IC Validator. Host: The server that runs the process. Choose the remote server from the list. To add a server that is not in the list, click + to the right of the Host list, and then type the host name. To remove a host from the list, select the host name in the list, and then click -. If a grid is set in the Job Monitor, the job executed from the Custom Compiler environment is queued on the grid.

Setting the Options in the Control Variables Tab

- The **Control Variables** tab lets you review and optionally modify current values for the runset control variables that are preferred for this color decomposition process. The control variables from the rule deck only appears in this tab if you set the `loadRunset` preference in the `pdk.xml` file, as follows:

```
<pveJob name="colorDecomposition" jobType="DRC" title="Color
Decomposition">
  <tool name="IC Validator">
    <preference name="LoadRunset">
      <value>true</value>
    </preference>
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/example.rs</value>
    </preference>
    <preference name="Options">
      <value>-turbo</value>
    </preference>
  </tool>
</pveJob>
```

You can also use Tcl to perform the definition and setup of the control variables (instead of the `pdk.xml` file). For example:

```
set s [db::getNext [db::getScopes pdk_pve]]
db::setPrefValue xtDRCICVControlVars -value {
  POLY_RULES_PP
    {description {Enable Poly Rules(pp)}
      value true group PP_TEST_GROUP mutable true
      visible true scope preProc valueType bool}
  METAL_RULES_PP
    {description {Check Metal Rules up to(pp)}
      enumList {M1 M2 M3 M4 M5 M6} default M4 value M4 group
} -scope $s
```

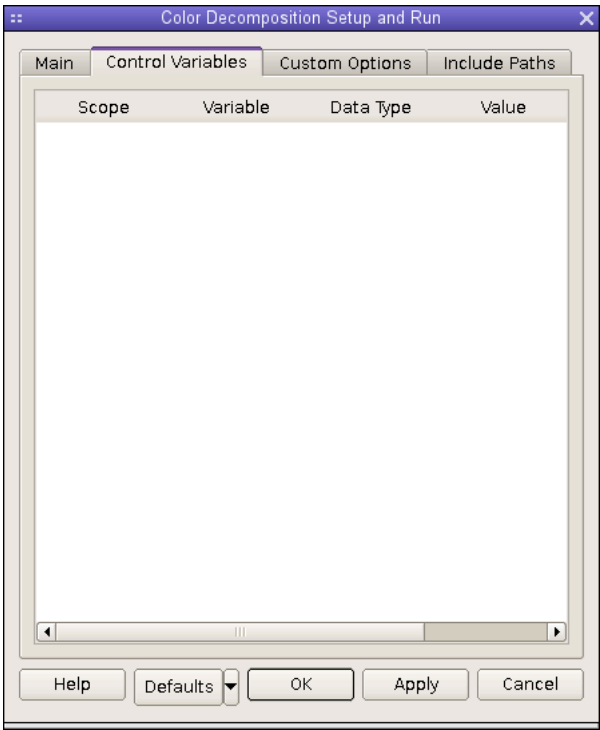
Table 3 Scope of Control Variables

Scope	Runset Presentation	Example
pxl	{name} : {type} = {value}	PREFIX : string = "METAL"
env	#pragma envvar default {type} {value}	#pragma envvar default OUTPUT_MODE "stream"
preProc	#define {name} {value}	#define FUNCTIONS_RS TRUE

Alternatively, you can set the `xtColorDecompositionICVLoadRunset` Tcl preference to *true*.

To modify a control variable:

- Double-click a **variable**, and then edit the field.

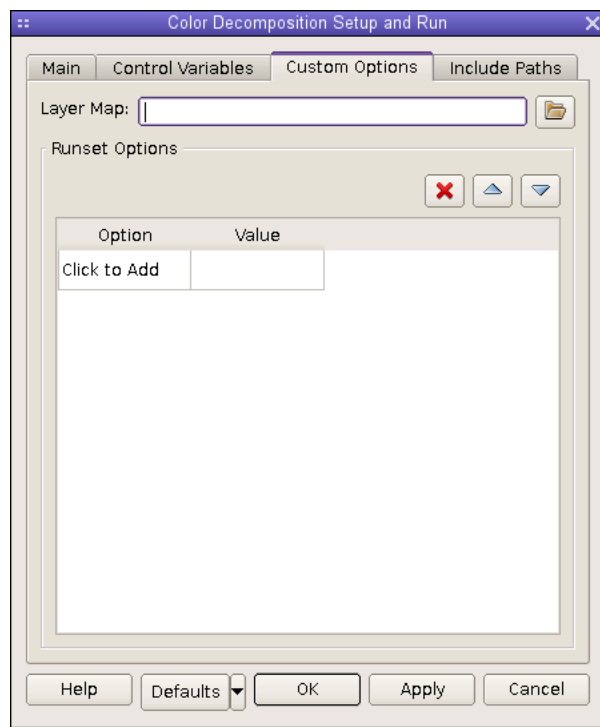


Setting the Options in the Custom Options Tab

The **Custom Options** tab lets you add or modify existing custom options that are accepted by IC Validator. Note that the options are not validated by the Custom Compiler tool.

Existing custom options from the rule deck appear in this tab only if you set the `loadRunset` preference to *true* in the `pdk.xml` file. For more information, see [Setting the Options in the Control Variables Tab](#).

To add custom options:



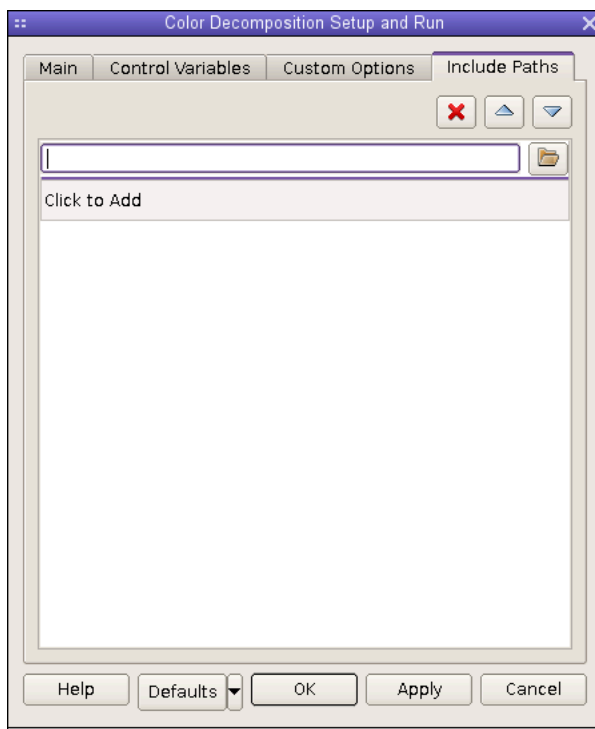
1. Adjacent to the **Layer Map** box, click  to locate and select the layer mapping file. For information about the layer mapping file, see [Obtaining the Layer Mapping File](#).
2. Click a blank **Option** box to add a new row.
3. Double-click the **Option** box and type an IC Validator option name. Be sure to use the IC Validator syntax.
4. Double-click the corresponding **Value** box and type the value you want to assign to the custom option.

5. To move the cursor to an option above or below the current option, click  or click .
6. To delete an option, select the option, and then click .

Setting the Options in the Include Paths Tab

The **Include Paths** tab lets you add and delete paths to files that are needed for the color decomposition process.

To add or delete a path:



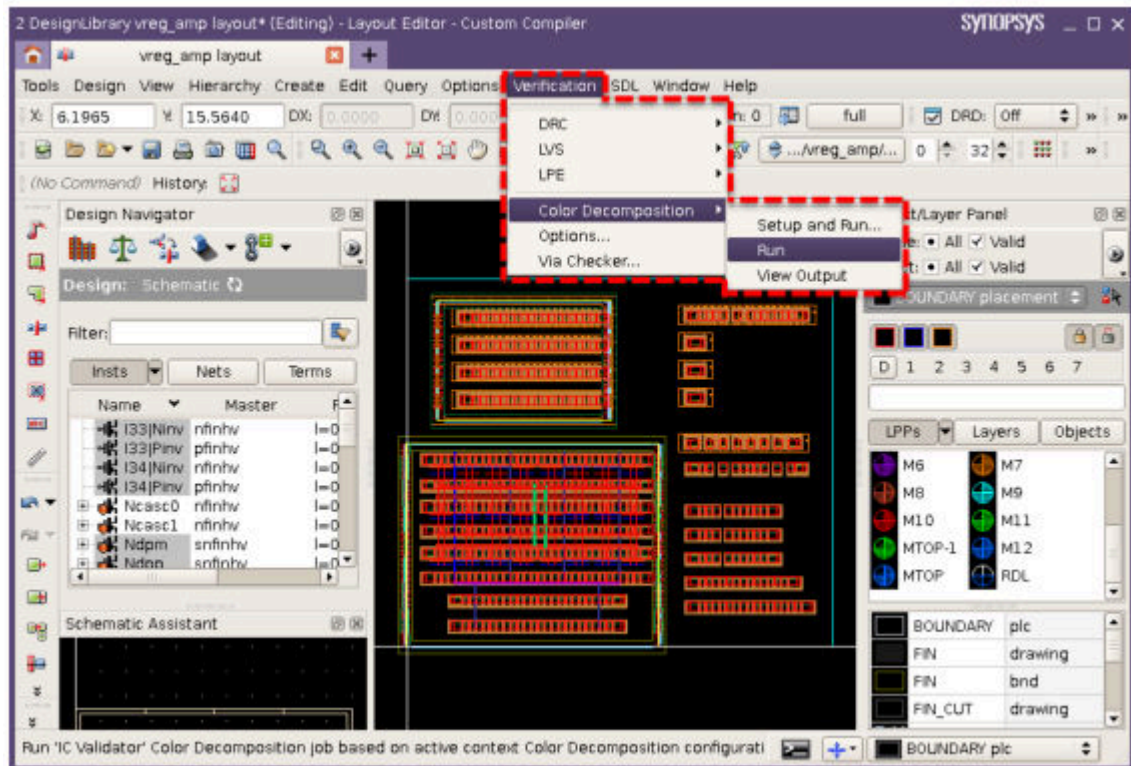
1. Click a blank **Option** box to add a new row.
2. Click a blank **Option** box, and then click  to locate and select the file.
3. To move the cursor to an option above or below the current option, click  or click .
4. To delete an option, select the option, and then click .

Starting the Color Decomposition Process

To start the color decomposition process:

- In the **Color Decomposition Setup and Run** dialog box, click **Apply** or **OK**.

Alternatively, in the Layout Editor, choose **Verification > Color Decomposition > Run**.



The color decomposition process starts. During the process:

- The Layout Editor checks to see if the design is saved. If it is not, the Layout Editor prompts you to save the design before continuing.
- The design data is exported to a Stream file with the name you specified.
- IC Validator processes the Stream file, assigns colors to uncolored shapes, and generates an updated Stream file.

If color violations were encountered, IC Validator prompts you to fix the violations. For more information about fixing the violations, see the [IC Validator user documentation](#).

- The Layout Editor imports the modified Stream file. The imported design is stored under the same library as the original design and has the same cell name. The view name of the modified design will be <originalviewname>_color.
- If you selected the option, the Layout Editor creates an instance of the modified design and set the origin of the instance at (0, 0). If an instance of the modified design already exists, you will be prompted to delete the existing instance.

Monitoring the Color Decomposition Progress

You can monitor the progress of the color decomposition process using the Job Monitor. For instructions on opening the Job Monitor and viewing the running processes, see Job Monitor Tool.

To check the output, in the Job Monitor, right-click the job, and then choose **View Output**. The Text Viewer opens. Details of the running job are displayed in the Text Viewer. For more information on how to use the Text Viewer, see Text Viewer Tool.

8

Preparing for Physical Verification and Extraction

This chapter describes the files you must create for physical verification and extraction purposes.

This section covers the following topics:

- [Before You Start](#)
- [Creating the Device Mapping File](#)
- [Creating the Output Layer Mapping File](#)

Before You Start

The following mapping files are required for physical verification and extraction purposes:

- Device-Map File: Used to map the devices extracted by StarRC.
- Output Layer-Map File: Used to map the Layout versus Schematic (LVS) runset layers onto the technology file LPPs.

Following are the prerequisites for physical verification and extraction:

- Obtain the Design Rule Check (DRC) runset from the foundry.
- Obtain the LVS runset from the foundry.
- Obtain the StarRC runset from the foundry.

Follow these rules while creating a device mapping file or an output layer mapping file:

- Each mapping relation must appear on a new line.
- Each attribute in a mapping relation must be separated with a space.
- Lines that should be treated as a comment must be preceded by two forward slashes (//).

Note:

Before starting the physical verification and extraction jobs, Custom Compiler verifies that at least one back-end license is checked out.

Creating the Device Mapping File

The purpose of a device mapping file is to map the devices extracted by StarRC with the ivpcells that should appear for each device in the StarRC view produced after running StarRC.

The device mapping file must contain the following data columns in a table-like structure, where each row defines relationship between the extracted device to the corresponding device in the iPDK, Where the column `devModel` in the following example is the extracted device name and the column `cellName` is the name of the device in the iPDK:

```
//Fields:
//devModel devLib      cellName dev_view  termOrder magAttrib  mapProp
//-----
N          reference90RF  n_4t      ivpcell   D G S B    MAG=1.5    PROPMAP
w=wtot
P          reference90RF  p_4t      ivpcell   D G S B    MAG=1.5    PROPMAP
w=wtot
...
```

Creating the Output Layer Mapping File

The purpose of an output layer mapping file is to map the LVS runset layers to the technology file layer purpose pairs (LPPs) for use in the StarRC view produced.

The output layer mapping file must contain the following data columns in a table-like structure, where each row defines relationship between the LVS runset layer name and the corresponding layer name and layer purpose in the Technology file:

```
//Format:
//LVS_Runset_Layer  Techfile_Layer/LPP  Techfile_Layer/LPP  Techfile_Layer/LPP
//-----
poly               POLY1   drawing   POLY1   net      POLY1   subnode
metall1            METAL1  drawing   METAL1  net      METAL1  subnode
metal2             METAL2  drawing   METAL2  net      METAL2  subnode
...
```

9

Design Rule Checking Using Integrated Tools

Verify the expected behavior of layout components by performing design rule checks and layout-versus-schematic comparisons. Extract device parasitics to identify devices, including resistors, capacitors, and inductors, that are not present in the design schematic. This chapter describes how to use the Custom Compiler Verification interface to run integrated DRC tools.

This chapter describes the following:

- [Checking for Compliance With Design Rules \(DRC\)](#)
- [Setting Verification Process Node Options](#)
- [Using the Integrated Tools for Design Rule Checking](#)
- [Using Custom DRC Jobs for Design Rule Checking](#)
- [Rerunning Design Rule Checks](#)
- [Running Multiple PVE Jobs](#)
- [Viewing Design Rule Check Output](#)
- [Debugging the Design Using DRC Results](#)

For voltage DRC, see Annotating Voltage Data for VDRC.

Checking for Compliance With Design Rules (DRC)

You perform design-rule checking (DRC) to evaluate the design for compliance with layout fabrication rules. DRC results show compliance with width, spacing, enclosure, and other requirements. DRC runs can verify the correctness of the design mask set.

The Custom Compiler DRC process operates on OpenAccess or Stream format data. You can optionally stream the DRC information to GDSII, and then run the DRC process on the `.gds` file.

You can use the Custom Compiler Verification interface to run one of the following integrated DRC tools:

Integrated DRC Tool or Method	See...
IC Validator and Hercules	Using the Integrated Tools for Design Rule Checking
Mentor Graphics Calibre	Using the Integrated Tools for Design Rule Checking To use the Calibre Interactive tool to perform DRC and verification, see DRC and LVS Using the Calibre Interactive Tool
Custom DRC method	Using Custom DRC Jobs for Design Rule Checking

You can also rerun DRCs, view output, and debug the design. For information, see:

- [Rerunning Design Rule Checks](#)
- [Viewing Design Rule Check Output](#)
- [Debugging the Design Using DRC Results](#)

To run voltage DRC, see [Annotating Voltage Data for VDRC](#).

Setting Verification Process Node Options

Before you run any DRC tool, you need to set physical verification and extraction options to specify the manufacturing process node to associate with design rule checking, layout versus schematic comparison, and parasitic extraction runs.

In addition, make sure you have the required files for physical verification and extraction. See [Preparing for Physical Verification and Extraction](#).

For information about parasitic extraction, see [Design Rule Checking Using Integrated Tools](#).

Note:

See [Default Interfaces](#) and [Custom Jobs](#) for more information.

To set physical verification and extraction options:

1. Begin with an open design layout or schematic.
2. Choose **Verification > Options**.

(Choosing **Tools > Verification** adds the **Verification** menu to the window menu bar.)

3. Choose the **Process**, and its **Variant**, from the lists. See [Process and Variant Selection](#).
4. (Optional) If changes have been made to the PDK, click the **Reload** button to apply those changes.

Using the Integrated Tools for Design Rule Checking

You can use the IC Validator, Hercules, or Calibre tool for design rule checking on the Custom Compiler environment.

You can set and run the design rule check, and then rerun it using the same or slightly varied settings for iterative development. Use the IC Validator VUE, Hercules VUE tool or the Calibre RVE tool to evaluate results.

 [Show me](#) an example of how to report electromigration violations.

1. To set up and run an integrated design rule check, you can choose any one of the following options:

(Choose **Tools > Verification** to add the **Verification** menu to the window menu bar).

- In the Layout Editor, choose **Verification > DRC > Run**.
- Click the  option in the toolbar.

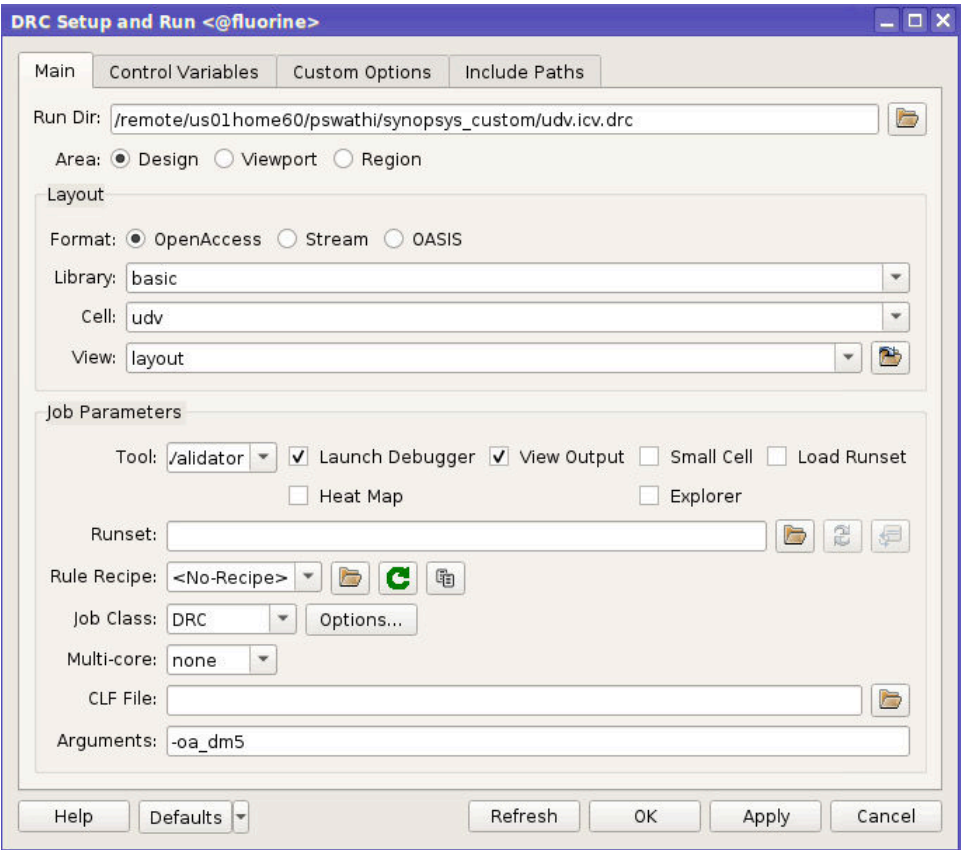
Note:

If you use the valid preferences from the .xml file, then the DRC run starts automatically with the saved preferences.

If the preferences are not defined correctly, then the **DRC Setup and Run** dialog box opens and the following error message appears:

```
Error: Cannot proceed job because the following errors:
xtDRCICVRunsetFile "" is not specified.
```

- In the Layout Editor, choose **Verification > DRC > Setup and Run**



2. In the **DRC Setup and Run** dialog box in the **Job Parameters** section, choose the tool you want to use from the **Tool** list.

The dialog box reconfigures to reflect available options for this tool based on the current runset.

Continue by entering input requirements for this tool on the **DRC Setup and Run** dialog box **Main** tab.

3. On the **Main** tab, enter settings as described in [Table 4](#).

Table 4 IC Validator, Hercules, Calibre DRC Setup and Run Settings (Main Tab)

Settings	Description
Run Dir	You can accept the current run directory or browse to choose a different directory with the <directory>/<cellname>.icv.drc runset or <directory>/<cellname>.hercules.drc runset or <directory>/<cellname>.calibre.drc runset. For more information on the Run Dir setting and supported tokens, see Table 5 .

Settings	Description
Area	Area on which to run the DRC check as design, viewport, or region, specified by coordinates of its rectangle, for example: [x1, y1, x2, y2] [-20, -20, 20, 20]. To automatically insert coordinates here for a selection area, click the interactive selection button, and then select the area in the design. Region supports the select window , which specifies rectangular regions for DRC operations to be processed within.
Layout Format	For OpenAccess format, additionally specify the Library, Cell, and View. For the Stream or OASIS format, set the following: ◦ Export Stream or Export OASIS: Exports the layout cellview used as input to the File you specify. ◦ Config: (Available only if you selected Export Stream or Export OASIS) Opens the Export Stream or Export OASIS dialog box. To set the options in the dialog box, see Exporting Design Data in GDSII Stream Format or Exporting Design Data in OASIS Format in the <i>Custom Compiler Environment User Guide</i>. The options you specify are integrated with PVE (preferences in the <code>pdk.xml</code> configuration).
Job Parameters	Tool is IC Validator, Hercules, or Calibre. Launch Debugger launches IC Validator or Hercules VUE tools, or the Calibre RVE tool when the process completes, if violations are found. Heat Map opens the IC Validator DRC Error Heat Map. Use the <code>xtDRCICVHeatMap</code> preference to enable this mode. View Output opens a Text Viewer window with the process log. For IC Validator only, Small Cell overrides automatic small cell determination and skips runset optimizations for distributed processing during runset compilation that could dominate the overall time for small cells. If the input design is too large, the IC Validator tool exits with an error. For Calibre only, Hierarchical runs a hierarchical design rule check. Explorer opens IC Validator DRC Explorer standalone mode. Use the <code>xtDRCICVExplorer</code> preference to set this mode. igraph runs the IC Validator tool in the igraph mode. Use the <code>xtDRCICVIgraph</code> preference to set this mode. The default is FALSE.

Settings	Description
	Load Runset loads values from the user-supplied runset into the setup dialog when the runset value changes. Use the <code>xtDRCICVLoadRunset</code> preference to set this mode. Load Runder loads automatically the values from the existing <code>prefs.xml</code> file inside the specified Run Dir (if any) into the Setup dialog. Use the <code>xt<pveJob>AutoPrefs</code> preferences, for example, <code>xtDRCAutoPrefs</code>, to set this mode. The default is true. Runset sets the path to the runset file with rules governing the run. Browse to select the file. Click Reload  to reload the runset. Click  to open the runset with a text viewer. Rule Recipe lists the available recipe files. ◦ Select  to choose an existing recipe file. ◦ Select  to reload a recipe file. ◦ Select  to edit an existing recipe file. See Creating and Modifying Recipes. Note: In DRC Setup and Run you can Create, Rename, Remove and Save recipes only, however you cannot Manage Recipes. Supports queue by Job Class > Options. Combines settings for individual analysis tools. You can configure existing job classes or user-defined job classes. Available options are: Simulation, Netlisting, RunMultiplePvejobs, DRC, LVS, LPE, PEX, PERC, and DFM. For details, see Adding Job Classes With <code>xt::createJobClass Tcl</code> in the <i>Custom Compiler Environment User Guide</i>. Multi-core allows you to select the number of CPUs to run the design rule checks. The default is NONE. CLF File is a text file representing the command-line options. Set the following tool-specific preference: <code>xt<pveJob><tool>CLFFile</code>. For example, <code>xtDRCICVCLFFile</code>. Options configures the job class in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i>. Arguments accepts command-line options for this tool, if any. Run Summary displays DRC results as either CLEAN or NOT CLEAN. This is a read-only option. It is visible only when you set the <code>xtDRC<pveJob><tool>ShowRunSummary</code> preference to true. When this option is set to true, the View Output option is disabled.

Settings	Description
Refresh	Checks if the design name is same as Library, Cell View (LCV). Note: if the design name is not the same as LCV, the LCV in the dialog box is updated to the current design name.  Show me an example of how to refresh the LCV.
Defaults	Creates and loads the default settings set in the dialog box.

The results directory for the PVE tool is **Run Dir**. Its value is defined by the `xt<pveJob>RunDir` preference, which is defined in the cellview preference scope of the active design context. If the `xt<pveJob>RunDir` preference is not defined, the run directory is defined by calling the `xt::getRunDirName` Tcl command. The default implementation of this procedure returns a string of the format:

```
<xtRunDirBase>/<xt<pveJob>RunDirNameTemplate>
```

Where: `<xtRunDirBase>` is the value of the `xtRunDirBase` preference and `xt<pveJob>RunDirNameTemplate` is the value of the `xt<pveJob>RunDirNameTemplate` preference -- a token substitution string. The allowed tokens in `xt<pveJob>RunDirNameTemplate` are as follows:

Table 5 Supported Tokens

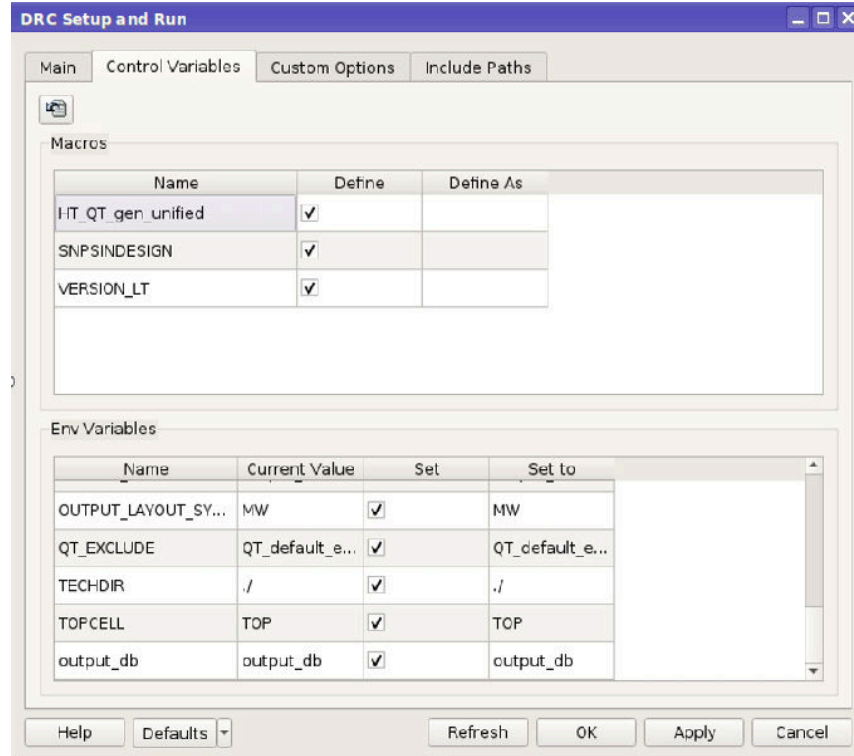
Token name	Token code	Substitution value
Library	%l	Library name
Cell	%c	Cell name
View	%v	View name
Job	%j	PVE Job name
Tool	%t	PVE Tool name
User	%u	User name

If the information required by some of the tokens is not available at the time the run directory name is constructed by `xt::getRunDirName`, the token is replaced by the string, `unknown<tokenName>`.

4. The control variables are loaded into the **Control Variables** tab using runset files compatible with the IGraph mode or using the legacy mode.

- To load control variables from IGraph compatible runset files,

- Click the **Load Control Variables**  option to load the control variables file from the runset file.



Select the settings for IC Validator as described in [Table 6](#):

Table 6 IC Validator DRC Setup and Run Settings (Custom Variables Tab)

Settings	Description
Macros	Lists the macro in the runset file. The following fields are available: ▪ Name: Displays the name of the macro. ▪ Define: Selects the macro defined in the runset file as the default. The option is selected if the macro is defined in the runset file. ▪ Define As: Displays the value of the macro.
Env Variables	Lists the environment variables in the user-defined runset file. The following fields are available: ▪ Name: Displays the name of the environment variable ▪ Current Value: Displays the value of the environment variable. ▪ Set: Selects the environment variable by default. To set the value of an environment variable to match the original setting of the runset file, deselect the Set option. ▪ Set to: Displays the Current Value of the environment variable as default. To add a value of the environment variable, deselect the Set option.

Note:

You can use the IGraph compatible version of **Control Variables** tab only with V-2023.12-SP2 or later versions of the IC Validator tool

- In the **Control Variables** tab, you can review, optionally modify, and reset the current values for DRC runset control variables as preferred for this DRC run. The control variables are preset for a PDK. To modify a preset value, click with the alternate (non-selection) mouse button, and edit the entry field.

To reset the control variables to the default, click the **Reset** button. The default is imported from the `pdk.xml` file.

- Preprocessor, environment, and PXL variables are supported on the **Control Variables** tab in legacy mode only. Variable settings are stored and recalled from the PVE run directory. They can be ordered, grouped, and displayed with special option types, depending on their type (such as Boolean, integer, file input, and so on).

- To use the **Control Variables** tab in the legacy mode:
 - Use the `pdk.xml` file. You can review, optionally modify, and reset the current values for DRC runset control variables as preferred for this DRC run. The control variables are preset for a PDK by default.

To modify a preset value, click with the alternate (non-selection) mouse button and edit the entry field. To reset the control variables to the default click the

Reset  button. The default is imported from the `pdk.xml` file.

In the legacy mode only, preprocessor, environment, and PXL variables are supported in the **Control Variables** tab.

- Add the `controlVarsFile` XML tag in the `pdk.xml` file to point to a variable definition file. For example:

```
<pveJob name="DRC" jobType="DRC">
  <preference name="ViewOutput" mutable="false">
    <value>true</value>
  </preference>
  <tool name="IC Validator">
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc.rs</value>
    </preference>
    <preference name="LoadRunset">
      <value>true</value>
    </preference>
    <controlVarsFile>./snps_reference40nm/pdk_controlVars.xml</controlVarsFile>
  </tool>
</pveJob>
```

Define the variables in the variable definition file with the XML syntax. For example:

```
<switch name="POLY_RULES_PP" scope="preProc" valueType="bool" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>Enable Poly Rules(pp)</description>
  <value>true</value>
</switch>

<switch name="METAL_RULES_PP" scope="preProc" valueType="enum" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>Check Metal Rules up to(pp)</description>
  <value>M4</value>
  <enumValues enumList="M1 M2 M3 M4 M5 M6" default="M4"/>
</switch>

<switch name="ICV_DRC_DIR_pp" scope="preProc" valueType="string" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>directory containing ICV rule decks(pp)</description>
  <value>$PDK_INSTALL_DIR/icv/drc</value>
</switch>
```


The tool issues the following information message when the Control Variables file is available in the pdf.xml file:

```
Information: Found Control Variables file in pdk.xml.
proceeding to use legacy control variable table. Recommend
for switching to new Control Variable table format.
```

Note:

If the file is not specified with the `controlVarsFile` XML tag, the tool issues the following information message:

```
Information: No Control Variables file found in
pdk.xml, switching to New Control Variables table for
mat.
```

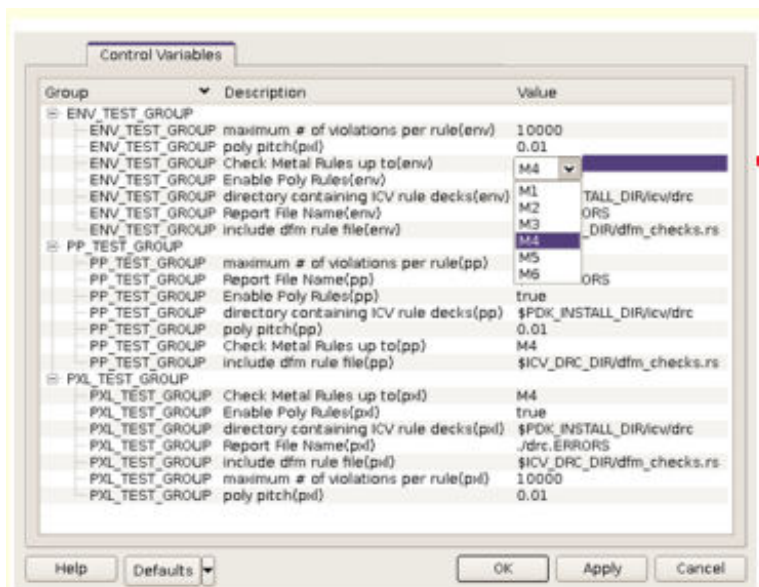
- You can also use Tcl commands to define and setup the control variables (instead of the `pdk.xml` file). For example:

```
set s [db::getNext [db::getScopes pdk_pve]]
db::setPrefValue xtDRCICVControlVars -value {
  POLY_RULES_PP
    {description {Enable Poly Rules(pp)}
     value true group PP_TEST_GROUP mutable true
     visible true scope preProc valueType bool}
  METAL_RULES_PP
    {description {Check Metal Rules up to(pp)}
     enumList {M1 M2 M3 M4 M5 M6} default M4 value M4 group
} -scope $s
```

Table 7 Scope of Control Variables

Scope	Runset Presentation	Example
pxl	{name} : {type} = {value}	PREFIX : string = "METAL"
env	#pragma envvar default {type} {value}	#pragma envvar default OUTPUT_MODE "stream"
preProc	#define {name} {value}	#define FUNCTIONS_RS TRUE

- After you define these variables and set them (from either the `pdk.xml` or Tcl), ensure that the ICV ruledeck is written correctly to consume these variables. See the following figure .



- To use the variable correctly, prepare the ruledeck as follows:

When you execute IC Validator, the tool writes the variables into the ICV ruledeck at the beginning of the file. To do prepare the rule deck, check if the original ruledeck has some of the variables defined with default, then wrap the variables into an #if statement, so the variables become active only when a conditional variable is `true`. (The variable can be `false` in the Custom Compiler variable definitions because the variables are obtained from the GUI. If the variable is `false`, the variable is executed from outside of the tool's environment. The variables are not expected to be set by the GUI.)

When you change a variable setting (as shown with a red arrow in the previous figure), the new value is written to the top of the ICV ruledeck, before IC Validator execution (as shown with a red arrow in the following figure).

```
METAL_RULES_pxl : string = "M4";
#define MAX_VIOLATION_COUNT_pp 10000
POLY_RULES_pxl : boolean = true;
#pragma envvar default MAX_VIOLATION_COUNT_env 10000
#define REPORT_FILE_NAME_pp ./drc.ERRORS
#pragma envvar default POLY_PITCH_env 0.01
ICV_DRC_DIR_pxl : string = "$PDK_INSTALL_DIR/icv/drc";
#pragma envvar default METAL_RULES_env M4
#define POLY_RULES_PP
#pragma envvar default POLY_RULES_env true
REPORT_FILE_NAME_pxl : string = "./drc.ERRORS";
#define ICV_DRC_DIR_pp $PDK_INSTALL_DIR/icv/drc
#pragma envvar default ICV_DRC_DIR_env $PDK_INSTALL_DIR/icv/drc
```

- For the Calibre tool only, on the **Control Variables** tab, review, optionally modify, and reset the current values for DRC runset control variables as preferred for this DRC

run. The control variables are preset for a PDK. To modify a preset value, click with the alternate (non-selection) mouse button, and edit the entry field.

To reset the control variables to the default, click the **Reset** button. The default is imported from `pdk.xml` file.

Preprocessor, environment, SVRF, and TVF variables are supported on the **Control Variables** tab. Variable settings are stored and recalled from the PVE run directory. They can be ordered, grouped, and displayed with special option types, depending on their type (such as Boolean, integer, file input, and so on).

To setup the control variables:

a. Use either the `pdk.xml` or `Tcl`.

- Add the XML tag `controlVarsFile` in the `pdk.xml` file to point to a variable definition file. For example:

```
<pveJob name="DRC" jobType="DRC">
  <tool name="Calibre" default="true">
    <preference name="RunsetFile" valueType="path">
      <value>./PDK/calibre/drc.svrf</value>
    </preference>
    <preference name="LoadRunset">
      <value>true</value>
    </preference>
    <preference name="Options" visible="false">
      <value>-nowait</value>
    </preference>
    <controlVarsFile>./pdk_calibre_controlVars.xml</controlVarsFile>
  </tool>
</pveJob>
```

Then define the variables in that variable definition file with XML syntax. For example:

```
<switch name="MAX_VIOLATION_COUNT_svrf" scope="svrf"
  valueType="integer" visible="true" mutable="true"
  group="SVRF_TEST_GROUP">
  <description>maximum # of violations per
  rule(svrf)</description>
  <value>10000</value>
</switch>

<switch name="POLY_PITCH_svrf" scope="svrf" valueType="float"
  visible="true" mutable="true" group="SVRF_TEST_GROUP">
  <description>poly pitch(svrf)</description>
  <value>0.01</value>
</switch>

<switch name="POLY_RULES_tvrf" scope="tvrf" valueType="bool"
  visible="true" mutable="true" group="TVF_TEST_GROUP">
```

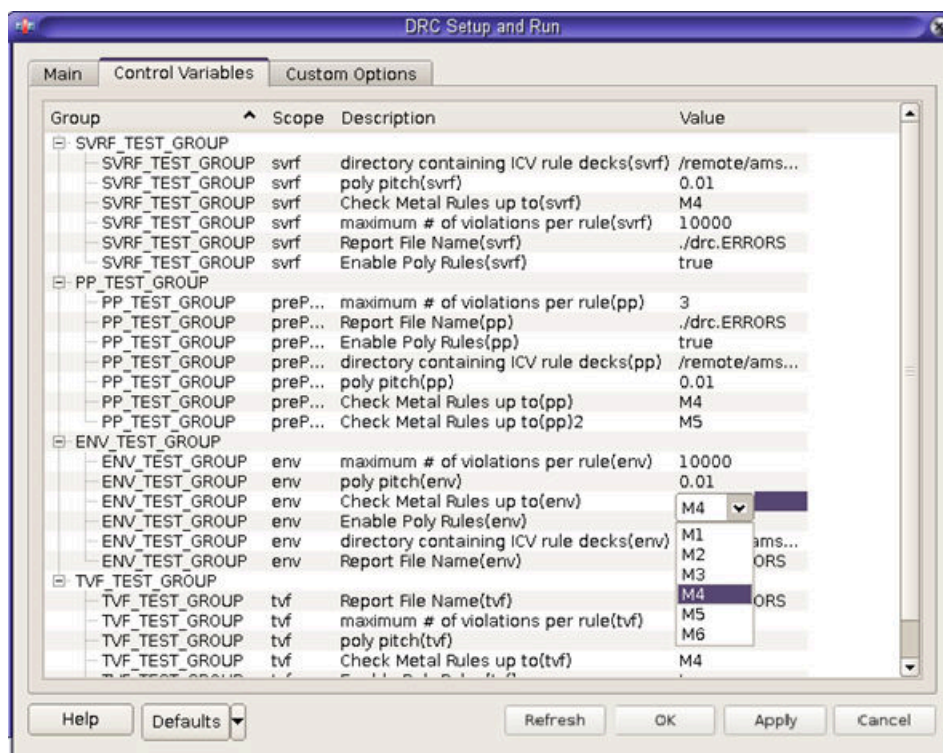
```
<description>Enable Poly Rules(tvf)</description>
<value>true</value>
</switch>

<switch name="METAL_RULES_tvf" scope="tvf" valueType="enum"
  visible="true" mutable="true" group="TVF_TEST_GROUP">
  <description>Check Metal Rules up to(tvf)</description>
  <value>M4</value>
  <enumValues enumList="M1 M2 M3 M4 M5 M6" default="M4"/>
</switch>
```

- Use Tcl to perform the definition and setup of the control variables (instead of the pdk.xml file). For example:

```
set s [db::getNext [db::getScopes pdk_pve*]]
db::setPrefValue xtDRCCalibreControlVars -scope $s -value{
  POLY_RULES
    {description {Enable Poly Rules}
     value true
     group group1
     mutable true
     visible true
     scope preProc
     valueType bool
    }
  POLY_PITCH
    {description {poly pitch}
     value 0.01
     group group2
     mutable true
     visible true
     scope tvf
     valueType float
    }
}
```

- b. After you define these variables and set them up (from either the pdk.xml or Tcl), ensure that the Calibre ruledeck is written correctly to consume these variables, as shown in the following figure.



- c. Calibre DRC or LVS ruledeck references the variables which is defined in the previous steps, but does not override the variables. All control variables are automatically written to the ruledeck in correct SVRF and TVF syntax.
6. For the IC Validator and Hercules tools, on the **Custom Options** tab, choose settings in [Table 8](#).

[Table 9](#) lists the runset options from the current runset as found by the IC Validator tool. You can edit the value for any option you see by changing it here on the **Custom Options** tab. You can add additional runset options here by choosing **Click to Add**, and entering a valid IC Validator option using the correct syntax and value.

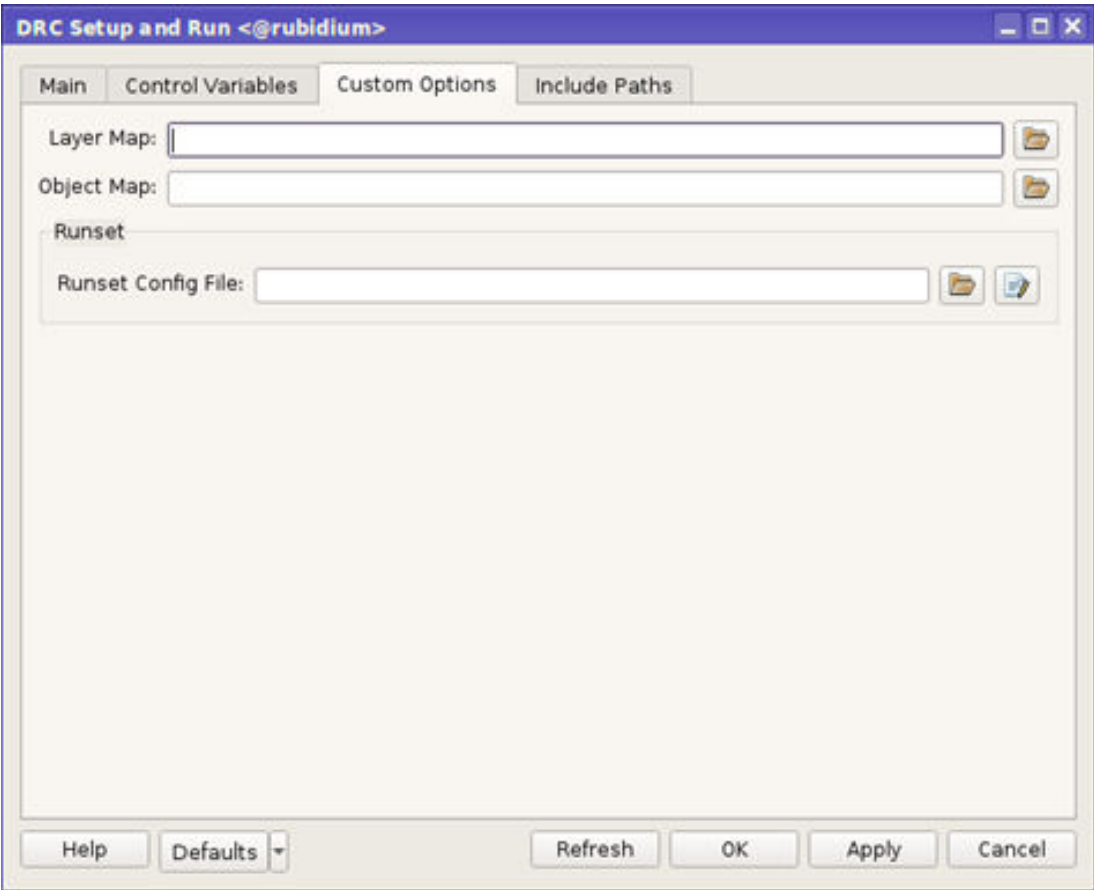


Table 8 IC Validator, Hercules DRC Setup and Run Settings (Custom Options Tab)

Settings	Description
Layer Map	For OpenAccess format, sets the file to map OpenAccess LPP numbers and traditional GDSII stream layers/data types.
Object Map	Specifies the object map to be used during design rule checking.
Runset	Runset Config File specifies a runset configuration file that is passed to an IC Validator or Hercules command using the <code>-runset_config</code> option. Use the <code>xt<pveJob>ICVRunsetConfigFile</code> preference to set this mode. Edit Runset Config edits the runset configuration file. ◦ If the file does not exist, create a file using the Create File Prompt dialog box ◦ If the file is empty, create a file with the default name <code>runset_config.rs</code> using the Create File Prompt dialog box

Settings	Description
	Runset Options specifies the supported IC Validator or Hercules runset options. Custom Compiler does not populate the Runset Options table while loading a runset file. To display the Runset Options table, set <code>xt<DRC>ICVShowRunsetOptions</code> preference to true if <code>xt<pveJob>ICVRunsetOptions</code> is not empty. For Hercules only, additional options to append to the runset. See the Hercules documentation for option details.

Table 9 IC Validator Setup and Run Settings (Custom Options Tab) Runset Options

IC Validator Function	
IC Validator Argument	Option Displayed for this Function and Argument
pex_runset_report_file	
file	PEX_RUNSET_REPORT_FILE_FILE
error_options	
error_limit_per_check	ERROR_OPTIONS_ERROR_LIMIT_PER_CHECK
pcell_list	ERROR_OPTIONS_PCELL_LIST
create_vue_output	ERROR_OPTIONS_CREATE_VUE_OUTPUT
report_layout_errors	ERROR_OPTIONS_REPORT_LAYOUT_ERRORS
hierarchy_auto_options	
application	HIERARCHY_AUTO_OPTIONS_APPLICATION
flow	HIERARCHY_AUTO_OPTIONS_FLOW
optimizations	HIERARCHY_AUTO_OPTIONS_OPTIMIZATIONS
no_explode_with_text	HIERARCHY_AUTO_OPTIONS_NO_EXPLODE_WITH_TEXT
hierarchy_options	
delete	HIERARCHY_OPTIONS_DELETE
explode	HIERARCHY_OPTIONS_EXPLODE
explode_all	HIERARCHY_OPTIONS_EXPLODE_ALL
flatten	HIERARCHY_OPTIONS_FLATTEN

IC Validator Function

IC Validator Argument	Option Displayed for this Function and Argument
no_explode	HIERARCHY_OPTIONS_NO_EXPLODE
incremental_options	
exclude_window	INCREMENTAL_OPTIONS_EXCLUDE_WINDOW
exclude_window_by_cells	INCREMENTAL_OPTIONS_EXCLUDE_WINDOW_BY_CELLS
exclude_window_by_cells_delta	INCREMENTAL_OPTIONS_EXCLUDE_WINDOW_BY_CELLS_DELTA
select_window	INCREMENTAL_OPTIONS_SELECT_WINDOW
clip_window	INCREMENTAL_OPTIONS_CLIP_WINDOW
window_error_filter	INCREMENTAL_OPTIONS_WINDOW_ERROR_FILTER
layout_grid_options	
resolution	LAYOUT_GRID_OPTIONS_RESOLUTION
check_45	LAYOUT_GRID_OPTIONS_CHECK_45
check_90	LAYOUT_GRID_OPTIONS_CHECK_90
openaccess_options	
view	OPENACCESS_OPTIONS_VIEW
layer_mapping_file	OPENACCESS_OPTIONS_LAYER_MAPPING_FILE
generate_pin_text	OPENACCESS_OPTIONS_GENERATE_PIN_TEXT
pin_text	OPENACCESS_OPTIONS_PIN_TEXT
merged_view_list	OPENACCESS_OPTIONS_MERGED_VIEW_LIST
prototype_options	
large_cells	PROTOTYPE_OPTIONS_LARGE_CELLS
symmetry	PROTOTYPE_OPTIONS_SYMMETRY
text_options	
delete_text	TEXT_OPTIONS_DELETE_TEXT
replace_text	TEXT_OPTIONS_REPLACE_TEXT

IC Validator Function

IC Validator Argument	Option Displayed for this Function and Argument
replace_text_characters	TEXT_OPTIONS_REPLACE_TEXT_CHARACTERS
use_exploded_text	TEXT_OPTIONS_USE_EXPLODED_TEXT
edtext	TEXT_OPTIONS_EDTEXT
net_prefix	TEXT_OPTIONS_REPLACE_NET_PREFIX
layout_power	TEXT_OPTIONS_LAYOUT_POWER
layout_ground	TEXT_OPTIONS_LAYOUT_GROUND

7. For the Calibre tool only, on the **Custom Options** tab, set DRC custom options as described in [Table 10](#).

Table 10 Calibre DRC Setup and Run Settings (Custom Options Tab)

Settings	Description
Layer Map	For OpenAccess format. Set the file that maps OpenAccess LPP numbers with traditional GDSII stream layers/data types.
Output Runset	Specify the output runset by accepting the listed runset or browsing to use a new top-level control file containing the database inputs, control variables, custom options, and primary rules. Alternatively, you can use a local runset you have edited. To do this, click the Edit SVRF button to the right of the browse button. In the default text editor, edit the runset, and save the changes. The presence of an edited runset deselects the Override Local Runset option, allowing the DRC process to automatically use the edited version. Override Local Runset overrides the manually-edited runset. Check this box to discard manual edits made to the output runset and use Edit SVRF to regenerate the runset before running Calibre. Deselect this box to pass the manually-edited runset to Calibre. Override Primary Runset overrides primary rules file SVRF commands with top-level control file SVRF commands set using this dialog box. Choose this option to reduce the number of unnecessary specification statements. By default, the Calibre tool exits with an error when it finds duplicate specification statements.
Results Database	Accept or browse to save the Calibre DRC Results database file in a new location.

Settings	Description
Summary Report	Accept or browse to save the Calibre DRC Summary Report file in a new location.

8. For the IC Validator tool only, on the **Include Paths** tab, set DRC include paths for directories whose include files you want to pass to IC Validator.
 - a. In this table, select a row, and click the mouse.
 - b. Browse to select the include directory.
 - c. Repeat for each include path you want to add.
 - d. (Optional) Delete an include path by selecting it, and then clicking the Delete button, a red **x**.
9. Click **OK** to start the design rule check.
10. (Optional) If you did not choose to view output on the **Main** tab, and you want to view it now, see [Viewing Design Rule Check Output](#).
11. For the IC Validator and Hercules tools only, complete the flow by [Debugging the Design Using DRC Results](#).

Note:

The following warning message appears in the console when the OpenAccess version of Custom Compiler is higher than the OpenAccess version on IC Validator:

```
Warning: OA version of Custom Compiler <CC OA overion> is  
higher that IC Validator <ICV OA version>.  
This may lead to OA API compatibility errors to evaluvate  
PyCells in the design.  
Please proceed with GDS input for DRC run to avoid [liboa]  
errors.
```

Using Custom DRC Jobs for Design Rule Checking

Use a custom job for a DRC run to apply selected settings from a Process Design Kit (PDK).

To use a custom job for a DRC run:

1. In the design layout window, choose **Verification > <custom_job_name> Setup and Run**, where *custom_job_name* is the name of the job created for your use by your CAD support group.

(Choosing **Tools > Verification** adds the **Verification** menu to the window menu bar.)

2. In the **Setup and Run** dialog box for this custom job, specify settings for the design rule check.

For assistance with settings, refer to [Using the Integrated Tools for Design Rule Checking](#).

3. Click **OK** to begin design rule checking.
4. Continue by [Viewing Design Rule Check Output](#).
5. Complete the flow by [Debugging the Design Using DRC Results](#).

Rerunning Design Rule Checks

Rerun a design rule check after running it one time. You can use the same tool, and settings, and you can vary the runset or reload a changed runset.

To rerun a DRC:

1. In the design layout window, consider your choices.
 - To rerun the DRC with the same tool, settings, and runset, you can choose **Verification > DRC > Run**.

Alternatively, you can start the run by clicking the **DRC Run** button on the command history toolbar, or the Layout Verification toolbar at the lower left.
 - To rerun the DRC with the same tool, and settings, but a different runset, choose **Verification > DRC > Setup and Run**. Then, on the **Main** tab under **Job Parameters**, browse to select a new runset file.
 - To rerun the DRC with the same tool, and settings, and a runset file whose contents have changed, choose **Verification > DRC > Setup and Run**. Then, on the **Main** tab under **Job Parameters** reload the file by clicking the **Reload Runset** button.
2. Continue the flow by [Viewing Design Rule Check Output](#).
3. Complete the flow by [Debugging the Design Using DRC Results](#).

Running Multiple PVE Jobs

Run multiple PVE jobs simultaneously from a single dialog to accommodate multiple DRC ruledecks.

To run multiple PVE jobs, choose **Verification > Run Multiple**.

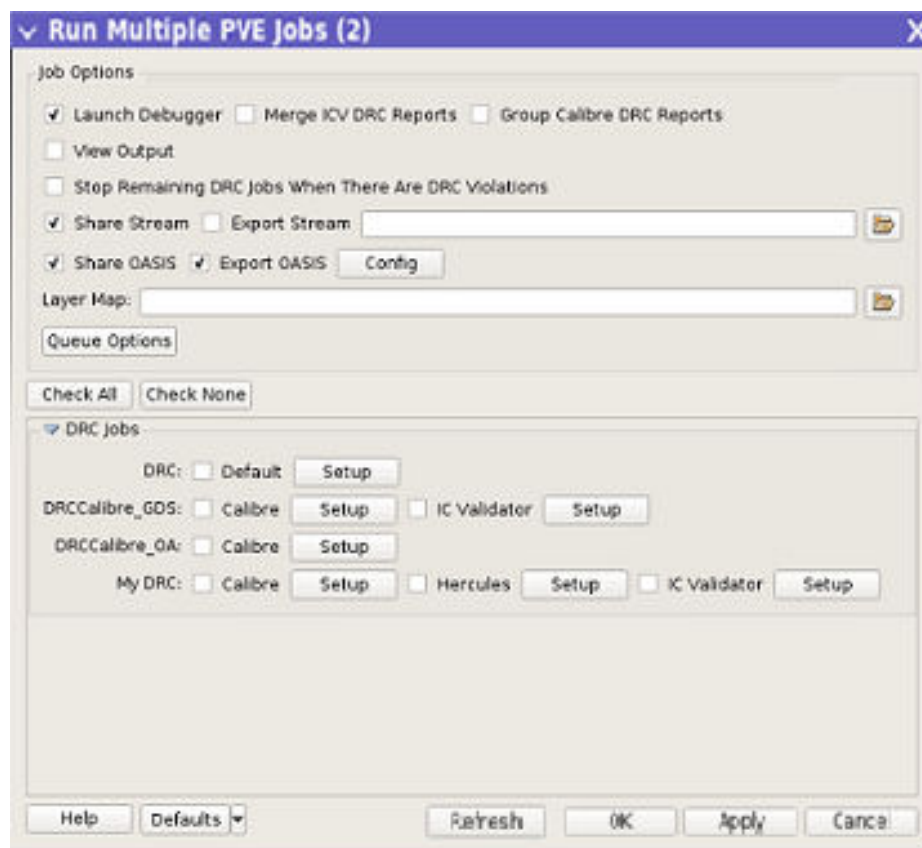


Table 11 Running Multiple PVE Job Options

Job Option	Description
Launch Debugger	Overwrites the corresponding Launch Debugger option in each job. Check this option to launch the debugger after the job completes.
Merge ICV DRC Reports	Merges multiple ICV DRC results in a single VUE database.
Group Calibre DRC Reports	Groups multiple Calibre DRC results in a single RVE database.
View Output	Overwrites the corresponding View Output option in each job. Check this option to open the view output for each running job.

Table 11 Running Multiple PVE Job Options (Continued)

Job Option	Description
Stop Remaining DRC Jobs When There Are DRC Violations	If DRC jobs contain violations while running multiple PVE DRC jobs from Run Multiple PVE Jobs , this option stops all other DRC jobs that are still running.
Share Stream/Share OASIS	To share the input stream and OASIS data, check the Share Stream and Share OASIS options. Use the Layer Map and Config options to Export Stream and Export OASIS data.
Queue Options	Set up concurrent jobs under the RunMultiplePveJobs class in the Job Management Options dialog box.
Check All	Checks all jobs.
Check None	Unchecks all jobs.
DRC Jobs	Multiple jobs classified by DRC job type.
Defaults	If there is no tool section from the PVE job in the <code>pdk.xml</code> file, the default tool is used. This is the user-specified tool previously selected in the Setup and Run dialog.
Refresh	Checks if the design name is same as LCV. Note: if the design name is not the same as LCV, the LCV in the dialog box is updated to the current design name.
Setup	Launches Save Setting button in the Setup and Run dialog. Press this button to save the setting without running the job. Note: This button is shown only when the dialog is opened from the Setup button of the Run Multiple PVE Jobs dialog. There is no Save Setting button when invoked from the Verification menu.
Apply	Starts multiple jobs and the Defaults , OK , Apply , and Cancel buttons are disabled. Only after all jobs have started are the buttons enabled. Note: If you click OK or Apply while multiple jobs are running, you will see a warning message in the console. After specifying the correct setting for each job, click Apply or OK to start running multiple jobs. Open the job monitor to see the Status of those jobs.

Viewing Design Rule Check Output

View design rule check output to see logged operations, results, and errors.

To view design rule check output:

1. In the design layout, choose **Verification > DRC > View Output**. Alternatively, you can use the keyboard shortcut Shift+Ctrl+D.
2. In the Text Viewer, view results, and logged operations, on each tab.
3. After you view DRC run output, continue by [Debugging the Design Using DRC Results](#).

Debugging the Design Using DRC Results

Debug the design by viewing DRC output and using the IC Validator VUE, Hercules VUE, or Calibre RVE tools. Running a DRC one time initializes the debugging environment for that tool.

If you are using the IC Validator tool, you can debug results of multiple DRC processes for one or more designs simultaneously.

To debug the design using design rule check results:

1. (Optional) Begin by [Viewing Design Rule Check Output](#).

The Text Viewer tool opens with tabs showing the runset file, DRC results, any errors, and the DRC run log.

2. Open the debugging tool.
 - If you elected to launch the debugging tool when you set up the DRC check, it has opened automatically if violations were found.
 - Any time, you can choose **Verification > DRC > Debug**.
3. In the debugging window, identify and evaluate violations.

For IC Validator and Hercules VUE, you can use the **Edit-in-Place** option to review violations. You can select a cellView in the error list to highlight it in the design window.

You can perform cross-probing between the list of errors and the design.

For further instructions, refer to the IC Validator VUE, Hercules VUE, or Calibre RVE documentation.

After iterating the design until it achieves design compliance with layout fabrication rules, you can optionally continue with the layout versus schematic comparison process.

Note:

The same VUE window is updated when you run DRC multiple times on the same design.

A new VUE window opens when you open a new design.

10

Comparing Layout Versus Schematic (LVS)

Verify the expected behavior of layout components by performing design rule checks and layout-versus-schematic (LVS) comparisons. Extract device parasitics to identify devices, including resistors, capacitors and inductors, that are not present in the design schematic.

This chapter describes the following:

- [Verifying Layout Accuracy](#)
- [Enabling the Verification Interface](#)
- [Setting Verification Process Node Options](#)
- [Using Integrated Tools for LVS Comparisons](#)
- [Using Custom Jobs for LVS Comparisons](#)
- [Rerunning Layout Versus Schematic Comparisons](#)
- [Viewing LVS Comparison Output](#)
- [Debugging the Design Using LVS Results](#)
- [Comparing Layout Versus Layout \(LVL\)](#)
- [Comparing Netlist Versus Netlist \(NVN\)](#)

Verifying Layout Accuracy

You perform design layout-versus-schematic (LVS) comparison to verify that the layout accurately represents the electronic equivalent of the schematic. You can use the Custom Compiler Verification interface to run one of the following integrated LVS tools:

Integrated LVS Tool or Method	See...
IC Validator	Using Integrated Tools for LVS Comparisons
Hercules	Using Integrated Tools for LVS Comparisons

Integrated LVS Tool or Method	See...
Mentor Graphics Calibre	Comparing Layout Versus Schematic (LVS)
Custom LVS method	Using Custom Jobs for LVS Comparisons

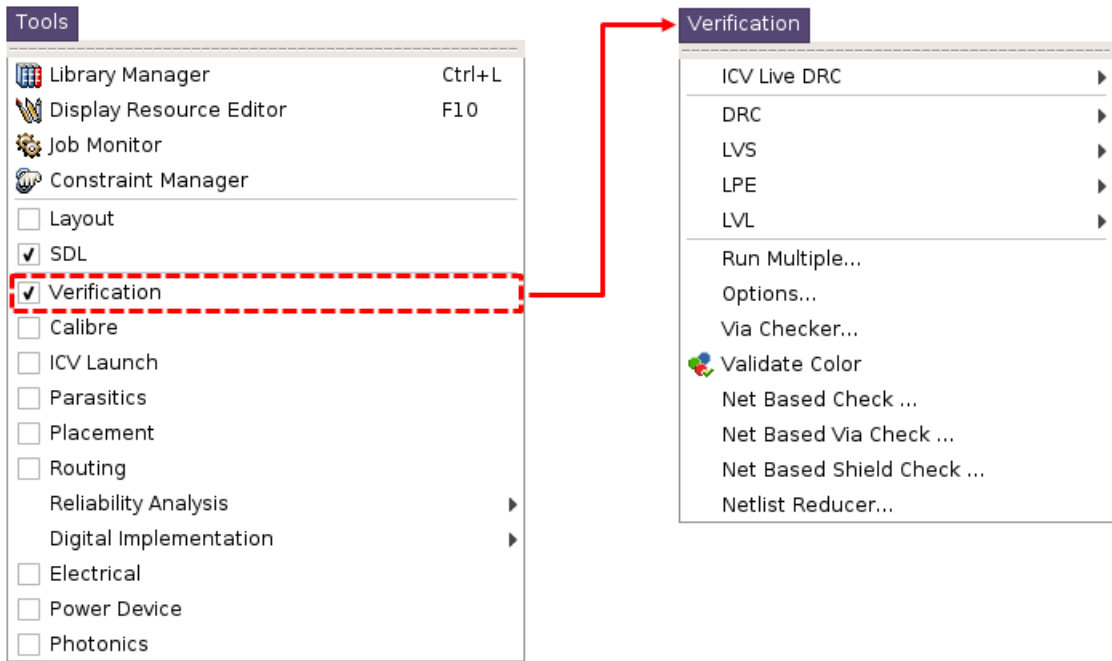
You can also rerun LVS comparisons, view output, and debug the design. For information, see:

- [Rerunning Layout Versus Schematic Comparisons](#)
- [Viewing LVS Comparison Output](#)
- [Debugging the Design Using LVS Results](#)

Enabling the Verification Interface

To enable the Custom Compiler Verification interface:

1. Begin with an open design layout or schematic.
2. In the Layout Editor, choose **Tools > Verification**. The **Verification** menu appears on the menu bar.



Setting Verification Process Node Options

Set physical verification and extraction options to specify the manufacturing process node to associate with design rule checking, layout versus schematic comparison, and parasitic extraction runs.

Before you start, make sure you have the required files for physical verification and extraction. See [Preparing for Physical Verification and Extraction](#).

For information about parasitic extraction, see [Comparing Layout Versus Schematic \(LVS\)](#).

Note:

See [Default Interfaces](#) and [Custom Jobs](#) for further information.

To set physical verification and extraction options:

1. In the Layout Editor, choose **Verification > Options**.
2. Choose the **Process** and its **Variant** from the lists. See [Process and Variant Selection](#).
3. (Optional) If changes have been made to the PDK, click the **Reload** button to apply those changes.

Using Integrated Tools for LVS Comparisons

You can use the IC Validator, Hercules, or Calibre tool for layout versus schematic comparisons. (To use the Calibre Interactive tool, see [DRC and LVS Using the Calibre Interactive Tool](#).)

You can set up and run the LVS comparison, and then rerun it using the same or slightly varied settings for iterative development. Use the IC Validator or Hercules VUE tool or the Calibre RVE tool to evaluate results.

If you use the valid preferences from the .xml file, then the LVS run starts automatically with the saved preferences.

 [Show me](#) how to use integrated tools for LVS comparisons.

1. To use an integrated tool for layout versus schematic comparisons, you can do any one of the following:
 - In the Layout Editor, choose **Verification > LVS > Run**.
 - Click the  option in the toolbar.

Note:

If you use the valid preferences from the .xml file, then the LVS run starts automatically with the saved preferences.

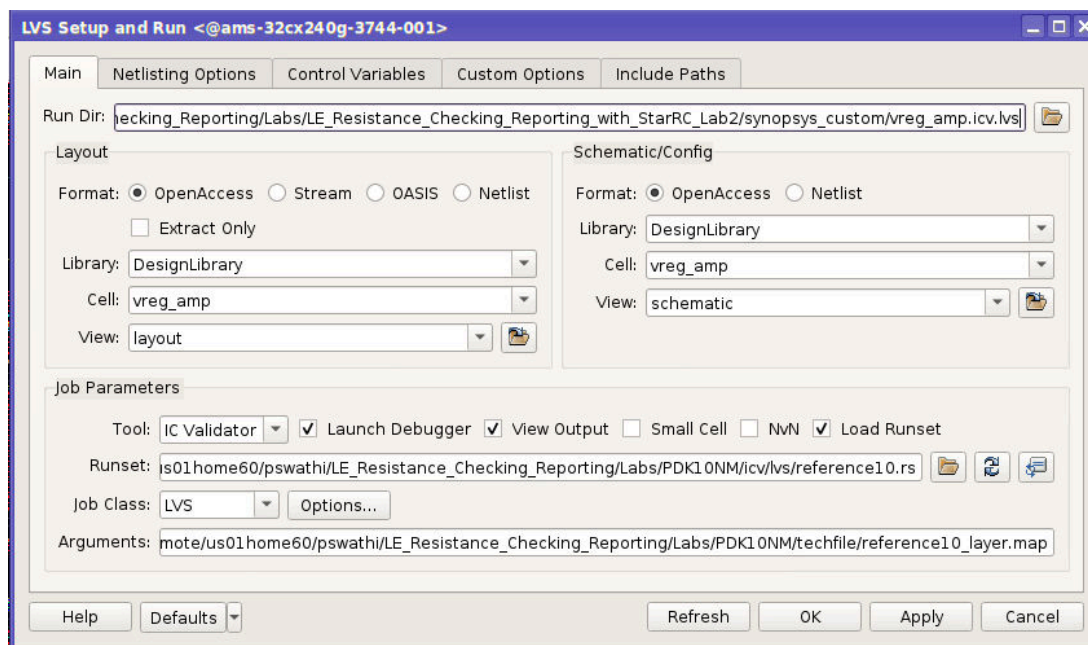
- In the Layout Editor, choose **Verification > LVS > Setup and Run**.

Note:

If you click the  option in the toolbar and the preferences are not defined correctly, then the **LVS Setup and Run** dialog box opens, and the following error message appears:

Error: Cannot proceed job because the following errors:
xtLVSIcVRunsetFile "" is not specified.

2. In the **LVS Setup and Run** dialog box, choose the tool you want to use from the **Tool** list. The dialog box reconfigures to reflect available options for this tool based on the current runset.



3. In the **Main** tab, enter settings for all tools as described in [Table 12](#).

Table 12 IC Validator, Hercules, Calibre LVS Setup and Run Settings (Main Tab)

Settings	Description
Run Dir	You can accept the current directory or browse to choose a different directory with the <directory>/<cellname>.icv.lvs.

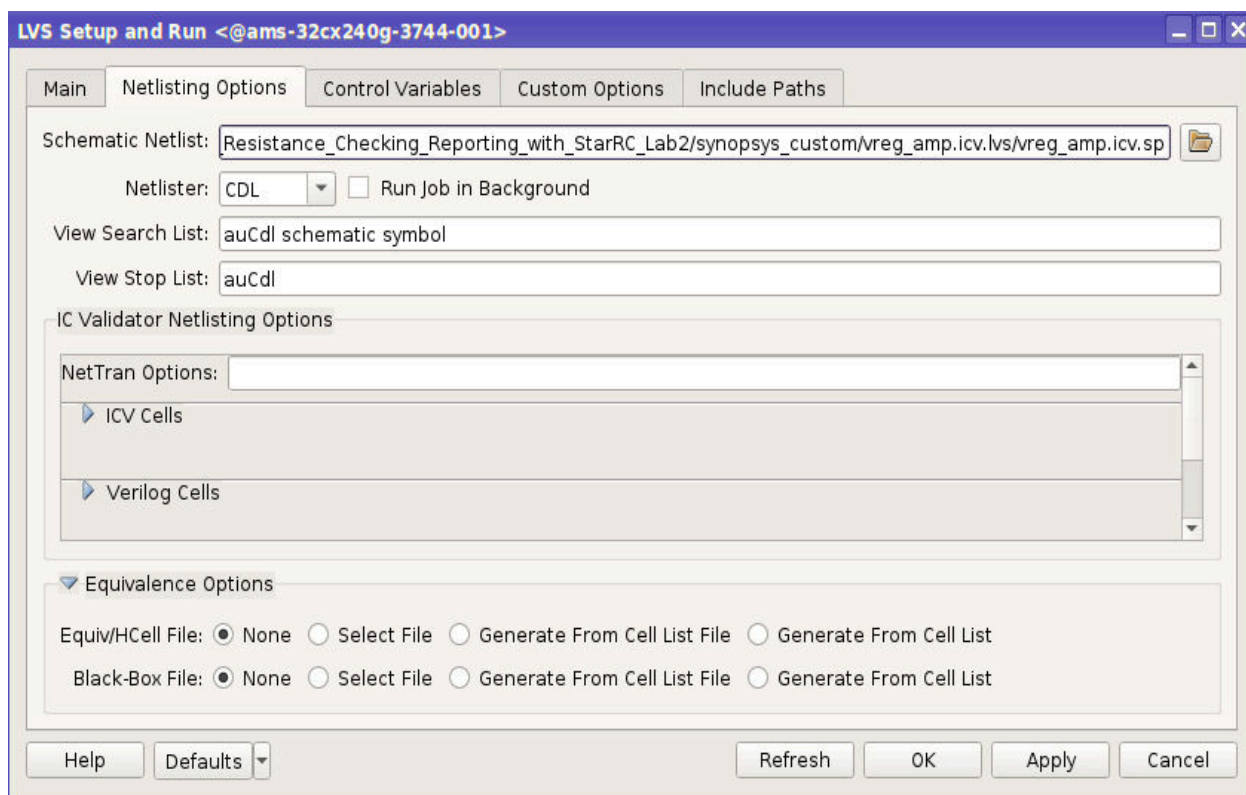
Settings	Description
Layout	For the OpenAccess format, additionally specify the Library, Cell, and View. Extract Only limits the LVS process to a netlist extraction and makes no LVS comparison. For the Stream or OASIS format, set the following: ◦ Extract Only: Limits the LVS process to a netlist extraction and makes no LVS comparison. ◦ Export Stream or Export OASIS: Exports the layout cellview used as input to the File you specify. ◦ Config: (Available only if you selected Export Stream or Export OASIS) Opens the Export Stream or Export OASIS dialog box. To set the options in the dialog box, see Exporting Design Data in GDSII Stream Format or Exporting Design Data in OASIS Format. The options you specify are integrated with PVE (preferences in the <code>pdk.xml</code> configuration). For the Netlist format, confirm the cell. The LVS process extracts a netlist from the layout and uses it as input for the comparison.
Schematic / Config	For OpenAccess format, additionally specify the Library, Cell, and View. For Netlist format, additionally browse to choose the netlist and the netlist format from the list, and then confirm the cell. The formats available for netlisting are: CDL, ICV, SPICE, VERILOG
Job Parameters	Tool lets you select a third-party verification tool: IC Validator, Hercules, or Calibre Launch Debugger launches the IC Validator or Hercules VUE tools, or the Calibre RVE tool when the process completes, if violations are found. View Output opens a Text Viewer window with the process log. (IC Validator only) Small Cell overrides automatic small cell determination and skips runset optimizations for distributed processing during runset compilation that could dominate the overall time for small cells. If the input design is too large, the IC Validator tool exits with an error. (IC Validator only) NvN enables the Netlist and Format fields in the Layout area so that you can compare a netlist against a reference netlist. See Comparing Netlist Versus Netlist (NVN). (Calibre only) Hierarchical runs a hierarchical layout versus schematic comparison.

Settings	Description
	Runset is the path to runset file with rules governing the run. If you want the tool to load the runset file when you start the comparison, select Load Runset. If you want to load the runset file before running the comparison, click . The Open Runset File  button opens the runset. Load Rundir determines if the <code>prefs.xml</code> file is automatically loaded. If set to false, the <code>prefs.xml</code> file is not loaded. The default is true. Use the <code>xtLVSAutoPrefs</code> preference to set this mode. Job Class combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see Adding Job Classes With <code>xt::createJobClass</code> Tcl in the <i>Custom Compiler Environment User Guide</i>. Options are the job settings to control batch processing limits, and to optionally set a queue engine in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i>. Arguments accepts command-line options for this tool, if any. Run Summary displays LVS results as either PASS or FAIL. This is a read-only option. It is visible only when you set the <code>xt<pveJob><tool>ShowRunSummary</code> preference to true. When this option is set to true, the View Output option is disabled.
Refresh	Checks if the design name is same as Library, Cell View (LCV). Note: if the design name is not the same as LCV, the LCV in the dialog box is updated to the current design name.  Show me an example of how to refresh the LCV.
Defaults	Creates and loads the default settings in the dialog box.

- In the **Netlisting Options** tab, enter settings for IC Validator, Hercules, and Calibre as described in [Table 13](#).

Note:

The top-level module is always created by default. You can control the top-level module creation using the `xtLVSNestlistTopAsModule` preference. It is best practice to set this prefix using the `pdk.xml` file, otherwise you must to set the preference in the run directory.



Note:

Set **Tool > Calibre** in the **Main** tab to see the **Calibre Additional Netlists** section in the **LVS Options** tab.

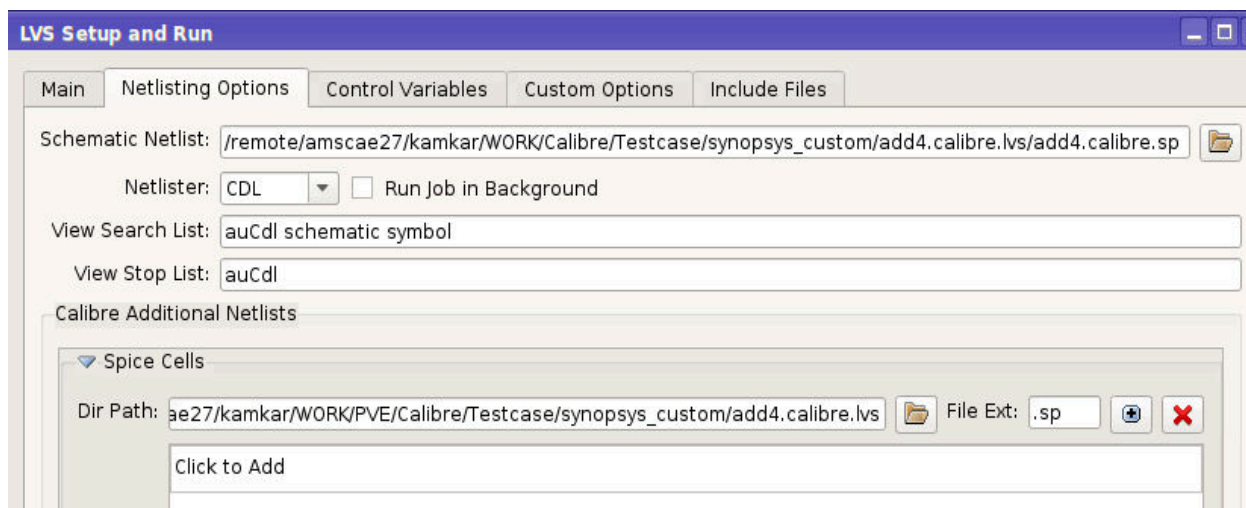


Table 13 IC Validator, Hercules, and Calibre LVS Setup and Run Settings (Netlisting Options Tab)

Settings	Description
Schematic Netlist	For the OpenAccess schematic/config format, you can accept the suggested path shown to the schematic netlist or browse to choose a different path.
Netlister	From the list, choose the netlister to generate a netlist from the schematic: CDL or HSPICE. Run job in Background: Runs the export netlist in the background. The default is false.
View Search List	Shows the list of views the LVS process will traverse during netlisting.
View Stop List	Shows the view stop list that terminates LVS traversal.
IC Validator or Hercules Netlisting Options	NetTran Options shows the netlist translator options for converting the netlist into the format recognized by IC Validator.
Calibre Additional Netlists	Provides an additional SPICE netlist during Calibre LVS.
Output Format	Specifies the IC Validator NetTran output format, with available options: ICV, SPICE, and VERILOG. The format is specified in the runset file. If Output Format is ICV, but the format supported in the runset file is SPICE, the following warning message appears: Currently, the Output Format selected is ICV which is not supported for the runset provided, please kindly change to SPICE
Equivalence Options	Using the Equiv/HCell File , choose from the following four settings in the : None , Select File , Generate From Cell List File , and Generate From Cell List . Using the Black-Box File , choose from the following four settings in the : None , Select File , Generate From Cell List File , and Generate From Cell List .

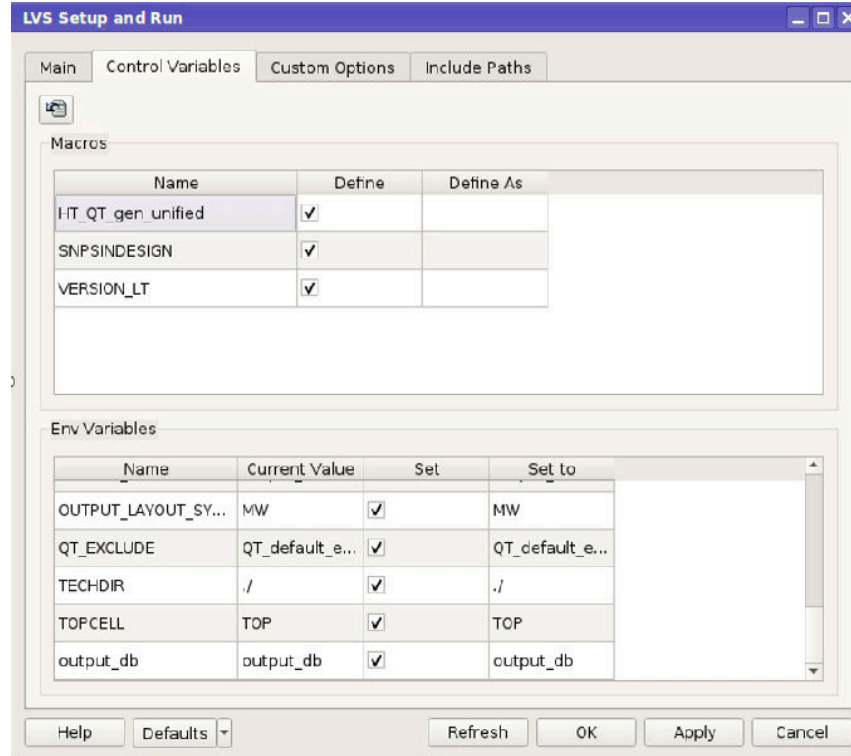
Table 14 *IC Validator LVS Setup and Run Settings (Netlisting Options Tab Multiple Inputs)*

Settings	Description
ICV Cells	Adds an input netlist in IC Validator format. You can specify the searching path in Dir Path and specify the file extension in File Ext. The default for Dir Path is the run directory and the default for File Ext. is .icv. Custom Compiler can run IC Validator using the <code>-icv</code> command-line option to specify these input netlists files in the ICV format. For example, <code>icv_nettran -icv <file> -outName <file> -outType <ICV SPICE SPICE_FLAT VERILOG></code> .
SPICE Cells	Adds an input netlist in SPICE format. You can specify the searching path in Dir Path and specify the file extension in File Ext. The default for Dir Path is the run directory and the default for File Ext. is .sp. Custom Compiler can run IC Validator using the <code>-sp</code> command-line option to specify these input netlists files in the SPICE format. For example, <code>icv_nettran -icv <file> -outName <file> -outType <ICV SPICE SPICE_FLAT VERILOG></code> .
VERILOG Cells	Adds an input netlist in Verilog format. You can specify the searching path in Dir Path and specify the file extension in File Ext. The default for Dir Path is the run directory and the default for File Ext. is .v. Custom Compiler can run IC Validator using the <code>-verilog</code> command-line option to specify these input netlists files in the Verilog format. For example, <code>icv_nettran -icv <file> -outName <file> -outType <ICV SPICE SPICE_FLAT VERILOG></code> .

5. The control variables are loaded into the **Control Variables** tab using runset files compatible with the IGraph mode or using the legacy mode.

- To load control variables from IGraph compatible runset files,

- Click the **Load Control Variables**  option to load the control variables file from the runset file.



Select the settings for IC Validator as described in [Table 15](#):

Table 15 IC Validator LVS Setup and Run Settings (Custom Variables Tab)

Settings	Description
Macros	Lists the macro in the runset file. The following fields are available: ▪ Name: Displays the name of the macro. ▪ Define: Selects the macro defined in the runset file as the default. The option is selected if the macro is defined in the runset file. ▪ Define As: Displays the value of the macro.
Env Variables	Lists the environment variables in the user-defined runset file. The following fields are available: ▪ Name: Displays the name of the environment variable ▪ Current Value: Displays the value of the environment variable. ▪ Set: Selects the environment variable by default. To set the value of an environment variable to match the original setting of the runset file, deselect the Set option. ▪ Set to: Displays the Current Value of the environment variable as default. To add a value of the environment variable, deselect the Set option.

Note:

You can use the IGraph compatible version of **Control Variables** tab only with V-2023.12-SP2 or later versions of the IC Validator tool

- In the **Control Variables** tab you can review, optionally modify, and reset the current values for LVS runset control variables as preferred for this LVS run. The control variables are preset for a PDK.

To modify a preset value, click with the alternate (non-selection) mouse button, and edit the entry field.

To reset the control variables to the default, click the **Reset** button. The default is imported from the `pdk.xml` file.

- The preprocessor, environment, and PXL variables are supported on the **Control Variables** tab. Variable settings are stored and recalled from the PVE run directory. They can be ordered, grouped, and displayed with special option types, depending on their type (such as Boolean, integer, file input, and so on).

- To use the **Control Variables** tab in the legacy mode:
 - Use the `pdk.xml` file. You can review, optionally modify, and reset the current values for LVS runset control variables as preferred for this LVS run. The control variables are preset for a PDK by default.

To modify a preset value, click with the alternate (non-selection) mouse button and edit the entry field. To reset the control variables to the default click the

Reset  button. The default is imported from the `pdk.xml` file.

In the legacy mode only, preprocessor, environment, and PXL variables are supported in the **Control Variables** tab.

- Add the `controlVarsFile` XML tag in the `pdk.xml` file to point to a variable definition file. For example:

```
<pveJob name="DRC" jobType="DRC">
  <preference name="ViewOutput" mutable="false">
    <value>true</value>
  </preference>
  <tool name="IC Validator">
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc.rs</value>
    </preference>
    <preference name="LoadRunset">
      <value>true</value>
    </preference>
    <controlVarsFile>./snps_reference40nm/pdk_controlVars.xml</controlVarsFile>
  </tool>
</pveJob>
```

Define the variables in the variable definition file with the XML syntax. For example:

```
<switch name="POLY_RULES_PP" scope="preProc" valueType="bool" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>Enable Poly Rules(pp)</description>
  <value>true</value>
</switch>

<switch name="METAL_RULES_PP" scope="preProc" valueType="enum" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>Check Metal Rules up to(pp)</description>
  <value>M4</value>
  <enumValues enumList="M1 M2 M3 M4 M5 M6" default="M4"/>
</switch>

<switch name="ICV_DRC_DIR_pp" scope="preProc" valueType="string" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>directory containing ICV rule decks(pp)</description>
  <value>$PDK_INSTALL_DIR/icv/drc</value>
</switch>
```

The tool issues the following information message when the Control Variables file is available in the pdf.xml file:

```
Information: Found Control Variables file in pdk.xml.
proceeding to use legacy control variable table. Recommend
for switching to new Control Variable table format.
```

Note:

If the file is not specified with the `controlVarsFile` XML tag, the tool issues the following information message:

```
Information: No Control Variables file found in
pdk.xml, switching to New Control Variables table for
mat.
```

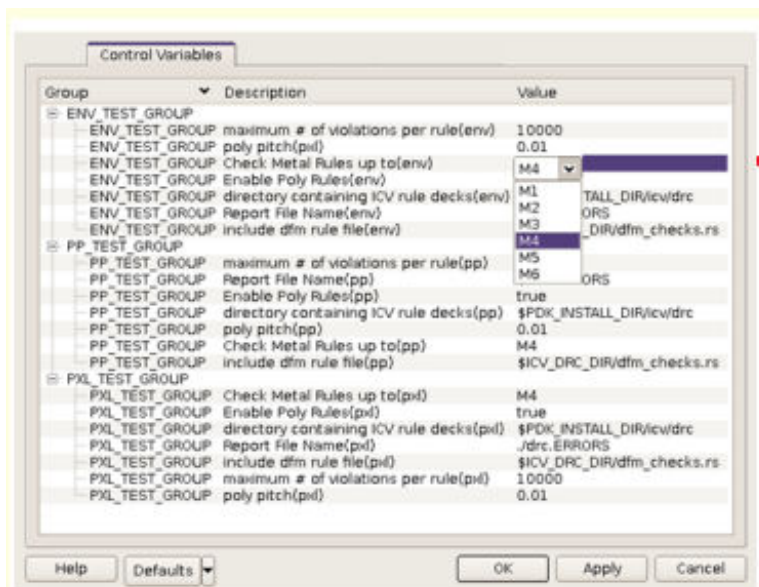
- You can also use Tcl commands to define and setup the control variables (instead of the `pdk.xml` file). For example:

```
set s [db::getNext [db::getScopes pdk_pve]]
db::setPrefValue xtDRCICVControlVars -value {
  POLY_RULES_PP
    {description {Enable Poly Rules(pp)}
     value true group PP_TEST_GROUP mutable true
     visible true scope preProc valueType bool}
  METAL_RULES_PP
    {description {Check Metal Rules up to(pp)}
     enumList {M1 M2 M3 M4 M5 M6} default M4 value M4 group
} -scope $s
```

Table 16 Scope of Control Variables

Scope	Runset Presentation	Example
pxl	{name} : {type} = {value}	PREFIX : string = "METAL"
env	#pragma envvar default {type} {value}	#pragma envvar default OUTPUT_MODE "stream"
preProc	#define {name} {value}	#define FUNCTIONS_RS TRUE

- After you define these variables and set them (from either the `pdk.xml` or Tcl), ensure that the ICV ruleddeck is written correctly to consume these variables. See the following figure .



- To use the variable correctly, prepare the ruledeck as follows:

When you execute IC Validator, the tool writes the variables into the ICV ruledeck at the beginning of the file. To do prepare the rule deck, check if the original ruledeck has some of the variables defined with default, then wrap the variables into an #if statement, so the variables become active only when a conditional variable is `true`. (The variable can be `false` in the Custom Compiler variable definitions because the variables are obtained from the GUI. If the variable is `false`, the variable is executed from outside of the tool's environment. The variables are not expected to be set by the GUI.)

When you change a variable setting (as shown with a red arrow in the previous figure), the new value is written to the top of the ICV ruledeck, before IC Validator execution (as shown with a red arrow in the following figure).

```
METAL_RULES_pxl : string = "M4";
#define MAX_VIOLATION_COUNT_pp 10000
POLY_RULES_pxl : boolean = true;
#pragma envvar default MAX_VIOLATION_COUNT_env 10000
#define REPORT_FILE_NAME_pp ./drc.ERRORS
#pragma envvar default POLY_PITCH_env 0.01
ICV_DRC_DIR_pxl : string = "$PDK_INSTALL_DIR/icv/drc";
#pragma envvar default METAL_RULES_env M4
#define POLY_RULES_PP
#pragma envvar default POLY_RULES_env true
REPORT_FILE_NAME_pxl : string = "./drc.ERRORS";
#define ICV_DRC_DIR_pp $PDK_INSTALL_DIR/icv/drc
#pragma envvar default ICV_DRC_DIR_env $PDK_INSTALL_DIR/icv/drc
```

- For all tools, for OpenAccess format, on the **Custom Options** tab enter settings as described in [Table 17](#).

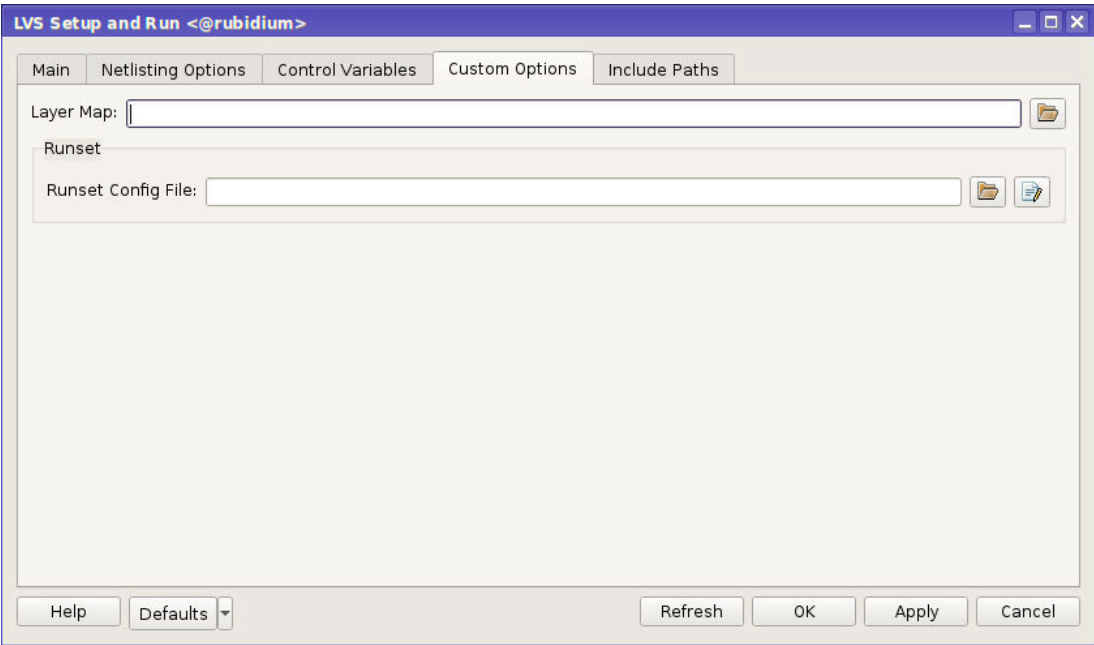


Table 17 LVS Setup and Run Settings (Custom Options Tab)

Settings	Description
Layer Map	For OpenAccess format only, accept or browse to choose the file mapping OpenAccess numbers and traditional GDSII stream layers/data types.
Runset	Runset Config File specifies a runset configuration file that is passed to an IC Validator or Hercules command using the <code>-runset_config</code> option. Use the <code>xt<pveJob>ICVRunsetConfigFile</code> preference to set this mode. Edit Runset Config edits the runset configuration file. ◦ If the file does not exist, create a file using the Create File Prompt dialog box ◦ If the file is empty, create a file with the default name <code>runset_config.rs</code> using the Create File Prompt dialog box Runset Options specifies the supported IC Validator or Hercules runset options. Custom Compiler does not populate the Runset Options table while loading a runset. To display the Runset Options table, set <code>xt<LVS>ICVShowRunsetOptions</code> preference to <code>xt<pveJob>ICVRunsetOptions</code> is not empty.

Note:

The **Layer Map** file is read from the `dbExportStreamLayerMapFile` preference when the `xtPVLayrMapFile` preference is not defined by the user, or found in the PDK configuration file.

7. For the IC Validator tool only, on the **Custom Options** tab, review and adjust settings for the IC Validator options now displayed from the current runset. Runset parsing is off by default. To switch on runset parsing and have the tool parse the ICV runset and add those values into this tab, use the `xtLVSICVLoadRunset` preference.
8. For the Calibre tool only, on the **Custom Options** tab, enter additional settings as described in [Table 18](#).

The screenshot shows the 'LVS Setup and Run' dialog box with the 'Custom Options' tab selected. The dialog has several tabs: Main, Netlisting Options, Control Variables, Custom Options (active), and Include Files. The settings are as follows:

- Layer Map:** (empty text field)
- Output Runset:** `lockWithTiming_Lab/synopsys_custom/opdecoder.calibre.lvs/opdecoder.lvs.custom_compiler.svrf` (with folder and file icons)
- ☒ Override Local Runset
- ☐ Override Primary Runset
- Report File:** `l_ExportingCustomBlockWithTiming_Lab/synopsys_custom/opdecoder.calibre.lvs/opdecoder.lvs.report` (with folder icon)
- ☐ Auto-match Cells by Name
- H-Cells File:** (empty text field)
- SVDB Directory:** `hg/CoDesign_ICC1_ExportingCustomBlockWithTiming_Lab/synopsys_custom/opdecoder.calibre.lvs/svdb` (with folder icon)
- ☒ Generate CCI Database
- Output Netlist:** `gn_ICC1_ExportingCustomBlockWithTiming_Lab/synopsys_custom/opdecoder.calibre.lvs/opdecoder.net` (with folder icon)
- LVS Power Name:** (empty text field)
- LVS Ground Name:** (empty text field)
- ☒ Virtual Connect
 - Connect Net(s) By Name:** (empty text field)
 - Ignore Characters After:** ☒ None ☐ Colon ☐ Colon+Semicolon
 - Hierarchical Depth:** ☒ Primary ☐ All
 - Connection Report:** ☒ No ☐ Yes ☐ Unsatisfied
 - Maximum message:** ☒ All ☐ Count (with empty text field)

At the bottom, there are buttons for 'Help', 'Defaults' (with a dropdown arrow), 'Refresh', 'OK', 'Apply', and 'Cancel'.

Table 18 Calibre Custom Options

Settings	Description
Output Runset	Specify the output runset by accepting the listed runset or browsing to use a new top-level control file containing the database inputs, control variables, custom options, and primary rules.

Settings	Description
	Alternatively, you can use a local runset you have edited. To do this, click the Edit SVRF button to the right of the browse button. In the default text editor, edit the runset and save the changes. The presence of an edited runset deselects the Override Local Runset option, allowing the LVS process to automatically use the edited version. Override Local Runset Overrides any local runset you have edited and applies the options you set in this dialog box. Override Primary Runset Overrides primary rules file SVRF commands with top-level control file SVRF commands set using this dialog box. Choose this option to reduce the number of unnecessary specification statements. By default, the Calibre tool exits with an error when it finds duplicate specification statements.
Report File	Accept or specify a new location for the LVS report file used to capture the results of the verification.
Auto-match Cells by Name	Choose this option to match cells in the layout with cells in the schematic by name wherever possible.
H-Cells File	Accept or browse to select a different directory to store the file with the names of corresponding schematic and layout cells.
SVDB Directory	Accept or browse to save the LVS verification results database in a different directory.
Generate CCI Database	Choose this option to include in the LVS results database information for generating the Calibre Connectivity Interface used by parasitic extraction tools to access layout extraction results.
Output Netlist	Accept or browse to save the exported layout netlist in a different location during the circuit extraction step.
LVS Power Name	Export the specified LVS power name during the circuit extraction step using the preference <code>xtLVSCalibrePowerName</code> .
LVS Ground Name	Export the specified LVS ground name during the circuit extraction step using the preference <code>xtLVSCalibreGroundName</code> .
Virtual Connect	Select to compute and verify circuit connectivity. Connect Net(s) by Name: Type the net names for which you want to determine connectivity. The names are not case sensitive. Labels that have the same name are considered to be on the same net and have a virtual connection. Note: To connect nets by name, wildcard characters are supported by the Calibre tool. See the Mentor Graphics® Calibre® user documentation for more information on wildcard support.

Settings	Description
	Ignore Characters After: Select the method to use to calculate connections: ◦ None: For the entire net name. ◦ Colon: For the net name that appears before colons. ◦ Colon+Semicolon: For the net name that appears either before colons or before semicolons. Hierarchical Depth: Select whether to calculate connections for the top level only (Primary) or for all levels (All). Connection Report: Select whether to generate a connection report and what to report when a virtual connection is made by the connectivity extractor: ◦ No: Does not generate a report. ◦ Yes: Generates a full report. ◦ Unsatisfied: Reports only errors. Maximum Message: Select how to limit the number of messages shown in the report: ◦ All: Reports all virtual connection messages. ◦ Count: Specifies the maximum number (integer value) of virtual connection messages to report.

9. For the IC Validator tool only, on the **Include Paths** tab, add or delete LVS include paths.

a. In the table, double-clicking a blank row adds an include path. Double-clicking a populated row replaces that include path.

Cycle through the list using the arrow buttons.

To delete a path, click the **Delete** button.

b. Browse to choose the path you want to add.

c. Click **Select**.

d. Repeat for each include path you want to add or replace.

10. For the Calibre tool only, on the **Include Files** tab, add or delete LVS include paths.

a. In the table, double-clicking a blank row adds an include file. Double-clicking a populated row replaces that include file.

Cycle through the list using the arrow buttons.

To delete a file, click the **Delete** button.

b. Browse to choose the file you want to add.

- c. Click **Select**.
- d. Repeat for each include path you want to add or replace.
- 11. Click **OK** to start the layout versus schematic comparison.
- 12. Track the process by [Viewing LVS Comparison Output](#).
- 13. Complete the flow by [Debugging the Design Using LVS Results](#).

Note:

The following warning message appears in the console when the OpenAccess version of Custom Compiler is higher than the OpenAccess version on IC Validator:

```
Warning: OA version of Custom Compiler <CC OA overversion> is  
higher than IC Validator <ICV OA version>.  
This may lead to OA API compatibility errors to evaluate  
PyCells in the design.  
Please proceed with GDS input for DRC run to avoid [liboa]  
errors.
```

Using Custom Jobs for LVS Comparisons

Use a custom job for an LVS run to apply selected settings from a Process Design Kit (PDK).

To use a custom job for an LVS run:

1. In the design layout window, choose **Verification > *custom_job_name* Setup and Run**, where *custom_job_name* is the name of the job created for your use by your CAD support group.
2. In the **Setup and Run** dialog box for this custom job, specify settings for the LVS run.
For assistance with settings, refer to [Using Integrated Tools for LVS Comparisons](#).
3. Click **OK** to begin layout versus schematic comparison.
4. Continue by [Viewing LVS Comparison Output](#).
5. Complete the flow by [Debugging the Design Using LVS Results](#).

Rerunning Layout Versus Schematic Comparisons

Rerun an LVS comparison after running it one time. You can use the same tool and settings, and you can vary the runset or reload a changed runset.

To rerun a layout versus schematic comparison:

1. In the design layout window, consider your choices.
 - To rerun the LVS comparison with the same tool, settings and runset, you can choose **Verification > LVS > Run**.

Alternatively, you can start the run by clicking the **LVS Run** button on the command history toolbar or the Layout Verification toolbar at the lower left.
 - To rerun the LVS comparison with the same tool and settings, but a different runset, choose **Verification > LVS > Setup and Run**. Then, on the **Main** tab under **Job Parameters**, browse to select a new runset file.
 - To rerun the LVS comparison with the same tool and settings, but a runset file whose contents have changed, choose **Verification > LVS > Setup and Run**. Then, on the **Main** tab under **Job Parameters** reload the file by clicking the **Reload Runset** button.
2. Continue the flow by [Viewing LVS Comparison Output](#).

Complete the flow by [Debugging the Design Using LVS Results](#).

Viewing LVS Comparison Output

View LVS comparison output to see logged operations, results, and errors.

To view layout versus schematic comparison output:

1. If the **View Output** option was set and the Text Viewer is now open, skip this step. Otherwise, in the design layout, choose **Verification > LVS > View Output**. Alternatively, you can use the keyboard shortcut Shift+Ctrl+D.
2. In the Text Viewer, view results and logged operations on each tab.
3. After you view LVS run output, continue by [Debugging the Design Using LVS Results](#).

Debugging the Design Using LVS Results

Debug the design by viewing LVS output and using the IC Validator VUE or Hercules VUE tools. Running an LVS one time initializes the VUE environment for that tool.

To debug the design using LVS results:

If you are using the IC Validator tool, you can debug results of multiple LVS processes for one or more designs simultaneously.

1. In the design layout, consider these options.
 - If you elected to launch VUE when you set up LVS, VUE has opened automatically if violations were found.
 - If VUE is not open, you can choose **Verification > LVS > Debug**.
2. In the VUE window, identify and evaluate violations by cross-probing error listings with the affected design areas.
3. After iterating the design until it achieves LVS equivalence, optionally perform [Extracting Interconnect Parasitics](#).

Note:

The same VUE window is updated when you run LVS multiple times on the same design.

A new VUE window opens when you open a new design.

Comparing Layout Versus Layout (LVL)

This topic describes the following:

- [About LVL Comparisons](#)
- [Before You Start](#)
- [Running LVL Comparisons](#)
- [Viewing LVL Comparison Output](#)
- [Debugging the Design Using LVL Results](#)

About LVL Comparisons

A layout-versus-layout (LVL) comparison allows you to compare a streamed out layout view against the reference layout view to ensure the design has not been modified or corrupted. The comparisons are performed layer by layer based on the comparison method you select.

You can use the Custom Compiler Verification interface to perform IC LVL comparisons. The Verification interface lets you select the one of the following comparison methods:

- XOR
- NOT
- OUTSIDE

For the description of the comparison methods, see the [IC Validator user documentation](#).

The Custom Compiler Verification interface also lets you:

- View the LVL output. See [Viewing LVL Comparison Output](#).
- Debug the design. See [Debugging the Design Using LVL Results](#).

Before You Start

Before you start the LVL comparison, see:

- [Enabling the Verification Interface](#)
- [Setting Verification Process Node Options](#)

Running LVL Comparisons

To compare two layouts:

1. In the Layout Editor, choose **Verification > LVL > Setup and Run**. The **LVL Setup and Run** dialog box opens and displays the **Main** tab.

The screenshot shows the 'LVL Setup and Run' dialog box with the 'Main' tab selected. The dialog is titled 'LVL Setup and Run' and has a close button (X) in the top right corner. The 'Main' tab is active, showing the following settings:

- Run Dir:** dmCompiler_Overview_Lab1/synopsys_custom/vco.icv.lvl (with a folder icon button)
- Area:** ☒ Design ☐ Viewport ☐ Region
- Type:** ☒ XOR ☐ NOT ☐ OUTSIDE
- Export:**
- Layout 1:**
 - Format:** ☒ OpenAccess ☐ Stream ☐ OASIS
 - Layer Mapping/File:** ☒ None ☐ OA Layer Map ☐ ICV Layer File
 - Library:** DemoVco (dropdown)
 - Cell:** vco (dropdown)
 - View:** layout (dropdown) (with a folder icon button)
- Layout 2:**
 - Format:** ☒ OpenAccess ☐ Stream ☐ OASIS
 - Layer Mapping/File:** ☒ None ☐ OA Layer Map ☐ ICV Layer File
 - Library:** DemoVco (dropdown)
 - Cell:** vco (dropdown)
 - View:** layout (dropdown) (with a folder icon button)
- Job Parameters:**
 - Tool:** Validator (dropdown) ☒ Launch Debugger ☒ View Output
 - Job Class:** LVL (dropdown) - Arguments:** -oa_dm5 (text field)

At the bottom of the dialog, there are buttons for **Help**, **Defaults** (with a dropdown arrow), **Refresh**, **OK**, **Apply**, and **Cancel**.

2. In the **Job Parameters** area, choose the tool you want to use from the **Tool** list. The dialog box reconfigures to reflect available options for this tool based on the current runset.

3. Set the following:

Settings	Description
Run Dir	Accept the current directory, or specify a different directory with <code><directory>/<cellname>.icv.lvl</code> .
Area	Area on which to run the LVL check as design, viewport, or region, specified by coordinates of its rectangle, for example: [x1, y1, x2, y2] [-20, -20, 20, 20]. To automatically insert coordinates here for a selection area, click the interactive selection button, and then select the area in the design.
Type	Select the LVL comparison method: XOR , NOT , or OUTSIDE . See the IC Validator user documentation .
Export	Set the options for exporting OpenAccess layout views to the GDS or OASIS format. The same export options are applicable to both Layout 1 and Layout 2.
Layout 1 Layout 2	Select the format of Layout 1 and Layout 2: - OpenAccess: Specify the Library, Cell, and View. - Stream: Specify the Stream File and the Cell. You have the option of selecting Export Stream to export the layout cellview to a GDS file. - OASIS: Specify the OASIS File and the Cell. You have the option of selecting Export Oasis to export the layout cellview to an OASIS file. If you select Export Stream or Export OASIS, you can click Export: Config Stream or Export: Config OASIS to set export options. See Exporting Design Data in GDSII Stream Format or Exporting Design Data in OASIS Format. The options you specify are integrated with PVE (preferences in the <code>pdk.xml</code> configuration). Select the Layer Mapping File: None, OA Layer Map, ICV Layer File. For OA Layer Map and ICV Layer File, select the layer map or file, and the Library, Cell, and View information.
Job Parameters	Select the Tool to use for LVL comparison. Select Launch Debugger to display the integrated debugging tool when the comparison operation finishes if violations are found. Select View Output to open a Text Viewer window with the process log after the comparison operation finishes.

Settings	Description
	Choose a Job Class to combine settings for individual analysis tools. You can configure existing job classes or user-defined job classes. For details, see Adding Job Classes With <code>xt::createJobClass Tcl</code> in the <i>Custom Compiler Environment User Guide</i> .
	Click Options to specify job settings that control batch processing limits and optionally set a queue engine in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i> .
	Type the Arguments to that are accepted by the command for this integrated tool you are using.
Refresh	Checks if the design name is same as LCV. Note: if the design name is not the same as LCV, the LCV in the dialog box is updated to the current design name.

4. (Optional) To compare only a region in the two layouts:

a. Click the **Custom Options** tab.

b. Click .

c. On the layout canvas, draw a rectangle to define the region that you want to compare.

5. Click **OK** to start the LVL comparison.

Viewing LVL Comparison Output

View LVS comparison output to see logged operations, results, and errors.

To view layout versus schematic comparison output:

1. If the **View Output** option was set and the Text Viewer is now open, skip this step. Otherwise, in the design layout, choose **Verification > LVL > View Output**.
2. In the Text Viewer, view results and logged operations on each tab.

Debugging the Design Using LVL Results

Debug the design by viewing LVL output and using the IC Validator VUE or Hercules VUE tools. Running LVS one time initializes the VUE environment for that tool.

To debug the design using LVS results:

If you are using the IC Validator tool, you can debug results of multiple LVS processes for one or more designs simultaneously.

1. In the design layout, consider these options.
 - If you elected to launch VUE when you set up LVS, VUE has opened automatically if violations were found.
 - If VUE is not open, you can choose **Verification > LVL > Debug**.
2. In the VUE window, identify and evaluate violations by cross-probing error listings with the affected design areas.
3. After iterating the design until it achieves LVS equivalence, optionally perform [Extracting Interconnect Parasitics](#).

Comparing Netlist Versus Netlist (NVN)

This topic describes the following:

- [About NVN Comparisons](#)
- [Before You Start](#)
- [Running NVN Comparisons](#)
- [Viewing NVN Comparison Output](#)
- [Debugging the Design Using NVN Results](#)

About NVN Comparisons

A netlist-versus-netlist (NVN) comparison allows you to compare a netlist against a reference netlist to ensure net integrity. The comparisons are performed using the integrated IC Validator tool, which is accessed through the Custom Compiler Verification interface.

For information about the IC Validator, see the [IC Validator user documentation](#).

Before You Start

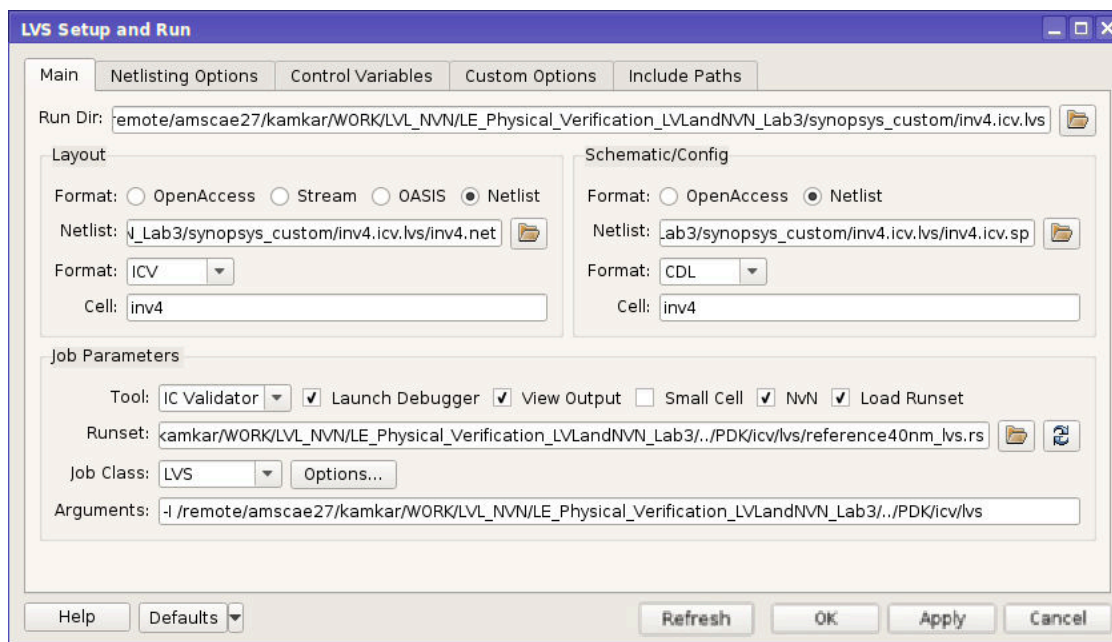
Before you start the NVN comparison, see:

- [Enabling the Verification Interface](#)
- (Optional) [Setting Verification Process Node Options](#)

Running NVN Comparisons

To compare a netlist against a reference netlist:

1. In the Layout Editor, choose **Verification > LVS > Setup and Run**. The **LVS Setup and Run** dialog box opens and displays the **Main** tab.



2. In the **Run Dir** area, accept the current directory, or specify a different directory with `<directory>/<cellname>.icv.lvs`.
3. In the **Job Parameters** area, set the following:

Settings	Description
Tool	Specifies the tool to use for the comparison. Choose IC Validator . The dialog box reconfigures to reflect available options for this tool based on the current runset.
Launch Debugger	Displays the integrated debugging tool when the comparison operation finishes if violations are found.
View Output	Opens a Text Viewer window with the process log after the comparison operation finishes.
Small Cell	(IC Validator only) Overrides automatic small cell determination and skips runset optimizations for distributed processing during runset compilation that could dominate the overall time for small cells. If the input design is too large, the IC Validator tool exits with an error.

Settings	Description
NvN	(IC Validator only) Enables the Netlist and Format fields in the Layout area so that you can compare a netlist against a reference netlist.
Runset	The Runset file. If you want the tool to load the runset file when you start the comparison, select Load Runset . If you want to load the runset file before running the comparison, click  .
Job Class	Combines settings for individual analysis tools. You can configure existing job classes or user-defined job classes. For details, see <i>Adding Job Classes With xt::createJobClass Tcl</i> in the <i>Custom Compiler Environment User Guide</i> . Click Options to specify job settings that control batch processing limits and optionally set a queue engine in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i> .
Arguments	Arguments that are accepted by the command for the integrated tool you are using.

4. In the **Layout** area, set the following:

Settings	Description
Format	The format to use for the comparison. Select Netlist .
Netlist	The path to the netlist file.
Format	The format of the netlist file.
Cell	The cell name.

5. In the **Schematic/Config** area, set the following:

Settings	Description
Format	The format to use for the comparison. Select Netlist .
Netlist	The path to the reference netlist file.
Format	The format of the reference netlist file.
Cell	The cell name.

6. (OpenAccess schematic format only) In the **Netlisting Options** tab, specify the NetTran Options (or netlist translator options) for converting the netlist into the format that IC Validator recognizes.

Note:

The top-level module is always created by default. You can control the top-level module creation using the `xtLVSNetlistTopAsModule` preference. It is best practice to set this prefix using the `pdk.xml` file, otherwise you will have to set the preference in the run directory.

7. If you want to view and modify runset control variables, see [Using Integrated Tools for LVS Comparisons](#).
8. In the **Custom Options** tab, specify the **Layer Map** file path. Also, review and adjust settings for the IC Validator options now displayed from the current runset. Runset parsing is off by default. To switch on runset parsing and have the tool parse the ICV runset and add those values into this tab, use the preference `xtLVSIcVLoadRunset`.
9. In the **Include Paths** tab, add any files that you want to the tool to process during the comparison operation.
10. Click **OK** to start the NVN comparison.

Viewing NVN Comparison Output

View NVN comparison output to see logged operations, results, and errors.

To view layout versus schematic comparison output:

1. If the **View Output** option was set and the Text Viewer is now open, skip this step. Otherwise, in the design layout, choose **Verification > LVS > View Output**.
2. In the Text Viewer, view results and logged operations on each tab.

Debugging the Design Using NVN Results

Debug the design by viewing NVN output and using the IC Validator VUE tool. Running an NVN one time initializes the VUE environment for that tool.

To debug the design using NVN results:

With the IC Validator tool, you can debug results of multiple NVN processes for one or more designs simultaneously.

1. In the design layout, consider these options.
 - If you elected to launch VUE when you set up NVN, VUE has opened automatically if violations were found.
 - If VUE is not open, you can choose **Verification > LVS > Debug**.
2. In the VUE window, identify and evaluate violations by cross-probing error listings with the affected design areas.
3. After iterating the design until it achieves NVN equivalence, optionally perform [Extracting Interconnect Parasitics](#).

11

FILL Cell Generation Using Integrated Tools

Verify the expected behavior of layout components by performing design rule checks, layout-versus-schematic (LVS) comparisons, and generating fill cells. Extract device parasitics to identify devices, including resistors, capacitors and inductors, that are not present in the design schematic.

This chapter describes the following:

- [Setting Verification Process Node Options](#)
- [Using Integrated Tools for FILL Cell Generation](#)
- [Using Custom Jobs for FILL](#)
- [Rerunning FILL](#)
- [Viewing FILL Output](#)

Setting Verification Process Node Options

Set physical verification and extraction options to specify the manufacturing process node to associate with design rule checking, layout versus schematic comparison, parasitic extraction, and Fill runs.

Before you start, make sure you have the required files for physical verification and extraction. See [Preparing for Physical Verification and Extraction](#).

For information about parasitic extraction, see [Using Integrated Tools for FILL Cell Generation](#).

Note:

See [Default Interfaces](#) and [Custom Jobs](#) for further information.

To set physical verification and extraction options:

1. In the Layout Editor, choose **Verification > Options**.
2. Choose the **Process** and its **Variant** from the lists. See [Process and Variant Selection](#).
3. (Optional) If changes have been made to the PDK, click the **Reload** button to apply those changes.

Using Integrated Tools for FILL Cell Generation

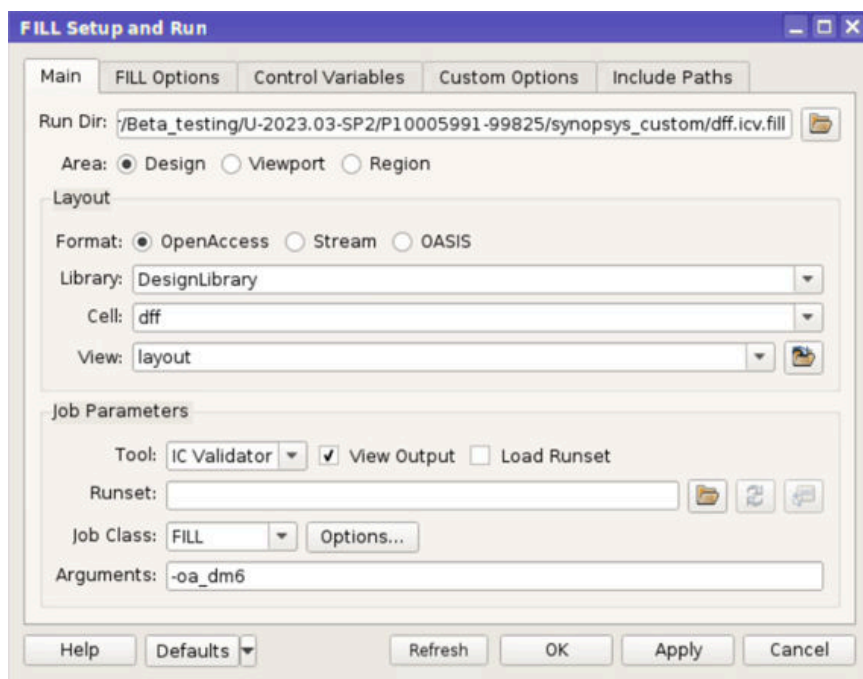
You can use the IC Validator tool for FILL on the Custom Compiler environment.

You can set up and run FILL, and then rerun it using the same or slightly varied settings for iterative development. Use the IC Validator VUE tool to evaluate results.

To set up and run integrated FILL:

1. In the design layout window, choose **Verification > FILL > Setup and Run**.

(Choosing **Tools > Verification** adds the **Verification** menu to the window menu bar.)



2. In the **FILL Setup and Run** dialog box under **Job Parameters**, choose the tool you want to use from the **Tool** list.

The dialog box reconfigures to reflect available options for this tool based on the current runset, and the tool uses the runset options shown here.

Continue by entering input requirements for this tool on the **FILL Setup and Run** dialog box **Main** tab.

3. On the **Main** tab, enter settings as described in [Table 19](#) .

Table 19 IC Validator Setup and Run Settings (Main Tab)

Settings	Description
Run Dir	You can accept the current run directory or browse to choose a different directory with the <directory>/<cellname>.icv.fill runset. For more information on the Run Dir setting and supported tokens, see Table 20.
Area	Area on which to run the FILL operation as Design, Viewport, or Region, specified by coordinates of its rectangle, for example: [x1, y1, x2, y2] [-20, -20, 20, 20]. To automatically insert coordinates here for a selection area, click the interactive selection button, and then select the area in the design. Region supports the select window , which specifies rectangular regions for FILL operations to be processed within.
Layout Format	For OpenAccess format, additionally specify the Library, Cell, and View. For the Stream or OASIS format, set the following: ◦ Export Stream or Export OASIS: Exports the layout cellview used as input to the File you specify. ◦ Config: (Available only if you selected Export Stream or Export OASIS) Opens the Export Stream or Export OASIS dialog box. To set the options in the dialog box, see Exporting Design Data in GDSII Stream Format or Exporting Design Data in OASIS Format in the <i>Custom Compiler Environment User Guide</i>. The options you specify are integrated with PVE (preferences in the <code>pdk.xml</code> configuration).
Job Parameters	Tool is IC Validator. View Output opens a Text Viewer window with the process log. Load Runset loads values from the user-supplied runset into the setup dialog when the runset value changes. Use the <code>xtFILLICVLoadRunset</code> preference to set this mode. Load Rundir loads automatically the values from the existing <code>prefs.xml</code> file inside the specified Run Dir (if any) into the Setup dialog. Use the <code>xt<pveJob>AutoPrefs</code> preferences, for example, <code>xtFILLAutoPrefs</code>, to set this mode. The default is true.

Settings	Description
	Runset sets the path to the runset file with rules governing the run. Browse to select the file. Click Reload  to reload the runset. Click  to open the runset with a text viewer. Supports queue by Job Class > Options. Combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. Available options are: FILL. For details, see Adding Job Classes With <code>xt::createJobClass Tcl</code> in the <i>Custom Compiler Environment User Guide</i>. Options configures the job class in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i>. Arguments accepts command-line options for this tool, if any.
Refresh	Checks if the design name is same as Library, Cell, View (LCV). Note: if the design name is not the same as LCV, the LCV in the dialog box is updated to the current design name.  Show me an example of how to refresh the LCV.
Defaults	Creates and loads the default settings in the dialog box.

The results directory for the PVE tool is **Run Dir**. Its value is defined by the `xt<pveJob>RunDir` preference, which is defined in the cellview preference scope of the active design context. If the `xt<pveJob>RunDir` preference is not defined, the run directory is defined by calling the `xt::getRunDirName Tcl` command. The default implementation of this procedure returns a string of the format:

```
<xtRunDirBase>/<xt<pveJob>RunDirNameTemplate>
```

Where: `<xtRunDirBase>` is the value of the `xtRunDirBase` preference and `xt<pveJob>RunDirNameTemplate` is the value of the `xt<pveJob>RunDirNameTemplate` preference -- a token substitution string. The allowed tokens in `xt<pveJob>RunDirNameTemplate` are as follows:

Table 20 Supported Tokens

Token name	Token code	Substitution value
Library	%l	Library name
Cell	%c	Cell name
View	%v	View name

Token name	Token code	Substitution value
Job	%j	PVE Job name
Tool	%t	PVE Tool name
User	%u	User name

If the information required by some of the tokens is not available at the time the run directory name is constructed by `xt::getRunDirName`, the token is replaced by the string, `unknown<tokenName>`.

4. In the **FILL Options** tab, enter settings for IC Validator, as described [Table 21](#).

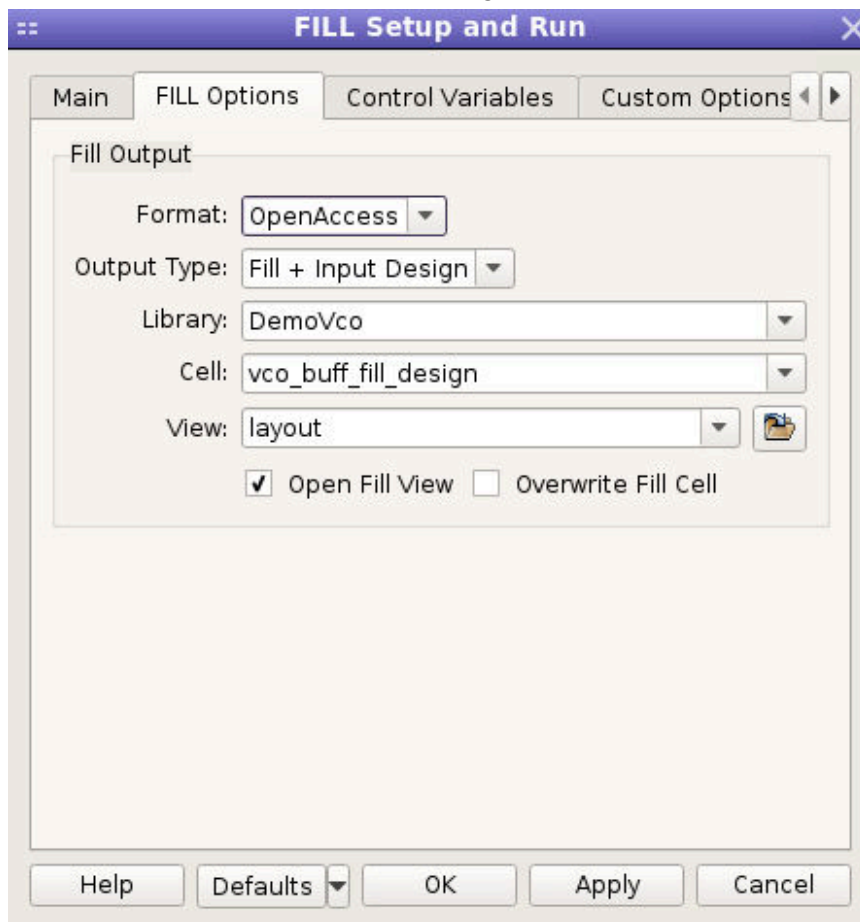


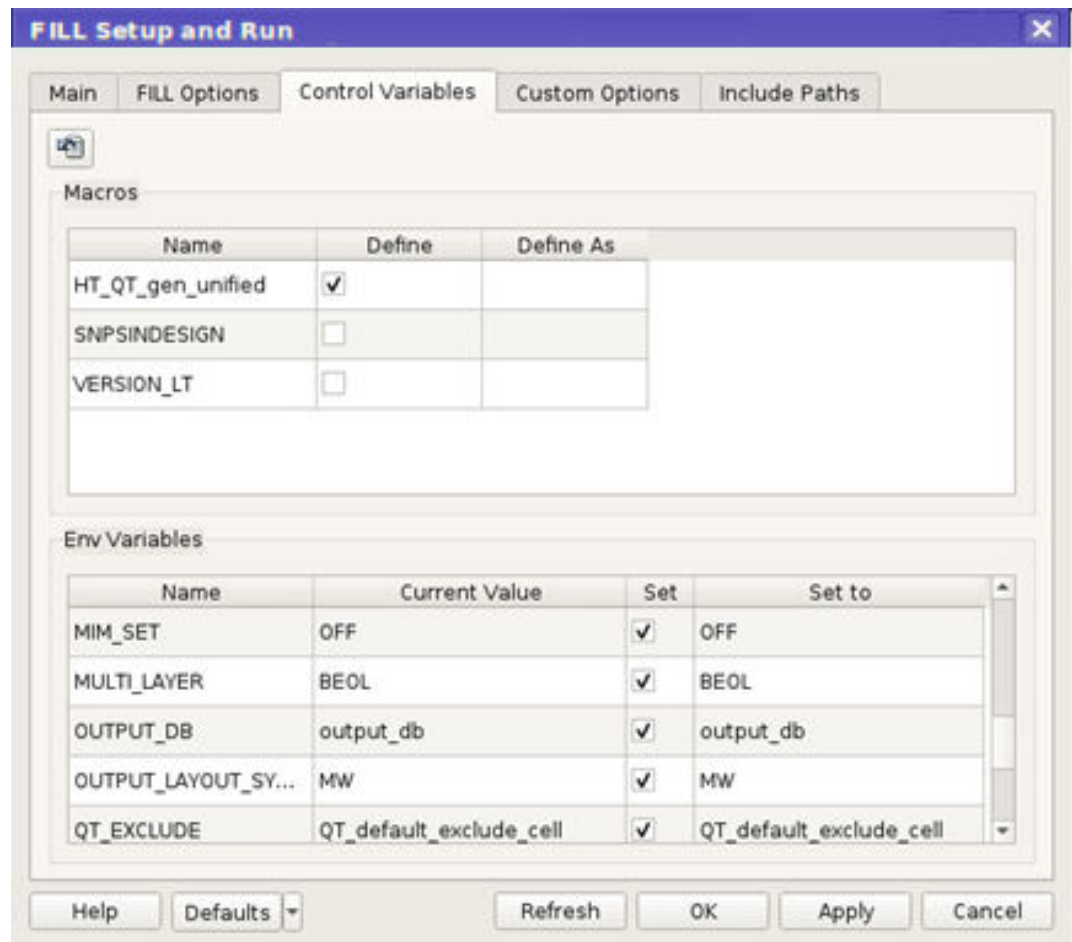
Table 21 IC Validator FILL Setup and Run Settings (FILL Options Tab)

Settings	Description
Fill Output Format	For OpenAccess format, additionally specify the Library, Cell, and View. For the Stream or OASIS format, the FILL operation checks for the presence of a stream or OASIS file in Run Dir. If these files do not exist, a warning message is issued.
Output Type	Specifies the output type with available options: ◦ Fill + Input Design: The default. Creates a fill cell in the output library with a copy of the input design, and instantiates the fill cell copied on that input design. ◦ Fill. Creates a new fill cell in the output library. ◦ Merge. Creates a fill cell and instantiates in the input design. The input design is backed up using <i>input_design_bak</i>.
Library	For OpenAccess output, the library is copied from the input library field. The library, cell, and view values are associated with the <code>xt<pveJob>OutputLCV</code> preference.
Cell	For OpenAccess output, the cell is copied from input cell field with the suffix, “_fill_design” or “_fill_only” depending on the output type.
View	See Library .
Open Fill View	If checked, opens fill cell in another layout window. It is only for Open Access output format, and it is checked by default. This checkbox is associated with the <code>xt<pveJob>ViewFillCell</code> .
Overwrite Fill Cell	If checked, overwrites the existing fill cell.

5. The control variables are loaded into the **Control Variables** tab using runset files compatible with the IGraph mode or using the legacy mode.

- To load control variables from IGraph compatible runset files,

- Click the **Load Control Variables**  option to load the control variables file from the runset file.



The screenshot shows the 'FILL Setup and Run' dialog box with the 'Control Variables' tab selected. The dialog has five tabs: Main, FILL Options, Control Variables, Custom Options, and Include Paths. The 'Control Variables' tab contains two sections: 'Macros' and 'Env Variables'.

Macros

Name	Define	Define As
HT_QT_gen_unified	☒	
SNPSINDESIGN	☐	
VERSION_LT	☐	

Env Variables

Name	Current Value	Set	Set to
MIM_SET	OFF	☒	OFF
MULTI_LAYER	BEOL	☒	BEOL
OUTPUT_DB	output_db	☒	output_db
OUTPUT_LAYOUT_SY...	MW	☒	MW
QT_EXCLUDE	QT_default_exclude_cell	☒	QT_default_exclude_cell

At the bottom of the dialog are buttons for Help, Defaults (with a dropdown arrow), Refresh, OK, Apply, and Cancel.

Select the settings for IC Validator as described in [Table 22](#):

Table 22 IC Validator FILL Setup and Run Settings (Custom Variables Tab)

Settings	Description
Macros	Lists the macro in the runset file. The following fields are available: ▪ Name: Displays the name of the macro. ▪ Define: Selects the macro defined in the runset file as the default. The option is selected if the macro is defined in the runset file. ▪ Define As: Displays the value of the macro.
Env Variables	Lists the environment variables in the user-defined runset file. The following fields are available: ▪ Name: Displays the name of the environment variable ▪ Current Value: Displays the value of the environment variable. ▪ Set: Selects the environment variable by default. To set the value of an environment variable to match the original setting of the runset file, deselect the Set option. ▪ Set to: Displays the Current Value of the environment variable as default. To add a value of the environment variable, deselect the Set option.

- To use the **Control Variables** tab in the legacy mode:
 - Use the `pdk.xml` file. You can review, optionally modify, and reset the current values for FILL runset control variables as preferred for this FILL run. The control variables are preset for a PDK by default.

To modify a preset value, click with the alternate (non-selection) mouse button and edit the entry field. To reset the control variables to the default click the

Reset  button. The default is imported from the `pdk.xml` file.

In the legacy mode only, preprocessor, environment, and PXL variables are supported in the **Control Variables** tab.

- Add the `controlVarsFile` XML tag in the `pdk.xml` file to point to a variable definition file. For example:

```
<pveJob name="DRC" jobType="DRC">
  <preference name="ViewOutput" mutable="false">
    <value>true</value>
  </preference>
  <tool name="IC Validator">
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc.rs</value>
    </preference>
    <preference name="LoadRunset">
      <value>true</value>
    </preference>
    <controlVarsFile>./snps_reference40nm/pdk_controlVars.xml</controlVarsFile>
  </tool>
</pveJob>
```

Define the variables in the variable definition file with the XML syntax. For example:

```
<switch name="POLY_RULES_PP" scope="preProc" valueType="bool" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>Enable Poly Rules(pp)</description>
  <value>true</value>
</switch>

<switch name="METAL_RULES_PP" scope="preProc" valueType="enum" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>Check Metal Rules up to(pp)</description>
  <value>M4</value>
  <enumValues enumList="M1 M2 M3 M4 M5 M6" default="M4"/>
</switch>

<switch name="ICV_DRC_DIR_pp" scope="preProc" valueType="string" visible="true" mutable="true" group="PP_TEST_GROUP">
  <description>directory containing ICV rule decks(pp)</description>
  <value>$PDK_INSTALL_DIR/icv/drc</value>
</switch>
```

The tool issues the following information message when the Control Variables file is available in the `pdf.xml` file:

```
Information: Found Control Variables file in pdk.xml.
proceeding to use legacy control variable table. Recommend
for switching to new Control Variable table format.
```

Note:

If the file is not specified with the `controlVarsFile` XML tag, the tool issues the following information message:

```
Information: No Control Variables file found in
pdk.xml, switching to New Control Variables table for
mat.
```

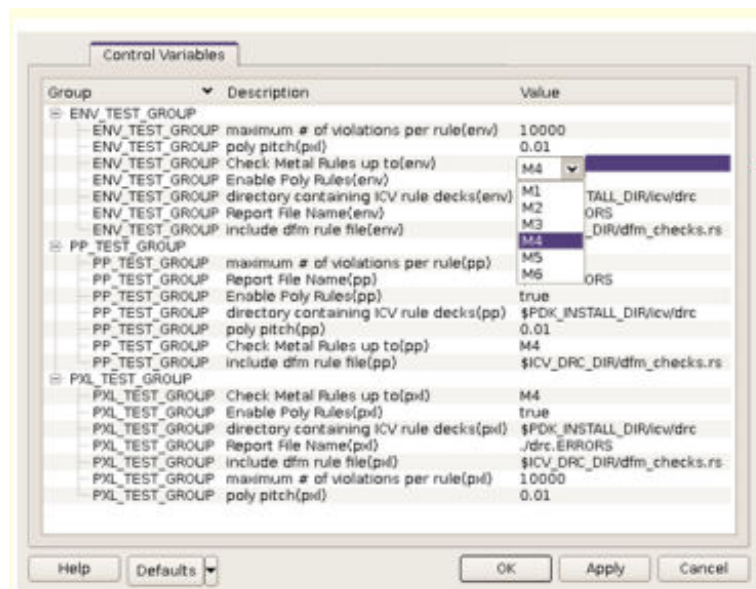
- You can also use Tcl commands to define and setup the control variables (instead of the `pdk.xml` file). For example:

```
set s [db::getNext [db::getScopes pdk_pve]]
db::setPrefValue xtDRCICVControlVars -value {
  POLY_RULES_PP
    {description {Enable Poly Rules(pp)}
     value true group PP_TEST_GROUP mutable true
     visible true scope preProc valueType bool}
  METAL_RULES_PP
    {description {Check Metal Rules up to(pp)}
     enumList {M1 M2 M3 M4 M5 M6} default M4 value M4 group
} -scope $s
```

Table 23 Scope of Control Variables

Scope	Runset Presentation	Example
pxl	{name} : {type} = {value}	PREFIX : string = "METAL"
env	#pragma envvar default {type} {value}	#pragma envvar default OUTPUT_MODE "stream"
preProc	#define {name} {value}	#define FUNCTIONS_RS TRUE

- After you define these variables and set them (from either the `pdk.xml` or Tcl), ensure that the ICV ruledeck is written correctly to consume these variables. See the following figure .



- To use the variable correctly, prepare the ruledeck as follows:

When you execute IC Validator, the tool writes the variables into the ICV ruledeck at the beginning of the file. To do prepare the rule deck, check if the original ruledeck has some of the variables defined with default, then wrap the variables into an `#if` statement, so the variables become active only when a conditional variable is `true`. (The variable can be `false` in the Custom Compiler variable definitions because the variables are obtained from the GUI. If the variable is `false`, the variable is executed from outside of the tool's environment. The variables are not expected to be set by the GUI.)

When you change a variable setting (as shown with a red arrow in the previous figure), the new value is written to the top of the ICV ruledeck, before IC Validator execution (as shown with a red arrow in the following figure).

```
METAL_RULES_pxl : string = "M4";
#define MAX_VIOLATION_COUNT_pp 10000
POLY_RULES_pxl : boolean = true;
#pragma envvar default MAX_VIOLATION_COUNT_env 10000
#define REPORT_FILE_NAME_pp ./drc.ERRORS
#pragma envvar default POLY_PITCH_env 0.01
ICV_DRC_DIR_pxl : string = "$PDK_INSTALL_DIR/icv/drc";
#pragma envvar default METAL_RULES_env M4
#define POLY_RULES_PP
#pragma envvar default POLY_RULES_env true
REPORT_FILE_NAME_pxl : string = "./drc.ERRORS";
#define ICV_DRC_DIR_pp $PDK_INSTALL_DIR/icv/drc
#pragma envvar default ICV_DRC_DIR_env $PDK_INSTALL_DIR/icv/drc
```



6. For the IC Validator tool, on the **Custom Options** tab, choose settings in [Table 24](#).

[Table 25](#) lists the runset options from the current runset as found by the IC Validator tool. You can edit the value for any option you see by changing it here on the **Custom Options** tab. You can add additional runset options here by choosing **Click to Add**, and entering a valid IC Validator option using the correct syntax and value.

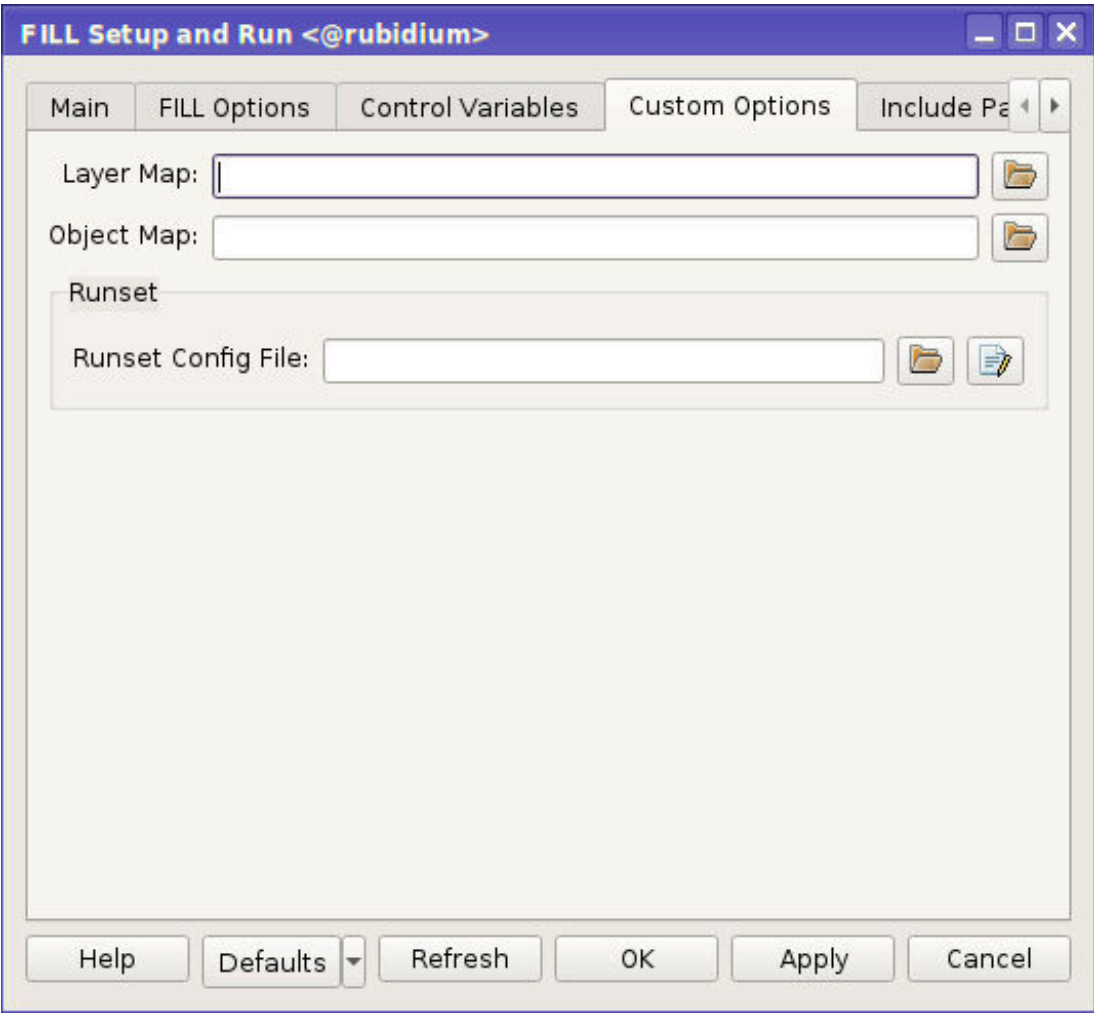


Table 24 IC Validator Fill Setup and Run Settings (Custom Options Tab)

Settings	Description
Layer Map	For OpenAccess format, sets the file to map OpenAccess LPP numbers and traditional GDSII stream layers/data types.
Object Map	Specifies the object map to be used during FILL.

Settings	Description
Runset	Runset Config File. Specifies a runset configuration file that is passed to an IC Validator command using the <code>-runset_config</code> option. Use the <code>xt<pveJob>ICVRunsetConfigFile</code> preference to set this mode. Edit Runset Config edits the runset configuration file. ◦ If the file does not exist, create a file using the Create File Prompt dialog box ◦ If the file is empty, create a file with the default name <code>runset_config.rs</code> using the Create File Prompt dialog box Runset Options specifies the supported IC Validator or Hercules runset options. Custom Compiler does not populate the Runset Options table while loading a runset file. To display the Runset Options table, set <code>xt<FILL>ICVShowRunsetOptions</code> preference to true if <code>xt<pveJob>ICVRunsetOptions</code> is not empty.

Table 25 IC Validator Setup and Run Settings (Custom Options Tab) Runset Options

IC Validator Function	
IC Validator Argument	Option Displayed for this Function and Argument
<code>pex_runset_report_file</code>	
file	PEX_RUNSET_REPORT_FILE_FILE
<code>error_options</code>	
error_limit_per_check	ERROR_OPTIONS_ERROR_LIMIT_PER_CHECK
pcell_list	ERROR_OPTIONS_PCELL_LIST
create_vue_output	ERROR_OPTIONS_CREATE_VUE_OUTPUT
report_layout_errors	ERROR_OPTIONS_REPORT_LAYOUT_ERRORS
<code>hierarchy_auto_options</code>	
application	HIERARCHY_AUTO_OPTIONS_APPLICATION
flow	HIERARCHY_AUTO_OPTIONS_FLOW
optimizations	HIERARCHY_AUTO_OPTIONS_OPTIMIZATIONS
no_explode_with_text	HIERARCHY_AUTO_OPTIONS_NO_EXPLODE_WITH_TEXT
<code>hierarchy_options</code>	
delete	HIERARCHY_OPTIONS_DELETE

IC Validator Function

IC Validator Argument	Option Displayed for this Function and Argument
explode	HIERARCHY_OPTIONS_EXPLODE
explode_all	HIERARCHY_OPTIONS_EXPLODE_ALL
flatten	HIERARCHY_OPTIONS_FLATTEN
no_explode	HIERARCHY_OPTIONS_NO_EXPLODE
incremental_options	
exclude_window	INCREMENTAL_OPTIONS_EXCLUDE_WINDOW
exclude_window_by_cells	INCREMENTAL_OPTIONS_EXCLUDE_WINDOW_BY_CELLS
exclude_window_by_cells_delta	INCREMENTAL_OPTIONS_EXCLUDE_WINDOW_BY_CELLS_DELTA
select_window	INCREMENTAL_OPTIONS_SELECT_WINDOW
clip_window	INCREMENTAL_OPTIONS_CLIP_WINDOW
window_error_filter	INCREMENTAL_OPTIONS_WINDOW_ERROR_FILTER
layout_grid_options	
resolution	LAYOUT_GRID_OPTIONS_RESOLUTION
check_45	LAYOUT_GRID_OPTIONS_CHECK_45
check_90	LAYOUT_GRID_OPTIONS_CHECK_90
openaccess_options	
view	OPENACCESS_OPTIONS_VIEW
layer_mapping_file	OPENACCESS_OPTIONS_LAYER_MAPPING_FILE
generate_pin_text	OPENACCESS_OPTIONS_GENERATE_PIN_TEXT
pin_text	OPENACCESS_OPTIONS_PIN_TEXT
merged_view_list	OPENACCESS_OPTIONS_MERGED_VIEW_LIST
prototype_options	
large_cells	PROTOTYPE_OPTIONS_LARGE_CELLS
symmetry	PROTOTYPE_OPTIONS_SYMMETRY

IC Validator Function

IC Validator Argument	Option Displayed for this Function and Argument
text_options	
delete_text	TEXT_OPTIONS_DELETE_TEXT
replace_text	TEXT_OPTIONS_REPLACE_TEXT
replace_text_characters	TEXT_OPTIONS_REPLACE_TEXT_CHARACTERS
use_exploded_text	TEXT_OPTIONS_USE_EXPLODED_TEXT
edtext	TEXT_OPTIONS_EDTEXT
net_prefix	TEXT_OPTIONS_REPLACE_NET_PREFIX
layout_power	TEXT_OPTIONS_LAYOUT_POWER
layout_ground	TEXT_OPTIONS_LAYOUT_GROUND

7. For IC Validator, on the **Include Paths** tab, set FILL include paths for directories whose include files you want to pass to IC Validator.
 - a. In this table, select a row, and click the mouse.
 - b. Browse to select the include directory.
 - c. Repeat for each include path you want to add.
 - d. (Optional) Delete an include path by selecting it, and then clicking the Delete button, a red **x**.
8. Click **OK** to start the FILL.
9. (Optional) If you did not choose to view output on the **Main** tab, and you want to view it now, see [Viewing FILL Output](#).

Note:

The following warning message appears in the console when the OpenAccess version of Custom Compiler is higher than the OpenAccess version on IC Validator:

```
Warning: OA version of Custom Compiler <CC OA overversion> is
higher than IC Validator <ICV OA version>.
This may lead to OA API compatibility errors to evaluate
PyCells in the design.
Please proceed with GDS input for DRC run to avoid [liboa]
errors.
```

Using Custom Jobs for FILL

Use a custom job for a FILL run to apply selected settings from a Process Design Kit (PDK).

To use a custom job for a FILL run:

1. In the design layout window, choose **Verification > <custom_job_name> Setup and Run**, where *custom_job_name* is the name of the job created for your use by your CAD support group.

(Choosing **Tools > Verification** adds the **Verification** menu to the window menu bar.)

2. In the **Setup and Run** dialog box for this custom job, specify settings for FILL.

For assistance with settings, refer to [Using Integrated Tools for FILL Cell Generation](#).

3. Click **OK** to begin FILL.
4. Continue by [Viewing FILL Output](#).

Rerunning FILL

Rerun FILL after running it one time. You can use the same tool, and settings, and you can vary the runset or reload a changed runset.

To rerun FILL:

1. In the design layout window, consider your choices.
 - To rerun FILL with the same tool, settings, and runset, you can choose **Verification > FILL > Run**.

Alternatively, you can start the run by clicking the **FILL Run** button on the command history toolbar, or the Layout Verification toolbar at the lower left.

- To rerun the FILL with the same tool, and settings, but a different runset, choose **Verification > FILL > Setup and Run**. Then, on the **Main** tab under **Job Parameters**, browse to select a new runset file.
 - To rerun the FILL with the same tool, and settings, and a runset file whose contents have changed, choose **Verification > FILL > Setup and Run**. Then, on the **Main** tab under **Job Parameters** reload the file by clicking the **Reload Runset** button.
2. Continue the flow by [Viewing FILL Output](#).

Viewing FILL Output

View FILL output to see logged operations, results, and errors.

To view FILL output:

1. In the design layout, choose **Verification > FILL > View Output**. Alternatively, you can use the keyboard shortcut Shift+Ctrl+D.
2. In the Text Viewer, view results, and logged operations, on each tab.

12

Physical Verification Flows Using the ICV Launch Tool

Access the ICV Launch tool from the command-line interface or Custom Compiler environment to run physical verification flows.

This chapter describes the following:

- [About Using the ICV Launch Tool](#)
- [Setting Up the ICV Launch Environment](#)
- [Displaying the ICV Launch Menu](#)
- [Displaying ICV Launch DRC](#)
- [Displaying ICV Launch LVS](#)
- [Displaying ICV Launch PERC](#)
- [Running the ICV Launch Tool](#)
- [Tracking Progress for ICV Launch Interactive Processes](#)
- [Debugging the Design Using the ICV Launch Results Viewing Environment](#)

About Using the ICV Launch Tool

You can access the ICV Launch tool from the Custom Compiler environment to set up and run the following:

Design Check	Description
ICV Launch DRC	Checks the design rule violations.
ICV Launch LVS	Compares the layout against the schematic to verify device connectivity.
ICV Launch PERC	Performs reliability verification tasks on physical IC designs.

Design Check	Description
ICV Launch VUE	IC Validator debugging tool to load IC validator results.

The workflow for using the ICV Launch tool is:

Step		See...
1	Set up the ICV Launch Environment.	Setting Up the ICV Launch Environment
2	Display the ICV Launch menu.	Displaying the ICV Launch Menu
3	Invoke the ICV Launch tool.	Running the ICV Launch Tool
4	Track the ICV Launch processes.	Tracking Progress for ICV Launch Interactive Processes
5	View the output and debug the design.	Debugging the Design Using the ICV Launch Results Viewing Environment

You can repeat the steps in the flow to rerun the DRC, LVS, PERC, or VUE processes as necessary.

To use the native Custom Compiler Verification interface to run third-party DRC and LVS tools, see [Design Rule Checking Using Integrated Tools](#) and [Comparing Layout Versus Schematic \(LVS\)](#).

Setting Up the ICV Launch Environment

To facilitate the ICV Launch DRC and LVS processes, you can set up the environment by setting the `ICV_HOME_DIR` environment variable to IC Validator version R-2020.09 or newer. To facilitate PERC runs, use IC Validator version U-2022.12.

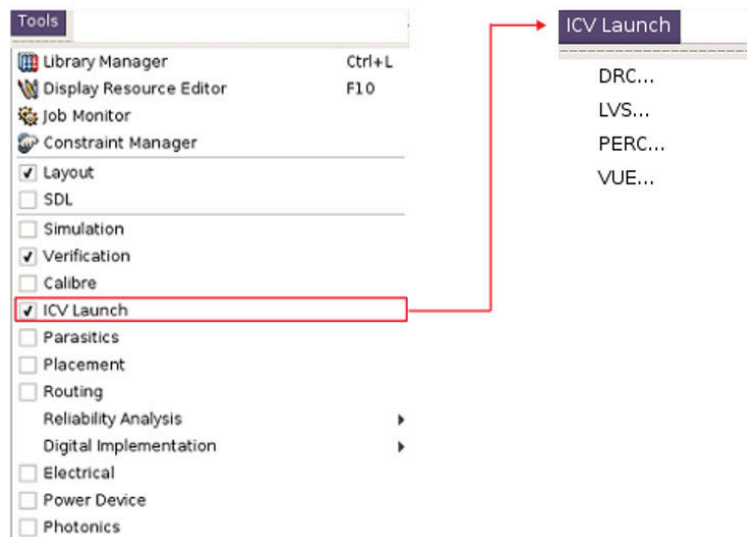
Displaying the ICV Launch Menu

You must display the **ICV Launch** menu to access the ICV Launch tool.

To display the **ICV Launch** menu:

- In the Layout Editor (design layout), choose **Tools > ICV Launch**. The **ICV Launch** menu appears on the menu bar.

You can run DRC, LVS, PERC, and VUE jobs in the **Layout** window.

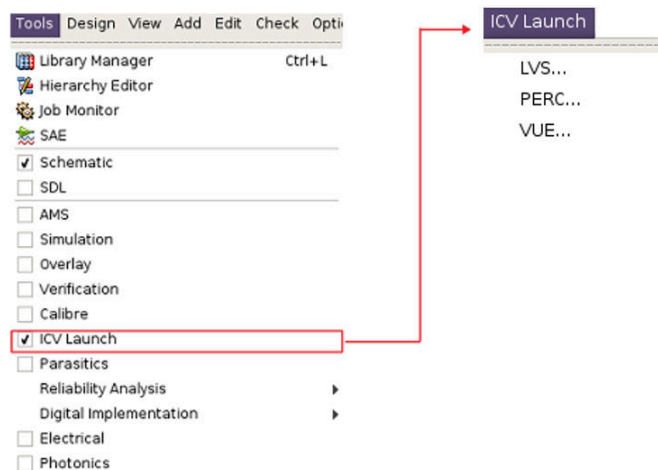


- You can also run a single input LVS, PERC or VUE job in the **Schematic** window.

Chapter 12: Physical Verification Flows Using the ICV Launch Tool

Displaying the ICV Launch Menu

In the Schematic Editor (design schematic), choose **Tools > ICV Launch**. The **ICV Launch** menu appears on the menu bar.

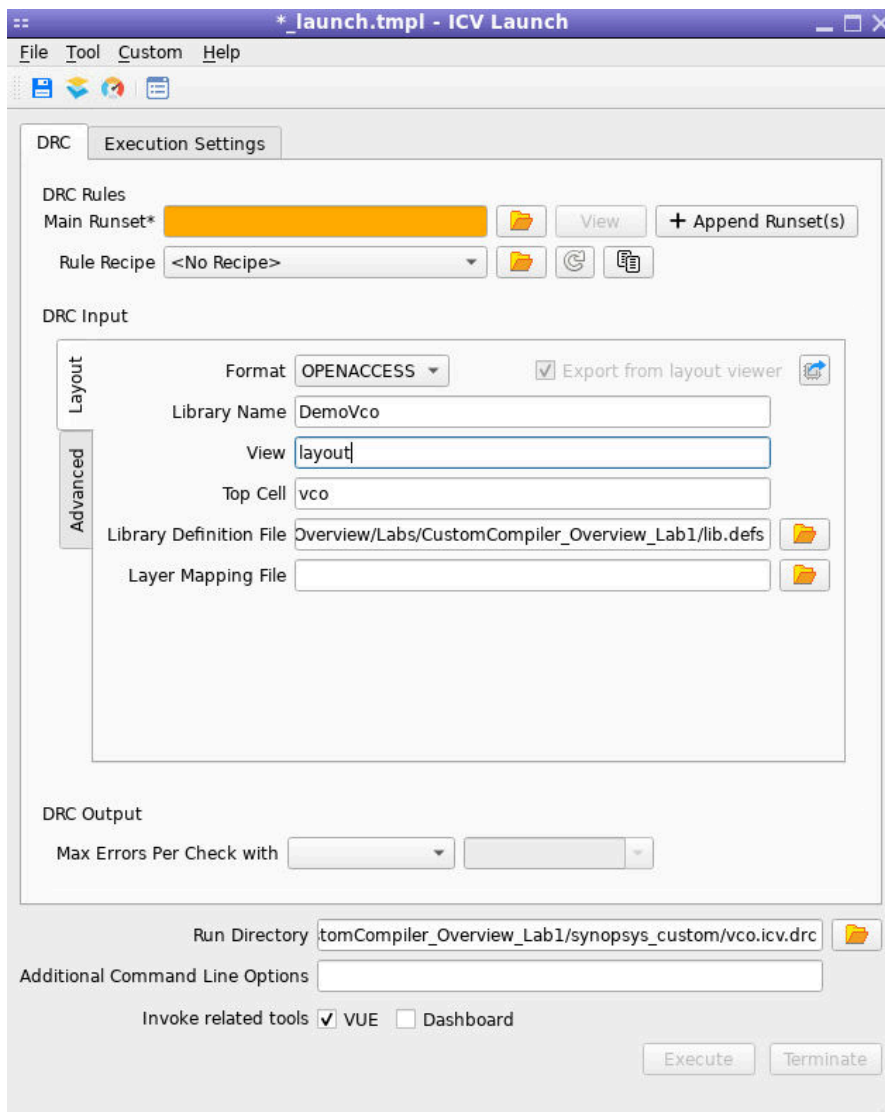


See the *IC Validator Launch User Guide* for more information on using ICV Launch.

Displaying ICV Launch DRC

Choose **ICV Launch > DRC** to launch the DRC dialog box.

Figure 10 ICV Launch DRC Dialog Box



The **ICV Launch > DRC** menu opens ICV Launch with the DRC tab preset options. The descriptions of these options are described in [Table 26](#).

Table 26 ICV Launch DRC Menu in Layout View

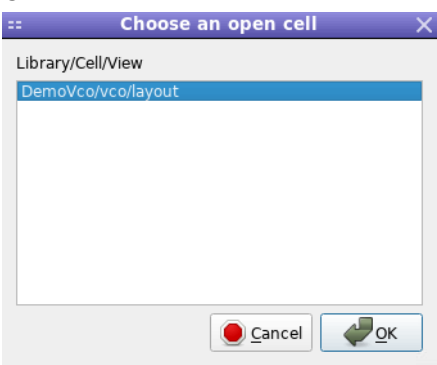
Menu item	Description
DRC Input	Layout Format is preset with the IC Validator DRC input format available options: GDSII, OASIS, OPENACCESS, GDSII.GZ, MILKYWAY, NDM, OASIS.GZ, and OASIS.OPT. The default is OPENACCESS. Export from layout viewer is off by default. Click  >  to query for the layout, view, or cell information of the designs that are currently open in the layout editor. You can select a design from the results returned in the Choose an open cell dialog box. ICV Launch updates the Library Name, View, and Top Cell fields. Library Name is preset with the library name of the current session. View is preset with the layout view of the current session. Top Cell is preset with the cell name of the current session. Library Definition File is preset with the path including the file name for the current session's library definition file. Layer Mapping File
DRC Output	Max Errors Per Check with specifies the maximum number of errors or error sampling rate, per function, stored in the error database and the LAYOUT_ERRORS file.
Run Directory	Preset with the run directory. The default is the current directory. <i>Current directory/Top Cell.icv.Job Name</i> For example, <i>./synopsys_custom/add4.icv.drc</i>
Additional Command Line Options	Specifies additional command-line options.

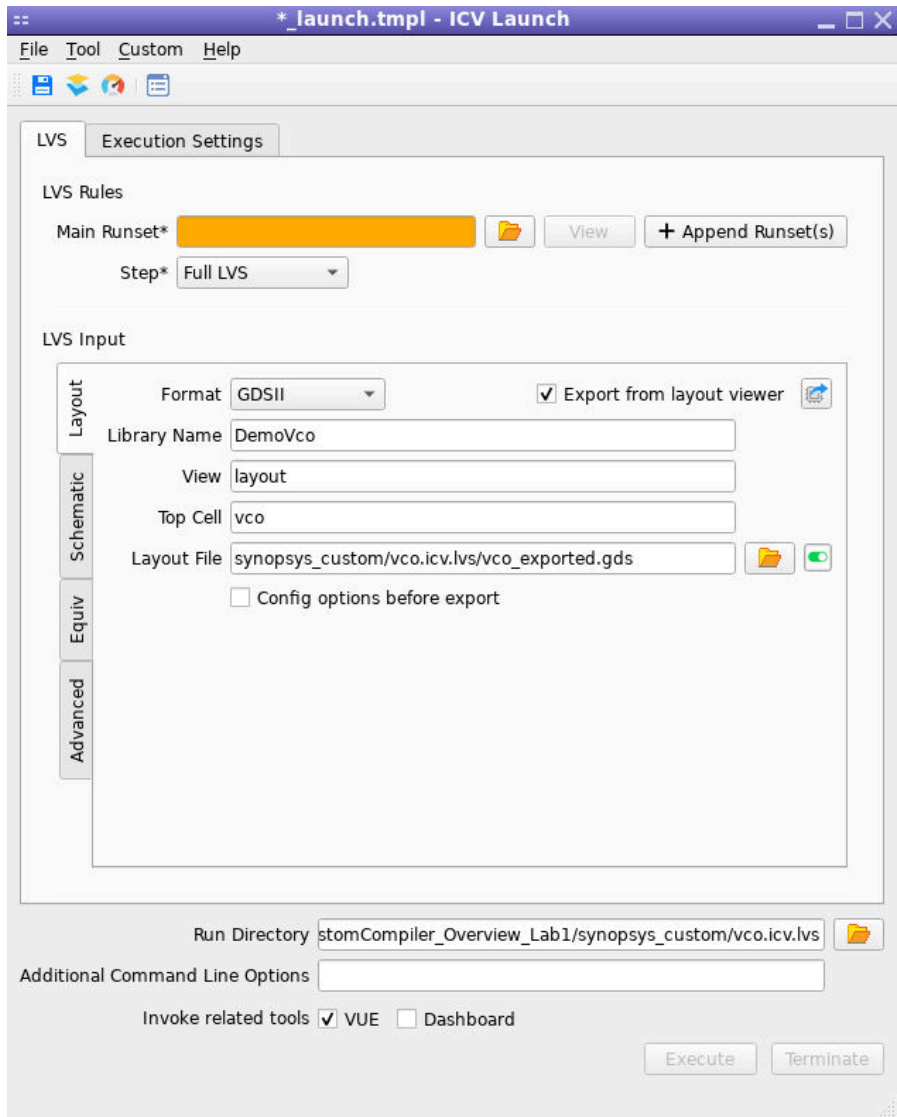
Table 26 *ICV Launch DRC Menu in Layout View (Continued)*

Menu item	Description
Invoke related tools	The selected tools (VUE/Dashboard) are automatically invoked after starting the IC Validator run. • To launch Dashboard, click . • To launch VUE, click . If an existing IC Validator VUE window is enabled, the VUE button brings the window to the foreground.
Custom GUI	Defines additional graphical entries to control the IC Validator execution. Click  to open the customization dialog box.

Displaying ICV Launch LVS

Choose **ICV Launch > LVS** in the Layout or Schematic window to launch the **LVS** dialog box.

Figure 11 ICV Launch LVS Dialog Box in Layout Window



The **ICV Launch > LVS** menu opens ICV Launch with the LVS tab preset options. The descriptions of these options are described in [Table 27](#) and [Table 28](#).

Table 27 LVS Menu in Layout View

Menu item	Description
LVS Rules	Step* is Full LVS, Extraction Only, or Compare Only.
LVS Input	Layout

Table 27 LVS Menu in Layout View (Continued)

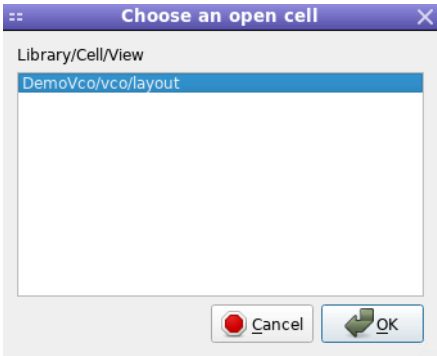
Menu item	Description
	Format is preset with the IC Validator DRC input format with available options: GDSII, OASIS, OPENACCESS, GDSII.GZ, MILKYWAY, NDM, OASIS.GZ, and OASIS.OPT. The default is OPENACCESS. Export from layout viewer is off by default. Click  > 
	to query for the layout, view, or cell information of the designs that are currently open in the layout editor. You can select a design from the results returned in the Choose an open cell dialog box. ICV Launch updates the Library Name, View, and Top Cell fields. Library Name is preset with the library name of the current session. View is preset with the layout view of the current session. Top Cell is preset with the cell name of the current session. Layout File is preset with the path including the file name for the current session's library definition file. Check Config options before export to export the layout.
Additional Command Line Options	Specifies additional command-line options.

Table 27 LVS Menu in Layout View (Continued)

Menu item	Description
Invoke related tools	The selected tools (VUE/Dashboard) are automatically invoked after starting the IC Validator run. • To launch Dashboard, click . • To launch VUE, click . If an existing IC Validator VUE window is enabled, the VUE button brings the window to the foreground.
Custom GUI	Defines additional graphical entries to control the IC Validator execution. Click  to open the customization dialog box.

Figure 12 ICV Launch LVS Dialog Box in Schematic Window

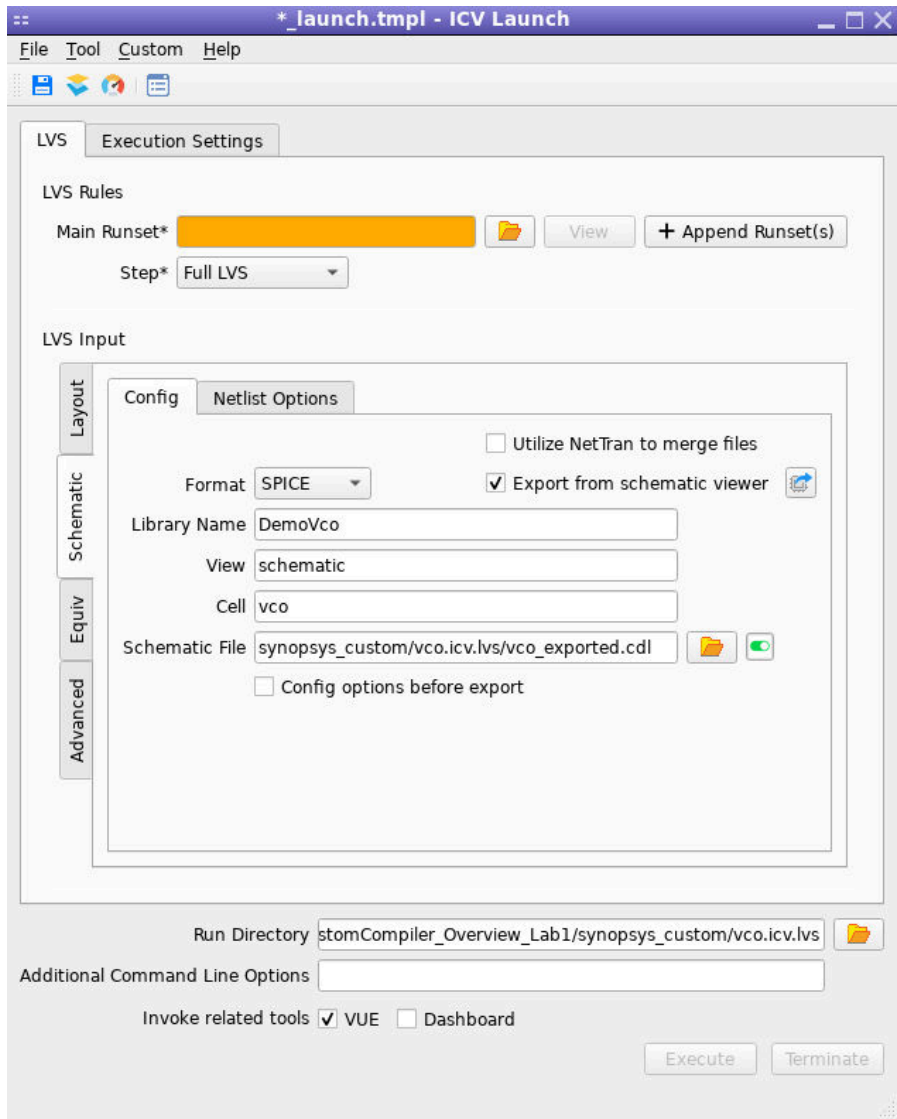
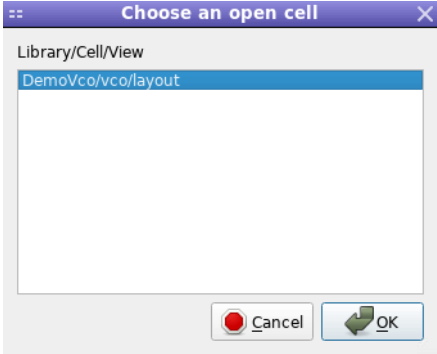


Table 28 LVS Menu in Schematic View

Menu item	Description
LVS Rules	Step* is Full LVS , Extraction Only , or Compare Only .
LVS Input	Schematic Format is preset with the IC Validator LVS input format with available options: SPICE, VERILOG, and ICV. The default is SPICE. Export from Schematic viewer is on by default.

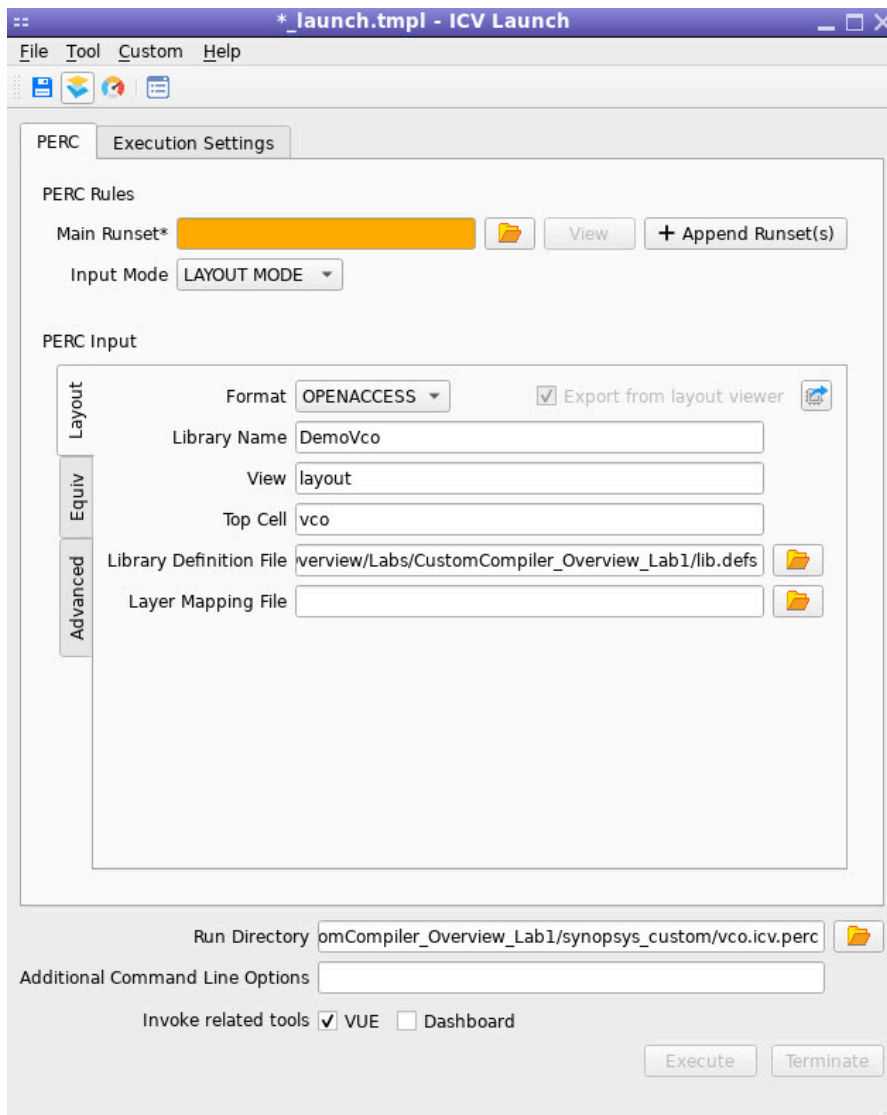
Table 28 LVS Menu in Schematic View (Continued)

Menu item	Description
	Click  > 
	to query for the layout, view, or cell information of the designs that are currently open in the layout editor. You can select a design from the results returned in the Choose an open cell dialog box. ICV Launch updates the Library Name, View, and Top Cell fields. Library Name is preset with th library name of the current session. View is preset with the schematic view of the current session. Cell is preset with the cell name of the current session. Schematic File is preset with the path including the file name for the current session's library definition file. Check Config options before export to export the schematic.
Run Directory	Preset with the run directory. The default is the current directory. <i>Current directory/Top Cell.icv.Job Name</i> For example, <i>./synopsys_custom/add4.icv.drc</i>
Additional Command Line Options	Specifies additional command-line options.
Invoke related tools	The selected tools (VUE/Dashboard) are automatically invoked after starting the IC Validator run.

Displaying ICV Launch PERC

Choose **ICV Launch > PERC** in the Layout or Schematic window to launch the **PERC** dialog box.

Figure 13 ICV Launch PERC Dialog Box in Layout Window



The **ICV Launch > PERC** menu opens ICV Launch with the PERC tab preset options. The descriptions of these options are described in [Table 29](#) and [Table 30](#).

Note:

See the *IC Validator Launch User Guide* for more information on the PERC options for ICV Launch.

Table 29 PERC Menu in Layout View

Menu item	Description
PERC Rules	Input Mode: LAYOUT MODE
PERC Input	Layout Format is preset with the IC Validator PERC input format with available options: GDSII, OASIS, OPENACCESS, GDSII.GZ, MILKYWAY, NDM, OASIS.GZ, and OASIS_OPT. The default is OPENACCESS. Export from layout viewer is on by default. Click  > Choose an open cell Library/Cell/View DemoVco/vco/layout Cancel OK
	to query for the layout, view, or cell information of the designs that are currently open in the layout editor. You can select a design from the results returned in the Choose an open cell dialog box. ICV Launch updates the Library Name, View, and Top Cell fields. Library Name is preset with the library name of the current session. View is preset with the layout view of the current session. Top Cell is preset with the cell name of the current session. Library Definition File is preset with the path including the file name for the current session's library definition file. Layout File is preset with the path including the file name for the current session's library definition file. Layer Mapping File maps OpenAccess (OA) layer-purpose names to GDSII or OASIS layer number and data types. Run Directory Preset with the run directory. The default is the current directory. Current directory/Top Cell.icv.Job Name For example, ./synopsys_custom/add4.icv.perc

Table 29 PERC Menu in Layout View (Continued)

Menu item	Description
Additional Command Line Options	Specifies additional command-line options.
Invoke related tools	The selected tools (VUE/Dashboard) are automatically invoked after starting the IC Validator run. • To launch Dashboard, click . • To launch VUE, click . If an existing IC Validator VUE window is enabled, the VUE button brings the window to the foreground.
Custom GUI	Defines additional graphical entries to control the IC Validator execution. Click  to open the customization dialog box.

Figure 14 ICV Launch PERC Dialog Box in Schematic Window

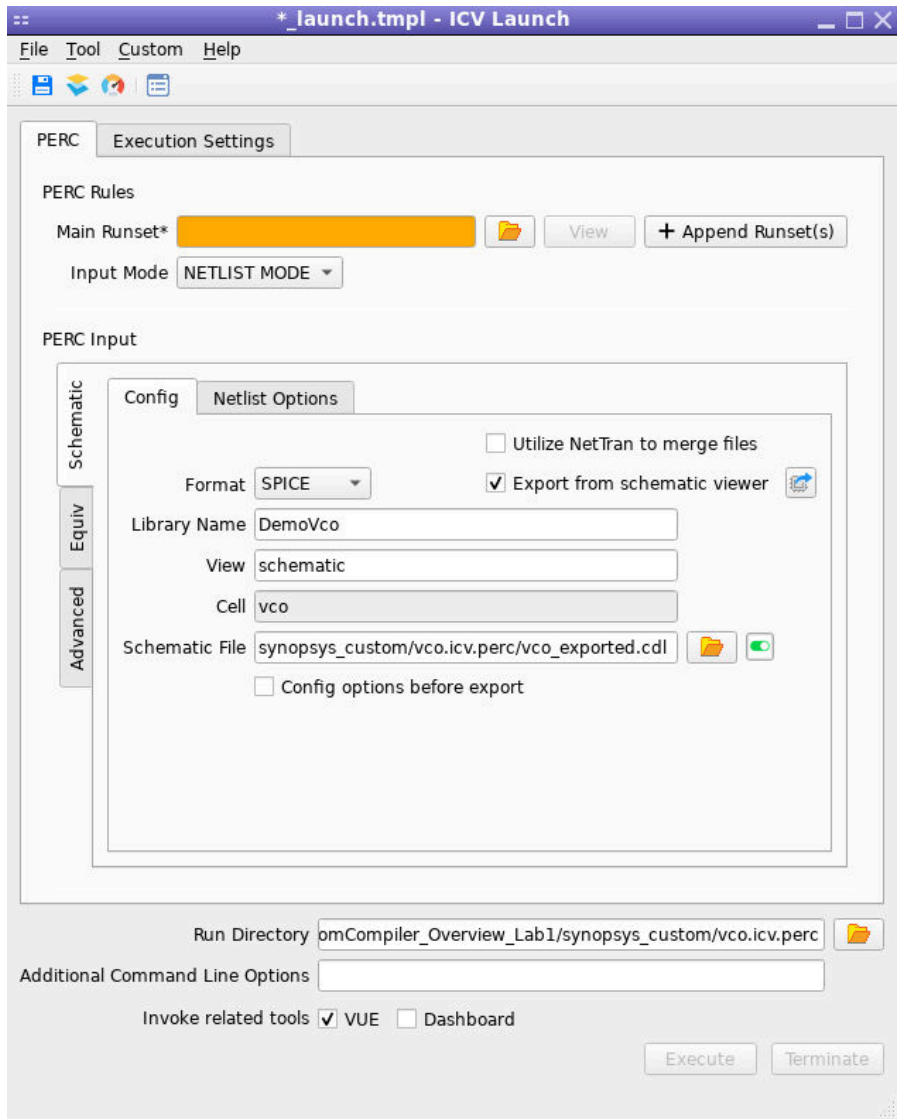
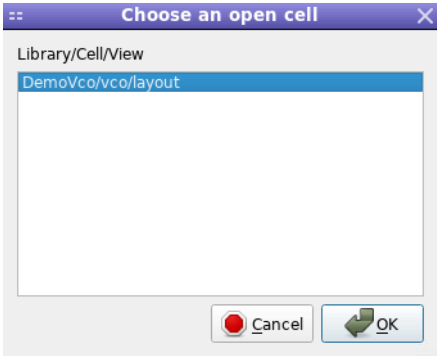


Table 30 PERC Menu in Schematic View

Menu item	Description
PERC Rules	Input Mode: NETLIST MODE
PERC Input	Schematic Format is preset with the IC Validator PERC input format with available options: SPICE, VERILOG, and ICV. The default is SPICE. Export from Schematic viewer is on by default.

Table 30 PERC Menu in Schematic View (Continued)

Menu item	Description
	Click  >  to query for the layout, view, or cell information of the designs that are currently open in the layout editor. You can select a design from the results returned in the Choose an open cell dialog box. ICV Launch updates the Library Name, View, and Top Cell fields. Library Name is preset with the library name of the current session. View is preset with the schematic view of the current session. Cell is preset with the cell name of the current session. Schematic File is preset with the path including the file name for the current session's library definition file. Check Config options before export to export the schematic. Run Directory Preset with the run directory. The default is the current directory. <i>Current directory/Top Cell.icv.Job Name</i> For example, <i>./synopsys_custom/add4.icv.perc</i> Additional Command Line Options Specifies additional command-line options. Invoke related tools The selected tools (VUE/Dashboard) are automatically invoked after starting the IC Validator run. • To launch Dashboard, click . • To launch VUE, click . If an existing IC Validator window is enabled, the VUE button brings the window to the foreground

Running the ICV Launch Tool

Note:

Before you start, see [Setting Up the ICV Launch Environment](#) and [Displaying the ICV Launch Menu](#).

To run the ICV Launch tool:

1. In the design layout or schematic, choose **ICV launch** > <tool_name> to run the specific ICV Launch check. See [About Using the ICV Launch Tool](#) for the list of supported ICV Launch checks.

The **ICV Launch** dialog box for the specific check opens. The library, cell, and view information for the currently opened design automatically populate the **Library Name**, **Top Cell**, and **View** boxes, as shown in [Table 31](#) and [Table 28](#).

For further instructions, refer to the IC Validator Launch documentation.

2. You can configure the **ICV Launch** dialog box with various settings and save them in a separate template file (for example, Template1.tmpl) by choosing **File > Save As** in the **ICV Launch** dialog box. This Template1.tmpl file invokes ICV Launch from the command line with the settings previously stored in it.

```
icv_launch -load Template1.tmpl &
```

Note:

You can simultaneously load multiple template files, that is Template_1.tmpl, Template_2.tmpl & Template_3.tmpl, with different settings for the same option using a space as delimiter.

```
icv_launch -load "Template_1.tmpl Template_2.tmpl  
Template_3.tmpl" &
```

ICV Launch is invoked with the first template name, Template_1, and the remaining templates are loaded incrementally. The later template overrides the value of an option if there is a different value present in the previous template file for the same option.

Tracking Progress for ICV Launch Interactive Processes

You can track the ICV Launch physical verification flow processes with the Custom Compiler Job Monitor.

To track the ICV Launch Interactive processes:

1. From the Custom Compiler home page, choose **Tools > Job Monitor**
2. In the Job Monitor, choose **View > Job Types > Interactive**.
3. Continue to [Debugging the Design Using the ICV Launch Results Viewing Environment](#).

Debugging the Design Using the ICV Launch Results Viewing Environment

When you check the **Start VUE after run** box, IC Validator VUE opens. You can debug the checks in the Layout and Schematic editors by viewing DRC and LVS output using the IC Validator VUE tool. Running a DRC or LVS one time initializes the debugging environment for that check.

In the debugging window, identify and evaluate violations.

For IC Validator VUE, you can use the **Edit-in-Place** option to review violations. To highlight an error, select a cellView in the error list in the design window. When you highlight an error, the Layout window appears.

You can perform cross-probing between the list of errors and the design.

For further instructions, refer to the IC Validator VUE documentation.

13

DRC and LVS Using the Calibre Interactive Tool

Access the Calibre Interactive tool from the Custom Compiler environment to run design rule checks and compare layout versus schematic processes.

This chapter describes the following:

- [About Using the Calibre Interactive Tool](#)
- [Setting Up the Calibre Interactive Environment](#)
- [Displaying the Calibre Menu](#)
- [Displaying the Calibre Setup Menu](#)
- [Displaying the Calibre View Menu](#)
- [Running the Calibre Interactive Tool](#)
- [Tracking Progress for Calibre Interactive Processes](#)
- [Debugging the Design Using the Calibre Results Viewing Environment](#)

About Using the Calibre Interactive Tool

You can access the Mentor Graphics Calibre Interactive tool from the Custom Compiler environment to set up and run the following:

Design Check	Description
Calibre nmDRC	Checks the design rule violations.
Calibre Fast XOR	Speeds up layout-vs-layout (LVL) comparisons to provide the same results as traditional XOR methodologies.
Calibre DFM	Checks the design for manufacturing optimization.
Calibre nmLVS	Renames Calibre LVS to Calibre nmLVS. See Calibre LVS.

Design Check	Description
Calibre PEX	Generates parasitic netlists and reports. It calculates parasitic resistance and capacitance in an IC layout and outputs a netlist for simulation.
Calibre PERC	Performs electrical checks to verify circuit reliability.
Calibre LVS	Compares the layout against the schematic to verify device connectivity.
Calibre RVE	Displays the results so that you can debug the design.
Calibre xACT	Calculates parasitic and coupling effects. This information is essential for driving static timing analysis tools.

The workflow for using the Calibre Interactive tool is:

Step	See...
1	Set up the Calibre Interactive Environment. Setting Up the Calibre Interactive Environment
2	Display the Calibre menu. Displaying the Calibre Menu
3	Run the Calibre Interactive tool. Running the Calibre Interactive Tool
4	Track the Calibre Interactive processes. Tracking Progress for Calibre Interactive Processes
5	View the output and debug the design. Debugging the Design Using the Calibre Results Viewing Environment

You can repeat the steps in the flow to rerun the DRC or LVS processes as necessary.

To use the native Custom Compiler Verification interface to run third-party DRC and LVS tools, see [Design Rule Checking Using Integrated Tools](#) and [Comparing Layout Versus Schematic \(LVS\)](#).

Setting Up the Calibre Interactive Environment

To facilitate the Calibre Interactive DRC and LVS processes, you can set up the environment by adding environment variables.

Note: If any library in the `lib.defs` file cannot be accessed by Calibre, an error occurs.

To set up the environment:

1. Set the `MGC_HOME` or `CALIBRE_HOME` environment variable to the Calibre installation path.
2. Set the `MGLS_LICENSE_FILE` environment variable to the license server.
3. (Optional) If your designs include Python PCells, set the `OA_HOME` environment variable to `$CNI_ROOT/plat_linux_gcc411_32/3rd/oa`.

Python PCell design requires the Ciranova Python application plug-in.

4. Use the `MGC_CALIBRE_DRC_RUNSET_FILE`, `MGC_CALIBRE_LVS_RUNSET_FILE`, and `MGC_CALIBRE_PEX_RUNSET_FILE` environment variables to tell Calibre Interactive to use a default runset template file, which seeds the values in the dialogs for iCalibre DRC, LVS, and PEX. The Lib/Cell/View information is always seeded from the opened layout, regardless of the runset template. The template itself supports Calibre Interactive's token substitution, for example, `%I=layoutCellName`, `%s = schCellName`, `%L=layout library name`, and so forth.
5. Use the `MGC_CALIBRE_VIEWER_MENU_CMDS` environment variable to set an optional user customization file location. For example,

```
setenv MGC_CALIBRE_VIEWER_MENU_CMDS your_customization_menu.tcl
```

Note:

The `CC_CALIBRE_SAVE_RUNSET_ON_RUN` environment variable is now replaced by the `xtCalibreSaveRunsetOnRun` preference.

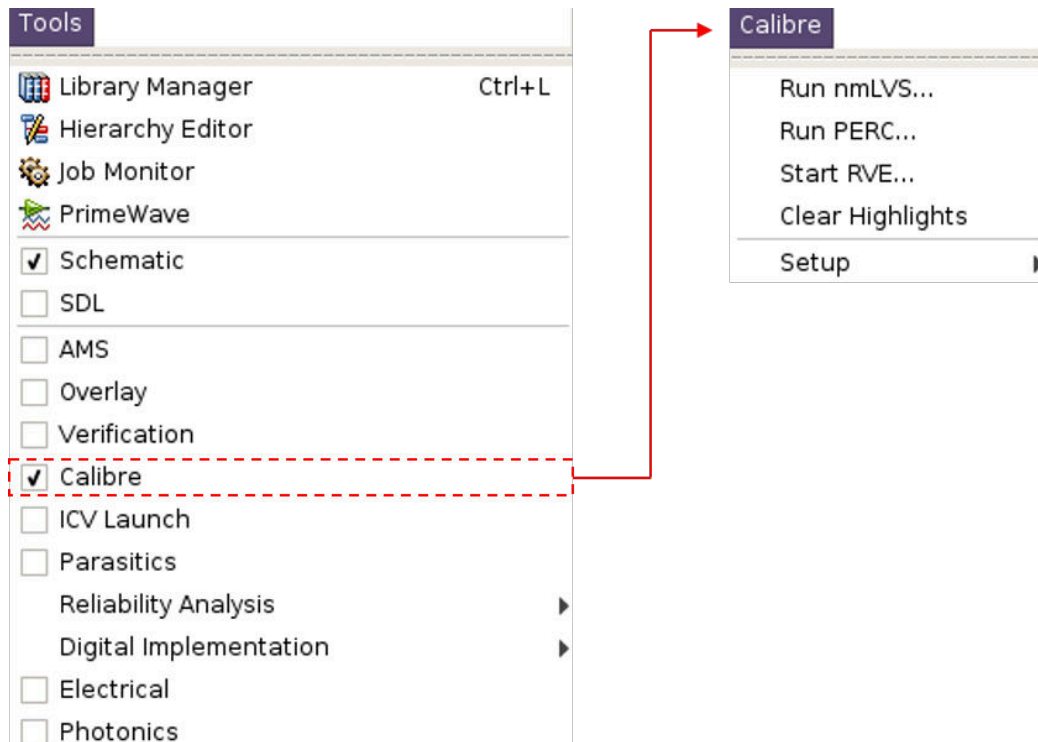
See [Preferences](#)

Displaying the Calibre Menu

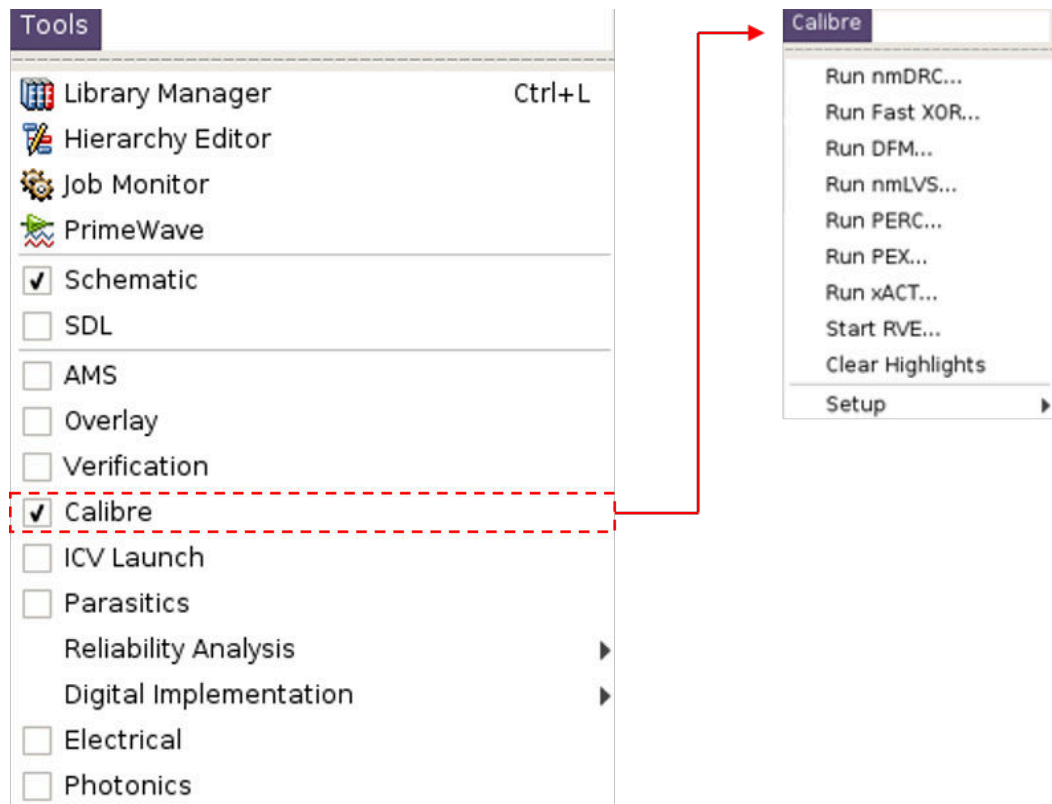
You must display the **Calibre** menu to access the Calibre Interactive tool.

To display the **Calibre** menu:

- In the Schematic Editor (design schematic), choose **Tools > Calibre**. The **Calibre** menu appears on the menu bar.



- In the Layout Editor (design layout), choose **Tools > Calibre**. The **Calibre** menu appears on the menu bar.



Displaying the Calibre Setup Menu

The **Calibre > Setup** menu in the **Layout** and **Schematic** windows is described in [Table 31](#).

Table 31 Calibre Setup Menu

Menu item	Description
Layout Export	Displays a dialog box that controls options used while exporting layout databases to Calibre.
Netlist Export	Controls options used while exporting schematic views to Calibre.
Calibre View	Sets Calibre View options for Calibre parasitic extraction.
Socket	Specifies the socket that Calibre Interactive and Calibre RVE use to communicate with Custom Compiler.

Table 31 Calibre Setup Menu (Continued)

Menu item	Description
Highlight Colors	Specifies the layout and schematic violation highlights with user-defined colors.

Exporting Netlist

To display the **Calibre Netlist Export Setup** dialog box:

- Choose **Setup > Netlist Export**. The **Calibre Netlist Export Setup** dialog box appears, which allows you to control the export of schematics to Calibre. It is optionally displayed before export layout during LVS from the Calibre Interactive tool.

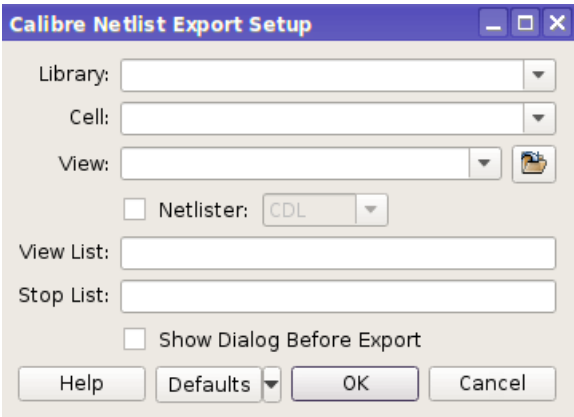


Table 32 Calibre Netlist Export Setup Options

Name	Description
Library	Sets the Library Name for the currently opened design.
Cell	Sets the Top Cell for the currently opened design.
View	Sets the View Name for the currently opened design.
Netlister	Specifies which tool performs netlisting.
View List	Controls hierarchy traversal during netlist export.
Stop List	Controls when expansion stops during hierarchy traversal for netlist export.
Show Dialog Before Export	Displays the Calibre Layout Export dialog box every time prior to export.

Exporting Layout

To display the **Calibre Layout Export Setup** dialog box:

- Choose **Setup > Layout Export**. The **Calibre Layout Export Setup** dialog box appears, which allows you to control the layout export from Custom Compiler to GDSII and OASIS formats, and change layout export settings for each run.

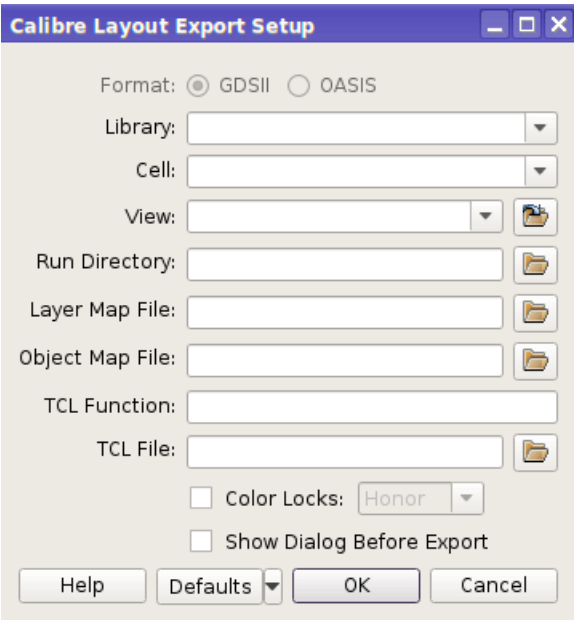


Table 33 *Calibre Layout Export Setup Options*

Name	Description
Library	Sets the Library Name for the currently opened design.
Cell	Sets the Top Cell for the currently opened design.
View	Sets the View Name for the currently opened design.
Run Directory	Specifies the run directory used during layout export. The default is the current directory.
Layer Map File	Specifies the layer map file to be used during layout export.
Object Map File	Specifies the object map file to be used during layout export.
TCL Function	Specifies the Tcl function to be executed before translation.
TCL File	Specifies the Tcl file to be sourced during translation.

Name	Description
Color Locks	Specifies which mapping lines in the layer mapping file are used to map shapes with a mask color attribute. ◦ Ignore. Specifies that all colored shapes are mapped with the line that has neither keyword. ◦ Strict. Specifies the anchor bit used to select either locked or unlocked from the layer map. ◦ Full. Specifies the layer mapping line to be used. ◦ Honor. The default. Specifies that when the layer map does not have a color mapping rule to match the color shapes, Custom Compiler tries to match the mapping rule without a color being specified. ◦ DisableColor. Specifies that all layer mapping lines with color attribute are ignored.
Format	Shows the GDSII or OASIS format for the exported database format. This setting cannot be changed from the Calibre Layout Export dialog box.
Show Dialog Before Export	Displays the Calibre Layout Export dialog box every time prior to export.

Specifying Calibre Violation Highlight Colors

You can specify the layout and schematic violation highlights with user-defined colors using the **Set Calibre Highlight Color** dialog box (choose **Calibre > Highlight Colors**).

Figure 15 Set Calibre Highlight Color Dialog Box

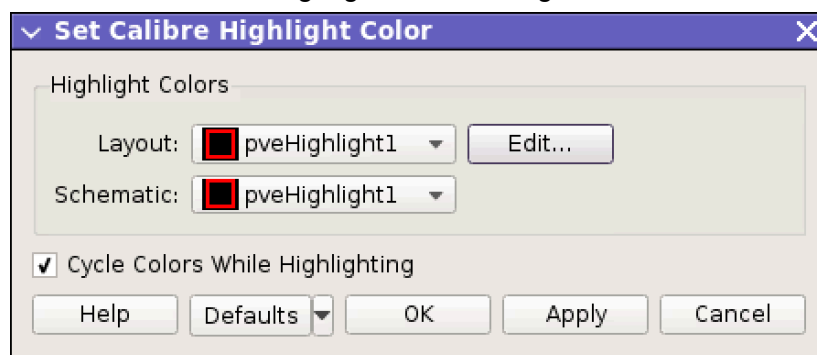
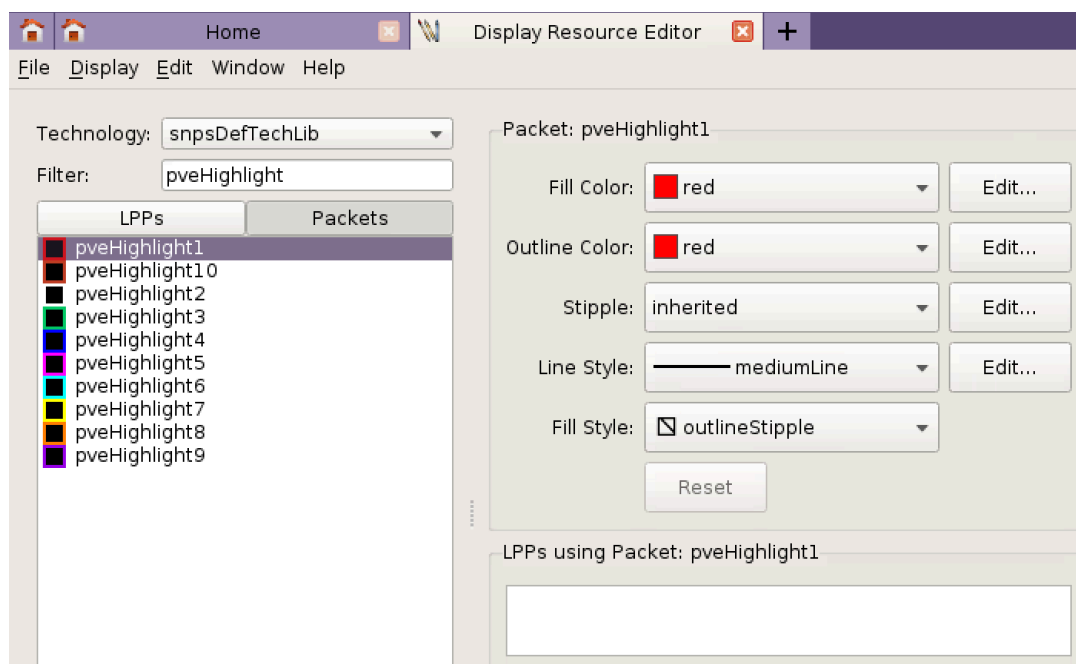


Figure 15 includes the following fields:

- **Highlight Colors:** Specify highlight colors for layout and schematic violations using 10 fixed highlight packets.

The values of current highlight packets are specified through the `xtCalibreLayoutHighlightPacket` and `xtCalibreSchematicHighlightPacket` preferences.

Click **Edit** to define the highlight color packets in the Display Resource Editor.



- **Cycle Colors While Highlighting:** If this option is checked, the current color will be changed after each violation is highlighted.

Initiating a Calibre Socket Server

You can initiate a socket server from Calibre using the **Setup Calibre Socket** dialog box (choose **Calibre** > **Socket**).

Figure 16 Setup Calibre Socket Dialog Box

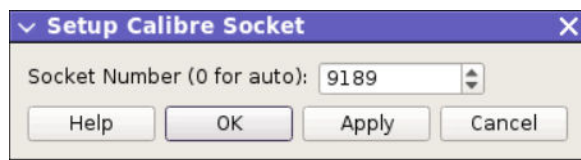


Figure 16 includes the following field:

- **Socket Number (0 for auto):** Specify a port number to start Calibre socket server. The default port number is 9189.

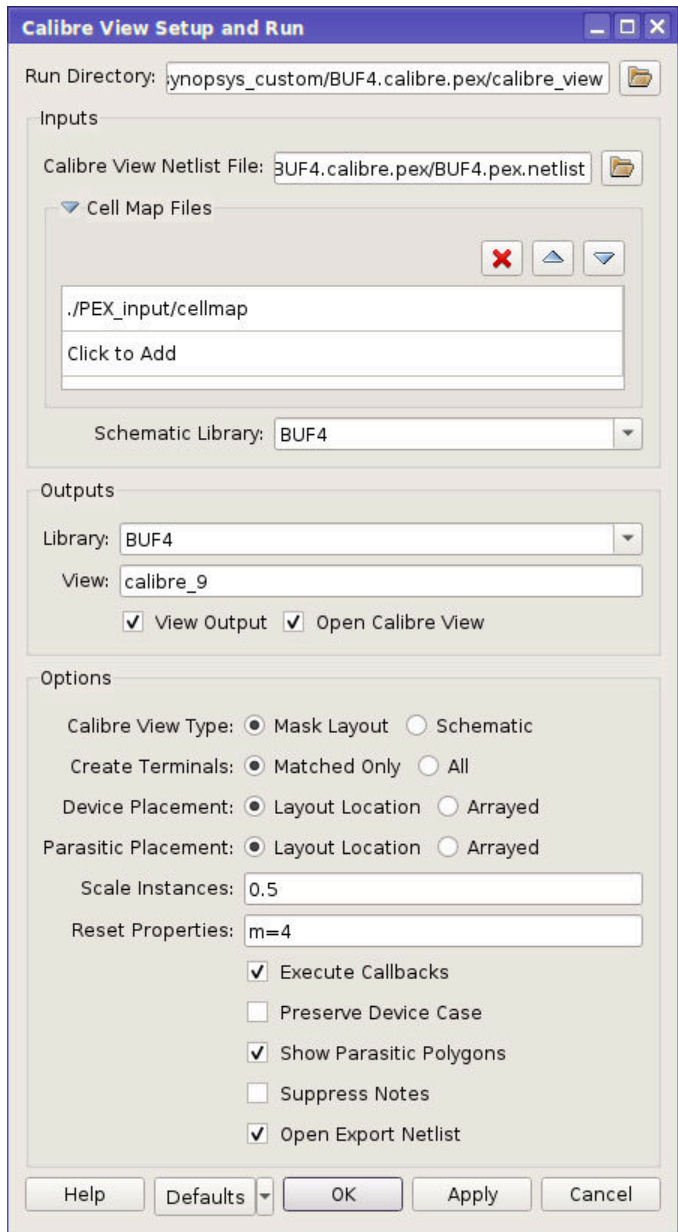
Displaying the Calibre View Menu

To invoke the **Calibre View Setup and Run** dialog box, choose **Tools > Calibre**. The **Calibre** menu appears on the menu bar. From the **Calibre** menu, choose **Calibre View**, as shown in Figure 17.

Figure 17 Calibre Menu



Figure 18 Calibre View Setup and Run



The **Calibre > Calibre View** menu in the **Layout** window is described in [Table 34](#).

Table 34 Calibre View Menu

Menu item	Description
Run Directory	Specifies the run directory.

Table 34 Calibre View Menu (Continued)

Menu item	Description
Calibre View Netlist File	Specifies CalibreView netlist file. The Calibre view netlist file is an output file format of Calibre PEX. You can run Calibre PEX with the CALIBREVIEW output format to generate the Calibre view netlist file.
Cell Map File	Specifies the cell map file. Multiple cell map file inputs are supported. The most recently loaded cell map file takes the highest priority. See the <i>Calibre Interactive User Guide</i> for more information on how to create a cellmap file.
Schematic Library	Specifies the schematic library name
Outputs	• Library: Specifies the output library name of the generated CalibreView. • View: Specifies output view name of the generated CalibreView. The default is “calibre”. • View Output: Specifies the output of the generated CalibreView. • Open Calibre View: Specifies whether to open the generated CalibreView.
Options	• Create View Type: Specifies the type of the generated CalibreView. The valid value is one of the enumerations: { Mask Layout Schematic }. The default is Mask Layout. • Create Terminals: Specifies whether to create global terminals when they are missing in the schematic design. Valid values are: { Matched Only All }. The default is Matched Only. • Device Placement: Specifies the placement type of intentional devices. Valid values are: { Layout Location Arrayed }. The default is Layout Location. • Parasitic Placement: Specifies the placement type of parasitic devices. Valid values are: { Layout Location Arrayed }. The default is Layout Location. This option is dependent on the Calibre View Type selection. • Scale Instances: Specifies the magnification factor for intentional devices and parasitics. Valid value is a positive floating-point number. The default is 1.0. This option is dependent on the Device Placement selection. • Reset Properties: Specifies properties to reset to a specific value. The format is a list of “propName=propValue”. For example, “ad=20 as=40”. The default is m=1. • Execute Callbacks: Specifies whether to execute CDF callback functions to override the property values. • Preserve Device Case: Specifies whether device mapping between CalibreView netlist and cellmap files is performed as case-sensitive. • Show Parasitic Polygons: Specifies whether to show polygons for the parasitic elements. This option is dependent on the Calibre View Type selection. • Suppress Notes: Specifies whether informational notes are output to the transcript. • Open Export Netlist: Specifies whether to open the Export Netlist dialog box after generating the Calibre View.

Running the Calibre Interactive Tool

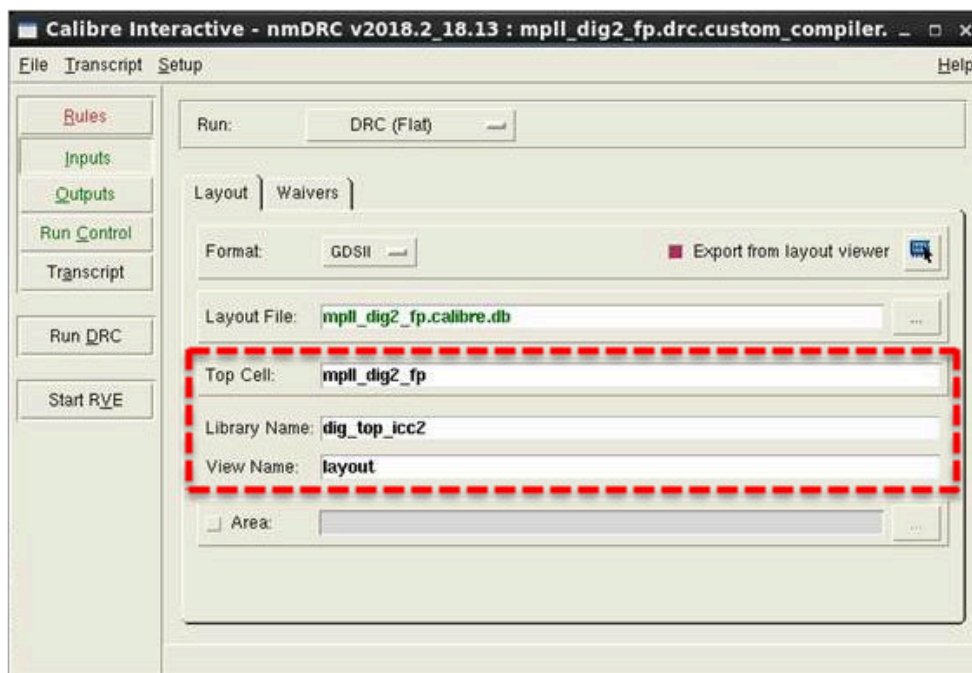
Note:

Before you start, see [Setting Up the Calibre Interactive Environment](#) and [Displaying the Calibre Menu](#).

To run the Calibre Interactive tool:

1. In the design schematic or layout, choose **Calibre** > <tool_name> to run the specific Calibre tool. See [About Using the Calibre Interactive Tool](#) for the list of supported Calibre tools.

The **Calibre Interactive** dialog box for the specific tool opens. The library, cell, and view information for the currently opened design automatically populate the **Library Name**, **Top Cell**, and **View Name** boxes. Use the `xtCalibreLoadRunset` preference to run the tool.



2. To switch to a different design, in the **Inputs** tab:

- a. Click **Choose an open layout** .

- b. Choose a design from the **Library/Cell/View** list, and click **OK**. The Library Name, Top Cell, and View Name boxes update and display the information of the chosen design.

3. For the remaining options and instructions for running the Calibre tool, see the Calibre Interactive user documentation.

Tracking Progress for Calibre Interactive Processes

You can track the Calibre Interactive DRC, PERC, DFM, and LVS processes with the Custom Compiler Job Monitor.

To track the Calibre Interactive processes:

1. In the design layout, choose **Tools > Job Monitor**.
2. In the Job Monitor, choose **View > Job Types > Interactive**.
3. Continue to [Debugging the Design Using the Calibre Results Viewing Environment](#).

Debugging the Design Using the Calibre Results Viewing Environment

Note:

Before you start, see [Setting Up the Calibre Interactive Environment](#) and [Displaying the Calibre Menu](#).

You can evaluate Calibre and Calibre Interactive DRC results in the Calibre Results Viewing Environment (RVE).

To debug the design:

1. In the design layout, open the RVE tool:
 - If you elected to launch Calibre RVE when you set up the DRC, RVE has opened automatically if violations were found.
 - If RVE is not open, you can choose **Calibre > RVE**.
2. (Optional) View error reports and output files in the Job Monitor.

3. (Optional) Double-click a rules violation to view the corresponding area in the design layout.

When you highlight an error, the Layout window appears.

4. (Optional) Search for missing instances in the hierarchy by choosing **Calibre > Highlights**.

Note:

The `xtCalibreRVENavigateMissingDevice` preference displays a warning message with the highest instance or lowest instance in the hierarchy. The default is set to `LOWEST`.

5. (Optional) Clear layout violation highlights from the design by choosing **Calibre > Clear Highlights**.

Note:

The focus remains on the RVE tool while navigating through violations. To shift the focus to the Custom Compiler tool, set the `xtCalibreRVERaiseOnHighlight` Tcl preference to `true`.

For additional information about using RVE, see the Calibre RVE documentation.

6. Continue iterating the design until it achieves design compliance with layout fabrication rules.

14

Extracting Interconnect Parasitics

Extract interconnect parasitics from the layout of a design. The parasitics are modeled as lumped elements that represent the unintended resistance, capacitance, and inductance that result from interconnect wiring of the physical layout.

You can extract interconnect parasitics with the Synopsys StarRC tool in the Custom Compiler environment.

The StarRC tool produces parasitic output, which can be in different formats: SPF, SPEF, SBPF, and OpenAccess (parasitic View).

You can run the StarRC interconnect parasitic extraction process after running device and connectivity extraction using one of the following LVS tools:

- Synopsys IC Validator
- Synopsys Hercules
- Mentor Graphics Calibre

This chapter covers setting input options for parasitic extraction, executing parasitic extraction, and viewing and analysis after parasitic extraction.

- [Before You Start](#)
- [Setting Options for Layout Parasitic Extraction](#)
- [Setting Up Partial Layout Extraction](#)
- [Running or Rerunning Parasitic Extraction](#)
- [Monitoring Parasitic Extraction Run Progress](#)
- [Analyzing Extracted Parasitic Views](#)
- [Exploring the GPD Parasitics Database](#)

Before You Start

This topic describes the prerequisites for using the StarRC simultaneous multicorner extraction process:

- [Defining the Process Corners](#)
- [Using the Correct PDK Library](#)

To choose the process corners for analysis, see [Setting LPE Extraction Options, Step 3](#).

Defining the Process Corners

To enable StarRC simultaneous multicorner extraction, you must define the process corners in the `pdk.xml` file. For each process corner definition, you must specify a `.nxtgrd` file, and optionally specify a `.nxtgrd.map` file. If you do not specify a `.nxtgrd.map` file, the tool will use the mapping file generated during LVS. You can define up to 15 process corners.

You can also define the process corners by setting the `xtLPEStarCornerDef` Tcl preference in the Console. The preference settings are retained only during the current session. To use the settings across sessions, save the preference settings in the `.synopsys_custom.tcl` file.

The `pdk.xml` file is stored in the same directory as the PDK library. Locate the file and add the following:

```
<preference name="xtLPEStarCornerDef">
  <value>
    cornerName1 {tcadFile1 mapFile1 [tempList1]}
    cornerName2 {tcadFile2 mapFile2 [tempList2]}
    ...
    cornerName3 {tcadFileN mapFileN [tempListN]}
  </value>
</preference>
```

Note:

Only the `tcadFile` is mandatory in the `pdk.xml` file.

For example:

```
<preference name="xtLPEStarCornerDef">
  <value>
    typical
      {$PDK_INSTALL_DIR/starrc/reference40_1P10M_typical.nxtgrd
      $PDK_INSTALL_DIR/starrc/reference40_1P10M_typical.nxtgrd.map}
    cmin
      {$PDK_INSTALL_DIR/starrc/reference40_1P10M_cmin.nxtgrd
      $PDK_INSTALL_DIR/starrc/reference40_1P10M_cmin.nxtgrd.map}
```

```

    rcmax
    {$PDK_INSTALL_DIR/starrc/reference40_1P10M_rcmax.nxtgrd
    $PDK_INSTALL_DIR/starrc/reference40_1P10M_rcmax.nxtgrd.map}
  </value>
</preference>

```

For a multicorn definition of different default operating temperatures, a sublist of the `xtLPEStarCornerDef` preference is added as follows:

```

<preference name="CornerDef">
  <value>
    typical {../PDK/starrc/reference40_1P10M_typical.nxtgrd "" {10 20
    30} }
    cmin {../PDK/starrc/reference40_1P10M_cmin.nxtgrd }
    rcmax {../PDK/starrc/reference40_1P10M_rcmax.nxtgrd "" {30 40 50} }
  </value>
</preference>

```

For a multicorn definition with several mapping files in the LVS folder, the `xtLPEStarCornerDef` preference can be defined as follows:

```

<preference name="xtLPEStarCornerDef">
  <value>
    corner1 {../PDK/starrc/corner1.nxtgrd \[LVS_RUNDIR\]/corner1.map}
    corner2 {../PDK/starrc/corner2.nxtgrd \[LVS_RUNDIR\]/corner2.map}
    corner3 {../PDK/starrc/corner3.nxtgrd \[LVS_RUNDIR\]/corner3.map}
  </value>
</preference>

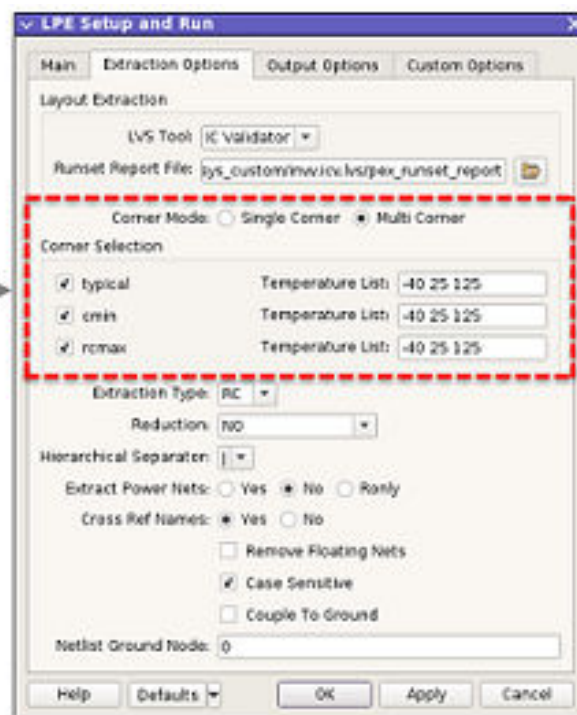
```

You can define preferences by removing the prefixes for custom LPE jobs.

Each `cornerName` in the file appears as a corner selection in the **LPE Setup and Run** dialog box. In the example shown, corner names in the XML file are `typical`, `cmin`, and `rcmax`. Notice that the same names appear as **Corner Selections** in the dialog box.

In the `pdk.xml` file:

```
<preference name="xtLPEStarCornerDef">
  <value> typical {... ..}
          cmin   {... ..}
          rcmay   {... ..}
</value>
</preference>
```



Using the Correct PDK Library

Make sure your design uses the correct PDK library with its updated `pdk.xml` file. To do this, before you start the Custom Compiler environment, set the `PDKLib` attribute to `true` for the PDK library in the `lib.defs` file:

```
DEFINE <lib_name> /<path>/<lib_name>
ASSIGN <lib_name> PDKLib true
```

If the `pdk.xml` file is not in the same directory as the PDK library, make sure the design references it.

```
ASSIGN <lib_name> PDKConfigPath /path/to/pdk.xml
```

For more information about `ASSIGN` statements, see [Controlling Library Availability](#).

Setting Options for Layout Parasitic Extraction

This section covers setting all options for parasitic extraction.

Before you start, make sure you have the required files for physical verification and extraction. See [Preparing for Physical Verification and Extraction](#).

In addition, set the following main options:

- [Setting LPE Main Options](#)
- [Setting LPE Extraction Options](#)
- [Setting LPE Output Options](#)
- [Setting LPE Custom Options](#)

Setting LPE Main Options

Set main options by entering information about the run directory and job parameters.

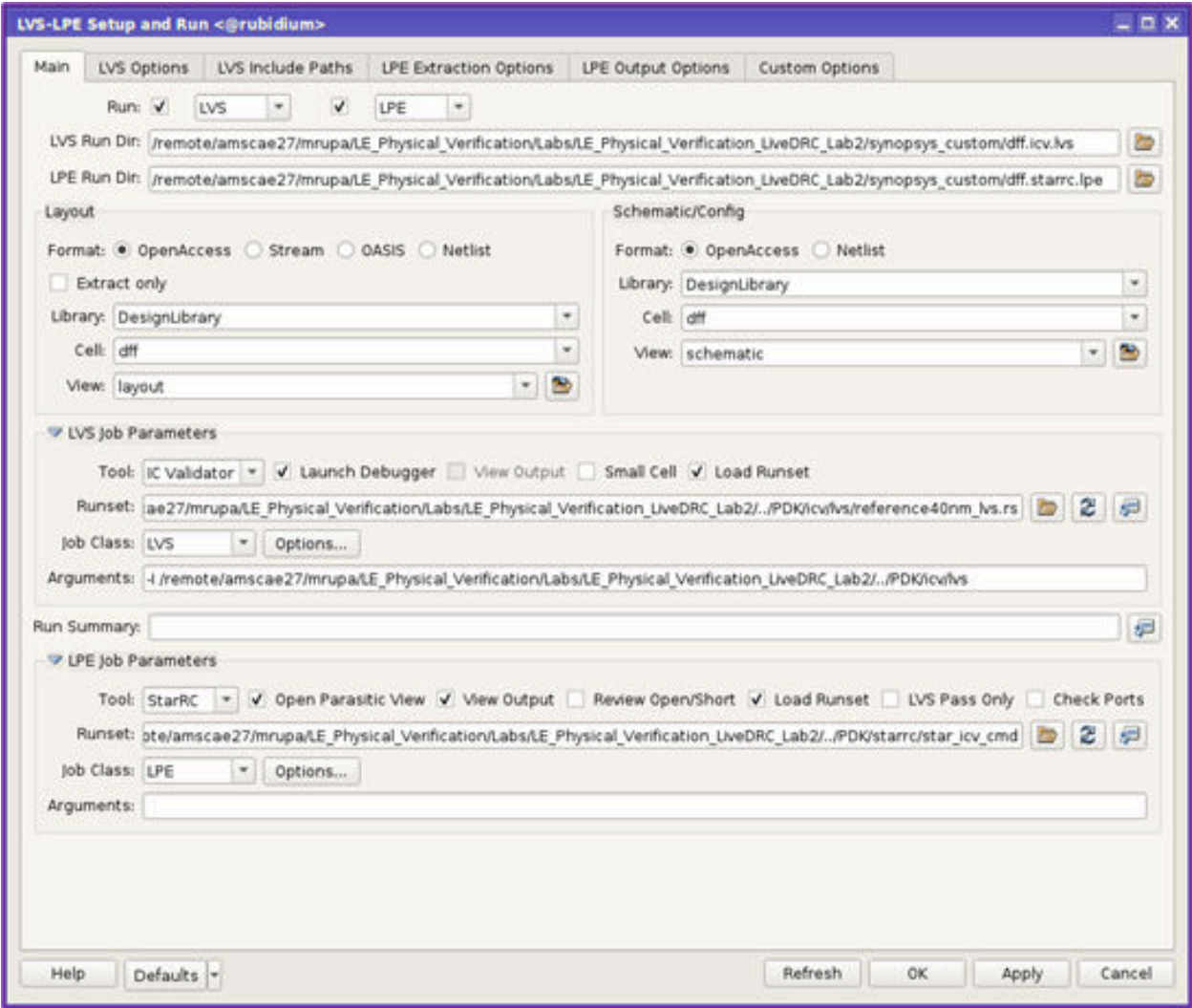
Note:

See [Default Interfaces](#) and [Custom Jobs](#) for more information.

To set main parasitic extraction options:

1. In the design layout, choose **Verification > LPE > Setup and Run** (Figure 19).

Figure 19 LPE Setup and Run Dialog Box



2. On the **Main** tab, enter settings as described in Table 35.

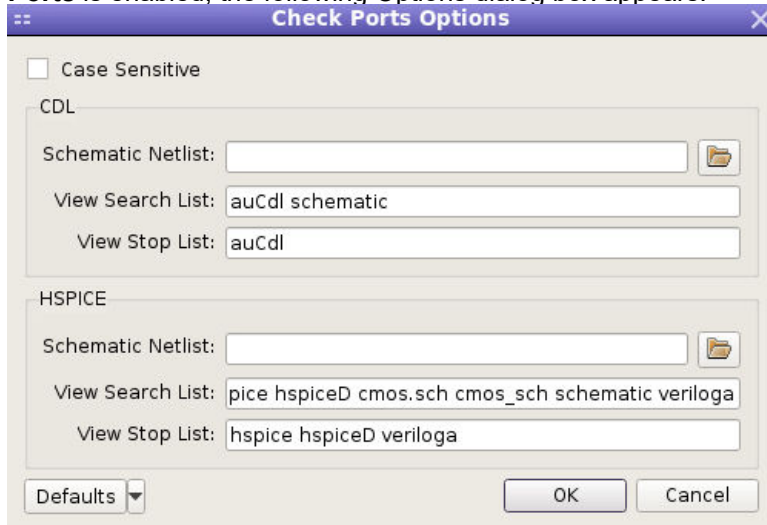
Table 35 LPE Setup and Run Settings (Main Tab) for All Tools

Settings	Description
Run Dir	You can accept the current run directory or browse to choose a different directory.

Settings	Description
Layout	You can accept the current Library , Cell , and View or browse to choose a different layout.
Schematic	You can accept the current Library , Cell , and View or browse to choose a different schematic.
Job Parameters	Tool is StarRC. View Output opens a Text Viewer window with the process log. On by default. Open Parasitic View opens the parasitic view or Netlist File in the Text Viewer based on the LPE Output Netlist format. On by default. Note: If the output format is NetName, SPF, SPEF, or STAR, the netlist file opens if it completes successfully. Review Open/Short reviews the open and short violations from the GPD database when the StarRC job is complete. Off by default. Use the <code>xtLPEStarReviewOpenShort</code> preference to set this mode. Load Runset Loads user-supplied values into the setup dialog. On by default. Load Rundir determines if the <code>prefs.xml</code> file is automatically loaded. If set to false, the <code>prefs.xml</code> file is not loaded. The default is true. Use the <code>xtLPEAutoPrefs</code> preference to set this mode.

Settings	Description
----------	-------------

Check Ports specifies whether to check ports between the CDL and HSPICE netlists before the preceding LPE job. When **Check Ports** is enabled, the following Options dialog box appears.



- **Case Sensitive:** Specifies whether to compare case-sensitive ports. This setting is disabled by default. Preference is `xtLPECheckPortsCaseSensitive`.

CDL. Optional setting used to provide a CDL netlist file. If the CDL **Schematic Netlist:** is empty, an information message appears and a CDL netlist file automatically generates a **View Search List** and **View Stop List**.

Preferences are `lpeCheckPortsCDLNetlistFile`, `xtLPECheckPortsCDLViewSearchList`, and `xtLPECheckPortsCDLViewStopList`.

HSPICE. Optional setting used to provide an HSPICE netlist file. If the CDL **Schematic Netlist:** is empty, an information message appears and a CDL netlist file automatically generates a **View Search List** and **View Stop List**.

- **Schematic Netlist:** Name is `lpeCheckPortsHSPICENetlistFile`.
- **View Search List:** Name is `lpeCheckPortsHSPICEViewSearchList`; preference is `xtLPECheckPortsHSPICEViewSearchList`.
- **View Stop List:** Name is `lpeCheckPortsHSPICEViewStopList`; preference is `xtLPECheckPortsHSPICEViewStopList`.

Runset is the path to the Star command file that contains the StarRC commands to configure the parasitic extraction run. The tool supports to parse the Tcl expression in the StarRC command

file. The **Reload**  button reloads the runset. The **Open Runset File**  button opens the runset.

Settings	Description
	Job Class combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see <i>Adding Job Classes With xt::createJobClass Tcl</i> in the <i>Custom Compiler Environment User Guide</i>. Options are the job settings to control batch processing limits, and to optionally set a queue engine in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i>. Arguments is the list of options for the extraction job that will go to the StarRC command line. Run Summary displays LVS Compare results as either PASS or FAIL and displays DRC Extraction Results as CLEAN or NOT CLEAN. This is a read-only option. It is visible only when you set the <code>xtLVSICV<pveJob><tool>ShowRunSummary</code> preference to true..

- To complete the option setting process for a new LPE run, continue by [Setting LPE Extraction Options](#).

Alternatively, if this LPE run requires no additional setting changes, you can continue by [Running or Rerunning Parasitic Extraction](#).

Setting LPE Extraction Options

To set extraction options for LPE:

- Begin by [Setting LPE Main Options](#).
- In the **Extraction Options** tab, enter settings for the tool supplying the LVS results.
 - For the IC Validator tool, refer to [Figure 20](#) and [Table 36](#).
 - For all Calibre tools, refer to [Figure 21](#) and [Table 37](#).

Figure 20 LPE Setup and Run: Extraction Options Tab for IC Validator

The screenshot shows the 'LVS-LPE Setup and Run' dialog box with the 'Extraction Options' tab selected. The 'Layout Extraction' section includes 'LVS Tool' set to 'IC Validator' and a 'Runset Report File' path. The 'Corner Mode' is set to 'Single Corner'. The 'Corner Selection' section has 'Corner' set to 'custom', with fields for 'TCAD GRD File', 'Mapping File', and 'Operating Temperature'. The 'Extraction Type' is 'RC', 'Analysis' is 'None', and 'Reduction' is 'LAYER'. The 'Hierarchical Separator' is '|'. 'Extract Power Nets' is 'No'. 'Cross Ref Names' is 'Yes', with checkboxes for 'Remove Floating Nets' (unchecked), 'Case Sensitive' (checked), and 'Couple To Ground' (checked). The 'Netlist Ground Node' is '0'. 'Skip Cells' is 'None', and 'Skip Cell Mapping File' is 'Auto Generate'. There are checkboxes for 'Remove Trivial Instance Ports' (unchecked) and 'Explode Trivial Instance Ports' (unchecked). At the bottom, there are tabs for 'Additional Options', 'Field Solver Options', and 'Inductance Options', and buttons for 'Help', 'Defaults', 'OK', 'Apply', and 'Cancel'.

Table 36 IC Validator LPE Setup and Run Settings (Extraction Options Tab)

Settings	Description
Layout Extraction	LVS Tool is IC Validator. See Table 38 for more information on available options for all LVS-tool related options.

Settings	Description
	Runset Report File is created by IC Validator containing pointers to the information StarRC needs to perform the parasitic extraction. This includes connection information, the location of LVS compare output, device pin and property information, and the location of the extract view. This option corresponds to the ICV_RUNSET_REPORT_FILE StarRC command.

Figure 21 LPE Setup and Run: Extraction Options Tab for Calibre

The screenshot shows the 'LVS-LPE Setup and Run' dialog box with the 'LPE Extraction Options' tab selected. The 'Layout Extraction' section includes 'LVS Tool' set to 'Calibre', 'Generate CCI Database' checked, 'Mask SVDB Directory' pointing to a path, and 'Query Command Type' set to 'Default'. The 'Corner Mode' is 'Single Corner'. The 'Corner Selection' section has 'Corner' set to 'custom', with fields for 'TCAD GRD File', 'Mapping File', and 'Operating Temperature'. The 'Extraction Type' is 'RC', 'Analysis' is 'None', 'Reduction' is 'LAYER', and 'Hierarchical Separator' is '|'. 'Extract Power Nets' is 'No', 'Cross Ref Names' is 'Yes', and 'Remove Floating Nets', 'Case Sensitive', and 'Couple To Ground' are all checked. 'Netlist Ground Node' is '0'. 'Skip Cells' is 'None', and 'Skip Cell Mapping File' is 'Auto Generate'. At the bottom, there are tabs for 'Additional Options', 'Field Solver Options', and 'Inductance Options', along with 'Help', 'Defaults', 'OK', 'Apply', and 'Cancel' buttons.

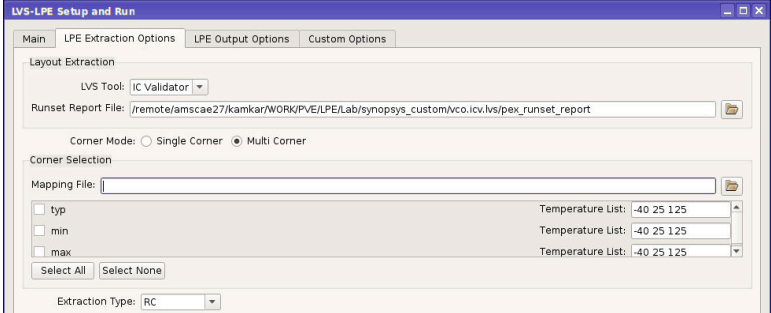
Table 37 Calibre or Calibre Interactive LPE Setup and Run Settings (Extraction Options Tab)

Settings	Description
Layout Extraction	LVS Tool is Calibre. See Table 38 for more information on available options for all LVS-tool related options. Generate CCI Database (on by default) tells StarRC to run the Calibre query server to create a CCI database before performing the parasitic extraction. Mask SVDB Directory is the mask SVDB database to use for generating input to StarRC. By default, this is the path the system uses: <code>./<cellname>.calibre.lvs/svdb</code>. Query Command Type offers the option of using the Default query command file for generating input to StarRC, or a Custom file you browse to choose. For details about the default file format, see the StarRC documentation.

3. Continue on the **Extraction Options** tab by entering options that apply to all LVS tools as described in [Table 38](#).

Table 38 LPE Setup and Run Settings for All LVS Tools (Extraction Options Tab)

Settings	Description
Corner Mode	(Available only if you defined the corners in the <code>pdk.xml</code> file or if you set the <code>xtLPEStarCornerDef</code> Tcl preference. See Before You Start.) Specifies the corner process: ◦ Single Corner: Runs a single extraction based on the Mapping File, TCAD GRD File, and Operating Temperature you specify. See Figure 20. ◦ Multi Corner: Runs multiple processes based on the Corner Selection and Temperature List that you specify. Use the <code>xtLPEStarCornerTemps</code> preference to specify the default temperatures for all corners.



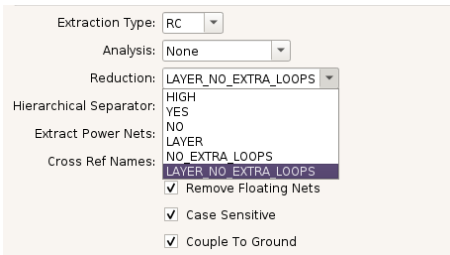
Corner Selection

TCAD GRD File corresponds to the StarRC `TCAD_GRD_FILE` command. This file typically comes from the foundry and describes the electrical characteristics of the metal interconnect layers in the foundry's process. The StarRC tool needs this information to properly extract interconnect parasitics from the layout, so that StarRC results match the physical and electrical characteristics of the foundry's process.

Mapping File maps TCAD process layers to corresponding layout database layers. This setting corresponds to the StarRC `MAPPING_FILE` command.

Note:
If a corner definition does not include a **MAPPING_FILE** command, a global mapping file must be defined in the top-level command file. If every corner includes a mapping file, a global mapping file is not necessary.

Operating Temperature sets the operating temperature. If the StarRC command `OPERATING_TEMPERATURE` is set in the input file, that value populates the field. If the value is outside the max and min values (specified in the StarRC manual), a warning appears and you cannot **Apply** the extraction options.

Settings	Description
Extraction Type	Sets R , C , RC , or NORC for the type of extraction to perform. This option corresponds to the StarRC <code>EXTRACTION</code> command and is set to RC by default.
Analysis	Specifies the analysis used to export specific sets of StarRC commands. Corresponding StarRC commands are listed in Table 41 . Use the <code>xt{Job}StarAnalysis</code> preference to specify the analysis.
Reduction	Sets the level of reduction for extracted parasitic elements during extraction to HIGH, YES, NO, LAYER, NO_EXTRA_LOOPS, or LAYER_NO_EXTRA_LOOPS. The StarRC default is YES and the Custom Compiler default is LAYER. Choosing LAYER restricts the reduction so that it does not occur across different conductor layers, facilitating correlation of parasitics with their underlying source. This option corresponds to the StarRC <code>REDUCTION</code> command. 
Hierarchical Separator	Sets the separator (/ or or . or :) to delineate hierarchy levels in the parasitic netlist. The default is . This option corresponds to the StarRC <code>HIERARCHICAL_SEPARATOR</code> command.
Extract Power Nets	Specifies whether to extract power nets, either Yes , No (the default), Ronly or Device Layers . This option corresponds to the StarRC <code>POWER_EXTRACT</code> command. If <code>POWER_NETS</code> need to be set, add that option to an input command file.
Cross Ref Names	Sets the naming convention to be used in the extracted output to either Yes (the default) or No . This option corresponds to the StarRC <code>XREF</code> command. (For Hercules, another option, Complete , is also available.) No tells StarRC to use net and instance names inferred from the layout design. Yes and Complete use information (and more complete information) from the schematic to infer net and instance names when producing the parasitic netlist or view.
Remove Floating Nets	(On by default.) Sets the LPE process to remove those nets with no connections from the extraction output. This is a Boolean option. This option corresponds to the StarRC <code>REMOVE_FLOATING_NETS</code> command.
Case Sensitive	(On by default.) Applies case-sensitive rules to net and cell names. This option corresponds to the StarRC <code>CASE_SENSITIVE</code> command.

Settings	Description
Couple to Ground	(On by default.) Sets parasitic coupling capacitances to be lumped to ground during extraction and netlisting. This option corresponds to the StarRC <code>COUPLE_TO_GROUND</code> command.
Netlist Ground Node	Sets the ground net name to be used for specifying capacitance to noncritical material or to the SUBSTRATE. Only one net name can be specified. The default is 0 if not specified. This option corresponds to the StarRC <code>NETLIST_GROUND_NODE_NAME</code> command.
Skip Cells	Skip Cells are defined using three settings: None, Select Cell List File, and Select Cells. - When None is selected, no cell is skipped during extraction. - When Select Cell List File is selected, all cells in the specified file, except the current cell are skipped during extraction. Cells can be separated by white space or a new line in the cell list file. An error message appears if the file path does not exist. - When Select Cells is selected, the cells specified on the GUI are skipped during extraction. Cells can be separated by white space, and this selection supports wildcard filters.
Additional Options	Specifies additional LPE Extraction Options in a separate dialog box, as shown in Figure 22, Figure 23, and Figure 24. See Table 43. The following command dependencies exist: <code>MAGNIFICATION_FACTOR</code> is shown if <code>MAGNIFY_DEVICE_PARAMS</code> is YES. <code>METAL_FILL_GDS_FILE</code> and <code>METAL_FILL_POLYGON_HANDLING</code> are shown if <code>GDS_LAYER_MAP_FILE</code> specifies a valid file path. <code>MERGE_INSTANCE_PORTS</code> and <code>XREF_FEEDTHRU_NETS</code> are shown if Cross Ref Names is Complete. <code>XREF_LAYOUT_INST_PREFIX</code>, <code>XREF_LAYOUT_NET_PREFIX</code>, and <code>XREF_USE_LAYOUT_DEVICE_NAME</code> are shown if Cross Ref Names is Yes. - The Resistance Field Solver (RFS) provides accurate resistance extraction for any shape of geometries, including non-manhattan shapes with many holes.

Settings	Description
Field Solver Options	Shows the Field Solver Options in a separate dialog box. See Table 39 .

FSCOMPARE_COUPLING_RATIO: 0.1

FSCOMPARE_COUPLING_THRESHOLD: 1.0e-15

FSCOMPARE_THRESHOLD: 3.0e-15

FSCOMPARE_OPTIONS:

FS_DP_STRING:

FS_EXTRACT_NETS

Net Names

Click to Add

Resistance Field Solver

RFS_EXTRACT_NETS

Net Names

Click to Add

RFS_LAYERS:

RFS_OPTIONS

PORT_MODE: LINE

PROBE_SIZE: 0.002um

MESH_SNAP_SIZE: 0.001um

REFINE: 0

Defaults

OK

Cancel

Settings	Description
Inductance Options	Shows the Inductance Extraction Options in a separate dialog box. The Inductance Options button, as shown in Figure 20 , is enabled if the output format is SPF . See Table 40 .

Table 39 *Field Solver Options*

Option	Description
FSCOMPARE_COUPLING_RATIO	Specifies the minimum ratio of the coupling capacitance to the total capacitance required on a particular net to make it eligible for comparison in an FSCOMPARE extraction. The valid value is a floating-point number, which is between 0.0 and 1.0. The default is 0.1.

Option	Description
FSCOMPARE_COUPLING_THRESHOLD	Specifies the minimum coupling capacitance required on a particular net to make it eligible for comparison in an FSCOMPARE extraction. The valid value is a floating-point number, which is greater than or equal to 0.0. The default is 1.0e-15.
FSCOMPARE_THRESHOLD	Specifies the minimum total capacitance required on a particular net to make it eligible for comparison in an FSCOMPARE extraction. The valid value is a floating-point number, which is greater than or equal to 0.0. The default is 3.0e-15.
FSCOMPARE_OPTIONS	Specifies field solver options, such as convergence goal and multiprocessing. No validation in this option.
FS_DP_STRING	Specifies the distributed processing method and job control parameters for field solver extraction runs. No validation in this option.
FS_EXTRACT_NETS	Specifies nets to be extracted by the field solver. This option is enabled if Analysis is not Field Solver.
Resistance Field Solver (RFS)	Provides accurate resistance extraction for any shape of geometries, including non-manhattan shapes with many holes.
RFS_EXTRACT_NETS	Specifies nets to be extracted by the Resistance Field Solver (RFS). All other nets are extracted by StarRC. When used with RFS_LAYERS, only nets in RFS_EXTRACT_NETS and layers in RFS_LAYERS are extracted by the RFS.
RFS_LAYERS	Specifies the database layers to be extracted by the RFS. All other layers are extracted by StarRC.
PORT_MODE	Specifies how to model ports. A valid value is one of the enumerations: {LINE LINEX LINEY SQUARE}. The default is LINE.
PROBE_SIZE	Specifies the edge length (in um) when PORT_MODE = SQUARE. The default is 0.002. This option is available if PORT_MODE = SQUARE.
MESH_SNAP_SIZE	Specifies mesh size, that is, the minimum distance (in um) between two mesh points in the x- and y-directions. The default is 0.001.
REFINE	Performs additional refinement on the mesh created by the RFS. The valid value is an integer between 0 and 10. The default is 0.

Table 40 Inductance Extraction Options

Option	Description
INDUCTANCE_MODE	Enables mutual inductance analysis as part of a transistor-level extraction and specifies the analysis mode. The valid value is one of the enumerations: { NO INDUCTANCE RELUCTANCE }. The default is NO.
INDUCTANCE_FREQUENCY	Specifies a frequency for mutual inductance analysis in a transistor-level extraction. The valid value is a floating-point number which is between 0.001 and 100. The default is 1.
INDUCTANCE_MIN_LENGTH	Specifies the minimum length for a resistor to be included in mutual inductance analysis performed as part of a transistor-level extraction. The default is 1. The valid value is a floating-point number.
INDUCTANCE_NINC	Specifies how finely to split up metal segments for mutual inductance analysis performed as part of a transistor-level extraction. By default, the StarRC tool automatically sets this parameter. The valid value is an integer number.
INDUCTANCE_REL_THRESHOLD	Specifies the minimum reluctance values to include in mutual inductance analysis performed as part of a transistor-level extraction. The default is 1. The valid value is a floating-point number.
INDUCTANCE_SELECT_BB	Specifies the bounding box and layer information for mutual inductance analysis performed as part of a transistor-level extraction. You can specify one bounding box and the layers or specify the name of a file that contains the information for multiple bounding boxes.
INDUCTANCE_SELECT_LAYER	Specifies layers for mutual inductance analysis performed as part of a transistor-level extraction. By default, the StarRC tool selects all layers for mutual inductance analysis.
INDUCTANCE_S_PARAM_FREQUENCY	Specifies a frequency range for mutual inductance analysis performed as part of a transistor-level extraction. This option is available if the output format is SPF .

Option	Description
INDUCTANCE_SELECT_S_BB	Specifies the bounding box and layer information for mutual inductance analysis of S parameters performed as part of a transistor-level extraction. You must specify an output file. You can specify one bounding box and the layers, or specify the name of a file that contains the information for multiple bounding boxes. This option is available if the output format is SPF and the <code>INDUCTANCE_S_PARAM_FREQUENCY</code> is specified.

Figure 22 Additional Extraction Options

Additional Extraction Options

COUPLING_ABS_THRESHOLD: 3e-15 F ▲

COUPLING_MULTIPLIER: 1

COUPLING_REL_THRESHOLD: 0.03

COUPLING_REPORT_FILE:

COUPLING_REPORT_NUMBER: 1000 ▲ ▼

COUPLING_THRESHOLD_OPERATION: AND ▼

DENSITY_BASED_THICKNESS: YES ▼

DENSITY_OUTSIDE_BLOCK:

DPT: NO ▼

EXTRACT_VIA_CAPS: NO ▼

FS_EXTRACT_NETS

Net Names

Click to Add

GDS_LAYER_MAP_FILE:

HN_NETLIST_MODEL_NAME

Model Names

Click to Add

Defaults ▼

OK Cancel

Figure 23 Additional Extraction Options (Continued)

Additional Extraction Options

HN_NETLIST_SPICE_TYPE

Types

Click to Add

IGNORE_CAPACITANCE: ALL

KEEP_VIA_NODES: NO

MAGNIFICATION_FACTOR:

MAGNIFY_DEVICE_PARAMS: YES

MERGE_INSTANCE_PORTS: NO

METAL_FILL_GDS_FILE:

METAL_FILL_POLYGON_HANDLING: IGNORE

MODEL_TYPE: LAYOUT

MOS_GATE_DELTA_RESISTANCE: NO

NETS

Net Names

Click to Add

PIN_CUT_THRESHOLD:

POWER_NETS:

POWER_REDUCTION: YES

PROBE_TEXT_FILE:

Defaults

OK Cancel

Figure 24 Additional Extraction Options (Continued)

Additional Extraction Options

REDUCTION_MAX_DELAY_ERROR: 1e-12

REMOVE_DANGLING_NETS: NO

Resistance Field Solver

RFS_EXTRACT_NETS

Net Names

Click to Add

RFS_LAYERS:

RFS_OPTIONS

PORT_MODE: LINE

PROBE_SIZE: 0.002 um

MESH_SNAP_SIZE: 0.001 um

REFINE: 0

RETAIN_CAPACITANCE_CAP_MODELS:

SHORT_PINS: YES

STAR_DIRECTORY: star

TRANSLATE_RETAIN_BULK_LAYERS: YES

WIDE_DEVICE_TERM_RESISTANCE: NONE

XREF_FEEDTHRU_NETS: NO

XREF_LAYOUT_INST_PREFIX: ld_

XREF_LAYOUT_NET_PREFIX: ln_

XREF_USE_LAYOUT_DEVICE_NAME: NO

Defaults OK Cancel

Table 41 Analysis Name Corresponding to StarRC Command

Analysis name	StarRC command
None (default)	

Analysis name	StarRC command
EM Analysis	REDUCTION: NO EXTRA_GEOMETRY_INFO: NODE RES POWER_REDUCTION: NO NETLIST_TAIL_COMMENTS: YES
Field Solver	FS_EXTRACT_NETS:*
Temperature-VX	TEMPERATURE_SENSITIVITY: YES

Table 42 Netlist Type Values and Extraction Type and Couple to Ground Option Dependencies

Possible Netlist Types	Couple To Ground is NO	Couple To Ground is YES
Extraction Type is RC	no_couple, Default, R, Cg, Cc, RCg, RCc	Default, R, Cg, RCg
Extraction Type is C	no_couple, Default, Cg, Cc	Default, Cg
Extraction Type is R	Default, R	Default, R
Extraction Type is NORC	Default, None	Default, None

Table 43 Additional Extraction Options

Option	Description
COUPLING_ABS_THRESHOLD	Specifies an absolute threshold for grounding coupling capacitors. The valid value is a floating-point number. The default is 3e-15. This option is enabled if Couple To Ground is disabled.
COUPLING_MULTIPLIER	Specifies a scaling factor to be applied for transferring coupling capacitances to ground. A valid value is a floating-point number which is greater than zero. The default is 1.
COUPLING_REL_THRESHOLD	Specifies the ratio of coupling capacitance to total capacitance used to determine whether to ground coupling capacitors. The valid value is a floating-point number which is between 0.0 and 1.0. The default is 0.03. This option is enabled if Couple To Ground is disabled.

Option	Description
COUPLING_REPORT_FILE	Specifies a file to report the coupling capacitance by net. The default is none.
COUPLING_REPORT_NUMBER	Specifies the number of nets reported in the coupling report file. The valid value is an integer number which cannot be less than 0. The default is 1000. This option is enabled if Couple To Ground is disabled.
COUPLING_THRESHOLD_OPERATION	Specifies the use of AND filtering or OR filtering of coupling thresholds. The valid value is one of these enumerations: {AND OR}. The default is AND. This option is enabled if Couple To Ground is disabled.
DENSITY_BASED_THICKNESS	Enables the calculation of density and thickness variation during extraction. The default is true. This option is available if DENSITY_BASED_THICKNESS is YES.
DENSITY_OUTSIDE_BLOCK	Specifies the pattern density outside the block, which affects the thickness variation and parasitic RC values. The valid value is a floating-point number from 0.0 to 1.0, inclusive.
DPT	Enables extraction for double or multiple patterning processes. The default is NO.
EXTRACT_VIA_CAPS	Performs a detailed via capacitance extraction. The valid value is one of these enumerations: {NO YES YES IGNORE_GATE_CONTACT_COUPLING}. The default is NO.
GDS_LAYER_MAP_FILE	Specifies the file which includes the mapping between the GDSII layer number and layer name in the design database.
HM_NETLIST_MODEL_NAME	Writes a simulation model name instead of the StarRC model name in the parasitic netlist. The syntax should be <code>ref_model_name SPICE_model_name</code> . The <code>ref_model_name</code> is the model name in the schematic or layout and the <code>SPICE_model_name</code> is the internal database is updated with this SPICE model name.
HN_NETLIST_SPICE_TYPE	Specifies StarRC netlist standard devices as SPICE .SUBCKT calls beginning with X.

Option	Description
IGNORE_CAPACITANCE	Prevents certain types of device-level capacitances from being extracted. The valid value is one of these enumerations: {ALL ALL RETAIN_GATE_DIFFUSION_COUPLING DIFF NONE}. The default is ALL.
KEEP_VIA_NODES	Defines via nodes as the original nodes that a via resistor connects. The valid value is one of these enumerations: {YES NO MAPPING FILE}. The default is NO.
MAGNIFICATION_FACTOR	Scales input data uniformly for all layers. The value should be a positive floating-point number. This option is enabled if MAGNIFY_DEVICE_PARAMS is YES.
MAGNIFY_DEVICE_PARAMS	Enables the controls of the MAGNIFICATION_FACTOR command behavior. The default is YES.
MERGE_INSTANCE_PORTS	Enables the ability to decrease the effective resistance connecting the schematic port to the rest of the net when set to YES. The default is NO. This option is enabled only when Cross Ref Names is Complete.
METAL_FILL_POLYGON_HANDLING	Specifies the treatment of metal fill polygons during extraction. The valid value is one of the enumerations: {IGNORE GROUNDED FLOATING AUTOMATIC}. The default is IGNORE. This option is enabled only if the GDS_LAYER_MAP_FILE option specifies an existing file.
MOS_GATE_DELTA_RESISTANCE	Enables the extraction of the gate resistance of MOS devices as a delta network. The default is NO.
METAL_FILL_GDS_FILE	Specifies one or more GDSII files that contain metal fill data. This option is enabled only when the GDS_LAYER_MAP_FILE option specifies an existing file.
MODEL_TYPE	Specifies whether the reference model comes from the layout or the schematic. The default is LAYOUT.
NETS	Specifies a list of nets to extract.
PIN_CUT_THRESHOLD	Takes a long port and splits it into multiple * P nodes. The value specifies the distance at which to cut a long port.

Option	Description
POWER_NETS	Specifies power nets for special treatment during extraction. The *, ?, and ! wildcards are accepted. Case sensitivity is determined by the CASE_SENSITIVE command.
POWER_REDUCTION	Specifies the amount and type of reduction to perform for extracted power nets. The valid value is one of the enumerations: {HIGH YES NO LAYER LAYER_NO_EXTRA_LOOPS}. The default is YES. This option is enabled if Extract Power Nets is not NO.
PROBE_TEXT_FILE	Specifies a file that contains simulation probe points to appear in the parasitic netlist.
REDUCTION_MAX_DELAY_ERROR	Specifies the acceptable net delay error tolerance during reduction. The valid value is a floating-point number. The default is 1e-12.
REMOVE_DANGLING_NETS	Specifies whether to identify dangling nets and reassign them to ground (effectively removing them). The default is NO.
RETAIN_CAPACITANCE_CAP_MODELS	Specifies model-specific plate-to-plate capacitance retention for capacitor devices.
SHORT_PINS	Specifies whether to short top-level pin ports that have multiple placements. The valid value is one of these enumerations: {YES NO MIXED}. The default is YES.
STAR_DIRECTORY	Specifies a directory that the StarRC tool creates for reports and intermediate files. The default is star.
TRANSLATE_RETAIN_BULK_LAYERS	Specifies how to treat mapped bulk layers during transistor-level extraction. The valid value is one of these enumerations: {YES NO ONLY ONLY_ALL_TO_CLOSEST_CONTACT}. The default is YES.
WIDE_DEVICE_TERM_RESISTANCE	Treats resistor terminals as line nodes or point nodes. The valid value is one of these enumerations: {RES NONE}. The default is NONE. Note: This option is not exported to an intermediate Star command file if the default is NONE.
XREF_FEEDTHRU_NETS	Enables the output of pure layout feedthrough nets and ports. This option is enabled only when Cross Ref Names is Complete. The default is NO.

Option	Description
XREF_LAYOUT_INST_PREFIX	Specifies a prefix for layout device instance names that were not cross-referenced to a schematic device by the LVS tool. This option is enabled only when Cross Ref Names is Yes. The default is "Id_".
XREF_LAYOUT_NET_PREFIX	Specifies a prefix for layout device instance names that were not cross-referenced to a schematic device by the LVS tool. This option is enabled only when Cross Ref Names is Yes. The default is "In_". Note: This option is not exported to an intermediate Star command file if the default is "Id_".
XREF_USE_LAYOUT_DEVICE_NAME	Specifies whether to use layout device names. This option is enabled only when Cross Ref Names is Yes. The default is NO.

4. Continue by [Setting LPE Output Options](#).

Alternatively, if this LPE run requires no additional setting changes, you can continue by [Running or Rerunning Parasitic Extraction](#).

Setting LPE Output Options

Set output options to determine StarRC output formats and related details. Use this procedure for results from all LVS tools.

To specify parasitic extraction output options:

- Begin by entering settings for the previous tabs in the **LPE Setup and Run** dialog box.
 - [Setting LPE Main Options](#)
 - [Setting LPE Extraction Options](#)
- On the **Output Options** tab ([Figure 25](#)), enter settings for the StarRC netlistDB phase as described in [Table 44](#).

Figure 25 LVS-LPE Setup and Run: LPE Output Options Tab

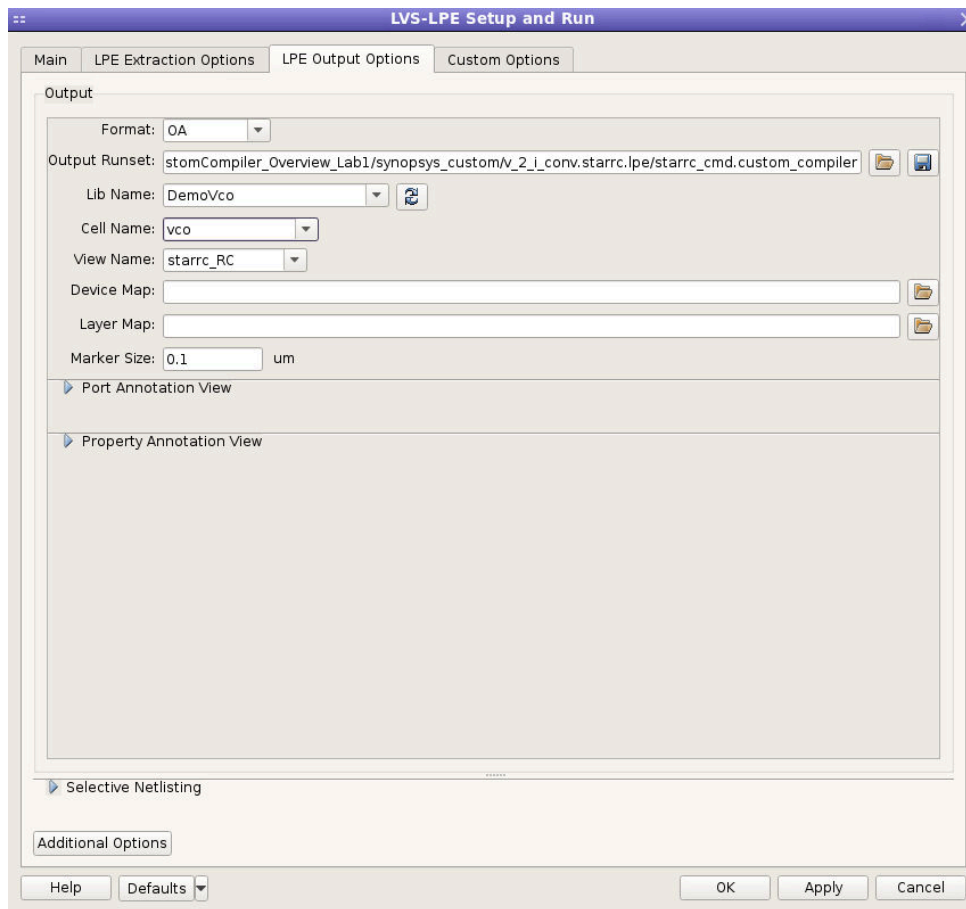
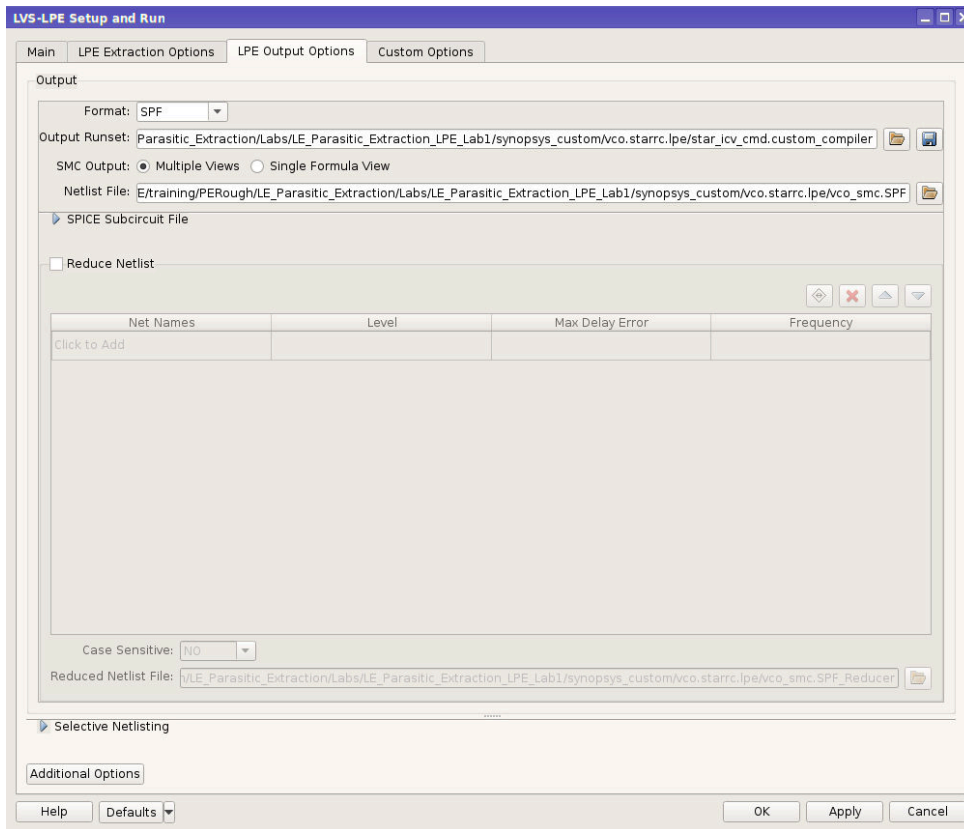
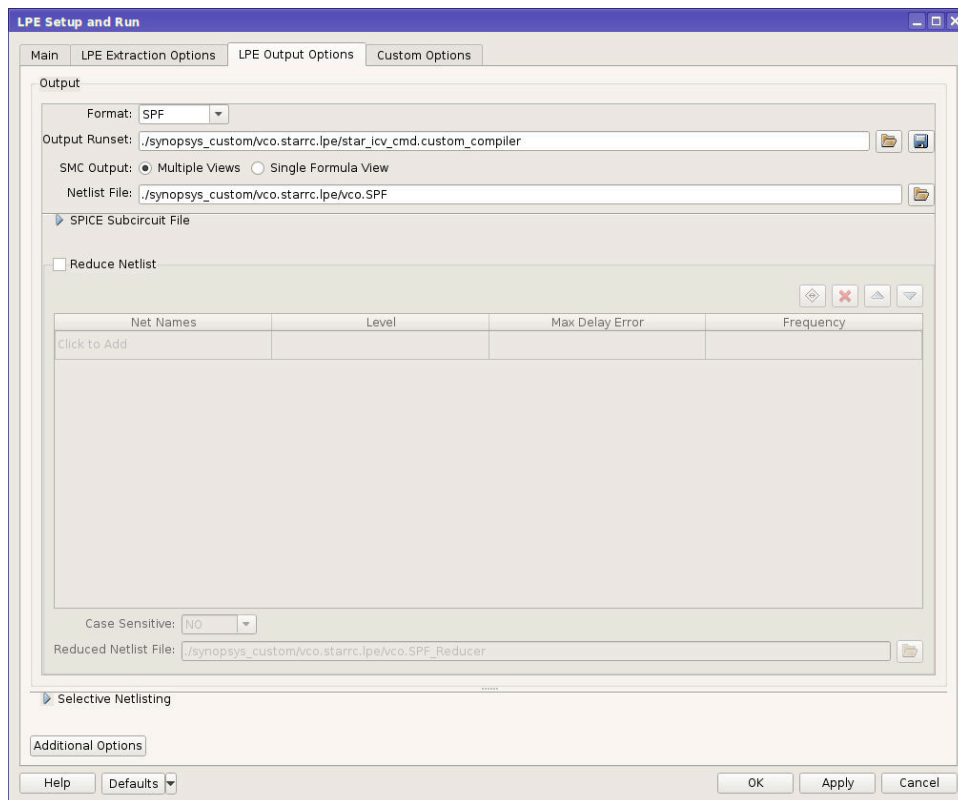


Figure 26 LVS-LPE Setup and Run: LPE Output Options Tab: SPF Format



- When the output **Format** is **SPF**, **SPEF**, **OA & SPF**, or **OA & SPEF** for the **Multiple Views**, **Reduce Netlist** is enabled, as shown in [Figure 27](#).

Figure 27 Reduce Netlist Enabled in LPE Setup and Run



4. In the **LPE Extraction Options** tab, when **Cross Ref Names** is set to
 - **Yes:** In the **LPE Output Options** tab, the **Selective Netlisting** has the **NET_TYPE: SCHEMATIC** command type enabled.
 - **No:** In the **LPE Output Options** tab, **Selective Netlisting** has the **NET_TYPE: LAYOUT** command type enabled.
 - **Complete:** In the **LPE Output Options** tab, the **Selective Netlisting** has the **NET_TYPE: SCHEMATIC** command type enabled again.

Chapter 14: Extracting Interconnect Parasitics Setting Options for Layout Parasitic Extraction

Figure 28 Cross Ref Names

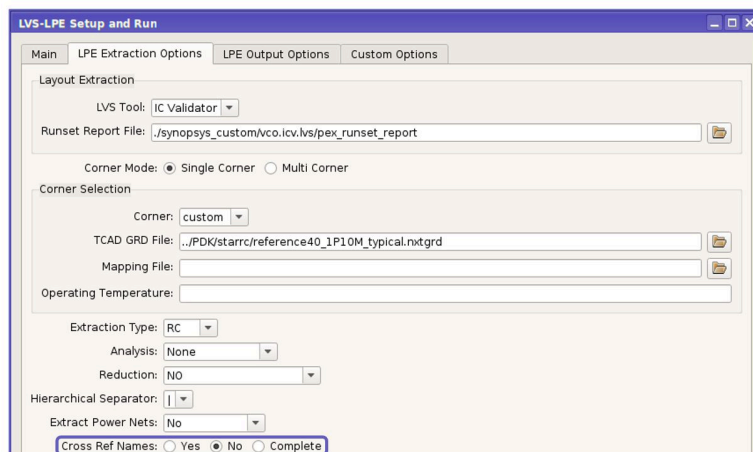


Figure 29 NET_TYPE: LAYOUT in Selective Netlisting

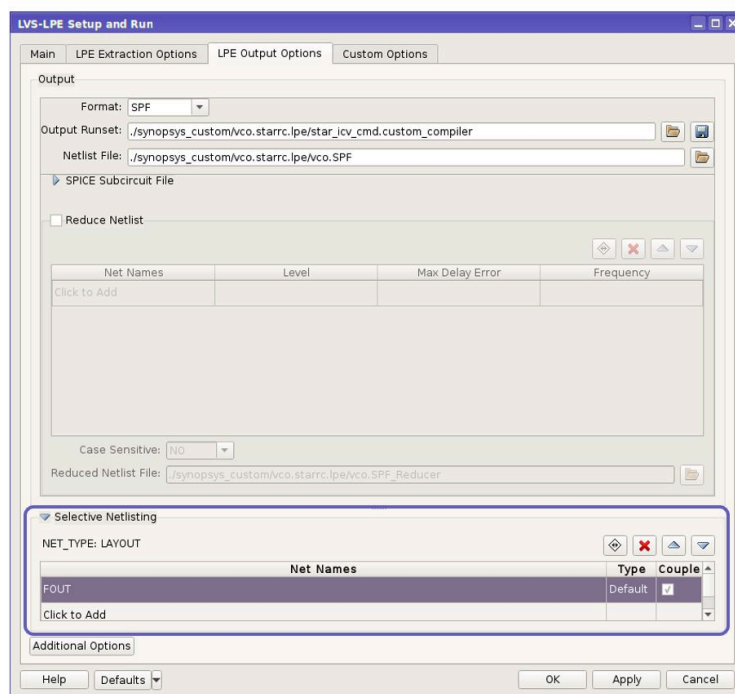


Figure 30 NET_TYPE: SCHEMATIC in Selective Netlisting

Selective Netlisting

NET_TYPE: SCHEMATIC

Net Names	Type	Couple
CIN	Default	☒
COUT	RCg	☒
Click to Add		

5. When set to **Single Formula View**, **Reduce Netlist** is disabled. See [Table 44](#) for more information on **Single Formula Views**.

When the output **Format** is **GPD**, you can specify the path of the output GPD directory, as shown in [Figure 31](#).

Figure 31 GPD Directory in LPE Output Options

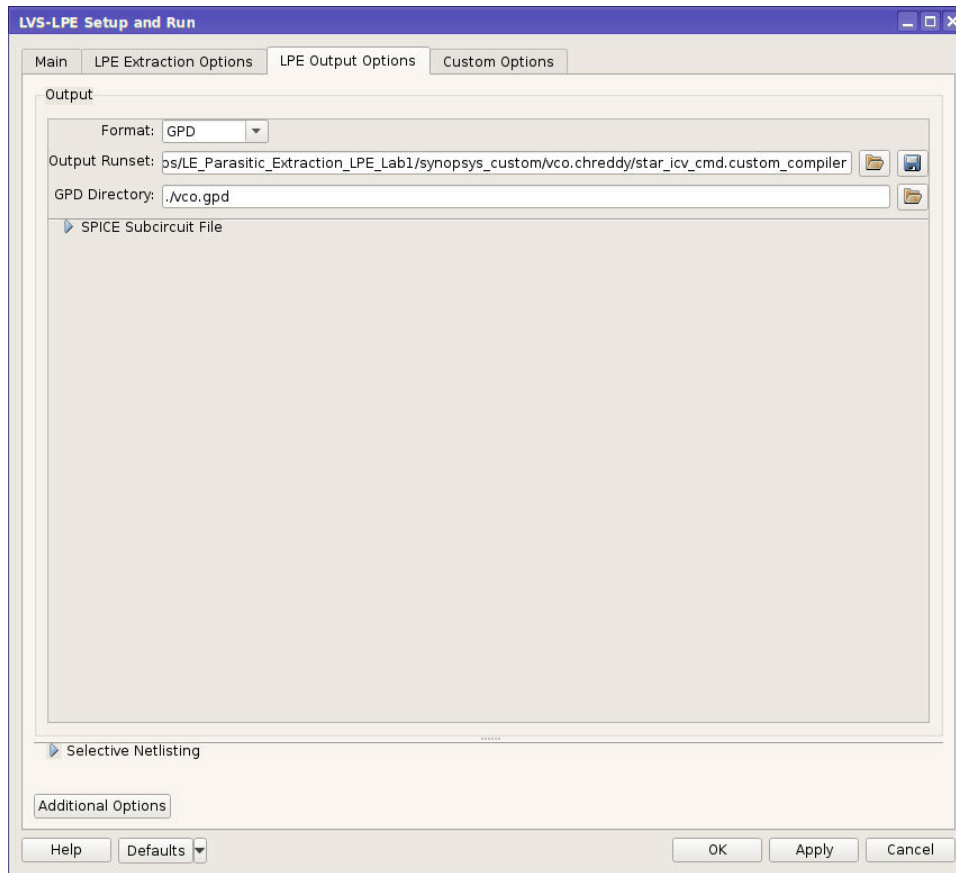


Table 44 LPE Setup and Run Settings (Output Options tab) for All LVS Tools

Settings	Description
Output	Format sets the output format to either OA, NetName, SPF, SPEF, STAR, OA & SPF, OA & SPEF, OA & STAR, or GPD. For only the OpenAccess format, additionally accept or browse to choose the View Name, Device Map, Layer Map, and Marker Size. For the other format options, you can also choose Netlist File, Use SPICE Subcircuit. The Skip-Cells Subcircuit setting creates .SUBCKT definitions for all of the cells in the Skip Cells setting. This setting is used primarily in the black-box flow. The setting of the Netlist File corresponds to the StarRC <code>NETLIST_FORMAT</code> command. The Netlist File field automatically populates on changes to the run directory, unless you manually modify the name in the field. The default name is <code>{RunDir}/{fileName}.{extension}</code>, where extension is the netlist format and fileName is set, in tokenized format, using the preference <code>xt{JobPrefId}OutputNetlistTemplate</code>. The preference <code>xtLPEDefaultOutputNetlistTemplate</code> can be used for the default LPE job. <pre>db::setPrefValue xtLPEDefaultOutputNetlistTemplate -value {%l.%c} -scope [de::getActiveContext]</pre> Possible tokens are %l for library, %c for cell, %v for view name, %u for user name, %j for the job name, %t for the tool. The GPD format controls to output a GPD (Galaxy Parasitic Database) which is the StarRC internal binary parasitic data. After generating a GPD, you can query a GPD using the following commands: <code>rc::readparasitics</code>, <code>rc::removeAnnotatedParasitics</code>, <code>rc::getGPDorners</code>, <code>rc::getCouplingCapacitors</code>, <code>rc::getGroundCapacitors</code>, <code>rc::getGPDLayers</code>, <code>rc::getNets</code>, <code>rc::getPins</code>, <code>rc::getPorts</code>, <code>rc::getCells</code>, <code>rc::readResistors</code>, <code>rc::reportGPDPProperties</code>, <code>rc::setGPDConfig</code>, <code>rc::reportGPDConfig</code>, and <code>rc::resetGPDConfig</code>. Output Runset specifies the output file of the new runset (command file) written from the Custom Compiler tool. The default file name is <code><name_of_the_runset_in_the_Main_tab>.custom_compiler</code> and the file is located in the LPE run directory. Specifying a relative path name gives the path relative to the directory in which this Custom Compiler session was started. To save current settings from this dialog box to a runset for later batch processing, click .

Settings	Description
	(Available only if you defined the corners in the <code>pdk.xml</code> file or if you set the <code>xtLPEStarCornerDef</code> Tcl preference. See Before You Start. You also must select the Multi Corner mode in the Setting LPE Extraction Options tab.) SMC Output specifies whether to generate the following for each selected corner: ◦ Multiple Views: Generates one view for each corner and corresponding operating temperature list you specified in the Setting LPE Extraction Options tab. ◦ Single Formula View: Generates a single view (OA view or SPF netlist) that contains resistance and capacitance values presented as formulas where corner names are variables. This view is limited to transistor-level extractions. Lib Name (for OpenAccess format) sets the library name. Cell Name (for OpenAccess format) sets the cell name. View Name (for OpenAccess format) sets the output view name (starrc by default). If you selected a corner in the Extraction Options tab, the options will include the <code>cornerName</code> options. Device Map (for OpenAccess format) sets the file to use for mapping device masters in StarRC to a symbol library in OpenAccess. Before you start, make sure you have the required files for physical verification and extraction. See Preparing for Physical Verification and Extraction. Layer Map maps StarRC layers to OpenAccess layers. This is used by StarRC netlister/OA writer to create the OA parasitic view and, in so doing, map the layers from StarRC database to OpenAccess PDK layer definitions for the Custom Compiler tool. Before you start, make sure you have the required files for physical verification and extraction. See Preparing for Physical Verification and Extraction. Marker Size (for OpenAccess format) sets the size of markers shown in the output parasitic view.

Settings	Description
	When the output format is set to SPF, SPEF, OA&SPF, or OA&SPEF, the tool enables Reduce Netlist for a specified list of Net Names, Level, Max Delay Error, and Frequency settings. The wildcards, * and ! are supported. The default is * (all nets). Case sensitivity of net names is determined by the Case Sensitive setting. The following optional keywords are supported: ◦ LEVEL (default = HIGH) ◦ MAX_DELAY_ERROR (default = 1e-13s) ◦ MAX_DELAY_ERROR_FREQUENCY See Running the Standalone Netlist Reducer for more information on the Netlist Reducer dialog. Note: The SPEF and SPF output formats are available for the Netlist Reducer. Case Sensitive specifies whether net names in the <code>REDUCTION_NETS</code> command are case-sensitive. Default is NO. Reduced Netlist File represents the reduced generated file destination. The default is <i>netlist_Reducer</i>.
Net Selection	Selects specified nets for parasitic netlisting, and is only applicable when the <code>XREF</code> command is not set to NO. The table of nets has columns for Net Names, Type, and Couple. In the Net Names net selection area, choose Click to Add to add nets by name. To delete a row in the net selection area, select it and click . The Net Names setting corresponds to the StarRC <code>NETLIST_SELECT_NETS</code> command, which determines which of the nets StarRC extracted should appear in the output netlist/parasitic view. Type corresponds to the StarRC <code>NETLIST_TYPE</code> command, which determines the type of parasitics to report on a net that is specified in <code>NETLIST_SELECT_NETS</code>: R (resistance), Cc (capacitance coupled to other nets), Cg (capacitance to ground only), RCg (both R and Cg), RCc (both R and Cc). Couple can be optionally set to specify whether coupling capacitance should be netlisted for other nets specified in this command (see the <code>no_couple</code> option of the <code>NETLIST_TYPE</code> command in the StarRC manual for details). The <code>NET_TYPE</code> label shows the type of net names. ◦ <code>NET_TYPE</code> is LAYOUT when Cross Ref Names is No. You must provide a net name from the LAYOUT netlist for Selective Netlisting. ◦ <code>NET_TYPE</code> is SCHEMATIC, when Cross Ref Names is Yes or Complete. You must provide a net name from the SCHEMATIC netlist for Selective Netlisting.

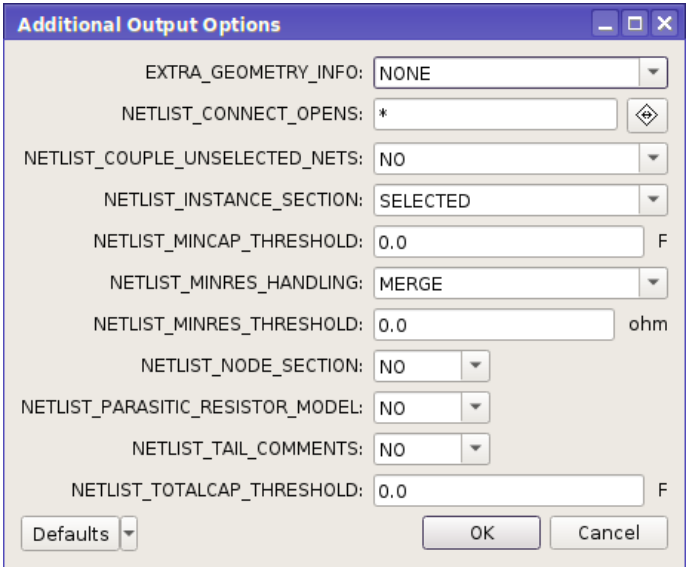
Settings	Description
Additional Options	Specifies additional LPE Output Options in a separate dialog box. See Table 45. 

Table 45 Additional Output Options

Option	Description
EXTRA_GEOMETRY_INFO	Reports the internal node bounding box information for resistors. The valid value is one of these enumerations: {RES NODE RES NODE NODE RES NONE}. The default is NONE.
NETLIST_CONNECT_OPENS	Specifies the net names to be connected by a shorting resistor if the net is found to be open during extraction. The default is all nets (*).
NETLIST_COUPLE_UNSELECTED_NETS	Specifies whether to include the coupling capacitances of unselected nets in the netlist. The valid value is one of these enumerations: {IDEAL COMPLETE NO}. The default is NO.
NETLIST_INSTANCE_SECTION	Specifies whether the instance section is included in the output netlist. The valid value is one of these enumerations: {YES NO SELECTED}. The default is SELECTED.

Option	Description
NETLIST_MINCAP_THRESHOLD	Sets the minimum capacitance allowed in the RC section of the netlist. The valid value is a floating-point number which cannot be less than 0. The default is 0.0.
NETLIST_MINRES_HANDLING	Specifies how a resistor is handled if it is less than or equal to the specified threshold. The valid value is one of these enumerations: {SHORT MERGE}. The default is MERGE. Ignore exporting this command if it is the default value.
NETLIST_MINRES_THRESHOLD	Specifies a threshold value. StarRC merges or shorts all resistances in the netlist less than or equal to the specified threshold. The valid value is a floating-point number which cannot be less than 0. The default is 0.0.
NETLIST_NODE_SELECTION	Generates the * S or *N statements in the output netlist. The valid value is one of these enumerations: { YES NO }. The default is NO. This option is available if output format is <i>SPF</i> or <i>SPEF</i> .
NETLIST_PARASITIC_RESISTOR_MODEL	Specifies whether to write parasitic resistor model names into the parasitic netlist. The default is NO.
NETLIST_TAIL_COMMENTS	Specifies whether to write geometric information about parasitic resistors as comments in the netlist. The default is NO.
NETLIST_TOTALCAP_THRESHOLD	Specifies the capacitance threshold that determines whether a net is reported in the netlist. The valid value is a floating-point number which cannot be less than 0. The default is 0.0.

6. Continue by [Setting LPE Custom Options](#).

If this LPE run requires no additional setting changes, you can alternatively continue by [Running or Rerunning Parasitic Extraction](#).

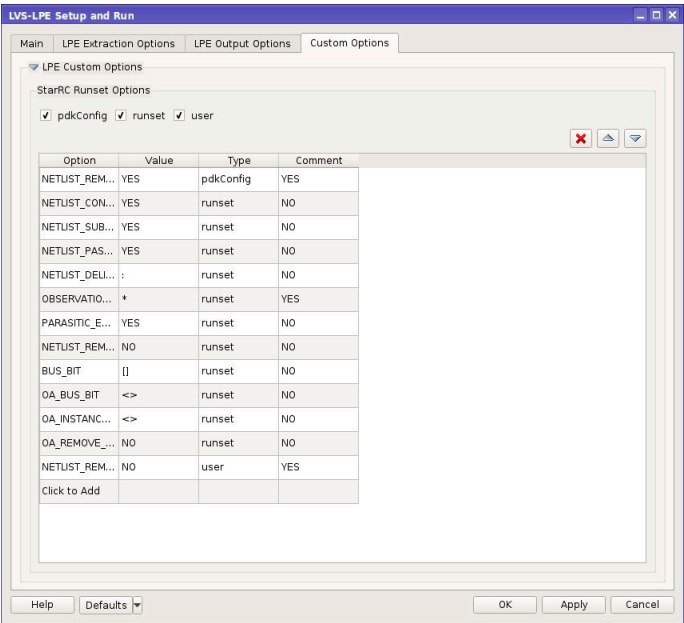
Setting LPE Custom Options

Set custom parasitic extraction options here for all LVS tool results. StarRC runset options that are independent of those in the LPE dialogs are loaded in the LPE Custom Options tab.

To set custom options for parasitic extraction:

1. Begin by specifying settings for the previous tabs in the **LPE Setup and Run** dialog box.
 - [Setting LPE Main Options](#)
 - [Setting LPE Extraction Options](#)
 - [Setting LPE Output Options](#)
2. On the **Custom Options** tab, specify settings accepted by StarRC.

Figure 32 LPE Setup and Run: Custom Options Tab

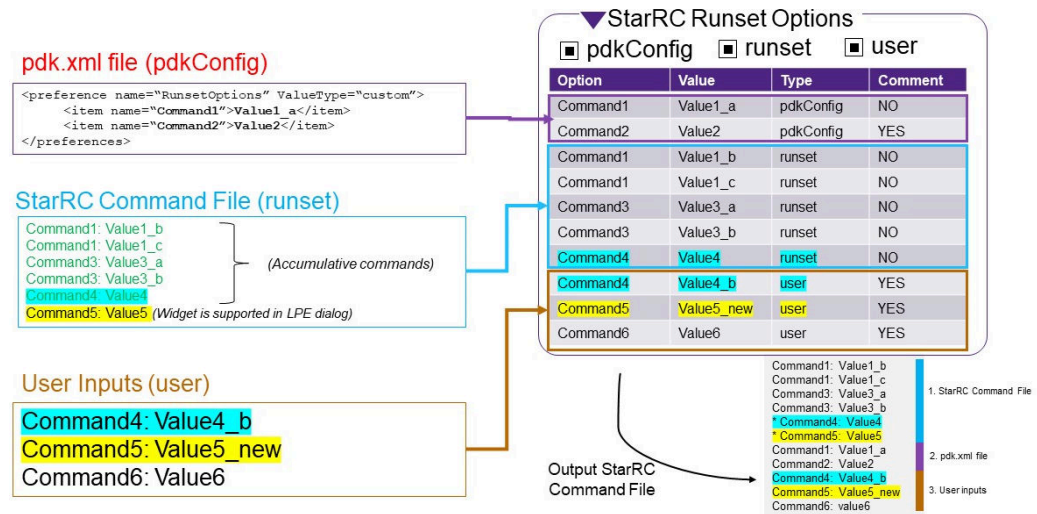


Use StarRC syntax. Entries are not validated by the Custom Compiler environment.

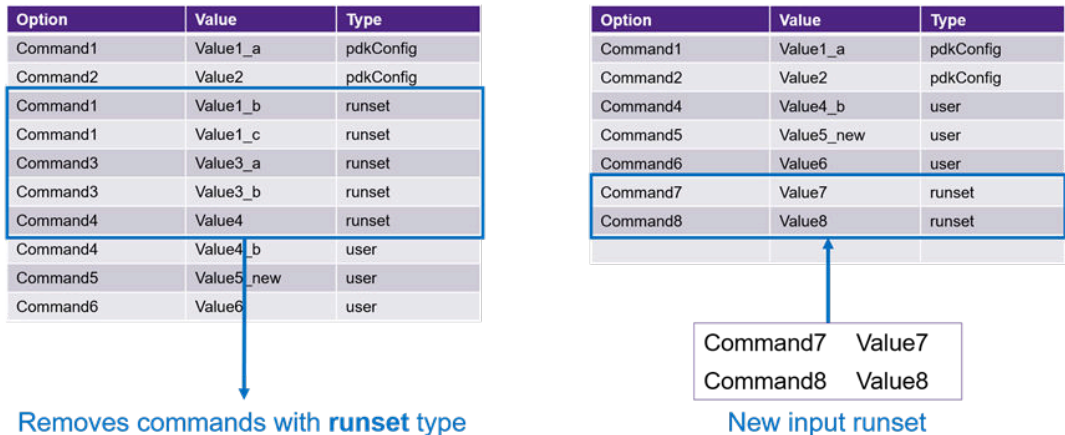
- a. Double-click the **Option** field.
- b. Enter a StarRC option name.
- c. Double-click the **Value** field.
- d. Enter a value.
- e. **Type** is derived from the origin of StarRC commands that are loaded in the **LPE Setup and Run** dialog box. You cannot edit the Type of StarRC command from the Custom Options table.

There are three command types:

- **pdkConfig**: Specified in the `pdk.xml` file. The default in the **Comment** column is **YES**.
- **runset**: Specified in the StarRC input runset. The default in the **Comment** column is **NO**.
- **user**: Manually loaded by the user. The default in the **Comment** column is **YES**.



3. Cycle through the entries using the arrow buttons.
4. Delete an entry by clicking the  button.
5. If you have specified settings on all four tabs in the **LPE Setup and Run** dialog box for a new LPE run, or if this run requires no additional setup changes, continue by **Running or Rerunning Parasitic Extraction**.
6. When you reload the new input runset, all of the commands with the **runset** type in the **Custom Options** table are removed and new commands are loaded, as shown in the following figure:



7. While exporting a command, Custom Compiler follows the order of **runset** > **pdkConfig** > **user**. If you comment the existing command using the **Comment** field, the following conditions are checked:
- If commands exist in multiple types:
Use the comment value of the type with the higher priority. If the comment value is yes, comment out all of the commands from the input runset that have the same command name.
 - If commands exist in only one type:
If **Type** is **runset** and the comment value is yes, comment out commands from the input runset that have the same command name and value.

Setting Up Partial Layout Extraction

Partial layout extraction (PLE) generates parasitic results with the following features:

- Does not require LVS to be clean
- Uses signoff tools
 - IC Validator for LVS (device extraction)
 - StarRC for parasitic extraction
- Shares the same signoff LVS and LPE setup
- Extracts
 - RC for routed nets
 - Stress/Layout Dependent Effect (LDE) parameters

For PLE to extract the correct results, the layout must be generated by Custom Compiler SDL. PLE uses the LVS and LPE flow first and then postprocesses the extracted SPF from LPE.

This section covers setting all options for partial layout extraction.

Before you start, make sure you have the required files for physical verification and extraction. See [Preparing for Physical Verification and Extraction](#).

In addition, set and configure the following options:

- [Setting PLE Main Options](#)
- [Setting PLE Options](#)
- [Configuring PLE](#)

Setting PLE Main Options

Set main options by entering information about the run directory and job parameters.

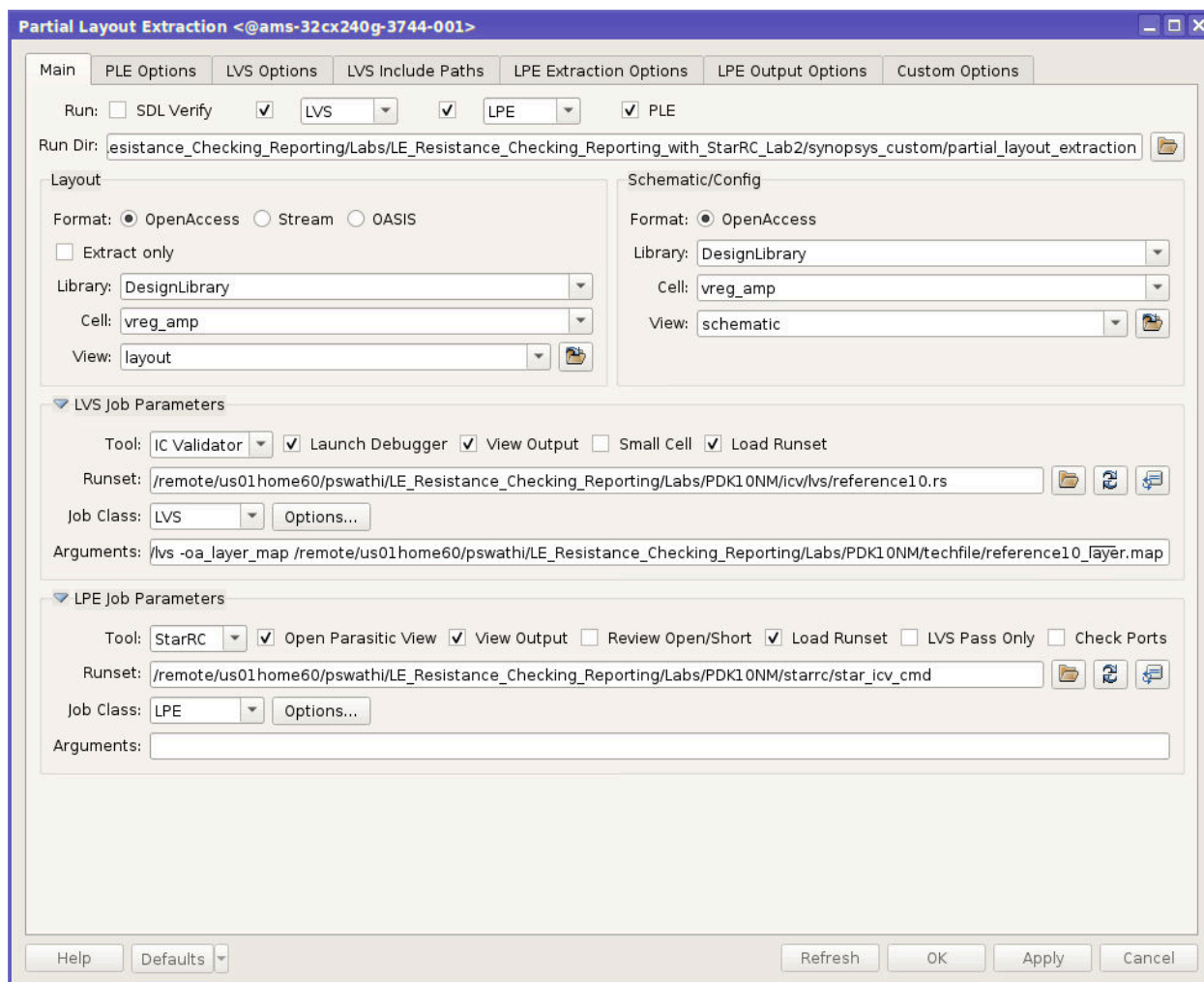
Note:

See [Default Interfaces](#) and [Custom Jobs](#) for more information.

To set the main partial layout extraction options:

1. In the design layout, choose **Verification > Partial Layout Extraction > Setup and Run** (Figure 33).

Figure 33 PLE Setup and Run Dialog Box



2. On the **Main** tab, enter settings as described in .

Table 46 PLE Setup and Run Settings (Main Tab)

Settings	Description
Run	Runs SDL Verify , LVS , LPE , PLE . Note: SDL Verify in PLE mode can now be run from Verification > Partial Layout Extraction > SDL Verify , and it uses the setup from the SDL Verify section of the Partial Layout Extraction dialog box.
Run Dir	You can accept the current run directory or browse to choose a different directory.
Layout	You can accept the current Library , Cell , and View or browse to choose a different layout.
Schematic	You can accept the current Library , Cell , and View or browse to choose a different schematic.
LVS Job Parameters	
Tool	IC Validator.
View Output	Opens a Text Viewer window with the process log. On by default.
Small Cell	(IC Validator only) Overrides automatic small cell determination and skips runset optimizations for distributed processing during runset compilation that could dominate the overall time for small cells. If the input design is too large, the IC Validator tool exits with an error.
Launch Debugger	Launches the IC Validator VUE tool when the process completes, if violations are found.
Load Runset	Loads user-supplied values into the setup dialog. On by default.
Runset	The path to the runset file with rules governing the run. The Reload  button reloads the runset. The Open Runset File  button opens the runset.
Arguments	Accepts command-line options for this tool, if any.
Job Class	Combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see <i>Adding Job Classes With xt::createJobClass Tcl</i> in the <i>Custom Compiler Environment User Guide</i> .
Options	The job settings to control batch processing limits, and to optionally set a queue engine in the Job Management Options dialog box. For details, see in the <i>Custom Compiler Environment User Guide</i> .

Settings	Description
LPE Job Parameters	
Tool	StarRC
View Output	Opens a Text Viewer window with the process log. On by default.
Open Parasitic View	Opens the parasitic view on completion. On by default.
Review Open/Short	Reviews the open and short violations from the GPD database when the StarRC job is complete. Off by default. Use the <code>xtLPEStarReviewOpenShort</code> preference to set this mode.
Load Runset	Loads user-supplied values into the setup dialog. On by default.
Load Rundir	Determines if the <code>prefs.xml</code> file is automatically loaded. If set to false, the <code>prefs.xml</code> file is not loaded. The default is true. Use the <code>xtLPEAutoPrefs</code> preference to set this mode.
LVS Pass Only	(disabled)
Check Ports	Specifies whether to check ports between the CDL and HSPICE netlists before the preceding LPE job. Setting LPE Main Options for more information.
Runset	The path to the Star command file that contains the StarRC commands to configure the parasitic extraction run. The tool supports to parse the Tcl expression in the StarRC command file. The Reload  button reloads the runset. The Open Runset File  button opens the runset.
Job Class	Combines settings for individual analysis tools. Users can configure existing job classes or user-defined job classes. For details, see <i>Adding Job Classes With xt::createJobClass Tcl</i> in the <i>Custom Compiler Environment User Guide</i> .
Options	The job settings to control batch processing limits, and to optionally set a queue engine in the Job Management Options dialog box. For details, see <i>in the Custom Compiler Environment User Guide</i> .
Arguments	The list of options for the extraction job that will go to the StarRC command line.

Settings	Description
Refresh	Checks if the design name is same as Library,Cell, View (LCV). If the design name is not the same as LCV, the LCV in the dialog box is updated to the current design name.  Show me an example of how to refresh the LCV.

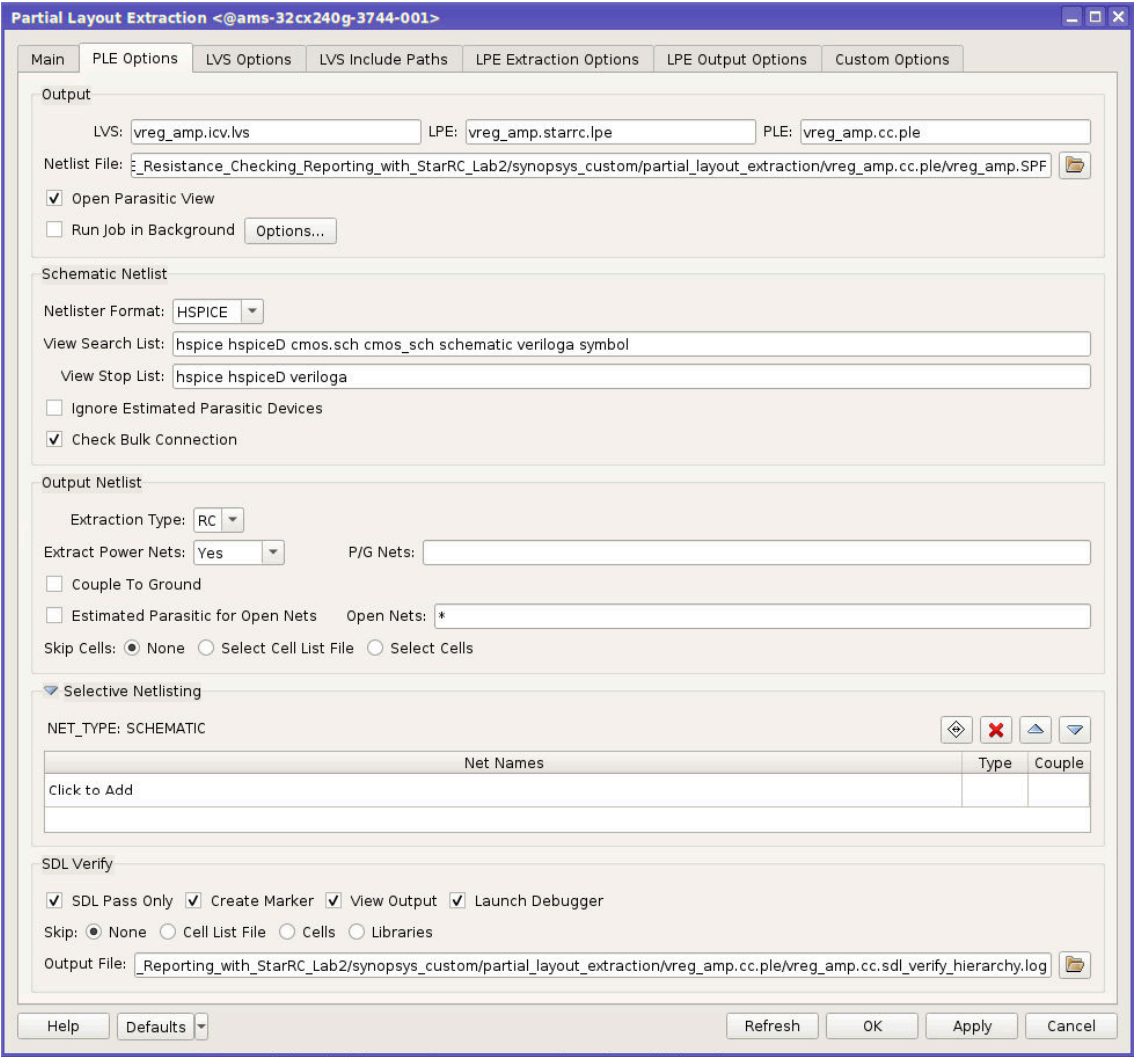
3. To complete the option setting process for a new PLE run, continue by [Setting PLE Options](#).

Setting PLE Options

To set the PLE options:

1. Begin by [Setting PLE Main Options](#).
2. In the **PLE Options** tab, the run directories for the LVS, LPE, and PLE jobs are set by default.

Figure 34 PLE Options Tab



3. On the **PLE Options** tab, enter settings as described in [Table 47](#).

Table 47 PLE Setup and Run Settings (PLE Options Tab)

Settings	Description
Output	
LVS, LPE, and PLE	Specifies the directory name of the selected net. The directories are located in the main run directory, as specified in the Main tab.

Settings	Description
Netlist File	Specifies the post-processed extracted SPF file, which is placed in the PLE run directory by default.
Open Parasitic View	Opens the post-processed extracted SPF file.
Run Job in Background	Runs the extraction in the background.
Options	The job settings to control batch processing limits, and to optionally set a queue engine in the Job Management Options dialog box. For details, see Job Monitor Tool in the <i>Custom Compiler Environment User Guide</i> .
Schematic Netlist	
Netlister Format:	HSPICE only.
View Search List and View Stop List	The options to control the PLE HSPICE netlist.
Ignore Estimated Parasitic Devices	Ignores any estimated parasitic devices in the schematic. Set the <code>lxPLEAdditionalParasiticDevices</code> preference to ignore these devices.
Check Bulk Connection	Controls the checking of bulk connections during PLE postprocessing.
Output Netlist	
Extraction Type	Specifies the type of extraction PLE performs. Valid values are R , C , RC .
Extract Power Nets	Determines whether the power nets are extracted. Valid values are Yes , NO , Ronly .
P/G Nets	Specifies which nets are power and ground nets.
Couple To Ground	Determines whether parasitic coupling capacitances are lumped to ground during extraction and netlisting.
Estimated Parasitics for Open Nets	Estimates parasitics for open nets in the top-level design. The estimated nets are listed in a file in the PLE run directory.
Open Nets	Lists the estimated open nets.
Skip Cells	When specified, Skip Cells do not need to be modified by PLE. For example, if you have cells that are completed but not created by SDL, PLE can skip these cells. To specify skip cells, see: Table 63
Selective Netlisting	

Settings	Description
Net Names	Selects the nets for parasitic netlisting. - To add the net, you can either type the name of the net in the field or select the net from the schematic by clicking the  button. - To delete the net name from the table, click . - You can select a group of connected nets by using a wildcard expression.
Type	Specifies the type of extraction performed on each net. The types of extraction available are Default, R, Cg, Cc, RCg, RCc. If Default is selected, then the available extraction options are Extraction Type and Couple To Ground of Output Netlist.
Couple	Specifies the coupling capacitances that are included in the selected net. To enable the Couple option, select R, Cg, or RCg in the Type option. If the Couple option is disabled, then the coupling capacitances are not retained from the Cc and RCc extraction types.

The **Selective Netlisting** option takes precedence over the **Output Netlist** option, as shown in the following examples:

- If the selected nets are in conflict with the nets specified in **P/G Nets**, the **Selective Netlisting** option extracts the nets.
- If the extraction specified in the **Type** option is different from the extraction specified in the **Extraction Type** option, then the type of extraction specified in the **Type** option is run.
- If the coupling capacitance specified in **Couple** is different from the coupling capacitance specified in **Couple To Ground**, then the **Couple** option is run

SDL Verify

SDL Pass Only	When specified, the LVS, LPE, or PLE run proceeds only when SDL Verify is clean.
Create Marker	When specified, exports violations as OA markers.
View Output	When specified, displays output in the Text Viewer.
Launch Debugger	When specified, opens the Error Viewer Assistant.
Skip	Controls whether to skip the verify target. See: Specifying the Verify Options
Output File	Displays the path of the output file.

Settings	Description
Refresh	Checks if the design name is same as the Library, Cell, and View fields. Note: If the design name is not the same as the Library, Cell, and View fields, the fields in the dialog box are updated to the current design name.  Show me an example of how to refresh the Library, Cell, View.
Defaults	Creates and loads the default settings in the dialog box.

Note:

- To use the `lx::generateEdtext` command and to use virtual connection, set the `lxPLEVirtualConnection` preference.
- To map the text and drawing layers, use the `lxPLETextLPPMapFile` preference. You must specify the path to the file that maps the text layer to the correct drawing layer and pin layer to create a port.

For more information, refer to: [Preferences](#)

4. To complete the option setting process for a new PLE run, continue by [Configuring PLE](#).

Configuring PLE

Partial layout extraction is based on the combined LVS and LPE extraction. Most functionality is supported in PLE, but there are differences that are specific to PLE. For example:

- PLE supports only the **RC Extraction Type** from the LPE Extraction Options tab. **R** and **C** are not supported.
- PLE disables the combining of parasitic coupling capacitances to ground during extraction and netlisting.

Figure 35 PLE Extraction Options Tab

Partial Layout Extraction

Main | PLE Options | LVS Options | LVS Include Paths | **LPE Extraction Options** | LPE Output Options | Custom Options

Layout Extraction

Configured LVS will be used

LVS Tool: IC Validator

Runset Report File: stomCompiler_Overview_Lab1/synopsys_custom/partial_layout_extraction/vco.icv.lvs/pex_runset_report

Corner Mode: ☒ Single Corner ☐ Multi Corner

Corner Selection

Corner: custom

TCAD GRD File: SDL/PartialLayout/STAR/.../output/PDK/starrc/reference40_1P10M_typical.nxtgrd

Mapping File:

Operating Temperature:

Extraction Type: RC

Analysis: None

Reduction: NO

Hierarchical Separator: |

Extract Power Nets: Yes

Cross Ref Names: ☒ Yes ☐ No ☐ Complete

☐ Remove Floating Nets

☒ Case Sensitive

☐ Couple To Ground

Netlist Ground Node: 0

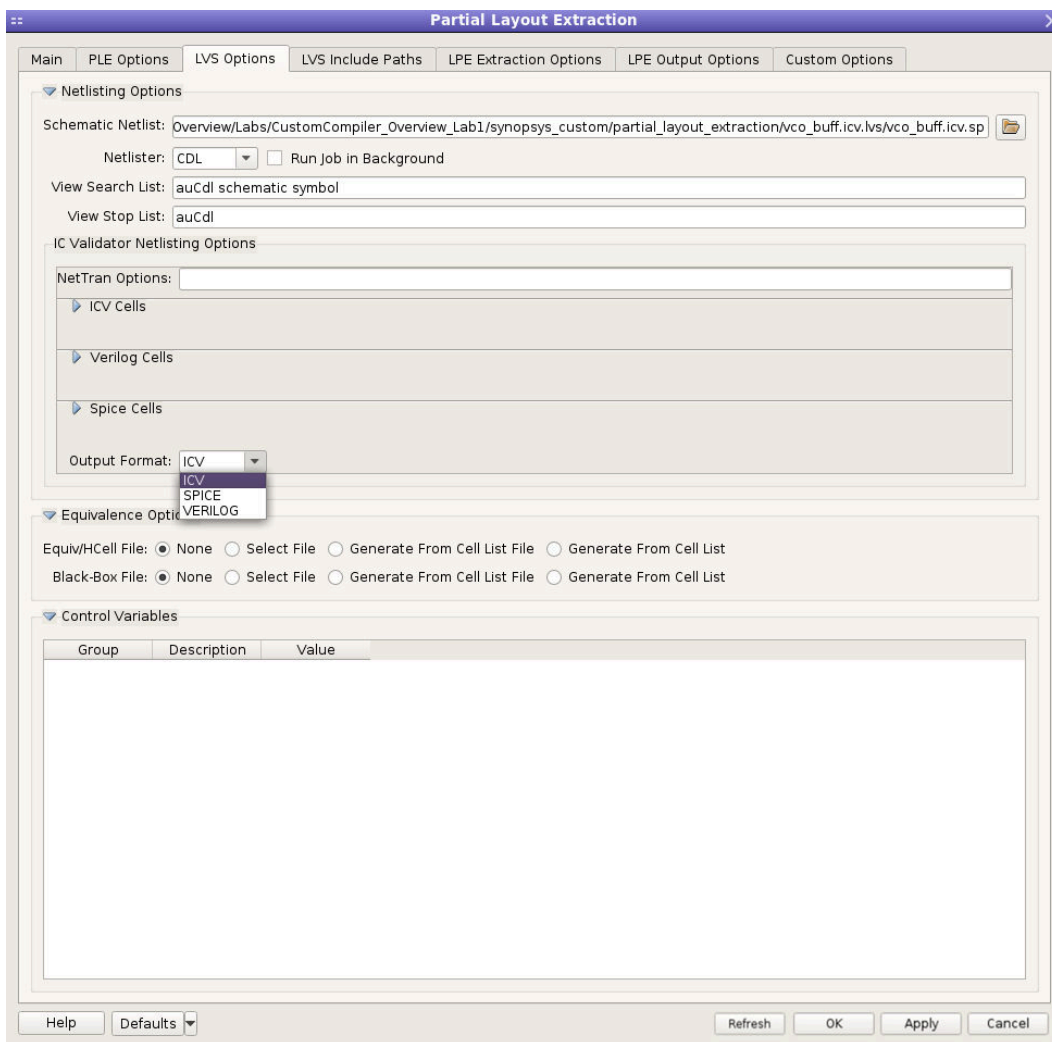
Additional Options

Help Defaults Refresh OK Apply Cancel

Note:

If you run LVS with a runset, which has **Output Format** set to **SPICE**, and the **Output Format** in the GUI is set to **ICV** (the default), as shown in [Figure 36](#), a warning message appears in the Custom Compiler console prompting you to change the **Output Format** to **SPICE**.

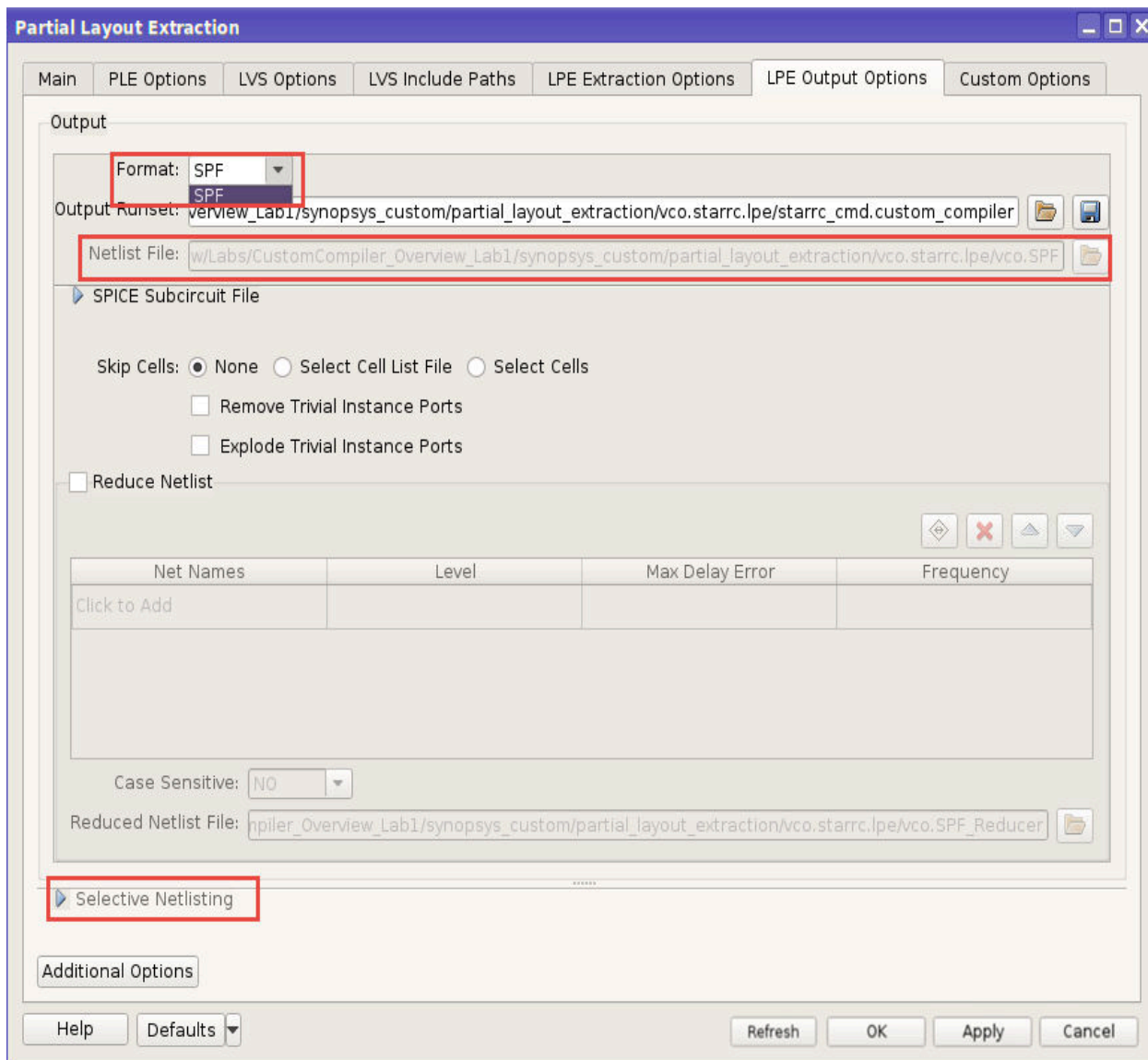
Figure 36 PLE LVS Options Tab



See [Table 13](#) for information on the **Equivalence Options** settings.

In the **LPE Output Options** tab, **Format** supports only the **SPF** mode, the output netlist file cannot be changed, and net selection is not supported.

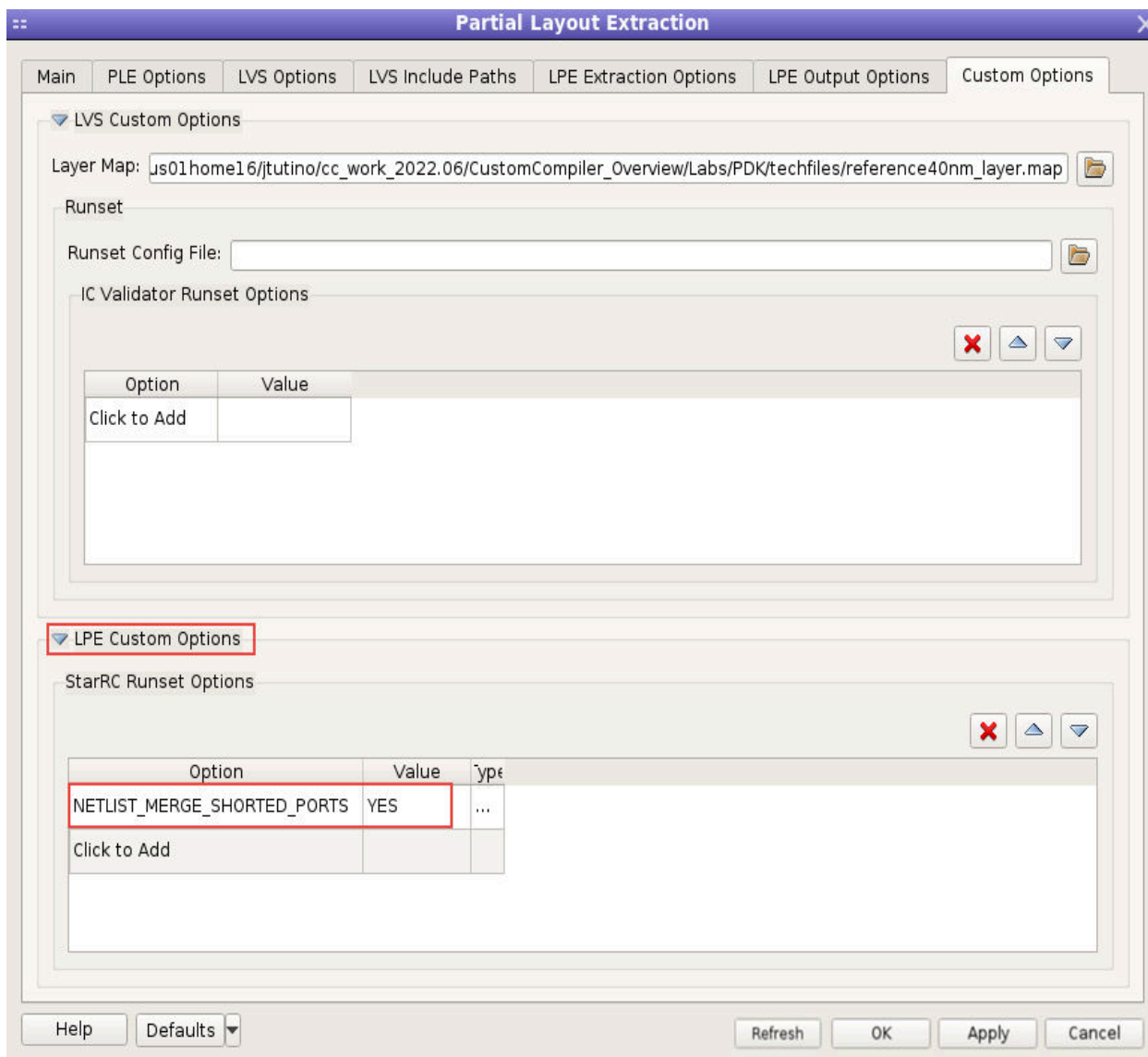
Figure 37 PLE Output Options Tab



Note:

To avoid PLE post-processing errors due to inconsistent ports, you must ensure the addition of a subcircuit file for all skip cells in the LPE run.

Figure 38 PLE Custom Options Tab



Note:

To support black-box cells for a PLE-LVS run, set the `ICV_ENABLE_PRESERVED_LVS_BLACK_BOX_CELLS` environment variable to true and the `extract_devices_in_black_box_cells` LVS option to true.

Note:

From the PLE **Custom Options** tab, if you set the `NETLIST_MERGE_SHORTED_PORTS` option to YES in the **LPE Custom Options** section, it is forced to NO and you will see a warning message in the console.

Running or Rerunning Parasitic Extraction

Run parasitic extraction after entering setup options.

Rerun parasitic extraction to use the current setup options.

To run or rerun parasitic extraction:

1. In the design layout window, perform the task related to the situation.
 - To run parasitic extraction for the first time, choose **Verification > Extract > Setup and Run**, specify settings (see [Setting Options for Layout Parasitic Extraction](#)), and then click **OK**.
 - To rerun the same extraction process again, you have two alternatives: either choose **Verification > Extract > Run** in the Layout Editor window or click the **Run StarRC LPE** button on the Layout Verification standard toolbar. By default, this toolbar is vertically aligned along the lower-left window edge.

If any settings are missing, the system shows blank entry fields with red borders. Enter settings and click **OK** again.

2. Continue by [Monitoring Parasitic Extraction Run Progress](#).

When you are ready, complete this flow by [Analyzing Extracted Parasitic Views](#).

 [Show me](#) how to use config views to perform post-layout simulation.

Using Calibre Interactive to Run Parasitic Extraction

You can also launch Calibre Interactive to run Calibre's parasitic extraction.

Note:

Consult the Calibre documentation for more details on configuring your environment to run Calibre if needed.

1. Make sure your environment is properly configured to run the Calibre Interactive tool:

Note: If any library in the `lib.defs` file cannot be accessed by Calibre, an error occurs.

To set up the environment:

- a. Set the `MGC_HOME` environment variable to point to the Calibre installation path.
- b. Use the environment variables `MGC_CALIBRE_DRC_RUNSET_FILE`, `MGC_CALIBRE_LVS_RUNSET_FILE`, and `MGC_CALIBRE_PEX_RUNSET_FILE` to tell Calibre Interactive to use a default runset template file, which seeds the values

in the dialogs for iCalibre DRC, LVS, and PEX. The Lib/Cell/View information is always seeded from the opened layout, regardless of the runset template. The template itself supports Calibre Interactive's token substitution, for example, %l=layoutCellName, %s = schCellName, %L=layout library name, and so forth.

2. In the Layout Editor, choose **Tools > Calibre** to display the Calibre menu.
3. Choose **Calibre > PEX** to run Calibre's parasitic extraction. The following figures show the various Calibre PEX set-up dialogs. Detailed documentation on the dialog settings and usage are in the *Calibre Interactive User Guide*.

Figure 39 Calibre Interactive - PEX: Rules

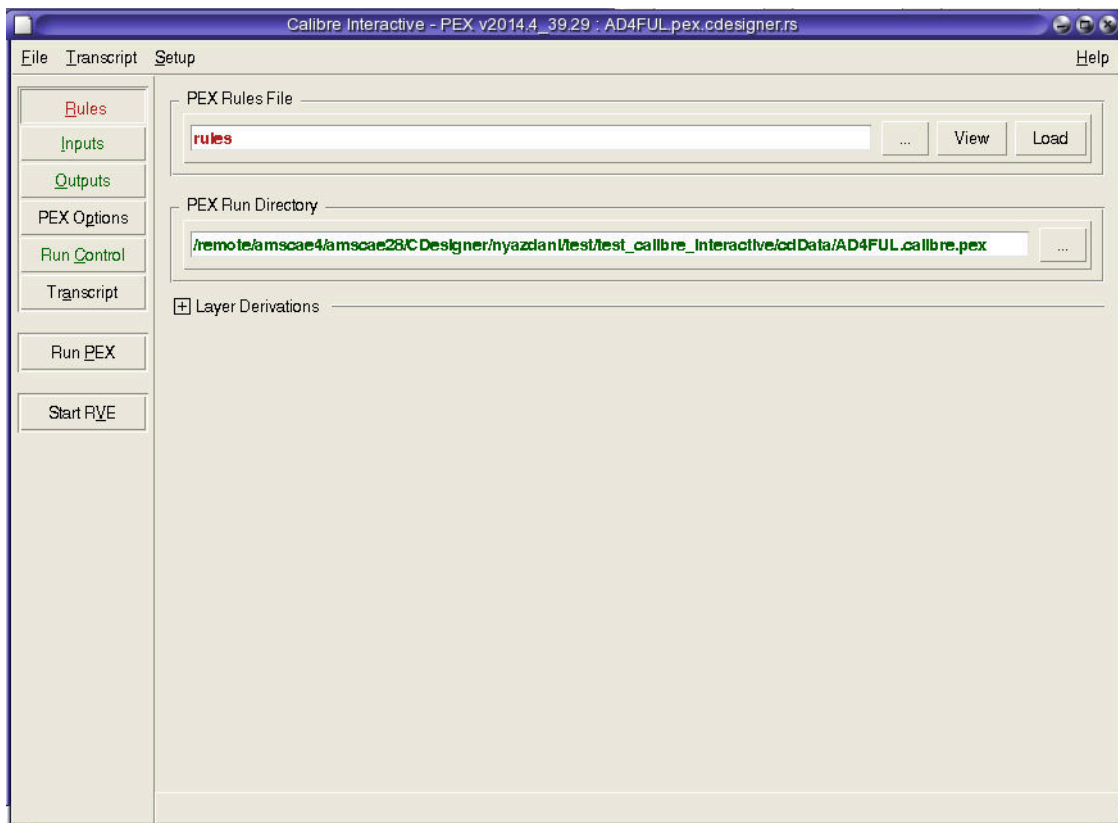


Figure 40 Calibre Interactive - PEX: Inputs

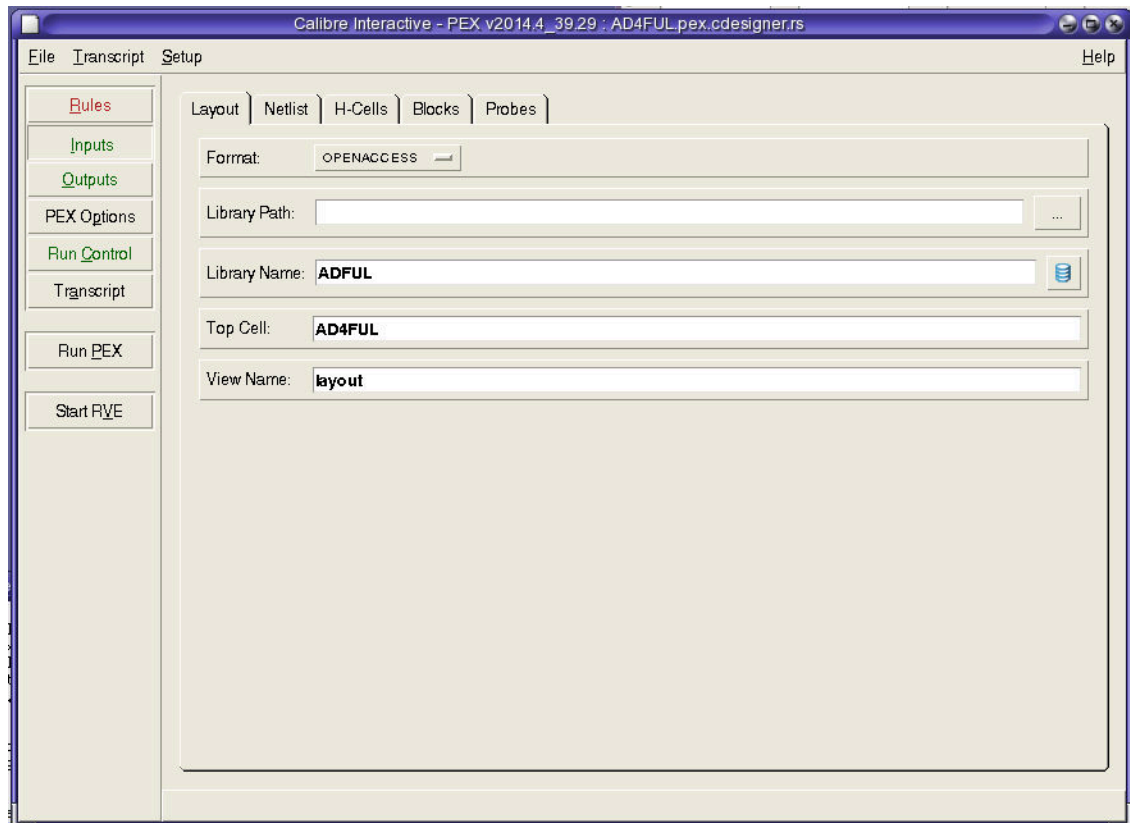
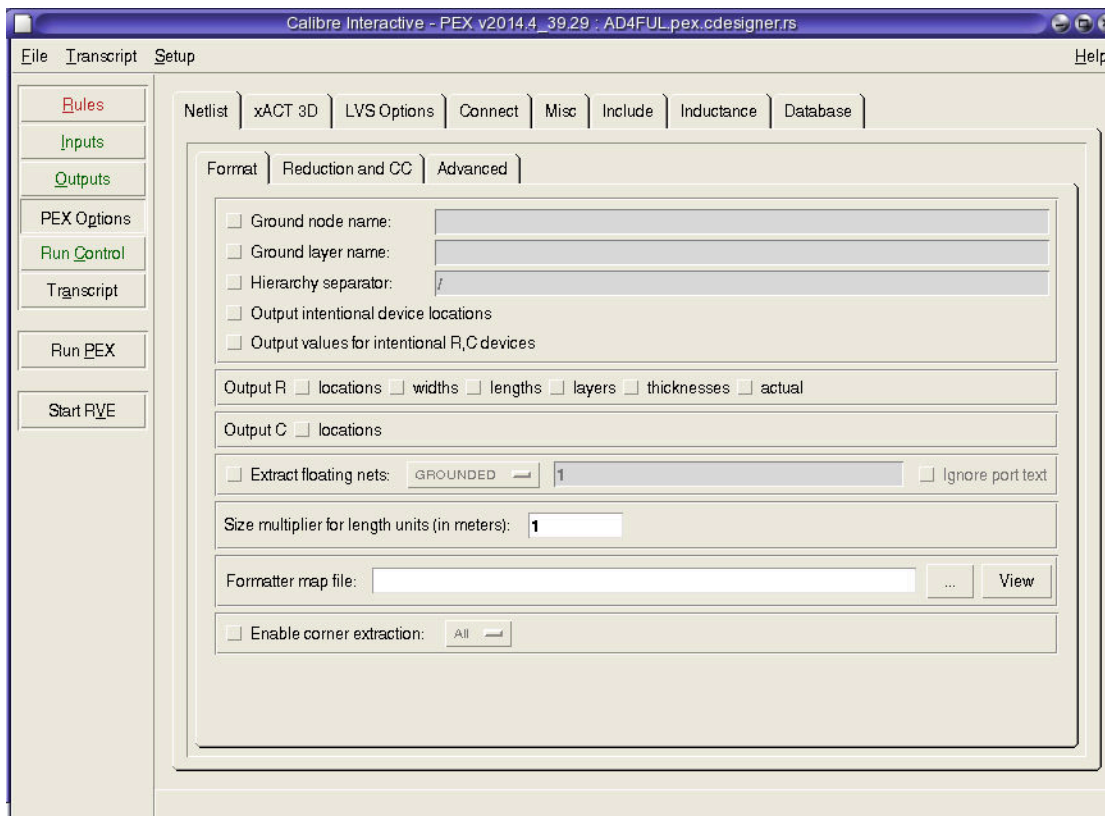


Figure 41 Calibre Interactive - PEX: PEX Options



Monitoring Parasitic Extraction Run Progress

View parasitic extraction output to track progress and see logged operations.

To view and monitor the files showing the progress and status of the StarRC process as its running:

1. To open the Job Monitor **View Output** window before the run begins, enable the **View Output** option under **Job Parameters** on the **LPE Setup and Run** dialog's **Main** tab.

Or, to open the Job Monitor **View Output** window while the run is in progress, choose **Tools > Job Monitor** from the console. Then in the Job Monitor window, right-click on the job of interest and select **View Output**.

Alternatively, you can use the keyboard shortcut Shift+Ctrl+D.

2. In the Text Viewer, view results and logged operations on each tab.
3. Complete this flow by [Analyzing Extracted Parasitic Views](#).

Analyzing Extracted Parasitic Views

Analyze the OA parasitic view to understand and evaluate the effects of parasitics on the design, in the context of the schematic and layout representations of the design.

This section covers Custom Compiler parasitic view analysis, which includes:

- [Initializing or Reinitializing Parasitic Source Views](#)
- [Viewing or Hiding Capacitance Annotations](#)
- [Reporting Crosstalk and Capacitance](#)
- [Setting Parasitic Query Options](#)
- [Querying Parasitic Views](#)
- [Querying All Nets in a Parasitic View](#)
- [Performing Parasitic Cross-Probing](#)
- [Clearing Cross Probes](#)
- [Running the Standalone Netlist Reducer](#)

Initializing or Reinitializing Parasitic Source Views

When an OA parasitic view is going to be analyzed in the context of its schematic representation, the **Initialize Parasitics** command associates the parasitic view to a schematic view. This is required, for example, to annotate parasitic elements from the parasitic view onto a schematic view, or to query the parasitics from a net in the schematic.

Initialization enables querying and cross-probing from the parasitic view to the schematic.

You can reinitialize source views to change the association to a new parasitic view.

To initialize a parasitic view with a schematic:

1. Begin by opening either the schematic or parasitic view.
2. Choose **Parasitics > Initialize** or **Parasitics > Reinitialize**.
(Choosing **Tools > Parasitics** displays the **Parasitics** menu.)
3. Enter settings as described in [Table 48](#).

Table 48 *Initialize Parasitics Settings*

Settings	Description
Extraction Tool	StarRC
Run Directory	Accept the default or browse to locate the directory to initialize for the parasitic source view.
Parasitic or Schematic View	Accept the current Library, Cell, and View or browse to select a different one. Association specifies whether to open the source view in a new window or not.

- Click **OK**.
- Individual parasitics can be queried for details about them. See Design Navigator Assistant for details on how to use the Design Navigator assistant to browse and understand design components.
- Continue by [Viewing or Hiding Capacitance Annotations](#) or [Setting Parasitic Query Options](#).

Viewing or Hiding Capacitance Annotations

View capacitance annotations (in schematics only) to visualize the net capacitance added to instance terminals. The capacitance annotations give the total net capacitance that is annotated to the schematic. This does not include coupling capacitors or resistors. This total net capacitance is essentially equivalent to what you would see in the *|NET section of the SPF netlist.

Note:

Prepare to perform this task, if necessary, by [Initializing or Reinitializing Parasitic Source Views](#).

Choose an option:

- To show annotations in the design schematic, choose **Parasitics > Annotate**.
- To hide annotations, choose **Parasitics > Hide**.

Reporting Crosstalk and Capacitance

With advanced nodes, designs with tighter spacing can pose signal integrity challenges. For example, lateral capacitance between adjacent wires can cause crosstalk.

To help you identify the sources of crosstalks, including parasitic coupling capacitance, you can use the following:

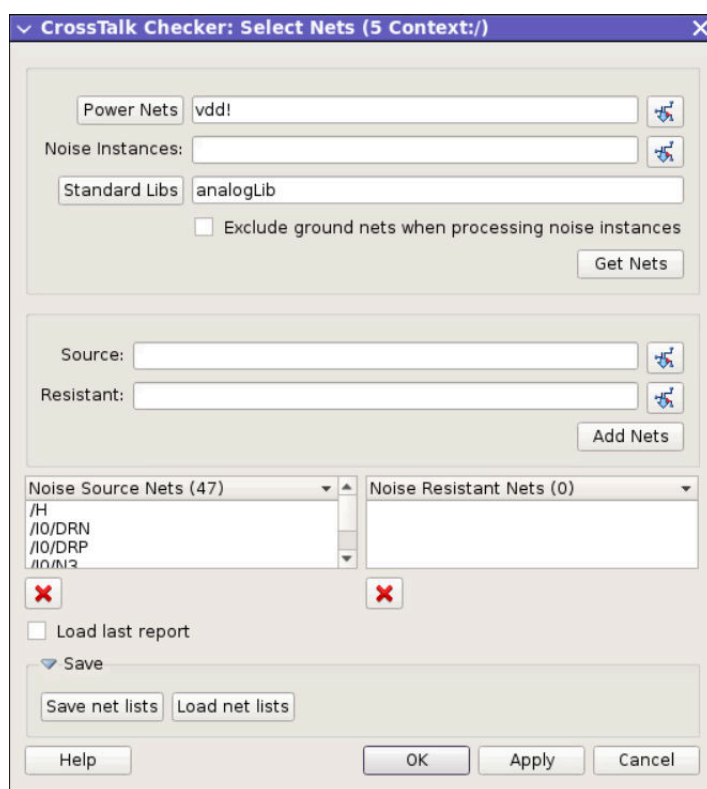
- [CrossTalk Checker](#)
- [CrossTalk Report](#)
- [Capacitance Report](#)

CrossTalk Checker

To check for crosstalk:

1. In the Layout Editor, choose **Tools > Parasitics**. The **Parasitics** menu appears on the menu bar.
2. Choose **Parasitics > CrossTalk Checker**. The **CrossTalk Checker** dialog box opens.
3. Set the **CrossTalk Checker** options.

If you do not specify any nets in the **CrossTalk Checker** dialog box, the checker queries all of the nets and opens the [Capacitance Report](#).



Options	Description
Power Nets	Displays the Select Digital Power Nets dialog box. Select the nets for which you would like to get noise probes. Click OK to finalize selection. Click  to select nets directly from the schematic.
Noise Instances	Displays and specifies the instances included power nets and ground nets for which you would like to get noise probes. Click  to select instances directly from the schematic.
Standard Libs	Displays the Select Standard Cell Libs dialog box. Select the libraries for which you would like to get noise probes. Enter a library name in Filter to narrow down the list. Click OK to finalize selection.
Exclude ground nets when processing noise instances	Check this option to exclude the ground nets from the specified noise instances for which you would like to get noise probes.
Get Nets	Retrieves the noise source nets and noise resistant nets from the schematic and displays them in the Noise Source Nets and Noise Resistant Nets lists.
Source	Click  to select nets directly from the schematic. Select nets by clicking on them in the schematic, or by clicking and dragging over an area to select all the nets within that area. Click Add Nets to add the noise probes to the Noise Source Nets list.
Resistant	Click  to select nets directly from the schematic. Select nets by clicking on them in the schematic, or by clicking and dragging over an area to select all the nets within that area. Click Add Nets to add the noise-resistant probes to the Noise Resistant Nets list.
Noise Source Nets	These nets are the noise source. To remove nets from the list, click the net and then click Delete .
Noise Resistant Nets	These nets are the noise resistant nets. To remove a net from the list, click the net and then click Delete . These nets are resistant to the noise.
Load Last Report	Check and click OK or Apply to load last report. The last report is saved automatically and this check box only appears when there is a saved report.

Options	Description
Save	Click the blue triangle to expand the Save section. Click Save net lists to save your current net lists to a file within the design. Click Load net lists to load previously saved net lists.

4. Click **OK**.

The **CrossTalk Report** dialog box opens and displays the results.

CrossTalk Report

The CrossTalk Report consists of the following:

Name	Description
Net	Type in a net name after filter to filter that net from the list. Type in a net name after exclude to exclude that net.
Pair	Type in a pair name after filter to filter that pair from the list. Type in a pair name after exclude to exclude that pair.
Cap (F)	Type in a minimum value after Min and a maximum value after Max to filter the Value (F) column. Scientific notation can be used.
Update Table	Updates the table to reflect the filters.
Hide/Show/Hide Columns	Right-click any of the column titles and click Hide to hide that column. Click Show/Hide Columns to display the Show/Hide Columns dialog box. Select the data columns you want to see.
PairID	The ID specific to different net pairs.
Aggressor	The nets creating crosstalk. These noise source nets are generally the digital nets.
Victim	The nets that cannot resist being affected by the crosstalk. These nets are generally the analog nets.
Value (F)	The total capacitance between the nets in farads.
Probe	Click on an Aggressor, Victim, or Value (F) and click Probe to display the item visually through object highlighting and zooming.
Clear Probes	Clears previous probes.
Cycle Colors	Check to cycle the probe colors. Uncheck to keep the probe colors the same.
Expand Nets	Shows the nets as partitioned and named in the extracted view.

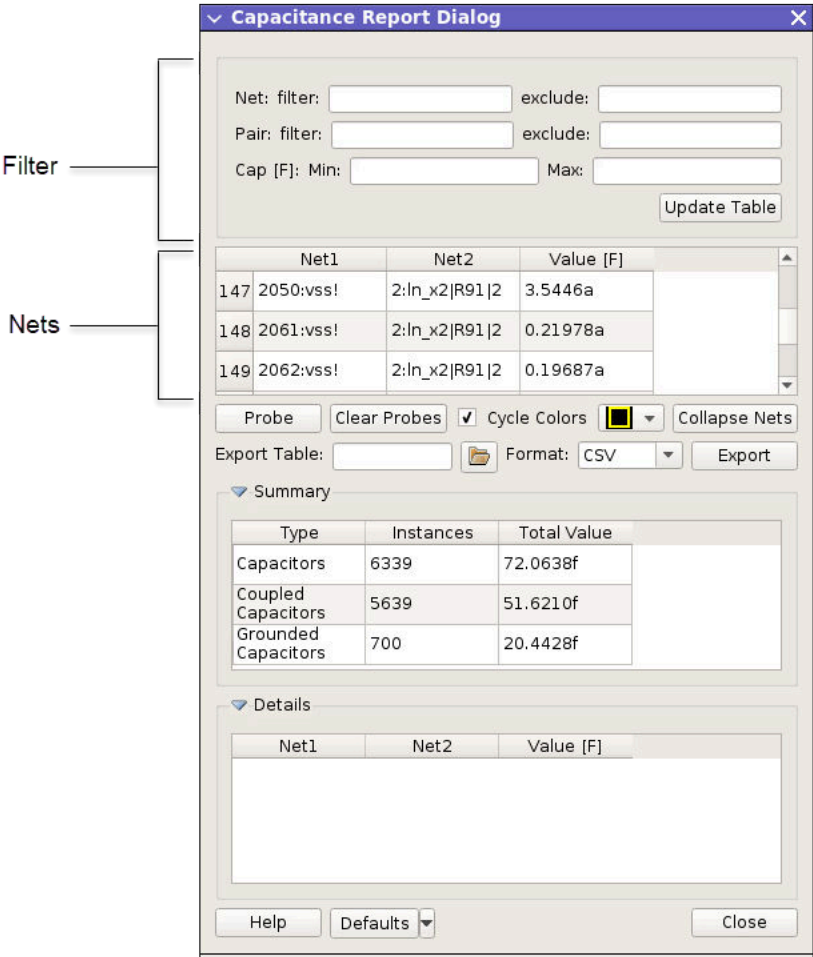
Name	Description
Export Table	Type in a path and file name for the table, or, browse to find a location and create a file name for the table. Select your format and click Export .
Summary	Click the blue triangle to expand the Summary section. This section reports the amount of Aggressor and Victim nets, and the total amount of both.
Details	Click the blue triangle to expand the Details section. Clicking on a net returns the expanded version of that net in the Details section and shows the names as reported in the extracted view.

Capacitance Report

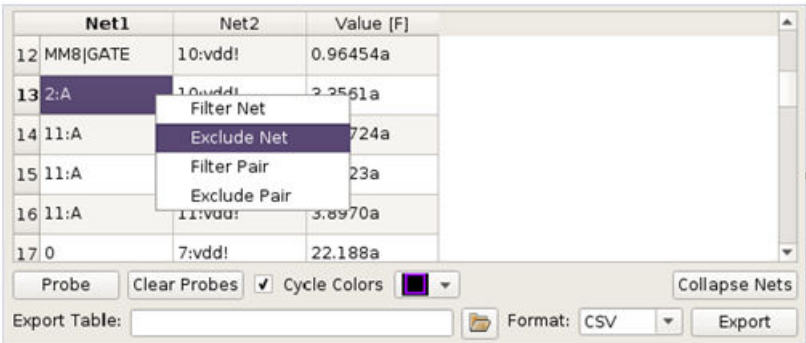
After you initialize or reinitialize the parasitic source views, you can view the Capacitance Report. The Capacitance Report lets you query nets and locate parasitic capacitors.

To view the Capacitance Report:

1. With the parasitic or schematic view open, choose **Parasitics > Parasitic Capacitance Report**. The **Capacitance Report** dialog box opens.



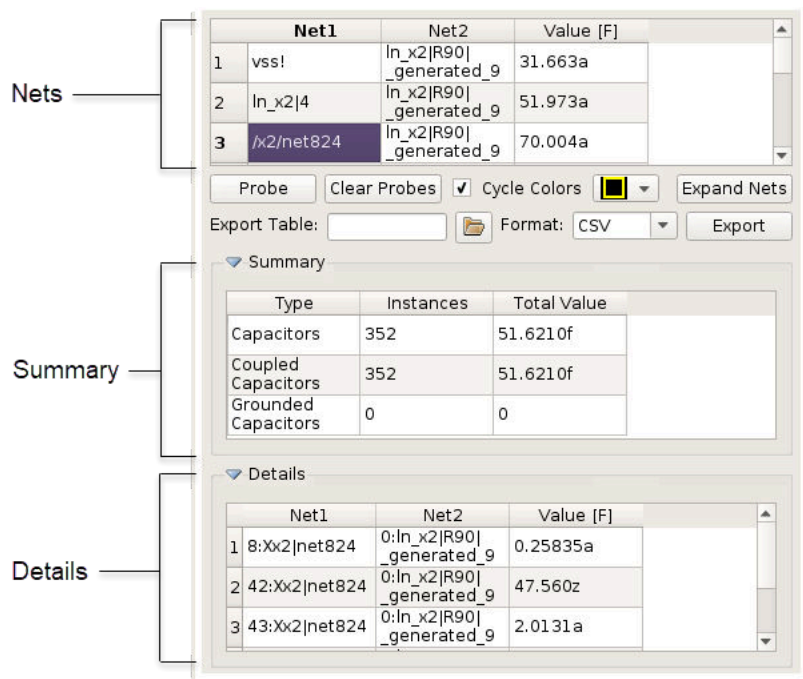
2. Specify the filters to use to customize the Capacitance Report. You can specify filters using one of the following methods:
- In the Filter section, specify the filter information.
 - In the Nets table, right-click the net name and choose a command: **Filter Net**, **Exclude Net**, **Filter Pair**, or **Exclude Pair**.



Name	Description
Net: filter Filter Net	Filter section Type the criteria to use to include specified nets in the report. Use a character space to separate consecutive net names. For example: netA netB. To exclude certain nets, type the exclusion criteria in the exclude box. You can use the wildcard character (*) to include or exclude multiple net names. Exclusion filters have priority over inclusion filters. In the following example, all netA and netB nets will be included in the report, except 3:netA and 7:netB. Net: filter: *:netA *:netB exclude: 3:netA 7:netB Nets table Right-click a net name and choose Filter Net. The net name appears in the Net:filter box. Right-click a net name and choose Exclude Net. The net name appears in the exclude box.

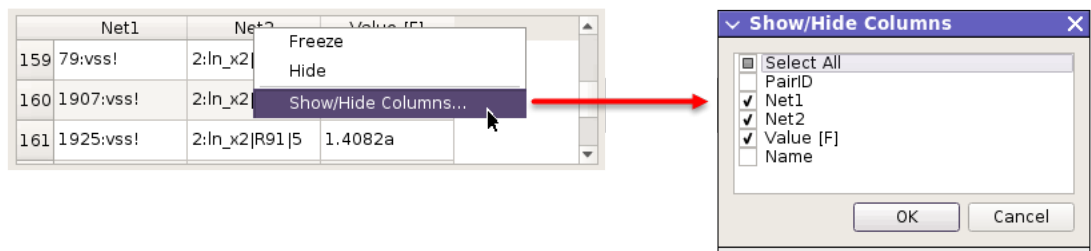
Name	Description
Pair: filter Filter Pair	Filter section Type the criteria to use to include pairs of nets in the report. To exclude certain pairs, type the exclusion criteria in the exclude box. Use a comma (,) to separate the names of the two net pairs. For example: netA netB, netY netZ You can use the wildcard character (*) to include or exclude multiple net pairs. Exclusion filters have priority over inclusion filters. In the following example, all netA and netB nets will be included in the report, except 3:netA and 7:netB. All netY nets and netZ nets will be included in the report, except 1:netY and 9:netZ. Pair: filter: *:netA *:netB, *:netY *:netZ exclude: 3:netA 7:netB, 1:netY 9:netZ Nets table Right-click a net name and choose Include Pair. The net names appears in the Pair:filter box. Right-click a net name and choose Exclude Pair. The net names appear in the exclude box.
Cap (F)	To specify a capacitance range, type a minimum (Min) and a maximum (Max) value. Only the nets that have capacitance within the specified range will be displayed in the report in the Value (F) column. You can use scientific notations, for example 2e-3. Note that the net name exclusion filters have priority over the Cap (F) filter.

3. Click **Update Table**. The filters you specified are applied. The Nets table updates. The **Summary** and **Details** tables refresh and display corresponding data.



Report Table	Description
Nets	Lists the nets based on the filters you specified. If you did not provide a filter, all nets are displayed. If the list of nets is long, click Collapse Nets to consolidate and group nets that have the same base net name. (Select the net name to view all the nets with the same base name in the Details section.) Click Expand Nets to list all nets in the Nets table.
Summary	Provides a summary of the report. • Type: Type of parasitic device: Capacitors (or total capacitors), Coupled Capacitors, and Grounded Capacitors. • Instances: Number of instances that contain the parasitic device type. Number of capacitors (total capacitors) = number of coupled capacitors + number of grounded capacitors. The instance counts change depending on the filters you specified. • Total Value: Total capacitance of the particular parasitic device. Value of capacitors (total capacitors) = value of coupled capacitors + value of grounded capacitors.
Details	Displays all the nets with the same pairID based on the net you selected in the Nets table. If you selected a net that does not have a pairID, nothing is displayed in the Details table.

4. To show or hide columns in the **Summary** or the **Details** table, right-click a column title and choose one of the following:
- **Hide**: Hides the selected column.
 - **Show/Hide Columns**: Opens the **Show/Hide Columns** dialog box so that you can select multiple columns to show or hide.



Column	Description
PairID	Identifies a group of nets that have the same base net name.
Net1/2	Names of the nets for which parasitic capacitance is evaluated.
Value	Capacitance value.
Name	Name of the device instance in OA. The same name appears in the Design Navigator Insts tab.

5. To freeze a column so that it remains visible while you scroll horizontally to other columns, right-click the table column and choose **Freeze**. If you freeze a column other than the first column, the frozen column will become the first column.
6. To examine a net or net pairs on the layout canvas, in the Nets table, select the net and click **Probe**. The selected net is highlighted on the layout canvas. The command also pans and zooms in so that the net is displayed within the viewport.

To use different colors when highlighting different nets, select **Cycle Colors**, and choose the starting color. Clear the check box to use the same color for all highlighting.

To remove the highlighting on the layout canvas, click **Clear Probe**.

7. To export the report:

- a. Click the **Export Table** browser button () and specify the target location for the exported file.
- b. Choose the **Format** of the export file.
- c. Click **Export**.

8. When you are finished, click **OK** to close the **Capacitance Report** dialog box.

Setting Parasitic Query Options

Set query options to determine the behavior of net query operations. These options apply to the parasitic and schematic views that are currently active (initialized) and to any subsequent queries.

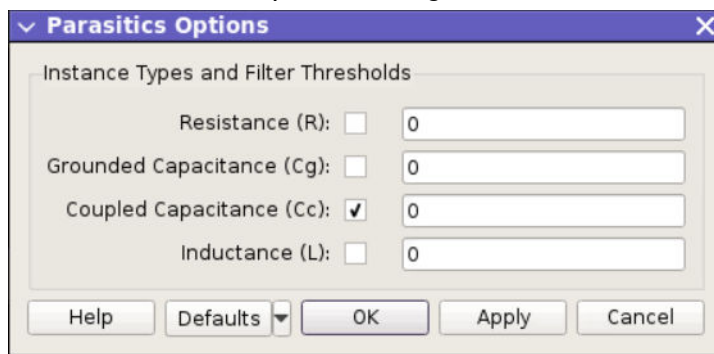
(Prepare to perform this task, if necessary, by [Initializing or Reinitializing Parasitic Source Views](#).)

If you intend to perform net queries directly from the Custom Compiler parasitic view, it is not necessary to initialize parasitics first; simply open these query options from the **Parasitics** menu in the Layout Editor window of the parasitic view and set the options. They will be applied to the next query from that context.

To set parasitic query options:

1. In the design schematic or parasitic view, choose **Parasitics > Options**.

Figure 42 Parasitics Options Dialog Box



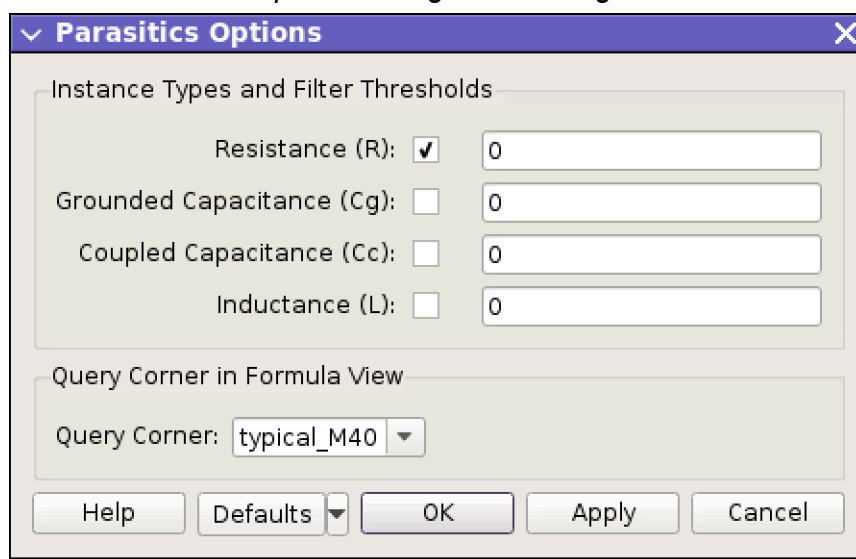
2. Choose the parasitic element types that should be included in subsequent net queries.
3. Set filter thresholds for **Parasitic Resistance**, **Grounded Capacitance**, **Coupled Capacitance**, and **Inductance**.

4. (Single formula view only) Specify the corner in the **Query Corner** selection field for querying the parasitics in the single formula extracted view from StarRC.

The **Query Corner in Formula View** section is available when enabling the dialog box from the single formula parasitic view or from the schematic view with the single formula parasitic view. See [Figure 43](#).

Parasitic query is recalculated the capacitance or resistance with the specified corner and then the parasitics with the calculated value is reported to [Capacitance Report](#) box and [CrossTalk Report](#) dialog box.

Figure 43 Parasitics Options Dialog Box for Single Formula View



The dialog box is titled "Parasitics Options" with a close button (X) in the top right corner. It contains two main sections. The first section, "Instance Types and Filter Thresholds", has a background of light gray and contains four rows of controls: "Resistance (R):" with a checked checkbox and a text box containing "0"; "Grounded Capacitance (Cg):" with an unchecked checkbox and a text box containing "0"; "Coupled Capacitance (Cc):" with an unchecked checkbox and a text box containing "0"; and "Inductance (L):" with an unchecked checkbox and a text box containing "0". The second section, "Query Corner in Formula View", also has a light gray background and contains a "Query Corner:" label followed by a dropdown menu showing "typical_M40". At the bottom of the dialog are five buttons: "Help", "Defaults" (with a small dropdown arrow), "OK", "Apply", and "Cancel".

5. Click **OK**.
6. Continue by [Querying Parasitic Views](#) or [Querying All Nets in a Parasitic View](#).

Querying Parasitic Views

Queries can find all parasitics on a net, all coupled capacitances between two nets, parasitic resistance between two points (instance terminals or subnode markers) on a net, and all parasitics on all nets.

This process generates reports in a tabular form, where columns can be sorted, probes can be added for visualization, and an overall summary can be seen at the bottom of the dialog. After initialization, you can initiate queries from the schematic or parasitic view.

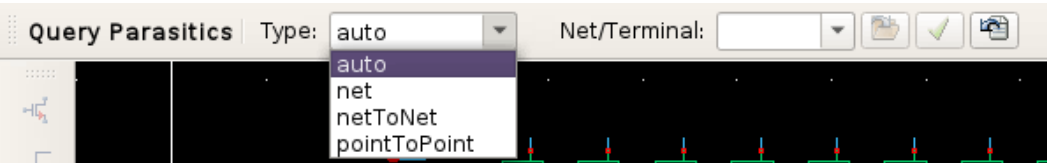
Note:

Although this procedure is regularly performed using a parasitic view, you can also perform it from the schematic view. If done from the schematic, you will need to start by [Initializing or Reinitializing Parasitic Source Views](#).

 [Show me](#) an example of how to query parasitics.

To query an extracted parasitic view:

1. In the design parasitic view, choose **Parasitics > Query**



2. In the command option toolbar, specify query options.

[Table 49](#) lists and describes query options.

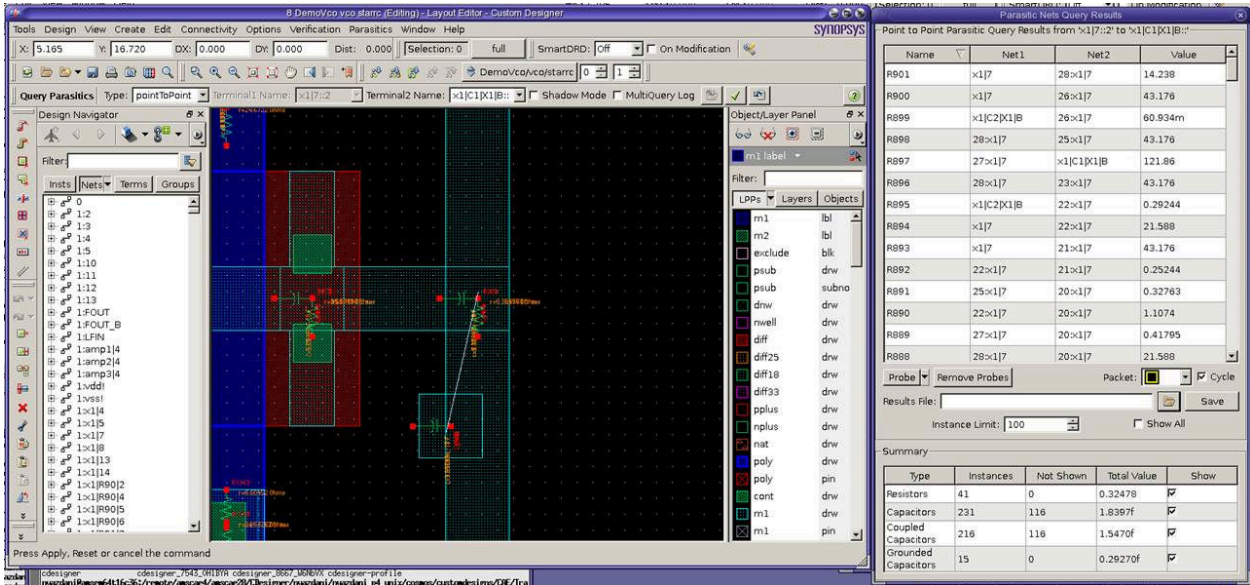
Table 49 Query Parasitics Options

Query Type	Accepts	Description
Auto		Selects the appropriate query for the following cases: ◦ If you select one net then the name of the net is displayed in the Net/Terminal field. ◦ If you select two nets then the names of the nets are displayed in Net/Terminal field and net2Name field. ◦ If you select two terminals then the names of the terminals are displayed in Net/Terminal field and terminal2Name field. For overlapping objects, select the required object using the drop down of Net/Terminal field. Click Apply to run the query. It is set as the default.
net	A net name	Generates a report listing all parasitic elements for this net when you select a shape in the extracted view.
netToNet	Two net names	Generates a report listing all parasitics between these two nets, including coupling capacitance, when you select multiple shapes that are on different nets. Note that your extraction options should have been run with consideration for coupling capacitance.

Query Type	Accepts	Description
pointToPoint	Two terminal names	Creates a listing of parasitic elements between the two physical points (where the points represent subnodes of the same net separated by parasitic resistors) and computes the total resistance between the two points/subnodes. For instructions, see the step that follows.

3. (Optional) For point-to-point query only, follow these steps:
- a. Set the type of query to **pointToPoint**.
 - b. On the canvas, click a coordinate point in proximity to the location of the first instance terminal you want to use. Zooming to obtain device identification visually is not required.
- You can use the Design Navigator Assistant to browse and locate subnodes. The system finds all terminals and subnodes in proximity to the coordinate point and attaches a flight line to your cursor (Figure 44).

Figure 44 Point to Point Parasitic Query



- c. In the **Terminal1 Name** list box, choose the exact terminal or subnode marker from the list.
- Use the available shortcut keys 1 and 2 to scroll between the overlapping active objects in the layout canvas to help you click the object of interest when multiple objects are overlapping.

- d. Set the second terminal by clicking in different canvas locations until the system displays a name in the **Terminal2 Name** field, indicating terminals connected to the same net.
 - e. Set the second terminal by choosing from the list box.
4. Run the query by clicking the green check mark (circled in red in [Figure 44](#)).
 5. In the **Point to Point Parasitic Query Results** report, double-click a parasitic element name.

The system highlights a zoomed view of the parasitic element in the extracted view using the color packet visible in the query report dialog box.

The report is a tabular list of all parasitic elements between the points. You can click the column headers to sort the rows by net1 name, net2 name, or value. The report includes a summary showing resistance totals.

6. (Optional) To modify the color packet used, choose a new one from the **Packet** list box.
 7. (Optional) To cycle through the color packets for each succeeding selection in the report, click the **Cycle** option. (To use one color for all selections, deselect this option.)
 8. (Optional) To create probes for items in the report table, select the items and click the **Probe** button (or double-click the items).
- Use the **Probe** button menu to extend probe distribution to **All Selected** items, **Selected Names**, **Selected Nets**, or **All Names**.
9. (Optional) To remove probes added through this report, click **Remove Probes**.
 10. (Optional) To save these results in a file, enter a filePath in the **Results & File** field or browse to choose a file to overwrite. You can save the results one of three formats, comma separated value (.csv), XML (.xml), or as an ASCII table.

Querying All Nets in a Parasitic View

Query all nets in a parasitic view to generate a report listing parasitic elements for all the nets in the design.

To query all nets in a parasitic view:

1. Choose **Parasitics > Query All Nets**.

The report lists results and includes a summary showing totals for each type of parasitic device listed in the results table except resistors.

2. (Optional) Limit the number of instances shown in the table by choosing an **Instance Limit**, which will apply to each type of instance. Alternatively, show all results by clicking **Show All**.
3. To create probes for items in the report table, select the items and click the **Probe** button.

Use the **Probe** button menu to extend probe distribution to **All Selected** items, **Selected Names**, **Selected Nets**, or **All Names**.
4. To remove probes added through this report, click **Remove Probes**.
5. To save these results in a file, enter a filePath in the **Results File** field or browse to choose a file to overwrite.

Performing Parasitic Cross-Probing

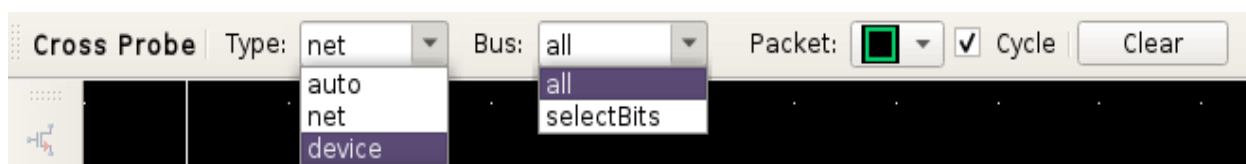
Perform cross-probing by opening the schematic and extracted views and probing, allowing the system to highlight extracted elements.

(Prepare to perform this task, if necessary, by [Initializing or Reinitializing Parasitic Source Views](#).)

 [Show me](#) an example of how to cross-probe.

To perform parasitic cross-probing:

1. In the design schematic or parasitic view, zoom to an area.
2. Select **Parasitics > Cross Probe**



3. In the command option toolbar, specify cross probe type

[Table 50](#) lists and describes Cross Probe type options.

Table 50 Cross Probe Type Options

Cross Probe Type	Description
Auto	Allows cross probe of a net or a device for the selected area. It is set as the default.
Net	Allows cross probe of a net.

Cross Probe Type	Description
Device	Allows cross probe of a device.

- To enable **Bus**, select **Auto** or **Net** in Cross Probe Type

[Table 51](#) lists and describes Bus types.

Table 51 Cross Probe Bus Options

Cross Probe Bus	Description
All	Allows cross probe of all bus bits in the selected bus net. It is set as the default.
selectBits	Allows cross probe of user-defined bus bits through the Select Bus Bits dialog box.

- To view extracted parameters for this element in the extracted view, open the Property Editor.

Clearing Cross Probes

Clear cross probes to remove them from all views.

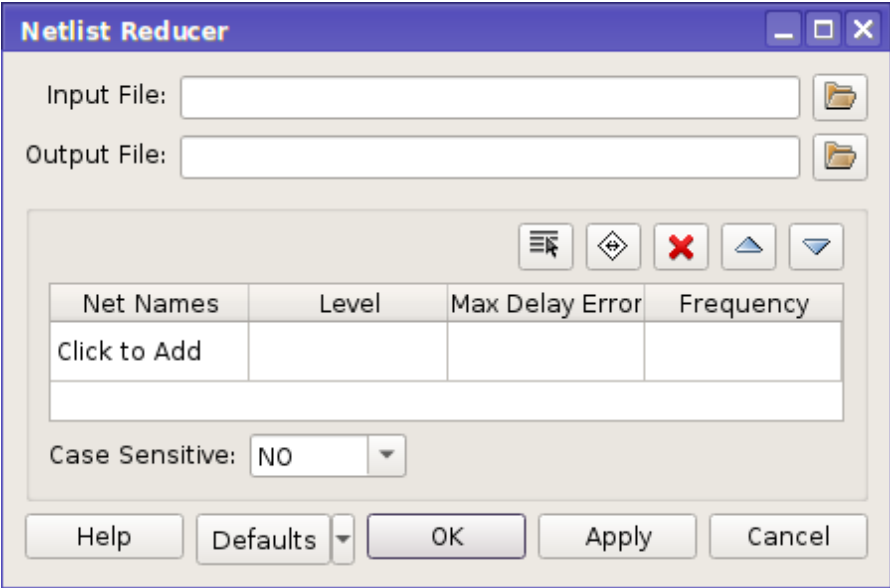
To clear cross probes:

- In the parasitic view, choose **Parasitics > Clear Cross Probes**.

Running the Standalone Netlist Reducer

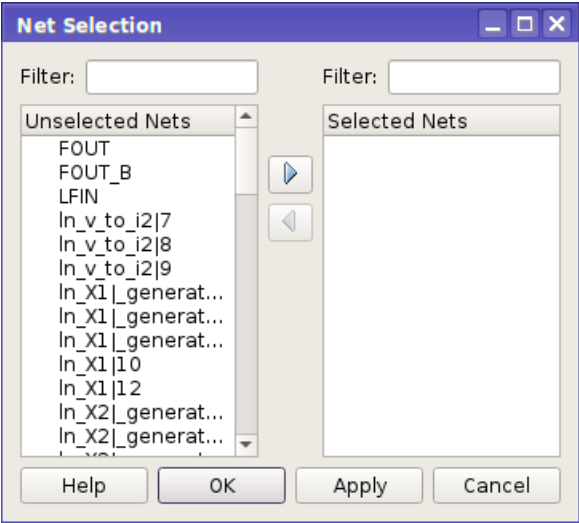
The Netlist Reducer utility reduces the existing SPEF or SPF netlist using the StarRC tool.

- In the Layout Editor, choose **Verification >Netlist Reducer**. The dialog box opens.
- Set the options in the **Netlist Reducer** dialog box.



Option	Description
Input File	Specifies the input file.
Output File	Specifies the reduced generated file destination.
Case Sensitive	Specifies whether net names in the <code>REDUCTION_NETS</code> command are Case Sensitive . The default is NO.

3. To select the required nets click  to open the **Net Selection** dialog box.



Select the nets required in the **Unselected Nets** field and click  to move into the **Selected Nets** field.

4. Click **Apply** and **OK** to add the selected nets to **Netlist Reducer** option.

See the **Reduce Netlist** options in [Table 44](#) for more information on specifying nets for reduction.

Exploring the GPD Parasitics Database

You can examine the parasitics results using the GPD database created directly from the StarRC tool. You can query the parasitic for:

- Net information
- Net-to-net capacitance or point-to-point parasitics

You can also:

- Highlight parasitics in a layout.
- Save the results to a file.

In the Layout Editor, choose **Tools > Parasitics** from the Console to display the **Parasitics** menu.

Figure 45 Parasitics Menu for the Layout Editor

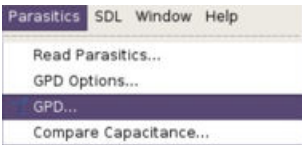


Table 52 GPD Parasitics Database Settings for the Layout Editor

Settings	Description
Read Parasitics	Specifies a path for retrieving parasitic data.
GPD Options	Sets the options for displaying information in the schematic and parasitic views.
GPD	Explores parasitic results.
Compare Capacitance	Compares capacitance value that is extracted from the layout with the value defined in the constraints for nets.

In the Schematic Editor, choose **Tools > Parasitics** from the Console to display the **Parasitics** menu.

Figure 46 Parasitics Menu for the Schematic Editor

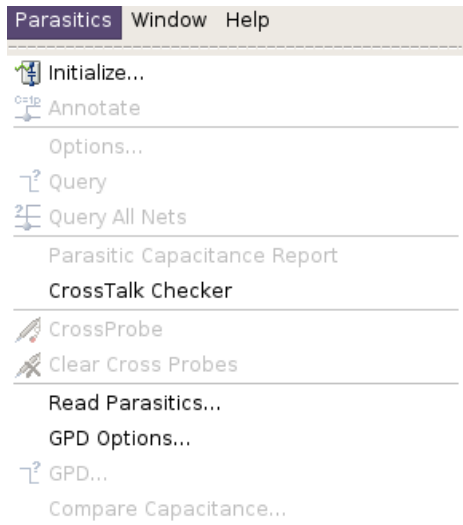


Table 53 GPD Parasitics Database Settings for the Schematic Editor

Settings	Description
Read Parasitics	Specifies a path for retrieving parasitic data.
GPD Options	Sets the options for displaying information in the schematic and parasitic views.

Read Parasitics

The **Read Parasitics** selection launches a dialog box when `xt::setGPDPATHForRead` is set.

Choose **Parasitics > Read Parasitics** to specify the path of the corresponding GPD for the current layout and layer map file to map the GPD (LVS) tech layer to an OA layer. By default, the OA layer map is disabled. After specifying the GPD path, you can query parasitics from the specified GPD.

Figure 47 Read Parasitics in the Layout View

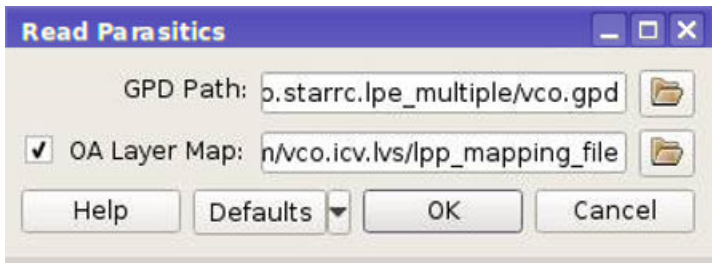
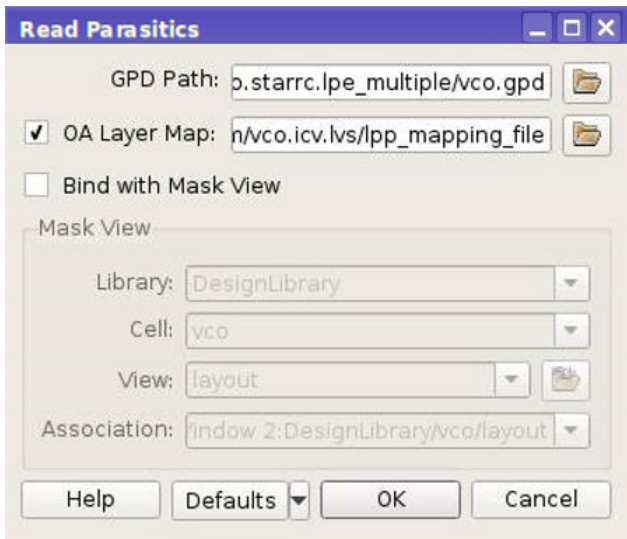


Table 54 Read Parasitics Options

Settings	Option Type	Description
GPD Path	String	Specifies the path to the GPD directory.
OA Layer Map	String	Specifies the layer map file to map the LVS (GPD) tech layer to the OA layer.

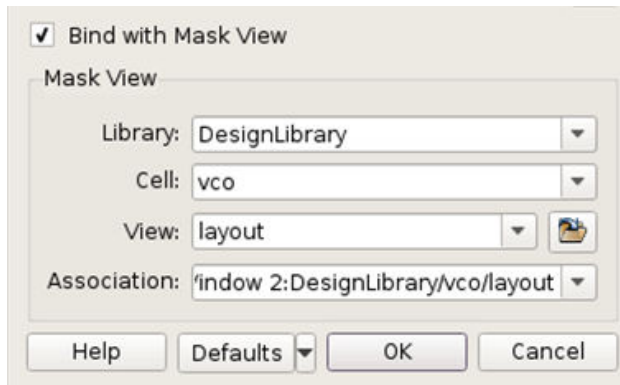
You can query the parasitic for a single maskLayout and schematic view design, or both simultaneously. When you choose **Read Parasitics** in the **Schematic** view, the parasitic will be associated with the schematic view design, as shown in [Figure 48](#).

Figure 48 Read Parasitics in the Schematic View



To query for both the maskLayout and schematic view, check the **Bind with Mask View** box to enable the **Mask View** options, as shown in [Figure 49](#).

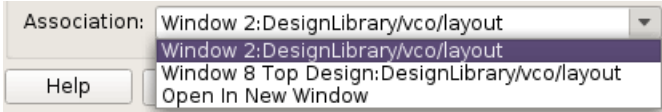
Figure 49 Bind With Mask View



From the **Mask View**, the **Library**, **Cell**, and **View** fields are automatically filled by searching the corresponding layout view type design. For example, if **Read Parasitics** is open in DesignLibrary, vco, or schematic designs, the layout design will be filled with the same information in DesignLibrary, vco, or layout designs. By default, the layout view is chosen. However, you can also choose the extracted views.

This association allows you to bind the maskLayout view design that is specified inside the **Mask View** with an open design, or open the maskLayout view design in a new window.

Table 55 GPD Options

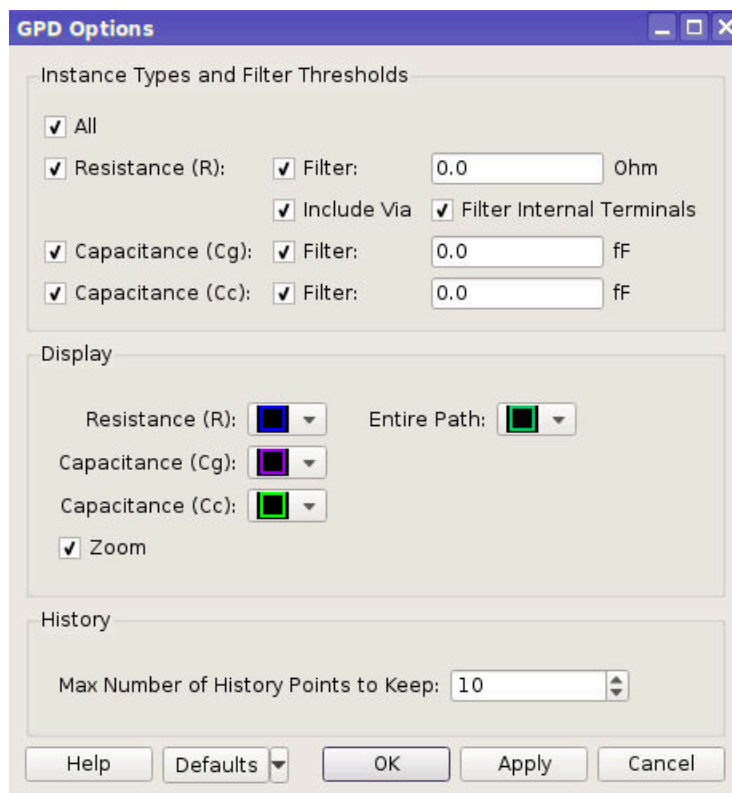
Settings	Description	Option Type
Bind with Mask View	Enabled when the Read Parasitics dialog box is open in the schematic design. Disabled when unchecked (the default).	Check
Library/Cell/View in Mask View	Allows you to choose the associated design in the GUI. By default, launching the dialog box from the schematic design sets the corresponding layout design.	
Association	If a design is open in an existing window, it will shown as the default design. If there is no open design, Open in New Window is the default.	Choice
		

GPD Options

The **GPD Options** set preferences that determine the behavior of the query commands.

Choose **Parasitics > GPD Options** to display the dialog box. You can specify parasitics (resistance, grounded capacitance, coupled capacitance) and a corresponding filtering threshold for query. You can also specify the color and stipple for highlighting parasitics. The options described in [Table 56](#) apply to the layout views currently active (initialized) and to any subsequent queries.

Figure 50 GPD Options



The GPD Options dialog box is divided into three main sections: Instance Types and Filter Thresholds, Display, and History.

Instance Types and Filter Thresholds

- ☒ All
- ☒ Resistance (R):
 - ☒ Filter: 0.0 Ohm
 - ☒ Include Via
 - ☒ Filter Internal Terminals
- ☒ Capacitance (Cg):
 - ☒ Filter: 0.0 fF
- ☒ Capacitance (Cc):
 - ☒ Filter: 0.0 fF

Display

- Resistance (R): [Blue icon]
- Entire Path: [Green icon]
- Capacitance (Cg): [Purple icon]
- Capacitance (Cc): [Green icon]
- ☒ Zoom

History

Max Number of History Points to Keep: 10

Buttons: Help, Defaults, OK, Apply, Cancel

Table 56 GPD Options

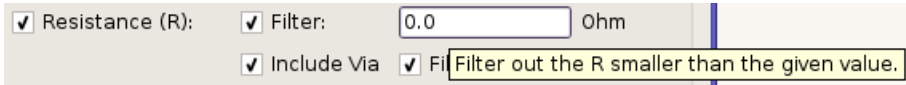
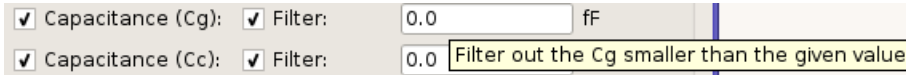
Settings	Description	Option Type
Instance Types and Filter Thresholds	ALL enables or disables queries for all parasitics (resistor, grounded capacitor, coupled capacitor, or pin capacitor). Use the <code>xtLPEGPDQueryAll</code> preference to set this type. The default is true.	Check
	Resistance (R) enables or disables query parasitics for resistors. Use the <code>xtLPEGPDQueryR</code> preference to set this type. The default is true. Tooltip: 	Check
	Filter (of Resistance) enables or disables the filter threshold to ignore those parasitic resistors with resistance smaller than the given value after the query resistance is enabled. Use the <code>xtLPEGPDFilterR</code> preference to set this type. The default is false.	Check
	Filter Threshold (of Resistance) enables or disables the filter threshold to ignore those parasitic resistors with resistance smaller than the given value after the query resistance is enabled. Use the <code>xtLPEGPDThresholdR</code> preference to set this type. The default is 0.	Float
	Include Via (of Resistance) indicates that the query of parasitic resistors must include via. Use the <code>xtLPEGPDQueryViaR</code> preference to set this type. The default is true.	Check
	Filter Internal Terminals (of Resistance) shows or hides internal terminals in a pull-down list and highlights the layout for a point-to-point query. Use the <code>xtLPEGPD2PFilterR</code> preference to set this type. The default is true.	Check
	Capacitance (Cg) enables or disables query parasitic for grounded capacitors. Use the <code>xtLPEGPDQueryCg</code> preference to set this type. The default is true. Tooltip: 	Check
	Filter (of Grounded Capacitance) enables or disables the filter threshold to ignore those grounded capacitors with capacitance smaller than the given value after the query grounded capacitor is enabled. Use the <code>xtLPEGPDFilterCg</code> preference to set this type. The default is false.	Check
	Filter Threshold (of Grounded Capacitance) defines the filter threshold to ignore grounded capacitors whose capacitance is smaller than the given value. Use the <code>xtLPEGPDThresholdCg</code> preference to set this type. The default is 0.	Float

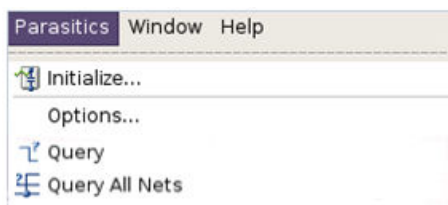
Table 56 GPD Options (Continued)

Settings	Description	Option Type
	Capacitance (Cc) enables or disables query parasitic for coupled capacitors. Use the <code>xtLPEGPDQueryCc</code> preference to set this type. The default is true. Tooltip: 	Check
	Filter (of Coupled Capacitance) enables or disables the filter threshold to ignore those coupled capacitors with capacitance smaller than the given value after the query coupled capacitor is enabled. Use the <code>xtLPEGPDFilterCc</code> preference to set this type. The default is false.	Check
	Filter Threshold (of Coupled Capacitance) defines the filter threshold to ignore coupled capacitors whose capacitance is smaller than the given value. Use the <code>xtLPEGPDThresholdCc</code> preference to set this type. The default is 0.	Float
Display	Resistor (R) defines the highlight marker to be used for parasitic resistors. The option choices are highlight1 – highlight30. The default is highlight4. Use the <code>xtLPEQueryGPDResHl</code> preference to set this type.	Highlight marker
	Entire Path defines the highlight marker to be used for the P2P entire path. The option choices are highlight1 – highlight30. The default is highlight3. Use the <code>xtLPEQueryGPDNetHl</code> preference to set this type.	Highlight marker
	Capacitance (Cg) defines the highlight marker to be used for parasitic grounded capacitors. The option choices are highlight1 – highlight30. The default is highlight5. Use the <code>xtLPEQueryGPDGCapHl</code> preference to set this type.	Highlight marker
	Capacitance (Cc) defines the highlight marker to be used for parasitic coupled capacitors. The option choices are highlight1 – highlight30. The default is highlight14. Use the <code>xtLPEQueryGPDCCapHl</code> preference to set this type.	Highlight marker
	Zoom enables or disables the viewpoint change for highlighting. If it is on, every time markers are highlighted the viewport on the canvas is changed to fit the current highlighted objects. Use the <code>xtLPEGPDQueryZoomFit</code> preference to set this type. The default is true.	Check
History	Max Number of History Points to Keep defines the maximum number of history records to keep. Use the <code>xtLPEGPDQueryHistoryLimit</code> preference to set this type. The default is 10.	Integer

GPD

In the Layout Editor, Choose **Parasitics > GPD** to query parasitic resistors and capacitors in the GPD database, customize display options, and view the heat map of queried parasitics.

Figure 51 GPD Query



Parasitic devices that are defined in the device mapping file are queried. The device mapping file refers to the `OA_DEVICE_MAPPING_FILE` property in the parasitic view. To configure the list of parasitic devices, set the `xtLPEQueryDevices` preference. The query order of parasitic devices is:

- Defined in the `OA_DEVICE_MAPPING_FILE`. If this file is not found or the referred file does not exist, this step is ignored.
- Specified in the `xtLPEQueryDevices` preference. If this preference is empty, this step is ignored.
- Default parasitic devices: cell name is `presistor`, `pcapacitor`, or `pinductor`.

A warning message appears if:

- The `OA_DEVICE_MAPPING_FILE` does not exist.
- The parasitic device in the `OA_DEVICE_MAPPING_FILE` does not exist.
- The specified library in the `OA_DEVICE_MAPPING_FILE` does not exist, and the parasitic device can be found in other libraries (same cell name and same view name).

Figure 52 GPD in Parasitics View

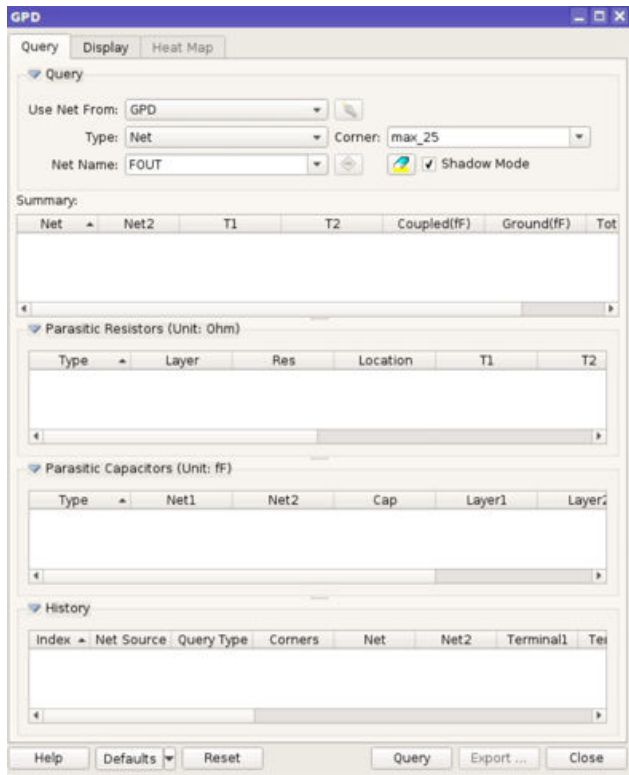


Table 57 GPD Query Tab

Settings	Description	Option Type
Query	There are four query types for the parasitics: All Nets, Net, NetToNet and PointToPoint. Note: The Net Name option for queries of type: Net, NetToNet and PointTo Point in GPD dialog box supports wildcard, partial matching, and regular expressions depending on the Filters option you choose in the General Options dialog box. You can filter terminal names for PointToPoint query using regular expressions when Filters: Regular Expression is selected in General Options dialog box.	
- All Nets	Queries parasitics for all nets at one time.	
- Net	Queries parasitics for multiple nets with multiple corners.	

Table 57 GPD Query Tab (Continued)

Settings	Description	Option Type
	Use Net From: Source of the net name list in the pull-down menu for query option choice: GPD or Design. Corner: Supports the corner name for the query and multiple select for corners in the pull-down menu.  Option choice: Corner names defined in the corresponding GPD. This field is hidden if GPD has only one corner. Option definition: Supports corner names for the query and multiple select for corners in pull-down menu.	Choice
	The clear highlight button  clears all highlights. The select net button  shows the Net Selection dialog. See Net Selection for more information.	
	Net: Net name for the query. Net names are defined in the current layout cell view. Shadow Mode enables shadow mode for the query highlight. The default is true.	Choice
- NetToNet	Net-to-net query for multiple corners See the Use Net From description in the Net setting for more information. See the Corner description in the Net setting for more information. See the clear highlight button in the Net setting. Net Name: Net name of the first net for the query. Net2 Name: Net name of the second net for the query. Net names are defined in the current layout cell view. See Shadow Mode description in the Net setting for more information.	Choice

Table 57 GPD Query Tab (Continued)

Settings	Description	Option Type
- PointToPoint	Queries resistance and delay between two points on one net for multiple corners.	
	See the Use Net From description in the Net setting for more information.	
	See the Corner description in the Net setting for more information.	
	See the clear highlight button in the Net setting.	
	Net Name of the device terminals for the query. Net names are defined in the current layout cell view.	Choice
	See the Shadow Mode description in the Net setting for more information.	
	Query For queries for point-to-point resistance or Elmore delay. Option Choice: Resistance & ElmoreDelay , Resistance , ElmoreDelay . The default is Resistance & ElmoreDisplay .	Choice
	List Resistors on Path queries the resistors on the path connecting those two specified terminals. The default is true.	Check
	Terminal1 Name is the name of the first terminal for the query.	Choice
	Terminal2 Name is the name of the second terminal for the query.	
	Option Choice: The names of the terminals defined in GPD that belong to the specified net. If Filter Internal Terminals in the Parasitics Options dialog box is enabled, all internal terminals are filtered. The reset terminal1 button  resets the pull-down list for Terminal1 and sets it as unselected.	
Heat Map Button	Click  to view the color map for queried results.	
Summary	Query results summaries are listed in tables. For the Net query type, the results summary contains the total coupled capacitance (Coupled), total ground capacitance (Ground), and total capacitance (Total Cap), as shown in Figure 52 .	

Table 57 GPD Query Tab (Continued)

Settings	Description	Option Type
	For the NetToNet query type, the results summary contains the total coupled capacitance (Coupled).	
	For the PointToPoint query type, the results summary contains the resistance (Res), Elmore delay (Delay)	
	If two or more corners are specified for the query, results for different corners are listed in the table	
- Parasitic Resistors	Hidden if query resistors is disabled. The columns in Parasitic Resistors are Type (wire or via), Layer, Resistance, Location, Terminal1, Terminal2, width/length (for wire), and area (for via). If two or more corners are specified for the query, the Resistance column is replaced by a column with a corner name for each specified corner. Each column can be sorted by selecting the column header. Select the cells to highlight and zoom fit to the corresponding objects of the parasitic item in the layout if the Zoom option is turned on.	
- Parasitic Capacitors	Hidden if query grounded capacitor, coupled capacitor, and pin capacitor are all disabled. The columns in Parasitic Resistors are Type (Coupling/Ground), Net1/Net2 Name, Capacitance, Layer1/Layer2 (layer of net1/net2) Location. If two or more corners are specified for the query, the Capacitance column is replaced by a column with a corner name for each specified corner. Each column can be sorted by selecting the column header. Select the cells to highlight and zoom fit to the corresponding objects of the parasitic item in the layout if the Zoom option is turned on.	
History	Use history is maintained in this section. You can select any item in the history table to recall the corresponding query results. The Browse button allows for the selection of a file for saving parasitic results. This field is enabled and the query results are saved in the specified file after the Export Result to File selection in the option dialog box is enabled. The file format is determined by the suffix as either XML (.xml), CSV (.csv), or ASCII (.ascii, or anything other than .xml and .csv).	

Button Bar

Table 57 GPD Query Tab (Continued)

Settings	Description	Option Type
	Reset clears the Summary , Parasitic Resistors , and Parasitic Capacitors sections.	
	Query provides corresponding parasitic data from GPD that is specified in the query section.	
	Export exports the queried parasitic results to a file. This button is disabled if both tables for Parasitic Resistors , and Parasitic Capacitors are empty.	

Table 58 GPD Display Tab

Settings	Description	Option Type
Display	Update the Display tab settings to the current query results. Options of Display:	
	Resistance (R) enables or disables query parasitics for resistors. Use the <code>xtLPEGPdQueryR</code> preference to set this type. The default is true.	Check
	Filter (of Resistance) enables or disables the filter threshold to ignore those parasitic resistors with resistance smaller than the given value after the query resistance is enabled. Use the <code>xtLPEGPdFilterR</code> preference to set this type. The default is false.	Check
	Filter Threshold (of Resistance) enables or disables the filter threshold to ignore those parasitic resistors with resistance smaller than the given value. Use the <code>xtLPEGPdThresholdR</code> preference to set this type. The default is 0.	Float
	Include Via (of Resistance) indicates that the query of parasitic resistors must include via. Use the <code>xtLPEGPdQueryViaR</code> preference to set this type. The default is true.	Check
	Filter Internal Terminals (of Resistance) shows or hides internal terminals in a pull-down list and highlights the layout for a point-to-point query. Use the <code>xtLPEGPdP2PFilterR</code> preference to set this type. The default is true.	Check

Table 58 GPD Display Tab (Continued)

Settings	Description	Option Type
	Capacitance (Cg) enables or disables query parasitics for grounded capacitors. Use the <code>xtLPEGPDQueryCg</code> preference to set this type. The default is true.	Check
	Filter (of Grounded Capacitance) enables or disables the filter threshold to ignore those grounded capacitors with capacitance smaller than the given value after the query grounded capacitor is enabled. Use the <code>xtLPEGPDFilterCg</code> preference to set this type. The default is false.	Check
	Filter Threshold (of Grounded Capacitance) defines the filter threshold to ignore grounded capacitors whose capacitance is smaller than the given value. Use the <code>xtLPEGPDThresholdCg</code> preference to set this type. The default is 0.	Float
	Capacitance (Cc) enables or disables query parasitic for coupled capacitors. Use the <code>xtLPEGPDQueryCc</code> preference to set this type. The default is true.	Check
	Filter (of Coupled Capacitance) enables or disables the filter threshold to ignore those coupled capacitors with capacitance smaller than the given value after the query coupled capacitor is enabled. Use the <code>xtLPEGPDFilterCc</code> preference to set this type. The default is false.	Check
	Filter Threshold (of Coupled Capacitance) defines the filter threshold to ignore coupled capacitors whose capacitance is smaller than the given value. Use the <code>xtLPEGPDThresholdCc</code> preference to set this type. The default is 0.	Float
	Resistor (R) defines the highlight marker to be used for parasitic resistors. The option choices are highlight1 – highlight30. The default is highlight4. Use the <code>xtLPEQueryGPDResH1</code> preference to set this type.	Highlight marker

Table 58 GPD Display Tab (Continued)

Settings	Description	Option Type
	Entire Path defines the highlight marker to be used for the P2P entire path. The option choices are highlight1 – highlight30. The default is highlight3. Use the <code>xtLPEQueryGPDNetH1</code> preference to set this type.	Highlight marker
	Capacitance (Cg) defines the highlight marker to be used for parasitic grounded capacitors. The option choices are highlight1 – highlight30. The default is highlight5. Use the <code>xtLPEQueryGPDGCapH1</code> preference to set this type.	Highlight marker
	Capacitance (Cc) defines the highlight marker to be used for parasitic coupled capacitors. The option choices are highlight1 – highlight30. The default is highlight14. Use the <code>xtLPEQueryGPDCCapH1</code> preference to set this type.	Highlight marker
	Refresh applies current settings in the Display section to both parasitic resistors and capacitors report tables and the highlighting over a layout canvas.	

Table 59 GPD Heat Map Tab

Settings	Description	Option Type
Heat Map	Customize the Heat Map tab options to view heat maps for parasitics listed in various sections of the Query tab. Choose different Heat Map options to show the histogram. Consider the filtering criteria in the Display tab. If a query is not made, it is reset, or the query type is set to NetToNet , this tab is inactive. The generated heat map is cleared when you perform another query.	
Histogram	is a graphical representation of parasitic distribution based on the value listed in the Query tab. The X-axis represents resistor and capacitor values and the Y-axis represents number of segments with respective parasitic values.	

Table 59 GPD Heat Map Tab (Continued)

Settings	Description	Option Type
	There are four Heat Map types: Path Resistance, Total Capacitance, Total Coupling Capacitance, and Total Ground Capacitance. Path Resistance displays the heat map based on the parasitic resistance values of each segment over the nets or path between two terminals. Available when Query type is All Nets, Nets, and PointToPoint. Total Capacitance, Total Coupling Capacitance, and Total Ground Capacitance display the heat map based on the capacitance values of nets listed in the Summary section. Available when the Query type is All Nets, and Nets. Use the <code>xtLPEGPDHeatMapType</code> preference to set the required heat map type in SMC mode. Min and Max values for the Histogram color scale (X-axis) are gathered from the lowest and highest values from the parasitics table. Color Levels represents the number of horizontal intervals in the Histogram scale. Use the <code>xtLPEGPDHeatMapLevel</code> preference to set the required color level in SMC mode. Corner consists of corners from the Query tab in the user-selected order. Use the <code>xtLPEGPDHeatMapCorner</code> preference to set the required corner in SMC mode.	

Net Selection

From the **Query** tab, click the net selection button  to select multiple nets. As shown in [Figure 53](#), the **Nets** column shows the nets elements that are not selected. The **Selected Nets** column shows the selected nets. You can use a regular expression to filter the required nets and select or unselect elements from the **Nets** and **Selected Nets** columns. You can use the **X** button to delete text from the **Filter** box.

Figure 53 Net Selection

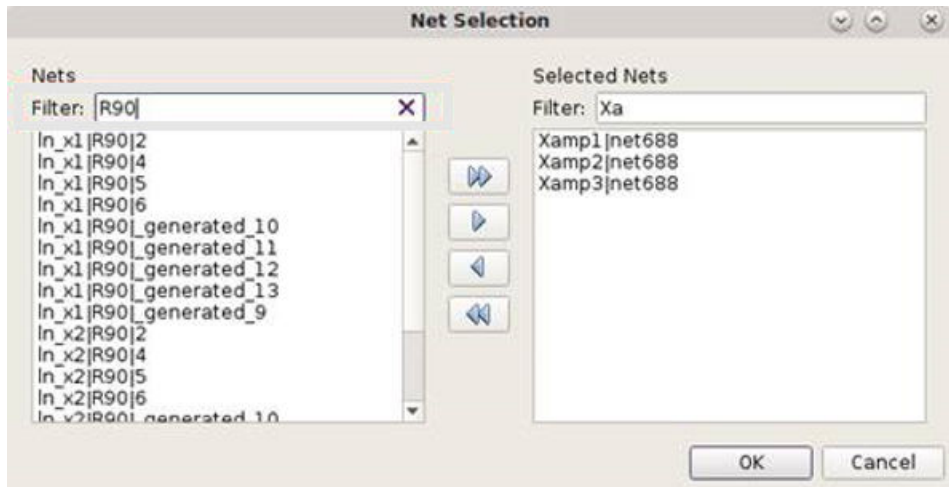


Table 60 Net Selection Options

Settings	Description
Select all 	Selects all nets in the Nets Column.
Select items 	Selects the chosen nets in the Nets column.
Unselect items 	Unselects the chosen nets in the Selected Nets column.
Unselect all 	Unselects all nets in the Selected Nets column.
OK	Refills the selected nets in the Query dialog.
Cancel	Closes the Query dialog and discards any changes.

Compare Capacitance

Choose **Parasitics > Compare Capacitance** to compare the parasitic capacitors of the imported GPD database and the extracted values against the routing capacitance constraints when they exist. Reporting on the extracted total capacitance for all nets is available.

Figure 54 Compare Capacitance Table With Constraints Defined

Unit: fF

Net	Value	Max	Min	Diff	Diff(%)
vss!	23.5341399953213	23.5	2.0	0.03	0.15 %
vdd!	25.0670631846463	25.3	2.0	-0.23	-0.92 %
PBIAS	3.73973653013396	3.8	--	-0.06	-1.59 %
outp1	2.68489225384184	2.7	--	-0.02	-0.56 %
outm1	2.47174988438358	2.5	--	-0.03	-1.13 %
NBIAS	3.46479717617017	3.6	--	-0.14	-3.76 %
outp3	1.63001743820507	1.6	--	0.03	1.88 %
outm3	1.6238074975367	1.6	--	0.02	1.49 %
outp2	1.04351237491567	1.05	--	-0.01	-0.62 %
outm2	1.04916225061288	1.05	--	-0.00	-0.08 %
vdd	0	--	--	--	--
FOUT_B	3.127318588673	3.1	--	0.03	0.88 %
FOUT	3.130838811839	3.1	--	0.03	0.99 %
LFIN	0.3232745347974	3.2	--	-2.88	-89.90 %

Close

Compare Capacitance constraints are defined in [Table 61](#).

Figure 55 Compare Capacitance Table With Undefined Constraints

Unit: fF

Net	typical_M40	typical_25	typical_125	Max	Min	Diff	Diff(%)
vss!	0	0	0	--	--	--	--
vdd!	0	0	0	--	--	--	--
PBIAS	3.91316164393012	3.91316164393012	3.91316164393012	--	--	--	--
outp1	2.70797261817243	2.70797261817243	2.70797261817243	--	--	--	--
outm1	2.49808390972889	2.49808390972889	2.49808390972889	--	--	--	--
NBIAS	3.60779183482623	3.60779183482623	3.60779183482623	--	--	--	--
outp3	1.65524053054679	1.65524053054679	1.65524053054679	--	--	--	--
outm3	1.647335836141	1.647335836141	1.647335836141	--	--	--	--
outp2	1.0714195673454	1.0714195673454	1.0714195673454	--	--	--	--
outm2	1.07249493077869	1.07249493077869	1.07249493077869	--	--	--	--

Close

Table 61 Compare Capacitance Constraints

Settings	Description
Net	The name of the net.
Value/Corner-Name	The extracted parasitic capacitance value for the net. If the maximum capacitance value of the net is defined in the constraint: If the capacitance value $\leq 95\% \times \text{Max value}$, the cell is filled with green. If $95\% \times \text{Max value} < \text{capacitance value}$ and $\text{capacitance} \leq \text{Max value}$, the cell is filled with yellow. If the capacitance value $> \text{Max value}$, the cell is filled with red.
Max	Maximum capacitance value for the net defined in the constraint. Will be "--" if the constraint value is undefined.
Min	Minimum capacitance value for the net defined in the constraint. Will be "--" if the constraint value is undefined.

Table 61 Compare Capacitance Constraints (Continued)

Settings	Description
Diff	Difference between the extracted parasitic capacitance (the maximum among all corners for a multicorn case) and the maximum value for the net. Will be "--" if the constraint value is undefined. If the maximum capacitance value of the net is defined in the constraint: The cell is filled with red if any cell for the capacitance value of this net is filled with red. The cell is filled with green if all cells for the capacitance value of this net are filled with green. Otherwise, the cell is filled with yellow.
Diff(%)	The difference between the extracted parasitic capacitance (the maximum among all corners for a multicorn case) and the maximum value for the net in percentage of the maximum value. Will be "--" if the constraint value is undefined. If the maximum capacitance value of the net is defined in the constraint, this cell is filled with the same color as the Diff column of the same net.

15

Comparison Layout Versus Schematic and Layout Parasitic Extraction

A flow to save setup time to run the LVS and LPE.

This chapter covers the following topic:

- [Using Integrated Tools for LVS and LPE Runs](#)

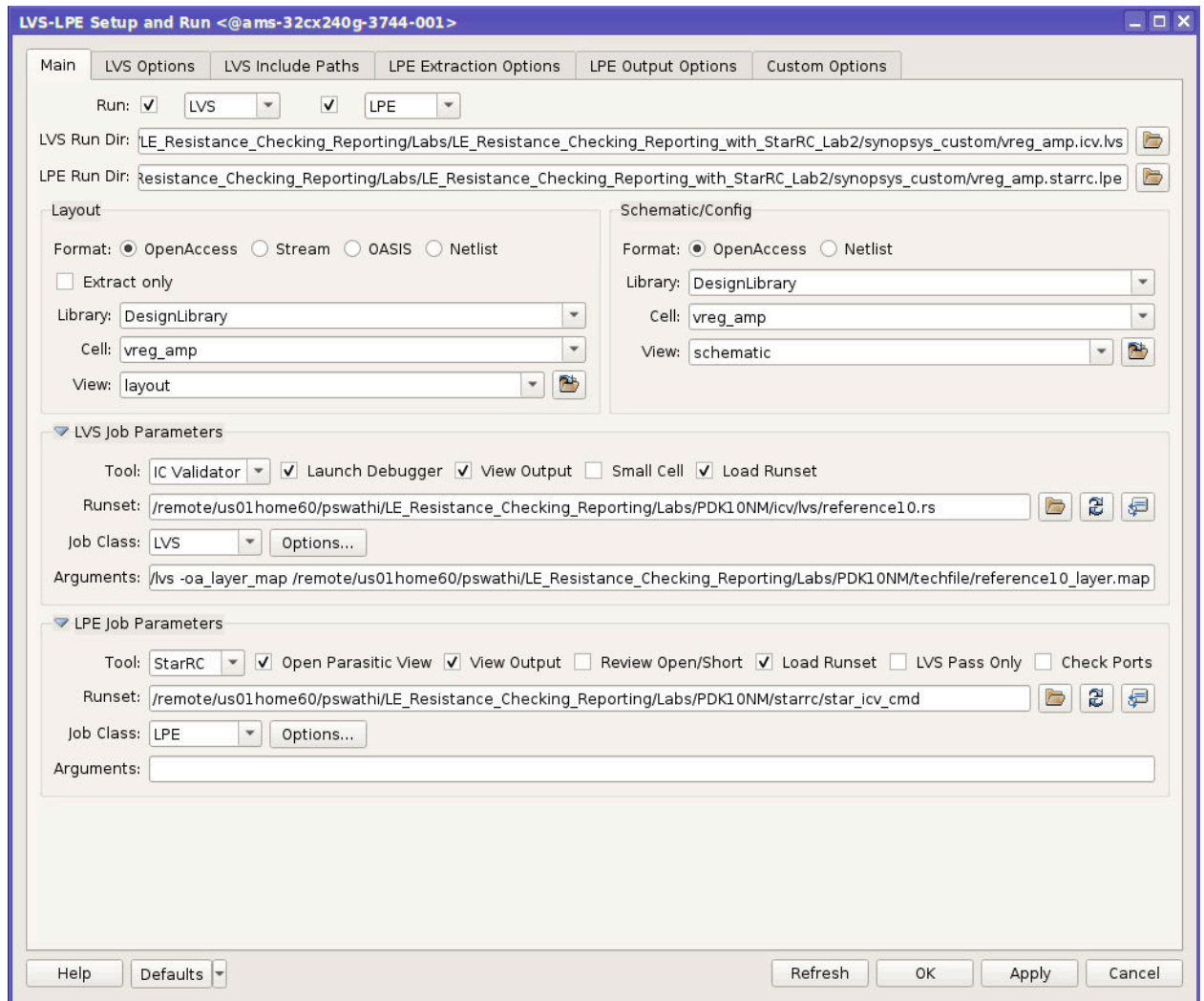
Using Integrated Tools for LVS and LPE Runs

You can run the IC Validator or Calibre tool for layout-versus-schematic comparisons and use the StarRC tool for parasitic extraction options from the same dialog box.

You can set up and run the LVS comparison and LPE using the following steps:

1. In the design layout window, choose **Verification > LPE > Setup and Run**.

(Choosing **Tools > Verification** adds the **Verification** menu to the window menu bar.)



2. In the **LVS-LPE Setup and Run** dialog box, you can enable or disable the **LVS** and **LPE** jobs from the **Run** option, as shown in the following **LVS** and **LPE** combination dialog box. The dialog box reconfigures to reflect available options based on the jobs selected.
3. LVS and LPE custom jobs are listed in the respective options in **Run** option when LVS or LPE custom jobs are defined in the `pdk.xml` file. For example, **customLVS1**, **customLVS2**, **customLPE1** and **customLPE2** are custom jobs defined in `pdk.xml` file. See [Configuring PDKs for DRC, LVS, and PVE](#) for more information.

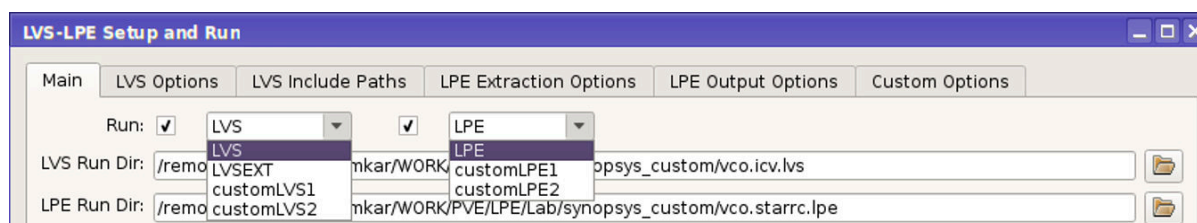
Note:

The **LVS** job, **LVSEXT**, is specific for extraction, which is available only from the **LVS and LPE** dialog box.

4. The following behaviors are supported for **LVS** and **LPE** job selection:

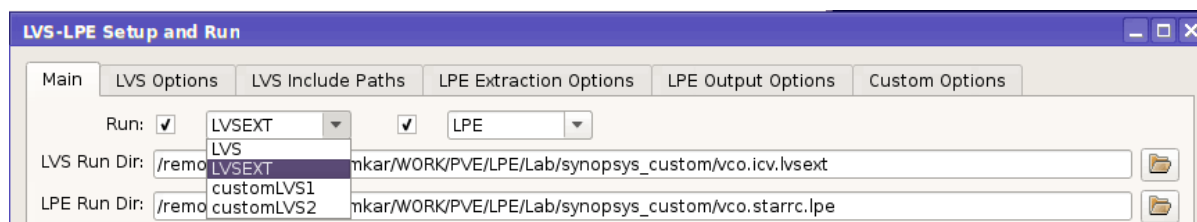
- When you change the **LVS** and **LPE** job, the current settings of the original job are cleared and the default settings of the new job are loaded.
- When you click the **Defaults** button, the initial values of both the current **LVS** and **LPE** jobs are loaded. For example, if the **LPE** job changes to **customLPE1**, press the **Defaults** button to revert to the initial value of the **customLPE1** job.

Use the `xt<lpeJob>EnableLVSTJob`, `xt<lpeJob>LVSJobName`, `xt<lpeJob>EnableLPEJob`, and `xt<lpeJob>LPEJobName` preferences to enable or disable and select the required **LVS** or **LPE** job.



Limitation: Backward compatibility is not supported when loading a preference value by selecting **Defaults > Load** in the **LPE Setup and Run** dialog box. You must update the preference names in the loaded preference xml file.

5. The **LVS** job for extraction, **LVSEXT** is supported, and available only from the **LVS** menu of the **LVS-LPE Setup and Run** dialog box.

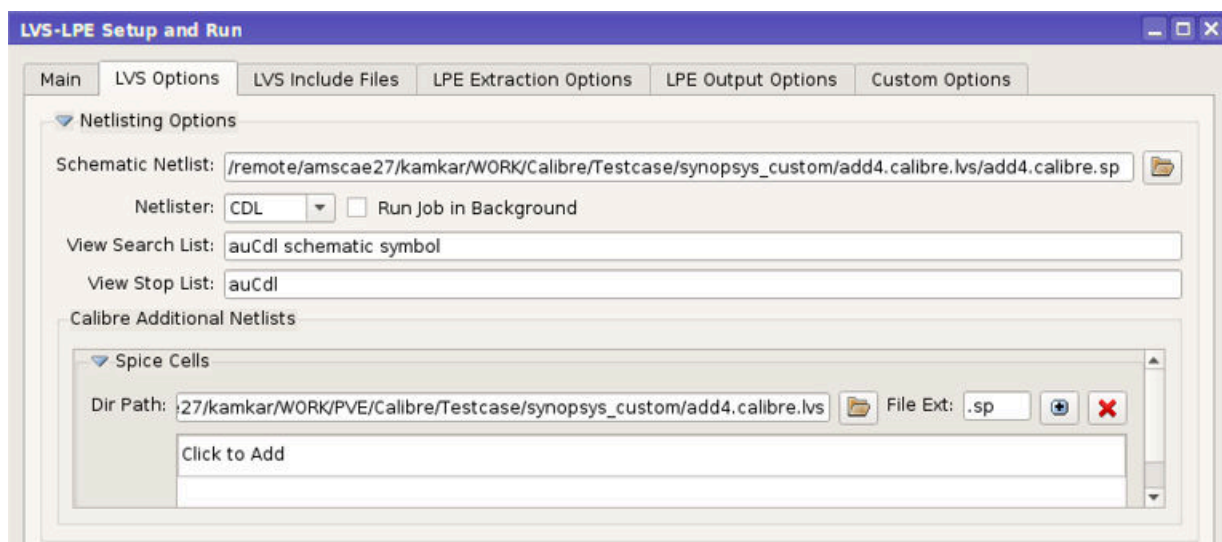


Use the `xtLVSTRunDir` preference for a default **LVS** job, and the `xtLVSEXTRunDir` and `xtLVSEXTICVRunsetFile` preferences to set the run directory and IC Validator runset file for the **LVSEXT** job, respectively. For information on using Tcl commands and preferences, see the *Custom Compiler Tcl Reference Manual*.

6. Use the **Main** tab, **LVS Options** tab, and **Custom Options** tab to set up the Layout Versus Schematic Comparison as described in [Using Integrated Tools for LVS Comparisons](#).

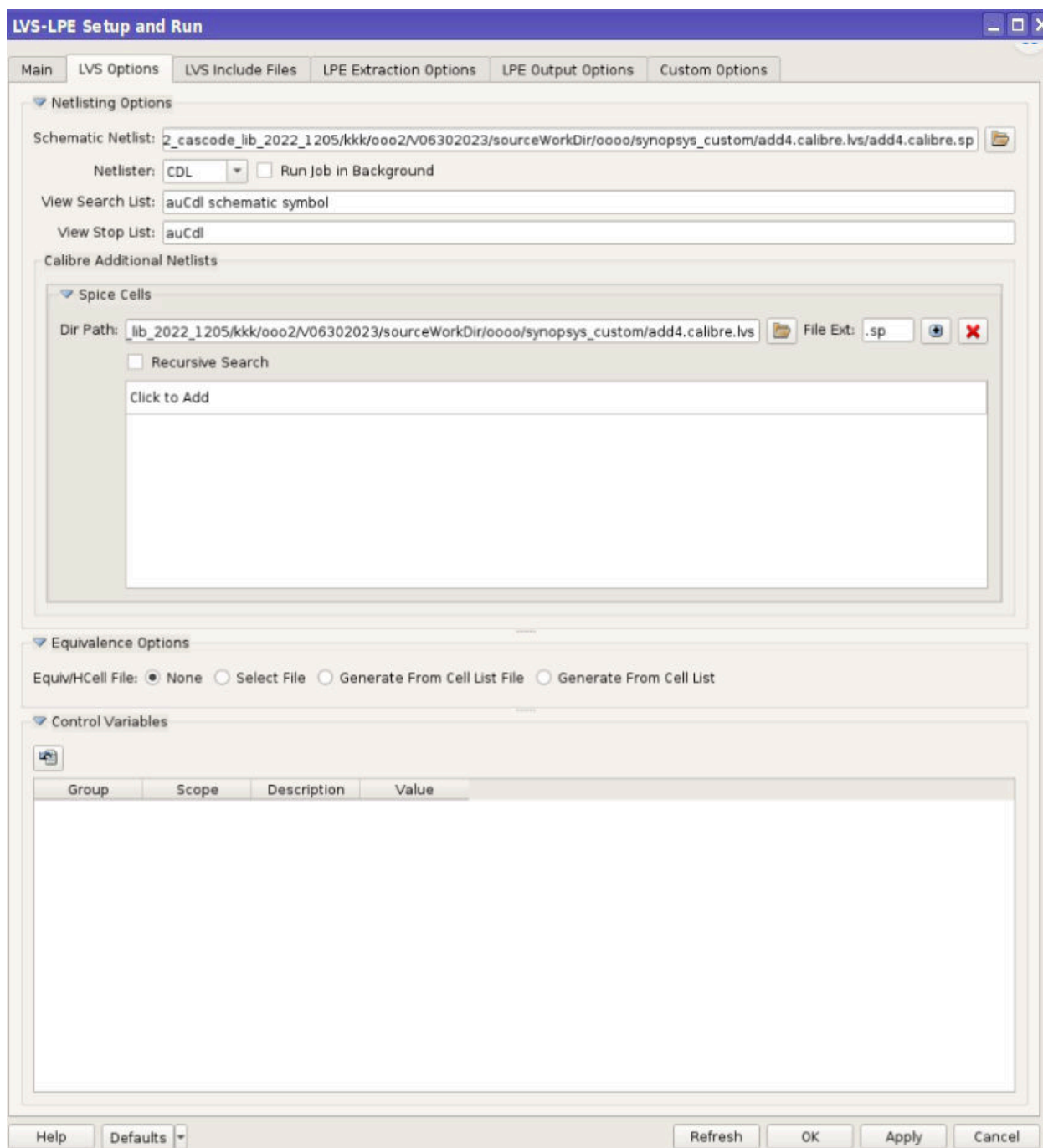
Note:

Set **Tool > Calibre** in the **Main** tab to see the **Calibre Additional Netlists** section in the **LVS Options** tab. See [Using Integrated Tools for LVS Comparisons](#) for more information.



7. For the IC Validator tool only, on the **LVS Include Paths** tab, add or delete LVS include paths.
 - a. In the table, double-click a blank row to add an include path. Double-click a populated row to replace that include path.
Cycle through the list using the arrow buttons.
To delete a path, click the **Delete** button.
 - b. Browse to choose the path you want to add.
 - c. Click **Select**.
 - d. Repeat for each include path you want to add or replace.
8. For the Calibre tool only, on the **LVS Include Files** tab, add or delete LVS include paths.
 - a. In the table, double-click a blank row to add an include file. Double-click a populated row to replace that include file.
Cycle through the list using the arrow buttons.
To delete a file, click the **Delete** button.
 - b. Browse to choose the file you want to add.
 - c. Click **Select**.
 - d. Repeat for each include path you want to add or replace.

9. Use the **Main** tab, **LPE Extraction Options** tab, **LPE Output Options** tab, and **Custom Options** tab to set up the extraction as described in [Setting Options for Layout Parasitic Extraction](#).
 - Use the **LVS Pass Only** setting in the **LPE Job Parameters** to stop the LPE run if LVS is not clean. This setting is **OFF** by default, and it is visible only when both LVS and LPE runs are enabled.
10. To open the **LPE Setup and Run** dialog box instead of the **LVS-LPE Setup and Run** dialog from the LPE menu, set the `xtEnableCombinedLVSLPEFlow` preference to false.
11. Set up the StarRC skip cells flow by using the Equivalence Options in the **LVS-LPE Setup and Run** dialog box **LVS Options** tab. For hierarchical designs, extract the top-level design for a parasitic view without extracting the lower-level blocks to reduce view-generation time.



Specify the following equivalence cells for the skip cells flow. The LVS run must have equivalence points to define the match between the layout and the schematic cells during LVS comparison.

In the **LVS Options** tab, you can define equivalence and black-box cells. The black-box LVS flow is supported only in the LVS-LPE combination dialog box and in the IC Validator tool. When both the equivalence and black-box functions are used, IC Validator handles the equivalence and black-box conflict behavior.

See [Table 62](#) for a list of equivalence options.

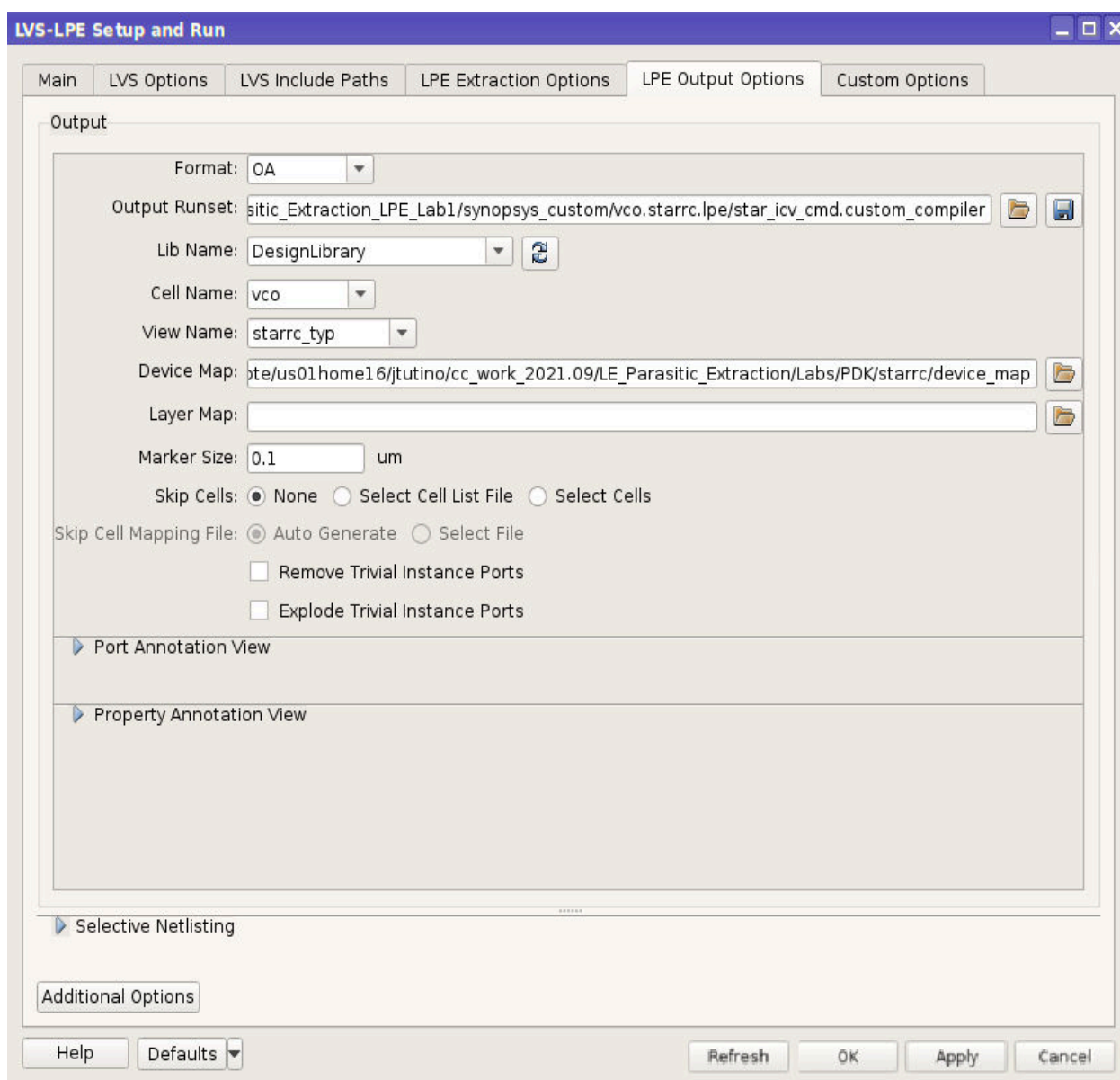
Table 62 *Equivalence Options*

Settings	Description
Equiv/HCell File	Define the Equiv/HCell file using four types: None, Select File, Generate From Cell List File, and Generate From Cell List. When None is selected, Equiv/HCell is not provided and LVS compares only the current cell. When Select File is selected, you can select the existing user-defined Equiv/HCell file. An error message appears if the file path does not exist. When Generate From Cell List File is selected, you can select a file that contains a cell list to compare to your LVS. Cells can be separated by white space or a new line in the cell list file. An error message appears if the file path does not exist. When Generate From Cell List is selected, you can define cells on the GUI to compare to your LVS. Cells can be separated by white space, and this selection supports wildcard filters.
Black-Box File	Define the black-box file using four types: None, Select File, Generate From Cell List File, and Generate From Cell List. When None is selected, the black-box file is not provided and LVS compares the entire hierarchy of the design. When Select File is selected, you can select the existing user-defined black-box file. An error message appears if the file path does not exist. When Generate From Cell List File is selected, you can select a file that contains a cell list to black-box to your LVS. Cells can be separated by white space or a new line in the cell list file. If an equivalence file was specified in the Equiv/HCell file, Custom Compiler combines the black-box and Equiv/HCell files into one file. An error message appears if the file path does not exist.

Settings	Description
	When Generate From Cell List is selected, you can define cells on the GUI to compare to your LVS. Cells can be separated by white space, and this selection supports wildcard filters. If an equivalence file was specified in the Equiv/HCell file, Custom Compiler combines the black-box and Equiv/HCell files into one file. This selection supports wildcard filters.

The height of **Netlisting Options**, **Equivalence Options** and **Control Variables** in the **LVS Options** tab can be adjusted to your preference.

12. From the **LPE Output Options** tab, you can specify **Skip Cells** settings for extraction. The LPE run uses these settings to define the cells that must be skipped while extracting parasitics.



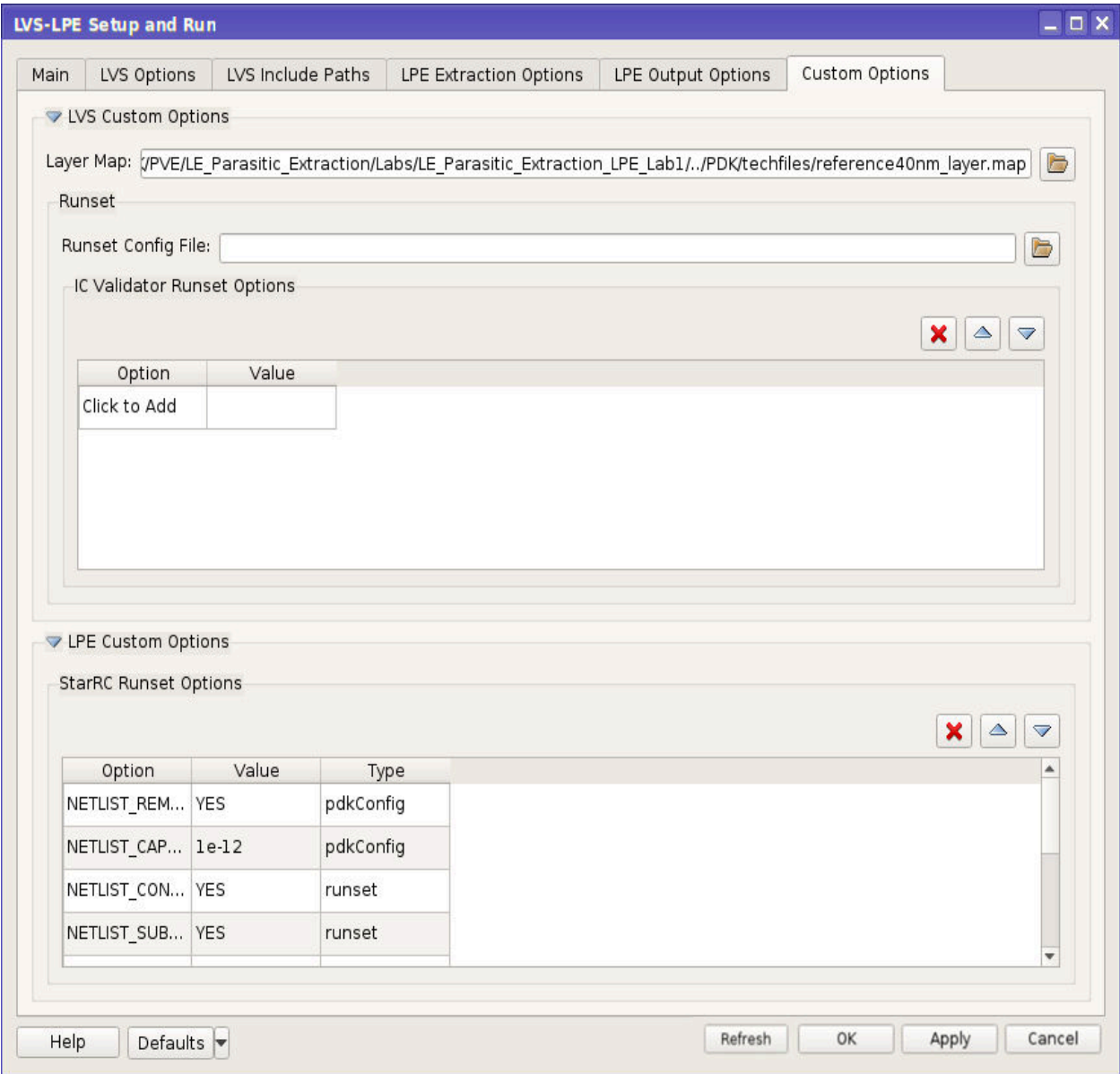
See [Table 63](#) for a list of skip cells settings.

Table 63 LPE Skip Cells Settings

Setting	Description
Skip Cells	Define the skip cells using three settings: None , Select Cell List File , and Select Cells . When None is selected, no cell is skipped during extraction.

Setting	Description
	When Select Cell List File is selected, all cells in the specified file, except the current cell are skipped during extraction. Cells can be separated by white space or a new line in the cell list file. An error message appears if the file path does not exist. When Select Cells is selected, the cells specified on the GUI are skipped during extraction. Cells can be separated by white space, and this selection supports wildcard filters.
Skip Cell Mapping File	Available only in OA output format. You must specify the Skip Cell Mapping File setting if Skip Cells is None. When Auto Generate is selected, a skip cell mapping file is automatically generated according to the selected Skip Cell setting. When Select File is selected, you can select the skip cell mapping file on the GUI. An error message appears if the file path does not exist. Remove Trivial Instance Ports removes skip-cell (black-box cell) ports that are not connected to devices. This setting is available only in Explode Trivial Instance Ports is disabled. Explode Trivial Instance Ports explodes skip-cell (black-box cell) ports that are not connected to devices. This setting is available only in Remove Trivial Instance Ports is disabled.

13. You can turn on the **LVS** option from the **Run** option on the **Main** tab to show the LVS **Custom Options**.



See [Table 64](#) for a list of LVS Custom Options.

Table 64 LVS Custom Options

Setting	Description
Layer Map	For OpenAccess format only, accept or browse to choose the file mapping OpenAccess LPP numbers and traditional GDSII stream layers/data types.

Setting	Description
Runset	Runset Config File specifies a runset configuration file that is passed to an IC Validator command using the <code>-runset_config</code> argument. Use the <code>xt<pveJob>ICVRunsetConfigFile</code> preference to preset this mode. Note: Syntax for the runset configuration file is same as the syntax for the IC Validator runset file. IC Validator inserts options and compare functions from the runset configuration file into the runset after the other options and compare functions (compare itself is modified since there can be only one). Options and compare functions are never removed from the runset, as most are designed to use the last setting. And, because options functions allow for incremental calls (specifying different arguments in different calls and using the last setting for each argument), it will feel like you are removing the runset function. Runset Options specifies the supported IC Validator runset options. Custom Compiler does not populate the Runset Options table while loading a runset. To display this table, set preference to: <code>xt<pveJob>ICVShowRunsetOptions</code>.

A

Configuring PDKs for DRC, LVS, and PVE

You can configure access to process design kits (PDKs) through the addition of process design kit configuration files. PDK configuration files are XML files you create and supply for designers running design rule checking, layout versus schematic, layout parasitic extraction, and additional custom processes on their designs.

This appendix covers:

- [Overview](#)
- [Tools Used](#)
- [PDK Identification](#)
- [Configuration File Elements](#)
- [Process and Variant Selection](#)
- [Default Interfaces](#)
- [Custom Jobs](#)
- [Attribute Versus Element Table](#)

Overview

A PDK configuration file lets you control file setup and the basic configuration of built-in dialogs. Tcl-based physical verification dialogs allow for the complete ability to customize the interface. You can also create custom jobs easily using the infrastructure of the DRC, LVS and LPE jobs.

Use of a prepackaged PDK greatly speeds up design development when one is available for the targeted foundry or process.

You have specific needs for physical verification setup. For ease of use, you typically need to provide metal stacks for project targeted processes, preset the process and rules file, control your ability to edit tool and rules files options, and need to be able to customize the graphical interface.

Custom Compiler tools address these needs through a PVE configuration infrastructure and built-in Tcl based graphical interface.

Identifying and locating the application-specific components and support files makes it possible for you to run your physical verification jobs on a newly created layout design with the click of a button without having to specify additional information.

This chapter describes:

- The attributes used to determine a PDK library
- How to set up and use a Process Design Kit (PDK) configuration file on the Custom Compiler environment
- How to configure the setup for Physical Verification and Extraction (PVE) jobs

For more about this, see [Configuring PDKs for DRC, LVS, and PVE](#).

Tools Used

To configure the PDK and set up for PVE, you need the following:

- Custom Compiler
- IC Validator
- StarRC

For version compatibility information, see the [Custom Compiler Tools Compatibility Table](#) on SolvNetPlus.

Note:

You must be logged in to SolvNetPlus for the link to connect directly to the article. If you are prompted to log in to SolvNetPlus upon clicking the link to the article, log in, then click the link again to reach the article.

For more about this, see [Configuring PDKs for DRC, LVS, and PVE](#).

PDK Identification

Two different attributes are used to automatically identify a PDK library. You can also use environment variables when using both these attributes.

These sections cover:

- [PDKLib attribute](#)
- [PDKConfigPath attribute](#)

PDKLib attribute

This is a Boolean attribute in the `lib.defs` file that you set to true for a PDK library. The location of the PDK library mentioned in the `lib.defs` file has to be defined relative to the configuration file location (see [Configuration File Elements](#)). A sample `lib.defs` file follows.

```
DEFINE reference40nm ./reference40nm
ASSIGN reference40nm libMode shared
ASSIGN reference40nm PDKLib true
```

PDKConfigPath attribute

This is an attribute in the `lib.defs` file that you can use to specify the location of the configuration file. This is useful if you want to store the configuration file outside of the PDK library. In that case, you use an absolute path to specify the location. Two examples for using this attribute in the `lib.defs` file follow.

```
DEFINE reference40nm ./reference40nm
ASSIGN reference40nm
    PDKConfigPath /
remote/disk1/pdkIntegTestcase/snps_reference40nm/current/pdk.xml

DEFINE reference40nm $ref40PDK/reference40nm
ASSIGN reference40nm PDKConfigPath $ref40PDK/pdk.xml
DEFINE reference40nm $ref40PDK/reference40nm
ASSIGN reference40nm PDKConfigPath $ref_40/current/pdk.xml
```

If a path is relative, it is considered relative to the library's parent directory.

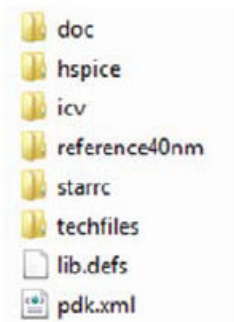
Configuration File Elements

Because each foundry has its own directory structure, it is not easy to generically locate application data from different foundries. So it is useful to specify the location of application specific data, such as runsets in a configuration file. A configuration file named `pdk.xml` can be found in the same directory as the example PDK library. This file contains additional information about file locations and settings for specific applications.

The configuration file is loaded by default if the library is identified as a PDK library (see [PDK Identification](#)). The `pdk.xml` example configuration file uses XML syntax to describe paths to runsets and common settings/options for the PVE verification runs.

By placing the configuration file in the directory containing the PDK library, the paths to runsets become relative to the directories found at the same level. Paths to process files, however, can be absolute or relative to the PDK library location. Typically, foundries provide the different runsets, model files, and the PDK library separately. All the relevant data that you gather from the foundry can be put together in a central location.

The following figure illustrates the recommended organization of PDK-related data.



These sections cover:

- [File Format Element](#)
- [Process Element](#)
- [Application Element](#)
- [Variant Element](#)
- [pveJob Element](#)
- [Tool Element](#)
- [Preference Element](#)
- [Action Element](#)
- [Value Element](#)
- [Item Element](#)

File Format Element

The `file-format` element is the document root and identifies the file format and version.

XML is chosen as the format for the configuration file because the Custom Compiler environment has an XML parser and some other applications use the XML syntax.

[Table 65](#) list and describes `file-format` element attributes.

Table 65 File-Format Element Attributes

Attribute	Type	Description
name	string	Name of the file format. The file format for a PDK Configuration File used by Custom Compiler integrations is <code>synopsysPDK</code> .
version	decimal	Version of the file format

Example:

```
<file-format name="synopsysPDK" version="1.0">
```

Process Element

The `process` element represents a foundry process node.

[Table 66](#) lists and describes `process` element attributes.

Table 66 Process Element Attributes

Attribute	Type	Description
name	string	Name that uniquely identifies the process node.
default	boolean	If true, the <code>xtPDKProcess</code> preference will be set to the name of the process node at startup. Your changes are kept persistent using the defaults button on the Physical Verification and Extraction Options dialog. Defaults to false. If no process default is specified, the one specified first in the configuration file <code>pdk.xml</code> is considered the default.

Example:

```
<process name="reference40nm" title="Sample pdk.xml">
```

Process selection can be performed using the **Physical Verification and Extraction Options** dialog box (see [Process and Variant Selection](#)).

Application Element

The application element represents a section that describes application specific data and settings.

Currently, this is used only for PVE applications, but is extensible to other applications.

The application name used to identify a PVE application section of a PDK Configuration File is `PVE`.

The application preference prefix for PVE is `xt`.

[Table 67](#) lists and describes `application` element attributes.

Table 67 Application Element Attributes

Attribute	Type	Description
name	string	Name that uniquely identifies the application section.
prePrefix	string	Two character lower-case preference prefix used to identify application specific preferences.

Example:

```
<application name="PVE" prefPrefix="xt">
```

Variant Element

The variant element represents a variation on a process node. Examples include metallization stacks and device voltage characteristics.

[Table 68](#) lists and describes `variant` element attributes.

Table 68 Variant Element Attributes

Attribute	Type	Description
name	string	Name that uniquely identifies the process variant.
default	Boolean	If true, the <code>xtPDKVariant</code> preference will be set to the name of the process variant at startup. Your changes are kept persistent via the defaults button on the Physical Verification and Extraction Options dialog. Defaults to false. If no variant default is specified, the one specified last in the configuration file <code>pdk.xml</code> is considered the default.

Example:

```
<variant name="1P10M" title="1-Poly 10-Metal" default="true">
```

The variant selection can be performed using the Physical Verification and Extraction Options dialog (see [Process and Variant Selection](#)).

pveJob Element

The `pveJob` element represents a physical verification or extraction job for running design checks and extracting parasitics in a specific process variant. All PVE integration job-related preferences are based on the job type, for example, `xtJobLayLCV`. Preference labels are created using the application preference prefix and the job name; for example `{xt DRC}`. All PVE integration job related actions (menu choices) will be created if they do not already exist for the job name; for example, `xtJobViewOutputs`.

Table 69 lists and describes `pveJob` element attributes.

Table 70 lists and describes the mappings between job types and window types.

Table 69 *pveJob Element Attributes*

Attribute	Type	Description
name	string	Alpha-numeric name that uniquely identifies the job. This should follow a camelBack notation for custom jobs; for example, <code>lpeSPF</code> . For default jobs, names are DRC, LVS, and LPE.
prefId	string	Job identifier used to encode job-specific preference and action names. Defaults to job name.
type	string	Job type by function and database inputs. Restricted to DRC, LVS, or LPE.
title	string	Job name as displayed in interface; for example, DRC, LVS, LPE (no name restrictions). Defaults to job name.
windowTypes	string	Specifies the window types for which actions related to this job should be made available. The value for this attribute is a space delimited list of window types. By default, actions are enabled in only those windows that support the design inputs required by the job as determined by the job type. Restricted to valid window types within the Custom Compiler environment.

Table 70 *Job Type to Window Type Default Mappings*

Job Type	Custom Compiler Window Types
DRC	leLayout
LVS	leLayout, seSchematic
LPE	leLayout

Example:

```
<pveJob name="density" jobType="DRC" title="Density"
  windowTypes="leLayout">
```

Tool Element

The `tool` element represents a physical verification or extraction tool capable of running a specific job. The tool name must match a registered `xtTool` object. If a direct match is not found, names are matched against tool aliases.

All PVE integration job-tool related preferences based on the job type will be created if they do not already exist for the tool name; for example, `xtJobToolRunsetFile`. Preference labels will be created using the application preference prefix, job, and tool names, such as `{xt DRC ICV}`. Tool elements that are direct descendants of an application element can be used to configure tool specific preferences for example, `xtToolRunsetFileSuffix`.

[Table 71](#) lists and describes `tool` element attributes.

Table 71 *Tool Element Attributes*

Attribute	Type	Description
name	string	Name that uniquely identifies the tool; for example, ICV, Hercules, StarRC.
default	boolean	If true, the <code>xtJobTool</code> preference will be set to the name of the tool at startup. Defaults to false.
tooltip	string	Changes the message that uniquely identifies the tooltip; for example, "Run ICV drc1 job".

Example:

```
<tool name="ICV" default="true">
```

Custom Compiler displays the default tooltip and shows its value if the **tooltip** attribute is not part of the tool definition. The following example shows the default tooltip for DRC1 (Run 'IC Validator' DRC1 job) and the user-specified tooltip for DRC2 (Run Custom DRC2 with ICV).

Example:

```
<pveJob name="customDRC1" jobType="DRC" title=DRC1">
  <action name="Run" icon="run_drc">
  </action>
  <tool name=IC Validator">
    <preference name="RunsetFile">
```



```

        <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drcl.rs</value>
    </preference>
</tool>
</pvejob>
<pveJob name="customDRC2" jobType="DRC" title=DRC2">
    <action name="Run" icon="/remote/amscae11/CDesigner/sandrin/WORK/PVE \
        LE_Physical_Verification_Lab1/myIcon.PNG">
    </action>
    <tool name="IC Validator" tooltip="Run Custom DRC2 with ICV">
        <preference name="RunsetFile">
            <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drcl.rs</value>
        </preference>
    </tool>
</pvejob>

```

Preference Element

The `preference` element represents a configuration setting in the PVE integration. Each preference must be defined as part of the PVE integration framework and must already exist. This does not preclude you from creating a preference in the Tcl startup scripts for Custom Compiler tools and then setting or configuring the preference from a PDK Configuration file. The PVE preferences available for the Custom Compiler environment are listed in the section covering integrated tool preferences in the *Custom Compiler Tcl Reference Manual*.

[Table 72](#) lists preference attributes.

[Table 73](#) lists preference naming rules.

Table 72 *Preference Element Attributes*

Attribute	Type	Description
name	string	Name that uniquely identifies the preference.
valueType	string	Preference value type. Restricted to path, string, float, integer, bool, or custom.
visible	Boolean	If true, the graphical representation of the preference is visible. Defaults to true.
mutable	Boolean	If true, the graphical representation of the preference is editable. Setting the visible attribute to false implies that the mutable attribute is false as well. Defaults to true.
required	Boolean	If true, the preference value cannot be null when Ok/Apply is pressed.

Table 73 Preference Naming Rules

Element Hierarchy (pref==preference)	Preference
application/pref	prefPrefix+prefName
application/variant/pref	prefPrefix+prefName
application/tool/pref	prefPrefix+toolName+prefName
application/variant/pveJob/pref	prefPrefix+jobName+prefName
application/variant/pveJob/tool/	prefPrefix+jobName+toolName+prefName
pref	fName

Example for preference `xtDensityICVRunsetFile` (prefPrefix+jobName+toolName+prefName):

```
<application name="PVE" prefPrefix="xt">
  <variant name="1P10M" title="1-Poly 10-Metal" default="true">
    <pveJob name="density" jobType="DRC" title="Density" \
      windowTypes="leLayout">
      <tool name="ICV">
        <preference name="RunsetFile" required="true" \
          valueType="path">
          <value>./icv/drc/reference40nm_mod_icv_drc.rs</value>
        </preference>
      </tool>
    </pveJob>
  </variant>
</application>
```

Example for preference `xtLVSLayDBformat` (prefPrefix+jobName+prefName):

```
<application name="PVE" prefPrefix="xt">
  <variant name="1P10M" title="1-Poly 10-Metal" \
    default="true">
    <pveJob name="LVS" jobType="LVS">
      <preference name="LayDBFormat" visible="true" \
        required="true" valueType="string">
        <value>OpenAccess</value>
      </preference>
    </pveJob>
  </variant>
</application>
```

Example for preference `xtLVSIcVOptions` (complete preference name specified):

```
<application name="PVE" prefPrefix="xt">
  <variant name="1P10M" title="1-Poly 10-Metal" default="true">
    <tool name="ICV">
      <preference name="xtLVSIcVOptions" required="true">
        <value>-sf SPICE -ndg -quiet=0</value>
      </preference>
    </tool>
  </variant>
</application>
```

```
</preference>
</tool>
</pveJob>
</variant>
</application>
```

The following table lists the preference names used for the preferences that are available through the tabs in the DRC, LVS and LPE **Setup and Run** dialog boxes.

Table 74 lists and describes custom preferences for tabs.

Table 74 Custom Preferences for Tabs

Preference Name	Tab Name	Tools
Variables	Control Variables	ICV, Hercules
IncludePaths	Include Paths	ICV
RunsetOptions	Custom Options	ICV, Hercules, StarRC

PDK configuration file preference values override the *user scoped* preference values and runset variable values obtained by parsing the runset. However, by default, run directory *scoped* preference values override PDK configuration file preference values.

To force the PDK configuration file preference values to override the *run directory scoped* preferences, set the *override* attribute to true for the specific preferences.

This can be useful for propagating any rule deck changes globally across all the end users.

Example:

```
<preference name="RunsetFile" visible="true" mutable="false"
  required="true" valueType="path" override="true">
  <value>./icv/lvs/reference10lm_lvs.rs</value>
</preference>
```

Action Element

The action element represents an action performed for a specific job. Each action must be defined as part of the PVE integration framework and already exist.

Table 75 lists and describes `action` element attributes.

Table 75 Action Element Attributes

Attribute	Type	Description
name	string	Alpha-numeric name that uniquely identifies the action. Restricted to Setup, Run and ViewOutputs.
binding	string	Key binding used to execute the action.
icon	string	File path to the icon used to represent the action in menus and toolbars. Specify the icon attribute in <code>pdk.xml</code> to view new toolbar icons.

When the icon is not defined for a given action, the tool uses the icon that corresponds to the **JobType**. In the following example, you see only the icons for jobs DRC1 and DRC3 because icons for the Run action of DRC2 were not described.

Example:

```
<pveJob name="customDRC1" jobType="DRC" title=DRC1">
  <action name="Run" icon="run_drc">
  </action>
  <tool name=IC Validator">
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc1.rs</value>
    </preference>
  </tool>
</pvejob>
<pveJob name="customDRC2" jobType="DRC" title=DRC2">
  <tool name=IC Validator">
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc1.rs</value>
    </preference>
  </tool>
</pvejob>
<pveJob name="customDRC3" jobType="DRC" title=DRC3">
  <action name="Run" icon="/remote/amscae11/CDesigner/sandrin/WORK/PVE \
    LE_Physical_Verification_Lab1/myIcon.PNG">
  </action>
  <tool name=IC Validator" tooltip="Run Custom DRC3 with ICV">
    <preference name="RunsetFile">
      <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc1.rs</value>
    </preference>
  </tool>
</pvejob>
```

Value Element

The `value` element is a simple text element that holds the value of a preference. Relative file path preference values are considered relative to the location of the PDK configuration file in which they are defined.

Example:

```
<preference name="RunsetFile" valueType="path">
  <value>./icv/drc/reference40nm_mod_icv_drc.rs</value>
</preference>
```

Item Element

The item element is used to hold a value for a custom preference type.

Item elements share the same visible, mutable, and required attributes as the preference, with the item attributes taking precedence.

Multiple item elements can exist for a preference.

[Table 76](#) lists and describes `item` element attributes.

[Table 77](#) lists and describes custom preferences for `item` element attributes.

Table 76 *Item Element Attributes*

Attribute	Type	Description
visible	boolean	If true, the graphical representation of the item is visible. Defaults to true.
mutable	boolean	If true, the graphical representation of the item is editable. Setting the visible attribute to false implies that the mutable attribute is false as well. Defaults to true.
required	boolean	If true, the item value cannot be null when OK or Apply is clicked.

The following attributes are added to the item element for representing values for custom preferences `xtJobToolVariables` and `xtJobToolRunsetOptions`.

Table 77 *Custom Preferences for Item Element Attributes*

Attribute	Type	Description
name	string	The name of a runset variable or option.

Table 77 Custom Preferences for Item Element Attributes (Continued)

Attribute	Type	Description
valueType	string	The value type of a runset variable. Restricted to boolean, integer, double, string.
scope	string	The scope of a runset variable. Restricted to define, environment, or variable.
section	string	The runset option section.

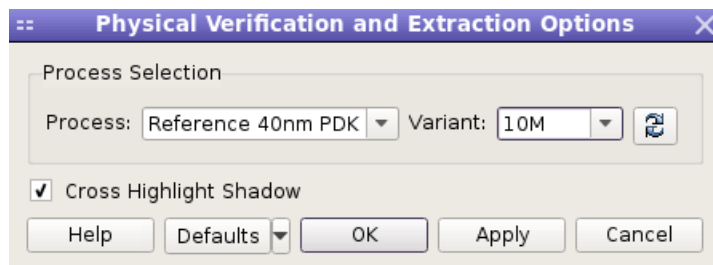
Example:

```
<preference name="Variables" valueType="custom" visible="true"
mutable="true" required="true">
  <item name="DENSITY_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
  <item name="DFM_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"true"</item>
</preference>
```

Process and Variant Selection

The **Physical Verification and Extraction Options** dialog box can be accessed from the **Verification > Options** menu available in the Layout Editor. This dialog box contains controls for selecting both a **Process** and a **Variant** from those defined in the PDK configuration file.

Figure 56 Physical Verification and Extraction Options Dialog Box



Parsing the contents of the PDK configuration file in the order in which the corresponding PDK library is opened populates the **Process** and **Variant** fields. If no PDK Configuration file is found, the **Process** and **Variant** fields are populated with the string Not Defined. The **Cross Highlight Shadow** check box can dim neighboring shapes when you click an error in the Calibre RVE or IC Validator VUE tools. By default, this option is false, and can be enabled by setting the `xtCrossHighlightShadow` preference to true. The refresh button available through the Physical Verification and Extraction Options dialog can be used to

reload the configuration. This can be useful when the configuration file is updated while the Custom Compiler environment is running.

Default Interfaces

The default interfaces use built-in menu entries and built-in dialogs. The built-in dialogs are coded in OPAL and Custom Compiler Tcl API is used for the graphical user interface. The fields in these dialogs correspond to preferences.

The default interfaces include three type of jobs: DRC, LVS, LVL, and LPE. The default interface also includes the Hercules, IC Validator, Calibre, and StarRC tools. The job-tool pair is defined when you start the Custom Compiler tool.

There are some hard-coded rules that control the validity of the job-tool association. To better control the job-tool customization, you can use preferences specific to each built-in pair and follow the naming convention as shown. Preference names are constructed using the preference naming rules described in [Table 73](#).

Example:

```
xtDRCICVRunsetFile  
prefPrefix+jobName+toolName+prefName
```

Where:

- `prefPrefix` = external tool prefix (for example.: `xt`)
- `jobName` = job's preference id (for example.: `DRC`)
- `toolName` = tool's preference id (for example.: `ICV`)
- `prefName` = preference id (for example.: `RunsetFile`)

The same infrastructure can be used as a template for custom jobs. See [Custom Jobs](#) for further information.

This naming convention defines a set of preferences for the exclusive use of the job-tool pair used to name them. The value of each of these preferences can be set through the `pdk.xml` file. Preference names are constructed using the preference naming rules listed and described in [Table 73](#).

Note:

The preferences for built-in jobs are always created (even if a `pdk.xml` file does not exist). The `pdk.xml` file can be used to set the initial values of preferences to something other than the defaults.

Some sections from the default DRC, LVS, and LPE jobs are included as examples below. Use the example data provided in this application note for the complete `pdk.xml` configuration file. See the README file for more details.

These sections cover:

- [DRC Interface](#)
- [LVS Interface](#)
- [LPE Interface](#)

DRC Interface

```
<?xml version="1.0"?>
<file-format name="synopsysPDK" version="1.0">
  <process name="Reference 40nm" default="true">
    <application name="PVE" prefPrefix="xt">
      <variant name="1P10M" title="1-Poly 10-Metal" default="true">

<pveJob name="DRC" jobType="DRC">

<tool name="ICV" default="true">

<preference name="RunsetFile" valueType="path">
<value>./icv/drc/reference40nm_mod_icv_drc.rs</value>
</preference>

<preference name="LoadRunset" visible="false">
<value>true</value>
</preference>

<preference name="Variables" valueType="custom"    visible="true"
  mutable="true" required="true">
  <item name="WIDTH_RULES" valueType="string" visible="true"
    mutable="true" scope="environment">"true"</item>
  <item name="AREA_RULES" valueType="string" visible="true" mutable="true"
    scope="environment">"true"</item>
  <item name="SPACING_RULES" valueType="string" visible="true"
    mutable="true" scope="environment">"true"</item>
  <item name="ENCLOSURE_RULES" valueType="string" visible="true"
    mutable="true" scope="environment">"true"</item>
  <item name="NWELL_RULES" valueType="string" visible="true"
    mutable="true" scope="environment">"true"</item>
  <item name="DIFF_RULES" valueType="string" visible="true" mutable="true"
    scope="environment">"true"</item>
  <item name="POLY_RULES" valueType="string" visible="true" mutable="true"
    scope="environment">"true"</item>
  <item name="RPOLY_RULES" valueType="string" visible="true"
    mutable="true" scope="environment">"true"</item>
```


Appendix A: Configuring PDKs for DRC, LVS, and PVE Default Interfaces

```
<item name="PPLUS_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="NPLUS_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="LOWVTN_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="LOWVTP_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="HIGHVTN_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="HIGHVTP_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="DIFF18_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="DIFF25_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="DIFF33_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="CONT_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"true"</item>
<item name="METAL1_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="METALX_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="VIAX_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"true"</item>
<item name="RESISTOR_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"true"</item>
<item name="VAR_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"true"</item>
<item name="DENSITY_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
<item name="DFM_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
</preference>
</tool>
</pveJob>
</variant>
</application>
</process>
</file-format>
```

LVS Interface

```
<pveJob name="LVS" jobType="LVS">

<preference name="LayDBFormat" visible="true" required="true"
valueType="string">
<value>OpenAccess</value>
</preference>
```

```
<preference name="SchDBFormat" visible="true" required="true"
  valueType="string">
<value>OpenAccess</value>
</preference>

<tool name="ICV">
<preference name="RunsetFile" visible="true" mutable="false"
  required="true" valueType="path" override="true">
<value>./icv/lvs/reference10lm_lvs.rs</value>
</preference>

<preference name="xtLVSICVOptions" required="true">
<value>-sf SPICE -ndg -quiet=0</value>
</preference>

<preference name="RunsetOptions" valueType="custom">
<item section="OPTIONS" name="IGNORE_CASE">FALSE</item>
</preference>
</tool>
</pveJob>
```

LPE Interface

```
<pveJob name="LPE" jobType="LPE">

<preference name="ViewOutput">
<value>true</value>
</preference>

<tool name="StarRC">
<preference name="RunsetFile" required="true">
<value>./snps_reference40nm/25FEB11_ICV201109/starrc/star_icv_cmd</value>
</preference>

<preference name="RunsetOptions" valueType="custom" visible="true"
  mutable="true">
  <item name="xref" visible="true" mutable="false">yes</item>
  <item name="oa_remove_spicecard_prefix" visible="true"
    mutable="true">no</item>

</preference>
</tool>
</pveJob>
```

Custom Jobs

The configuration file lets you create custom jobs using the default infrastructure for DRC, LVS, and LPE job types. The menu entries are loaded automatically and the job and tool preferences are created automatically. Custom job definitions for the current process and

variant as specified by the `xtPDKProcess` and `xtPDKVariant` preferences are added to the list of jobs in the Verification menu, sorted by type. Some sections from the custom DFM job are included as examples in the section that follows. You can obtain example data from your Synopsys Support representative for the complete `pdk.xml` configuration file.

PDK Processing for Custom Jobs

PDK processing is done in two stages:

1. At Tcl code loading, the `pdk.xml` file is read and a tree representing the XML information of the file is created in memory.
2. Next, the PDK tree is partially read a couple of times to extract the desired information.

Operations of interest, which are triggered each time a non-empty "Variant" node is read, can be summarized as follows:

- After PDK scope creation and Variant value update, creation of the custom jobs, if applicable, takes place.
- For each custom job created, a post-processing callback named `OnPostCreateJob()` is called and the following operations occur:
 - The general `createJobPrefs()` function is called, which creates the job-related preferences.
 - The tool specific `createJobPrefs()` function is then called and the job-tool related preferences are created.
- The PDK tree is traversed once again to update preferences values, tool attributes and job attributes (including custom jobs).

The same procedure that applies to defining custom jobs could be used for defining custom tools.

DFM Interface

```
<pveJob name="dfm" jobType="DRC" title="DFM">

<tool name="ICV">

<preference name="RunsetFile" valueType="path">
  <value>./icv/drc/reference40nm_mod_icv_drc.rs</value>
</preference>

<preference name="LoadRunset" visible="true"> <value>true</value>
</preference>
```

Appendix A: Configuring PDKs for DRC, LVS, and PVE Custom Jobs

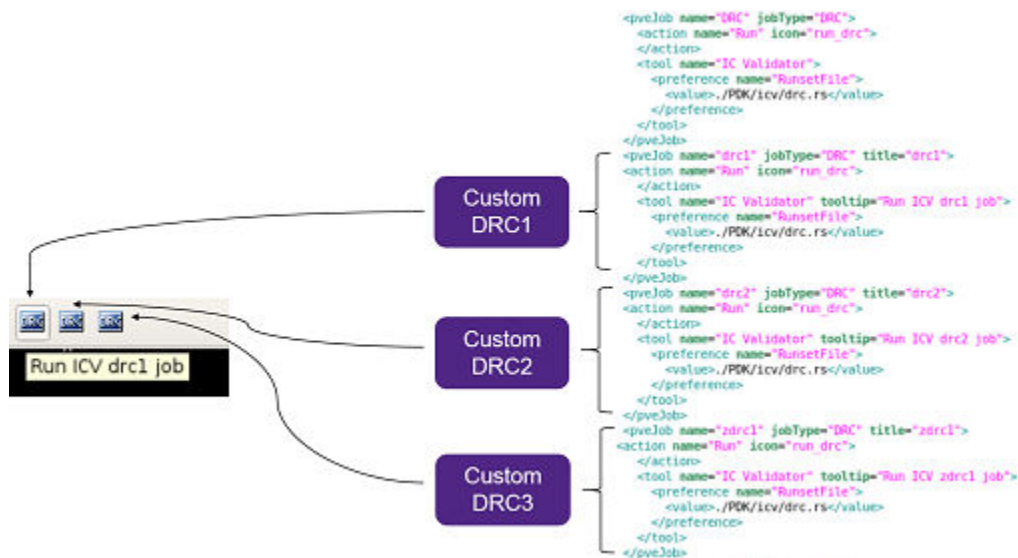
```
<preference name="Variables" valueType="custom" visible="true"
mutable="true" required="true">
  <item name="WIDTH_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="AREA_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
  <item name="SPACING_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="ENCLOSURE_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="NWELL_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="DIFF_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
  <item name="POLY_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
  <item name="RPOLY_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="PPLUS_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="NPLUS_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="LOWVTN_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="LOWVTP_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="HIGHVTN_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="HIGHVTP_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="DIFF18_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="DIFF25_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="DIFF33_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="CONT_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
  <item name="METAL1_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="METALX_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="VIAX_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
  <item name="RESISTOR_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="VAR_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"false"</item>
  <item name="DENSITY_RULES" valueType="string" visible="true"
mutable="true" scope="environment">"false"</item>
  <item name="DFM_RULES" valueType="string" visible="true" mutable="true"
scope="environment">"true"</item>
</preference>
```

```
</tool>
</pveJob>
```

PVE Toolbar

Specify the icon for the run using the `action` element:

```
<pveJob name="mydrc" jobType="DRC" title="My DRC">
  <action name="Run" icon="run_drc">
  </action>
```



You can specify the path of the icon or just the name of the icon. If you specify the name, you can set your search priority to find the icon you want to select:

1. Search for the file in the same directory as the `pdk.xml` file.
2. Search for the file in the working directory.
3. Use the default icon.

Specifying Tooltips for Icons on the PVE Toolbar

You can specify a tooltip in the `pdk.xml` file to change the message in the tooltip. For example,

- No tooltip on PVE jobs = default tooltip
- Default tooltip:
 - Default job: Run `<toolName> <jobType> job`
 - Custom job: Run `<toolName> <jobname> <jobType> job`

```
<application name="PVE" prefPrefix="xt">
  <variant name="10M" default="true">
    <pveJob name="DRC" jobType="DRC">
      <tool name="IC Validator" default="true">
        <preference name="RunsetFile">
          <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc.rs</value>
        </preference>
        <preference name="RunsetFile">
          <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc.rs</value>
        </preference>
      </tool>
    </pveJob>

    <pveJob name="drc1" jobType="DRC" title="drc1">
      <tool name="IC Validator" tooltip="Run ICV drc1 job">
        <preference name="RunsetFile">
          <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc1.rs</value>
        </preference>
      </tool>
    </pveJob>

    <pveJob name="drc2" jobType="DRC" title="drc2">
      <tool name="IC Validator">
        <preference name="RunsetFile">
          <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_drc2.rs</value>
        </preference>
      </tool>
    </pveJob>

    <pveJob name="zdrc1" jobType="DRC" title="zdrc1">
      <tool name="IC Validator">
        <preference name="RunsetFile">
          <value>$PDK_INSTALL_DIR/icv/drc/reference40nm_icv_zdrc1.rs \
        </value>
        </preference>
      </tool>
    </pveJob>
```

Attribute Versus Element Table

Table 78 Attribute Versus Element Table

Attribute	process	variant	pveJob	action	tool	preference	item
default	true default: false	true default: false			true default: false		
binding				<user-defi ned bindkey>			
icon / large icon				<absolute path relative to PDK> default: jobName			
scope							environm ent variable define
visible						default: true false	default: true false
mutable						default: true false	default: true false
valueType						path custom	path string double
required						true default: false	true default: false
name	<user-defi ned>	<user-defin ed>	<user-defin ed>	Run Setup View Output	ICV Hercules StarRC	<pre-exist> <pre-creat ed>	<runset-defi ned>
windowType			seSchematic le Layout				
title			<user-defin ed>				
jobType			DRC LVS LPE				

Table 78 Attribute Versus Element Table (Continued)

Attribute	process	variant	pveJob	action	tool	preference	item
prefId			<user-defin ed> default: jobName				